Consultancy Services for Preparation of Expressway Development Master Plan (EDMP) for Uganda

Contract NO. UNRA/SRVCS/16-17/00069

Traffic Survey Report

Final, 17th June 2020

Korea Expressway Corporation in Joint Venture with Kyong Dong Engineering Co., Ltd. and in association with Centre for Infrastructure Consulting Ltd.

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List of Acronyms

AADT	Annual Average Daily Traffic
ALS	Axle Load Survey
ATC	Automated Traffic Counts
BCR	Benefit to Cost Ratio
BRT	Bus Rapid Transit
CAPEX	Capital Expenditure
CBD	Central Business District
COMESA	Common Market for Eastern and Southern Africa
CPI	Consumer Price Index
CSI	Construction Sector Index
DRC	Democratic Republic of Congo
EAC	East Africa Community
EDMP	Expressway Development Master Plan
EIA	Environmental Impact Assessment
ESIA	Environmental and Social Impact Assessment
EU	European Union
FHWA	Federal Highway Administration
FF	Free Flow or Friction Factor
GDP	Gross Domestic Product
GIS	Geographic Information Systems
GKMA	Greater Kampala Metropolitan Area
GKMAPDF	Greater Kampala Metropolitan Area Physical Development Framework
GPS	Global Positioning System
GUI	Graphic User Interface
HBE_H	Home-Based Education, High-income
HBE_L	Home-Based Education, Low-income
HBE_M	Home-Based Education, Medium-income
HBO	Home-Based Other
HBW_H	Home-Based Work, High-Income
HBW_L	Home-Based Work, Low-Income
HBW_M	Home-Based Work, Medium-Income
HDM4	Highway Development and Management Model (version 4)
HIV/AIDS	Human Immunodeficiency Virus Infection/Acquired Immunodeficiency Syndrome
IP	Inter Peak
ITS	Intelligent Transportation System
JTC	Junction Turning Count
JTS	Journey Time Survey
KCC	Kampala City Council

KCCA	Kampala Capital City Authority
KDEC	Kyong Dong Engineering Co., Ltd.
KEC	Korea Expressway Corporation
KIIDP	Kampala Institutional and Infrastructure Development Project
Km	Kilometre
Km/h	Kilometres per hour
KUTIP	Kampala Urban Traffic Improvement Plan
LOS	Level of Service
LRT	Light Rail Transport
LU	Land Use
MCC	Manual Classified Counts
MoWHC	Ministry of Works, Housing and Communications
MoWT	Ministry of Works & Transport
MRT	Mass Rapid Transit
NDP	National Development Plan
NEF	Night Equivalent Factor
NHB	Non-Home Based
NPDP	National Physical Development Plan
NTMP	National Transport Master Plan
OD	Origin-Destination
OECD	Organisation for Economic Cooperation and Development
PCE	Passenger Car Equivalent
PDF	Physical Development Framework
PPP	Public Private Partnership
PT	Public Transport
RoW	Right of Way
RP	Revealed Preference
RUC	Road User Cost
SCF	Seasonal Correction Factor
SGR	Standard Gauge Railway
SP	Stated Preference
sqm	Square Meter
TAZ	Traffic Analysis Zone
TDM	Travel Demand Model
TOR	Terms of Reference
UBOS	Uganda bureau of statistics
UDB	Uganda Development Bank
UDC	Uganda Development Corporation
UGX	Uganda Shilling
UNBS	Uganda National Bureau of Standards

UNRA	Uganda National Roads Authority
USH	Uganda Shilling
VDF	Volume Delay Function
VHT	Vehicle Hours Travelled
VKT	Vehicle Kilometers Travelled
VOT	Value of Time
WTP	Willingness to Pay

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INTRODUCTION

1.1 Background

1.1.1 Roads Management in Uganda

The road transport infrastructure in Uganda comprises about 140,000 km of road, including about 20,544 km of national roads; 34,281 km of district roads; 10,108 km of urban roads; and 78,567 km of community access roads[1]. Road transport is by far the most dominant mode of transport in Uganda, carrying over 90% of passenger and freight traffic. Roads provide the only means of access to most of the rural communities and effective management of this asset is of vital importance to the Government of Uganda's strategy for economic development and poverty reduction. Within the road sector the national roads make up about 15% of the network but carry about 80% of the total road traffic. They also provide vital transport corridors linking the land-locked countries of Rwanda and Burundi, parts of Eastern DRC and Southern Sudan to the sea.

The Government of Uganda (GOU), through the Ministry of Works and Transport (MoWT) has overall responsibility for all roads in Uganda. The Uganda National Roads Authority (UNRA) is responsible for the maintenance and management of the national roads network, whilst the district and urban authorities are responsible for the management and maintenance of district and urban roads, respectively. UNRA's Mandate is to develop and maintain the national roads network, manage ferries linking the national roads network and control axle overloading.

1.1.2 Engagement of Consultant

Korea Expressway Corporation (KEC) in joint venture with Kyong Dong Engineering Company (KDEC) in association with Centre for Infrastructure Consulting (CIC) (collectively referred to as the Consultant), was commissioned by the Uganda National Roads Authority (UNRA) to prepare an Expressway Development Master Plan (EDMP) for Uganda.

1.1.3 Objectives of the Assignment

The objective of the study is to prepare an expressway development master plan EDMP) that would gradually cover the country on new alignments commencing with the highest trafficked sections of the primary national road network near Kampala.

This study will also determine the framework necessary for the progressive introduction of expressways in Uganda taking into account financing and operational modalities.

The anticipated benefits of this expressway system include but are not limited to the following:

[1] Ministry of Works and Transport database (2017) and UNRA data 2019

i. Provision of an integrated road network that will promote economic development of the country, by taking into consideration the immediate and future socio-economic development plans and policies of the Government of Uganda;

ii. Improved roadway capacity thus reducing congestion, user delays, vehicle emissions and vehicle operating costs;

iii. Increased mobility between viable demand and supply locations in the country;

iv. Facilitation of faster and safe uninterrupted flow of traffic, reduction in travel times and costs and improved accessibility; and

v. Supporting social and economic growth in Uganda;

vi. Maximizing the catchment for motorists by meeting the current and expected future transport needs of both passenger and freight and ensuring efficient utilization of assets and investments.

1.2 Purpose of the Traffic Survey

The traffic survey exercise responds to the ToR Section 4 – Scope of Work. The ToR required the Consultant to "*conduct necessary surveys for traffic counts, OD for each major category of vehicle, and travel speed. Prepare a nationwide OD matrix distributed by key freight types and major domestic and regional points of OD*".

Subsequent to the presentation of the Inception Report, the scope of the surveys was agreed to include the following surveys:

- Manual Classified Counts (MCC);
- Origin – Destination (OD) Surveys;
- Junction Turning Counts (JTC);
- Journey Time and Speed Data;
- Axle Load Surveys;
- Stated Preference Surveys;
- Public Transport Surveys;
- Traffic Accident Data;
- Ferry Surveys;
- Automated Traffic Counts (ATC); and
- Axle Load Surveys.

The objectives of each of the surveys is further discussed in the report. However, the overarching objective was to collect data through the surveys that would help formulate an expressway development master plan for Uganda.

1.3 Purpose of Report

The purpose of the report is to present the data collected during the traffic survey exercise for each survey type. For each survey type, the report discusses in brief the approach to the survey. The data collected for each survey type is then presented together with brief analysis to identify the key characteristics of the traffic flow captured within the survey.

It should be noted that the analysis of data presented in this report is meant to identify key traffic characteristics and is meant to be exhaustive to provide a comprehensive analysis of all existing traffic movement patterns.

1.4 Schedule of Traffic Surveys

The traffic survey exercise commenced on 22nd July 2019 and was fully completed on 31st January 2020. This period included a break of approximately one month between 27th August and 23rd September 2019 corresponding to school holidays and assignment reorganization.

A summary schedule of the surveys is presented in Table 1-1 below.

Table 0-1: Summary survey schedule

SUMMARY SURVEY SCHEDULE			
Survey Type	**No. Survey Sites**	**Survey Period**	**Description of Survey**
Manual Classified Counts (MCC)	146	22nd July – 05th November 2019	Total 7 days. 5 days 12-hours (7 am – 7 pm), 2 days 24-hours
Origin – Destination (OD) Surveys	171	22nd July – 05th November 2019	Total 2 days. 1 day for 24-hours (7am - 7am), and 1 day for 12-hours (7am - 7pm).
Junction Turning Counts (JTC)	5	23rd-September– 25th October 2019	Total 3 days, 12 hours (6 am – 6 pm)
Journey Time and Speed Data	36	16th October – 25th October 2019	Total 30 days, 1-3passes / Dir, for AM&PM
Axle Load Surveys	10	17th October – 20th October 2019	Total 3 days, 12 hours, (7 am – 7 pm)
Stated Preference Surveys	21	12th August – 27th October 2019	Total 3 days, 12 hrs (7 am – 7 pm)
Public Transport Surveys	20	19th September – 29th October 2019	Total 1 day, 12 hours (6 am – 6 pm)
Traffic Accident Data	18	14th October– 29th October 2019	Total 30 days

SUMMARY SURVEY SCHEDULE			
Survey Type	**No. Survey Sites**	**Survey Period**	**Description of Survey**
Ferry Surveys	10	14th October– 28th October 2019	Total 3 days, 12-hours (6 am – 6 pm)
Automated Traffic Counts (ATC)	7	20th October– 06th October 2019	Total 7 days, 24 hours (6 am – 6 am)
Axle Load Surveys	10	19th September – 29th October 2019	3 days, 12 hours (7am-7pm)

2 SCOPE OF SURVEYS

2.1 Traffic Survey Sites

2.1.1 Selection of Traffic Survey Sites

The proposed Expressway road network is intended to cover the entire country as much as possible. As such the project study area was the entire country. Traffic data collection was therefore required to cover the whole country. However, due to the always limited resources; time, manpower and funding, it is not feasible to collect data from all existing roads in the country.

Consideration was carefully taken of the nature of the project and the anticipated outcome using experienced traffic engineering judgement. The following was taken into account:

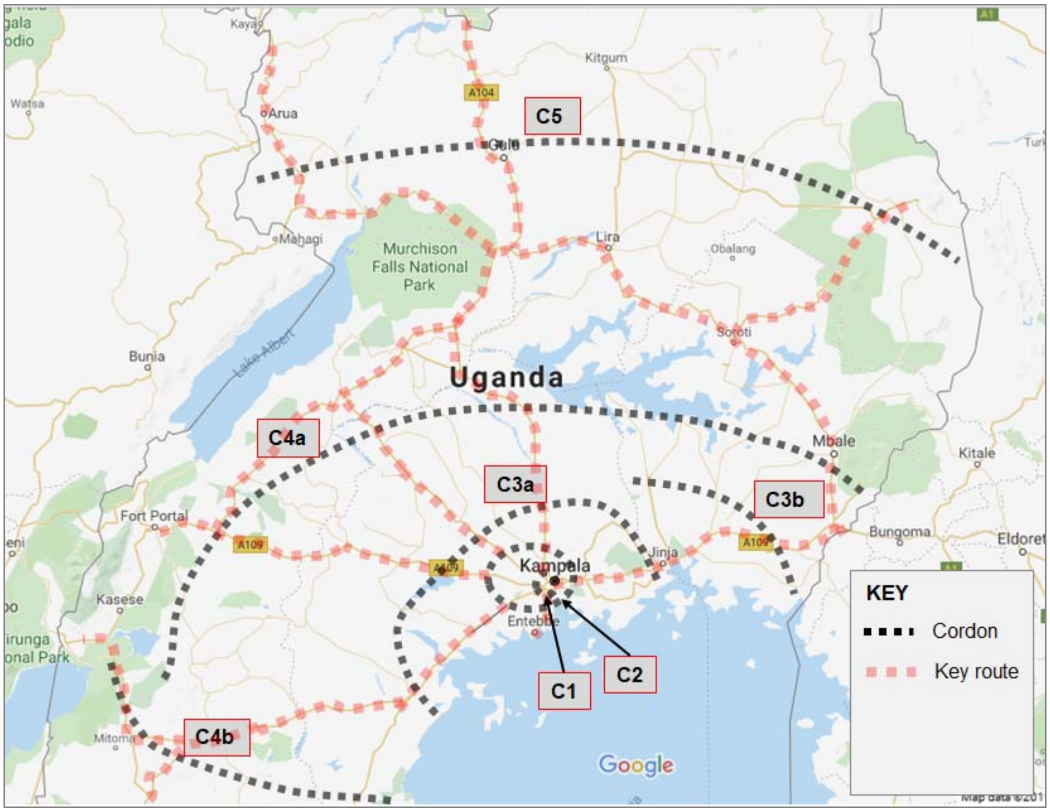
- The most popular origins and destinations of traffic within the road network;
- The potential routes of the expressways;
- The nature and density of the existing national road network;
- The existing and known future land use patterns and changes which would have an influence upon the proposed expressway network such as the proposed new cities, oil production in Western Uganda, border routes, etc.

Traffic surveys based on cordon lines (better referred to screen lines) were considered most appropriate. Imaginary cordon lines were setout across the entire road network moving from close to Kampala towards all parts of the country (Figure 2-1). The objective in a cordon line survey is to ensure that each trip crosses the line once. However, given the radial nature of the main routes, more than one cordon line crossed the same key route. Still, this has an advantage of ensuring that the detection of trips at the cordon survey sites is clear to avoid multiple counting of any single trip.

Cordon line surveys are also helpful in identifying the variation of traffic along a given route for example, the variation of traffic volume along the Kampala – Jinja highway as one moves from Kampala to Jinja.

Figure 2-1 below presents the cordon lines that were used to set out the traffic survey points.

Figure 2-1: Location of survey cordon lines



The cordon lines are described below:

- C1 – encircles the Kampala central along the Kampala Northern Bypass to the north, Kampala Jinja Road and Kibuye – Busega highway (Masaka Road);
- C2 – follows the expected route of the proposed Kampala Outer Beltway project. It encircles the Greater Kampala Metropolitan Area (GKMA) at a distance of approximately 20 Km from Kampala city centre, through Wakiso on Hoima Road, Matugga on Bombo Road, Seeta on the Kampala-Jinja highway, Nsangi on Masaka Road and Kajjansi on Entebbe Road.
- C3 – it is the line of an anticipated future second outer belt. The cordon line crosses commences at Masaka, cutting through Sembabule and Mityana, Busunju on Hoima Road, Luwero along the Kampala – Gulu highway and covers the immediate east of Jinja including Busia and Malaba border points. The cordon line runs at a distance of 70 – 200 Km from Kampala.
- C4 – covers the south west, west, central and eastern Uganda. The cordon line runs at an average distance of 200 – 400 Km from Kampala city.
- C5 – generally covers the north of the country. It runs at approximately 350 Km from Kampala in the northern region and covers the West Nile region, the northern and north eastern subregions of the country.

2.1.2 Inspection and Approval of Survey Sites

As part of the preparations for undertaking the traffic survey exercise, the Consultant prepared a Traffic Survey Proposal, a final version of which was presented to the Uganda National Roads Authority (UNRA) on 21st June 2020.

Prior to the commencement of traffic surveys at the identified survey sites, an inspection of the proposed survey sites was undertaken as part of the survey preparation process. The survey sites indicated in the Traffic Survey Proposal were thoroughly inspected to establish the following:

- Suitable locations for undertaking the survey to ensure the safety of both the survey personnel and the motorists;
- Convenience of the proposed survey station for enumerators;
- Minimisation of accident risks and delays to traffic especially for OD survey sites; and
- Rescheduling of surveys to fit with planning constraints, convenience and resource availability.

A joined-up review of the proposed surveys and proposed traffic modelling process was also undertaken to fully rationalise the importance of each proposed survey site.

The survey process described in this report therefore reflects the survey programme subsequent to these changes.

2.2 Location of Traffic Survey Sites

2.2.1 MCC and OD Surveys

The establishment of the cordon lines as described above, helped to systematically establish the traffic survey points. Figures 2-2, 2-3, 2-4, 2-5 and 2-6 show the approximate locations of the survey points used. Tables 2-1, 2-2, 2-3 and 2-4 provide the coordinates of the survey stations.

Figure 2-2: Location of Cordon 1 (C1) traffic survey sites

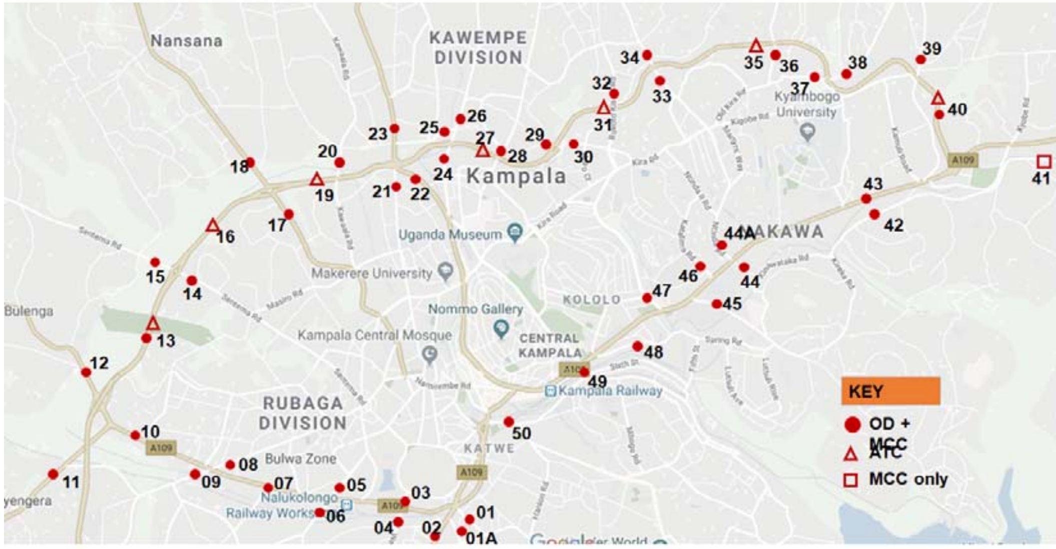


Table 2-1: Description and location of survey sites on Cordon 1 (C1)

DESCRIPTION AND LOCATION OF MCC AND OD SURVEY SITES – CORDON 1				
Site Ref.	**Highway / Link Name**	**Survey Type**	**Description of Site**	**Site Coordinates**
01	Mobutu Road	MCC & OD	Katwe Police	0.290673, 32.575600
01A	Salama Road	MCC & OD	Calendar Rest House	0.285704, 32.574774
02	Entebbe Road	MCC & OD	Kibuye Area	0.280650, 32.566231
03	Kalinda Rd	MCC & OD	Kalinda Rd	0.294576, 32.564423
04	Weraga Road	MCC & OD	Weraga Road	0.288085, 32.562599
05	Kabusu Rd	MCC & OD	Kabusu Rd, North	0.296818, 32.554118
06	Kabusu Rd	MCC & OD	Kabusu Road South	0.291314, 32.549955
07	Masaka Rd Ndeeba	MCC & OD	Ndeeba Opportunity Bank	0.295187, 32.548689
08	Wakaliga Rd North	MCC & OD	Petro City	0.301002, 32.537499
09	Wakaliga Rd South	MCC & OD	Family Loaf	0.298406, 32.529549
10	Masaka Rd Busega	MCC & OD	Quick Trip Petrol Stn	0.305455, 32.519442
11	Masaka Rd Kyengera	MCC & OD	Kalina Pub	0.299080, 32.508039
12	Mityana Rd	MCC & OD	Kitoogo Area	0.315527, 32.512427
13	Mityana-Sentema	MCC & OD	Mityana-Sentema	0.320012, 32.521890
14	Sentema Rd	MCC & OD	Sentema South	0.330655, 32.530473
15	Sentema Rd	MCC & OD	Sentema Nth	0.335000, 32.520699
16	Northern Bypass	MCC	Northern Bypass	0.333581, 32.528210
17	Hoima Rd South	MCC & OD	Namungoona	0.340879, 32.545912
18	Hoima Rd Nth	MCC & OD	Lubigi, Kampala Hoima Road	0.350524, 32.538788
20	Kawaala Rd	MCC & OD	Kawaala Rd	0.348475, 32.553808

21	Sir Apollo Kaggwa Road	MCC & OD	Sir Apollo	0.345213, 32.562820
22	Bombo Road North	MCC & OD	Makerere Kavule	0.346769, 32.567079
23	Bombo Road North	MCC & OD	Bwaise	0.355803, 32.562584
24	Gayaza Rd	MCC & OD	Kalerwe Market	0.345031, 32.571030
25	Tula Rd	MCC & OD	Tula Rd Scrap	0.354754, 32.571201
26	Gayaza Rd	MCC & OD	Kanyanya	0.356213, 32.572939
28	Kisingiri Street	MCC & OD	Kisingiri Rd O/Pass	0.351458, 32.579985
29	Kyebando Ring	MCC & OD	Kyebando Ring RD	0.353347, 32.579129
30	Kyebando-Kisalosalo	MCC & OD	Kyebando-Kisalosalo	0.352859, 32.590625
32	Bukoto Kisasi Rd	MCC & OD	Bukoto Kisasi O/P	0.360903, 32.598658
33	Kisasi-Ntinda Rd	MCC & OD	Kisasi-Ntinda Rd	0.363292, 32.606426
34	Kisasi-Kyanja Rd	MCC & OD	Kisasi-Kyanja Rd	0.367922, 32.603860
34	Kisasi-Kyanja Rd	MCC & OD	Kisasi-Kyanja Rd	0.367922, 32.603860
36	Mbogo Road	MCC & OD	Mbogo Rd O/P along Mbogo Road	0.370118, 32.624462
37	Naalya Road	MCC & OD	Nalya Road West	0.363439, 32.631545
38	Naalya Road	MCC & OD	Nalya Road East	0.363867, 32.635436
39	Namugongo Rd	MCC & OD	Namugongo Road O/P	0.365559, 32.648376
40	Bweyogerere - Nalya	MCC & OD	Northern Bypass	0.356779, 32.651061
41	Namanve Ind Park	MCC	Bweyogerere -Mukono Lower Road	0.350390, 32.667224
42	Kinawataka	MCC & OD	Kireka Road	0.341715, 32.642292
43	Kla-Jinja Hway	MCC & OD	Kyambogo	0.334609, 32.616881
44	Nakawa-Jinja Rd	MCC & OD	URA	0.332839, 32.619005
44A	Ntinda Road	MCC & OD	Ntinda Road MICD	0.334612, 32.616827
45	New Port Bell Rd	MCC & OD	MUBS	0.326986, 32.613302
46	Katalima Road	MCC & OD	Katalima Road	0.335609, 32.609976
47	Lugogo Bypass	MCC & OD	Lugogo Bypass	0.326840, 32.605141
48	Old Port Bell Road	MCC & OD	Old Port Bell Road	0.319329, 32.599589
49	Access Road	MCC & OD	Access	0.315844, 32.593938
50	Nsambya Rd	MCC & OD	Post Office	0.307334, 32.579817

Figure 2-3: Location of Cordon 2 (C2 North) traffic survey sites

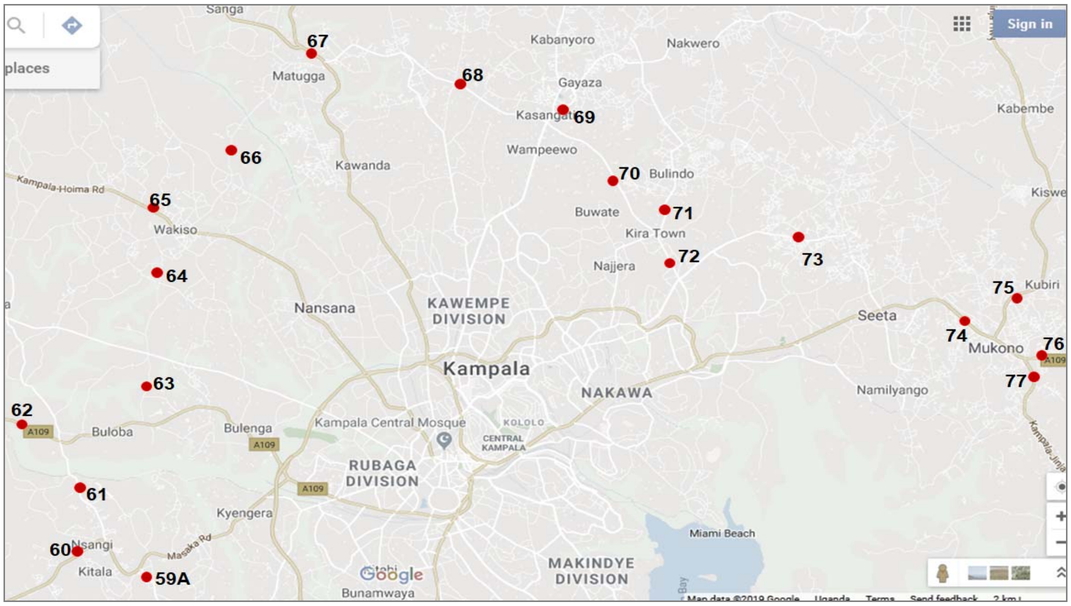


Figure 2-4: Location of Cordon 2 (C2 South) traffic survey sites

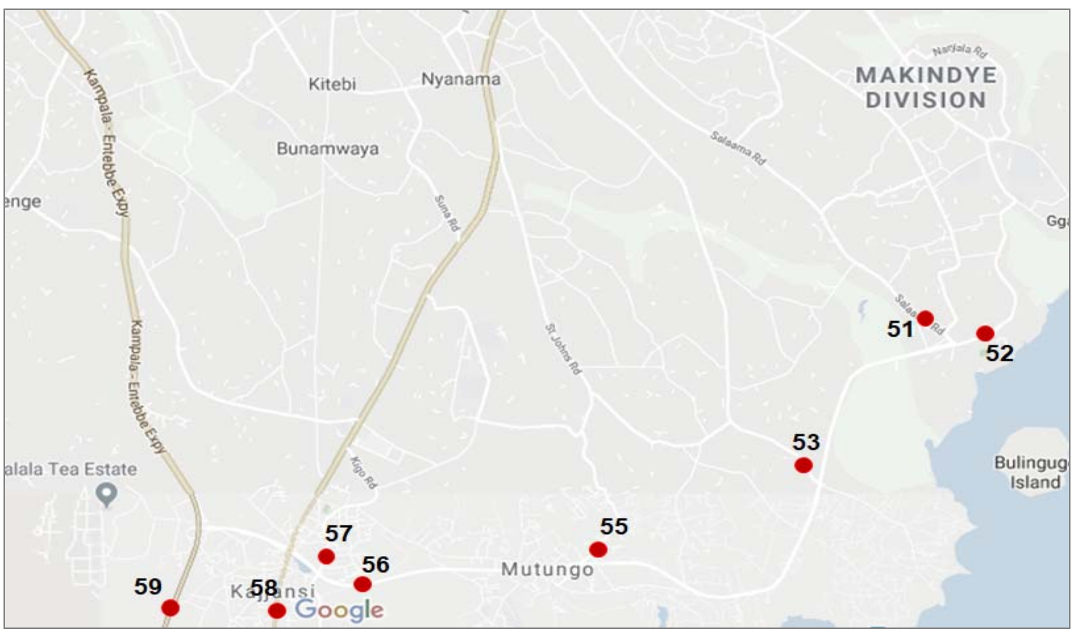


Table 2-2: Description and location of survey sites on Cordon 2 (C2)

DESCRIPTION AND LOCATION OF MCC AND OD SURVEY SITES - CORDON 2				
Site Ref.	**Highway / Link Name**	**Survey Type**	**Description of Site**	**Site Coordinates**
51	Salama Road	MCC & OD	Former Cloud Nine	0.241482, 32.619008
52	Wavamunno Rd	MCC & OD	Wavamunno Rd	0.239626, 32.622495
53	Busabala Road	MCC & OD	Busabala	0.222691, 32.603730
55	Lubugumu Road	MCC & OD	Lubugumu Road	0.211428, 32.577503

DESCRIPTION AND LOCATION OF MCC AND OD SURVEY SITES - CORDON 2				
Site Ref.	**Highway / Link Name**	**Survey Type**	**Description of Site**	**Site Coordinates**
56	Munyonyo Spur	MCC & OD	Munyonyo	0.208439, 32.546453
57	Kigo Road	MCC & OD	Kigo Road	0.210418, 32.545842
58	Entebbe Road	MCC & OD	Entebbe Dual Section	0.090099, 32.491886
59	Kampala Entebbe Expressway	MCC & OD	Kitende	0.115952, 32.512044
59A	Budo Road	MCC & OD	Budo Road	0.272468, 32.468676
60	Masaka Rd	MCC & OD	Maya	0.280159, 32.448202
61	Nsangi-Buloba Road	MCC & OD	Nsangi-Buloba	0.285838, 32.455233
62	Mityana Road	MCC & OD	Buloba, Mityana	0.322739, 32.436379
63	Buloba-Bukasa Rd	MCC & OD	Buloba-Bukasa	0.357924, 32.469542
64	Bukasa-Wakiso Road	MCC & OD	Bukasa-Wakiso	0.393462, 32.472860
65	Hoima Road	MCC & OD	Wakiso	0.402065, 32.476697
66	Wakiso-Matugga Rd	MCC & OD	Wakiso-Matugga Rd	0.401003, 32.485440
67	Bombo Road	MCC & OD	Matugga Gaz Fuel Station	0.464554, 32.523192
68	Matugga-Kasangati Road	MCC & OD	Matugga-Kasangati	0.440817, 32.596578
69	Gayaza Road	MCC & OD	Kasangati-Gayaza	0.441040, 32.605949
70	Kasangati-Kira Rd	MCC & OD	Kasangati-Kira	0.435823, 32.605117
71	Kira-Bulindo Rd	MCC & OD	Bulindo	0.399256, 32.640667
72	Kira-Kyaliwajjala Rd	MCC & OD	Kira-Kyaliwajjala	0.393285, 32.640646
73	Sonde Road	MCC & OD	Sonde	0.395449, 32.684640
74	Kampala -Jinja Road	MCC & OD	Near Kigunga	0.371954, 32.728287
75	Mukono-Kalagi Rd	MCC & OD	Mukono-Kalagi	0.375153, 32.760370
76	Kla-Jinja Hway	MCC & OD	Near Wantoni	0.351310, 32.765739
77	Katosi Rd	MCC & OD	Katosi Rd	0.348475, 32.762688

Figure 2-5: Location of Cordon 3 (C3) traffic survey sites

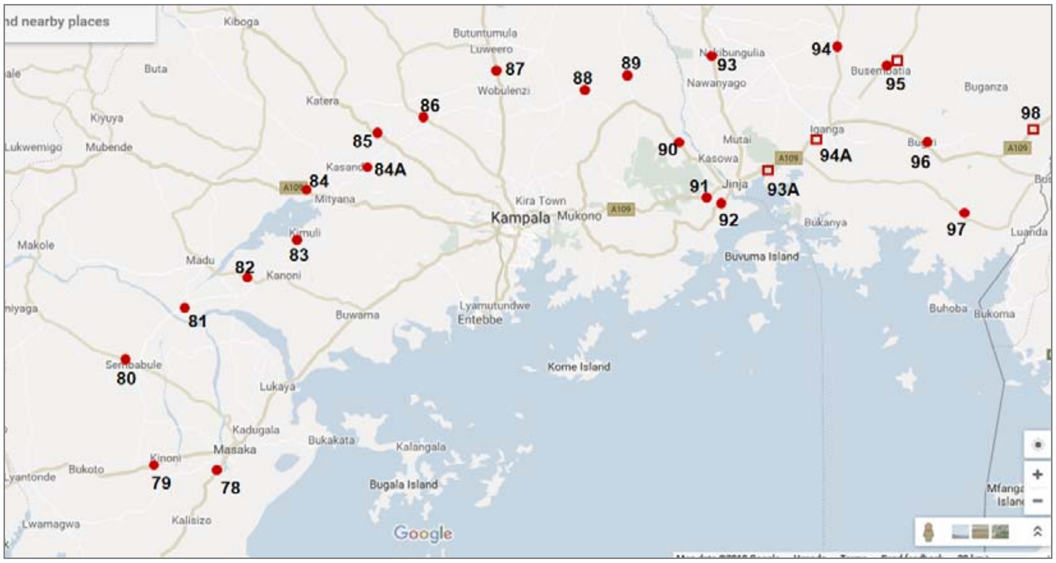


Table 2-3: Description and location of survey sites on Cordon 3 (C3)

DESCRIPTION AND LOCATION OF MCC AND OD SURVEY SITES - CORDON 3				
Site Ref.	**Highway / Link Name**	**Survey Type**	**Description of Site**	**Site Coordinates**
78	Masaka - Mbarara	MCC & OD	Kirimya Township	-0.371753, 31.715768
79	Masaka - Mbarara	MCC & OD	Kinoni, Masaka-Mbarara Rd	-0.353466, 31.555108
80	Villa Rd	MCC & OD	Near Sembabule	-0.087033, 31.469836
81	Sembabule-Kabulasoke	MCC & OD	Sembabule-Kabulasoke	-0.044441, 31.483455
82	Kanoni-Maddu	MCC & OD	Kanoni-Maddu	0.175392, 31.900612
83	Kanoni-Maddu	MCC & OD	Kanoni-Mityana	0.260097, 31.965118
84	Mityana Road	MCC & OD	Mityana – Myanzi	0.410743, 32.028316
84 A	Mityana-Busunju	MCC & OD	Sekanyonyi	0.498563, 32.132057
85	Hoima Rd	MCC & OD	Matte	0.573336, 32.192723
86	Semuto Rd	MCC & OD	Kapeka, Semuto Rd	0.664668, 32.259268
87	Kla-Gulu Hway	MCC & OD	Luwero, Kampala – Gulu Highway	0.806460, 32.506018
88	Wobulenzi-Zirobwe	MCC & OD	Near Zirobwe	0.678795, 32.690069
89	Galiraya-Kayunga	MCC & OD	Near Kayunga	0.697010, 32.910770
90	Kangulumira	MCC & OD	Kayunga-Kangulumira	0.595725, 33.014055
91	Kampala-Jinja Highway	MCC & OD	Njeru	0.412772, 33.135522
92	Nalubaale Rd	MCC & OD	Nyenga	0.413047, 33.182318
93	Jinja - Kamuli Rd	MCC & OD	Namulesa	0.494510, 33.216542
93 A	Jinja - Iganga	MCC	Jinja - Iganga	0.473575, 33.257824
94	Jinja - Kaliro Rd	MCC & OD	Near Kaliro	0.632684, 33.480546
94A	Musita-Iganga	MCC	Musita-Iganga	0.529569, 33.388857

DESCRIPTION AND LOCATION OF MCC AND OD SURVEY SITES - CORDON 3				
Site Ref.	**Highway / Link Name**	**Survey Type**	**Description of Site**	**Site Coordinates**
95	Tirinyi Road	MCC & OD	Namutumba	0.815870, 33.664797
96	Bugiri	MCC & OD	Jinja - Tororo Hway	0.605292, 33.687402
97	Musita-Lumino	MCC & OD	Musita-Lumino	0.525134, 33.391733
98	Busia Rd - Tororo	MCC	Busia Road	0.549275, 34.019232
119	Tororo-Mbale	MCC & OD	Tororo-Mbale	1.014194, 34.174336
142	Busia Boarder	MCC & OD	Busia Rd - Tororo	0.465118, 34.098315
143	Malaba Boarder	MCC & OD	Malaba Boarder	0.638549, 34.264929

Figure 2-6: Location of Cordons 4 and 5 (C4 and C5) traffic survey sites

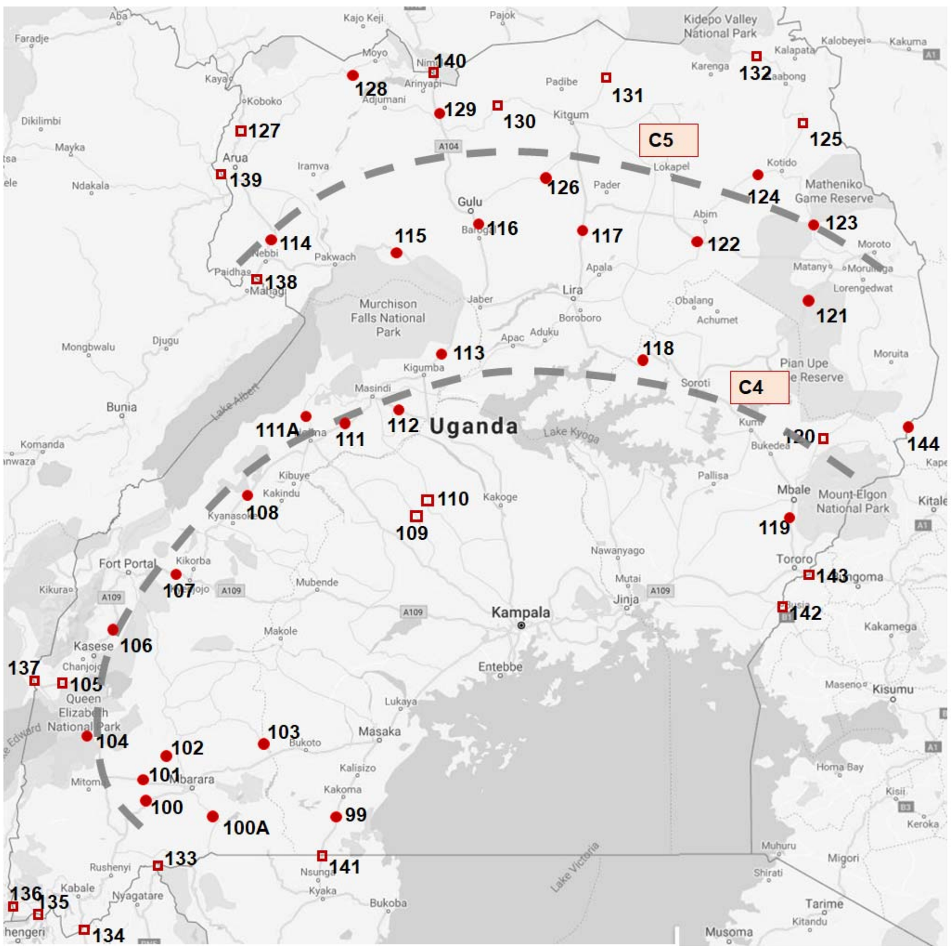


Table 2-4: Description and location of survey sites on Cordons 4 and 5 (C4 and C5)

DESCRIPTION AND LOCATION OF SURVEY SITES ON CORDONS 4 AND 5				
Site Ref.	**Highway / Link Name**	**Survey Type**	**Description of Site**	**Site Coordinates**
Cordon 4 – Mid Country & Western Region				
99	Masaka-Mutukula	MCC & OD	Near Mutukula	-0.998586, 31.417639
100	Mbarara-Ntungamo	MCC & OD	Mbarara-Ntungamo	-0.642561, 30.570337
100 A	Mbarara-Kikagata	MCC & OD	Mbarara-Kikagata	-0.640244, 30.675225
101	Mbarara-Bushenyi	MCC & OD	Mbaarar-Bushenyi	-0.579578, 30.390501
102	Mbarara-Ibanda	MCC & OD	Mbarara-Ibanda	-0.578657, 30.609883
103	Lyantonde-Kiruhura	MCC & OD	Lyantonde-Kiruhura	-0.415457, 31.116216
104	Ishaka-Katunguru	MCC & OD	Ishaka-Katunguru	-0.292987, 30.104367
105	Bwera - Mpondwe	MCC & OD	Bwera, Bwera – Mpondwe Rd	0.035206, 29.734605
106	Kasese-Fort Portal	MCC & OD	Kasese-Fort Portal	0.192190, 30.102165
107	Fort Portal - Kyenjojo	MCC & OD	Fort portal – Kyenjojo	0.620760, 30.627410
108	Kyenjojo-Hoima	MCC & OD	Kyenjojo-Hoima	1.359576, 31.279663
109	Kateera-Kyankwanzi	MCC & OD	Kateera-Kyankwanzi	0.703403, 32.041824
110	Ngoma-Kapeeka	MCC & OD	Ngoma-Kapeeka	0.832835, 32.236081
111	Hoima-Masindi	MCC & OD	Hoima-Masindi (Lot W)	0.620463, 30.629362
111A	Bulisa-Hoima	MCC & OD	Bulisa-Hoima	1.460976, 31.338734
112	Masindi-Kibangya	MCC & OD	Masindi-Kibangya	1.568050, 31.971597
113	Kigumba-Karuma	MCC & OD	Kigumba-Karuma	2.091945, 32.195992
118	Soroti-Kumi	MCC & OD	Soroti-Kumi	1.575409, 33.853673
119	Tororo - Mbale	MCC & OD	Tororo - Mbale	0.978157, 34.168724
120	Sironko Road	MCC & OD	Moroto - Mbale	1.146321, 34.195497
133	Mirama Hills Border	MCC	Mirama Hills Border Post	-1.033666, 30.460638
134	Katuna Border	MCC	Near Katuna Border Post	-1.424435, 30.011466
135	Cyahafi Border	MCC	Near Cyahafi Border Post	-1.335839, 29.737391
136	Bunagana Border	MCC	Near Bunagana Border Post	-1.294117, 29.606016
137	Mpondwe Border	MCC	Near Mpondwe Border Post	0.042339, 29.723972
144	Suam Border	MCC	Near Suam Border Post	1.217108, 34.716599
Cordon 5 – Northern Region				
114	Nebbi-Arua	MCC & OD	Nebbi - Arua	2.494334, 31.091114
115	Karuma-Pakwach	MCC & OD	Karuma-Pakwach	2.547125, 31.826231
116	Karuma -Gulu	MCC & OD	Karuma - Gulu	2.632568, 32.331442
117	Lira-Kitgum	MCC & OD	Lira-Kitgum	3.249089, 32.910044
121	Soroti-Moroto	MCC & OD	Soroti- Moroto	2.490282, 34.527947
122	Matany-Kilak	MCC & OD	Matany-Kilak	2.577189, 33.642365

DESCRIPTION AND LOCATION OF SURVEY SITES ON CORDONS 4 AND 5				
Site Ref.	**Highway / Link Name**	**Survey Type**	**Description of Site**	**Site Coordinates**
123	Moroto-Kotido	MCC & OD	Moroto-Kotido	2.518873, 34.524507
124	Abim - Kotido	MCC & OD	Abim - Kotido	3.004329, 34.082223
125	Kotido - Kabong	MCC & OD	Kotido - Kabong	3.014975, 34.080389
126	Gulu - kitgum	MCC & OD	Gulu - kitgum	2.820371, 32.343396
127	Arua - Koboko	MCC	Arua - Koboko	3.054507, 30.921233
128	Koboko - Moyo	MCC & OD	Koboko - Moyo	3.434504, 30.989237
129	Atiak - Nimule	MCC & OD	Atiak - Nimule	3.299421, 32.111235
130	Kitgum - Atiak	MCC & OD	Kitgum - Atiak	3.244852, 32.135562
131	Kitgum - Madi Opei	MCC & OD	Kitgum - Madi Opei	3.322660, 32.901827
132	Kabong - Apoka	MCC & OD	Kabong - Apoka	3.322660, 32.901827
138	Goli Border	MCC	Near Goli Border post	2.389214, 31.030323
139	Vurra Border	MCC	Near Vurra Border post	2.907638, 30.877134
140	Nimule Border	MCC	Near Nimule Border post	3.571259, 32.073669

2.2.2 Location of Junction Turning Counts

A limited number of junction turning counts (JTC) were undertaken in central Kampala. It was envisaged that even with the national nature of the model, some calibration would be required in and around Kampala. Figure 2-7 and Table 2-5 show the location of the junctions at which the surveys were undertaken.

Figure 2-7: Location of junction turning traffic survey sites

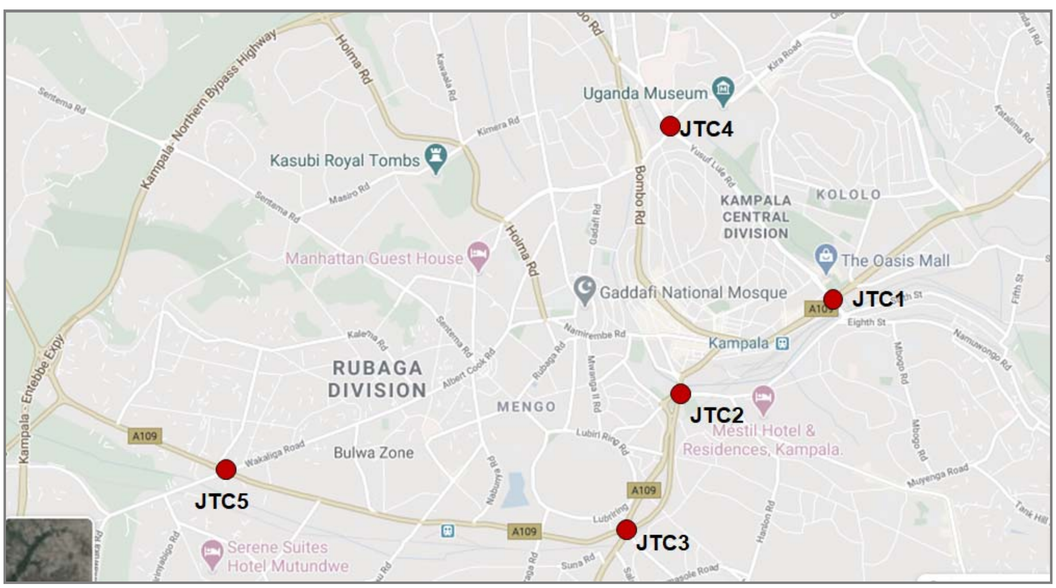


Table 2-5: Description of junction turning counts

DESCRIPTION AND LOCATION OF JUNCTION TURNING SURVEY SITES				
Site Ref.	**Highway / Link Name**	**Survey Type**	**Description of Site**	**Site Coordinates**
JTC1	Jinja Road / Access Road	JTC	Jinja Road (A109) / Access Road junction	0.316674, 32.593326
JTC2	Queensway/Nsambya	JTC	Clock Tower Junction, Queensway / Nsambya Rd	0.307457, 32.578022
JTC3	Entebbe Rd/Masaka Rd	JTC	Kibuye Juncion, Entebbe Rd/ Masaka Rd	0.309331, 32.580087
JTC4	Hajji Musa/Yusuf Lule	JTC	Mulago Junction, Hajji Musa/ Yusus Lule	0.333821, 32.576948
JTC5	Masaka / Wakaliga	JTC	Natete Junction, Masaka Wakaliga Rd	0.299724, 32.532813

2.2.3 Journey Time Survey Routes

An extensive survey of route journey times was also undertaken. Figure 2-8 and Table 2-6 present the location, start and end points of each route.

Figure 2-8: Location of journey time survey routes

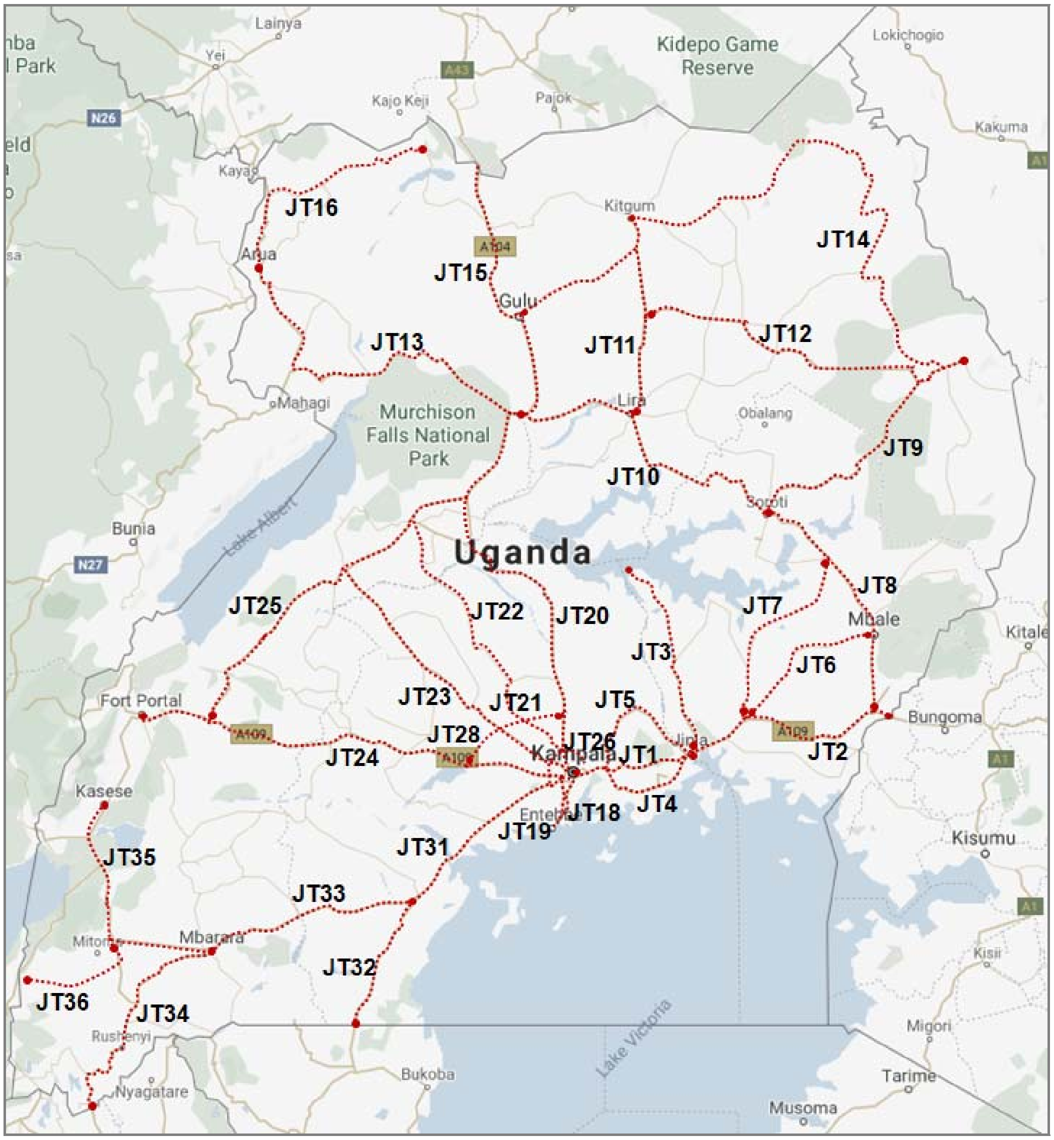


Table 2-6: Description of journey time survey routes

DESCRIPTION OF JOURNEY TIME SURVEY ROUTES				
Site Ref.	**Description**	**Start Coordinates**	**End Coordinates**	**Intermediate Points**
Eastern & Northern Region Routes				
JT1	Kampala - Jinja	0.333881, 32.618014	0.445048, 33.197575	• Spear Junction, Bweyogerere, Mukono, Lugazi, Jinja Java House
JT2	Jinja – Tororo - Malaba	0.445218, 33.197828	0.634604, 34.273316	• Jinja Java House, Iganga Bugiri, Busia, Tororo Mbale
JT3	Jinja - Kamuli - Bukungu	0.445218, 33.197828	1.437110, 32.868491	• Jinja, Buwenge, Kamuli Bukungu

DESCRIPTION OF JOURNEY TIME SURVEY ROUTES				
Site Ref.	**Description**	**Start Coordinates**	**End Coordinates**	**Intermediate Points**
JT4	Mukono - Nkokonjeru - Jinja	0.358366, 32.751225	0.445218, 33.197828	• Mukono, Kisoga, Nkokonjeru, Buikwe, Jinja
JT5	Mukono - Kayunga - Jinja	0.358405, 32.751221	0.445218, 33.197828	• Mukono, Kalagi, Kayunga, Jinja
JT6	Iganga - Budaka - Mbale	0.614273, 33.475648	1.071343, 34.174977	• Iganga, Namutumba, Budaka, Mbale
JT7	Iganga – Pallisa - Kumi	0.614273, 33.475648	1.488414, 33.935962	• Iganga, Kaliro, Pallisa, Kumi
JT8	Tororo - Mbale - Kumi - Soroti	0.688226, 34.179807	1.715815, 33.611102	• Tororo, Mbale, Kumi, Soroti
JT9	Soroti – Katakwi - Moroto	1.715815, 33.611102	2.533071, 34.657685	• Soroti. Katakwi, Moroto
JT10	Soroti – Lira - Karuma	1.715815, 33.611102	2.255022, 32.242856	• Soroti, Dokolo, Lira, Karuma
JT11	Lira - Kitgum	2.238024, 32.893785	3.298638, 32.882310	• Lira, Kilak, Kitgum
JT12	Kilak - Abim - Moroto	2.834093, 32.159512	2.533071, 34.657685	• Kilak, Abim, Moroto
JT13	Karuma – Pakwach – Arua	2.255283, 32.242742	3.014483, 30.914753	• Karuma, pakwach, Nebbi, Arua
JT14	Karuma- Gulu – Atiaka - Nimule	2.255283, 32.242742	3.591497, 32.064395	• Karuma, Gulu, Atiak, Nimule
JT15	Pakwach – Nebbi - Arua	2.461580, 31.506042	3.014483, 30.914753	• Moroto, Kotido, Kaabong, Karenga, Kitgum
JT16	Gulu - Kitgum	2.777053, 32.295405	3.292403, 32.883653	• Gulu, Kitgum
JT17	Arua – Koboko - Moyo	3.014483, 30.914753	3.652952, 31.726645	• Arua, Koboko, Moyo, Adjumani, Atiak
Central and Western Region Routes				
JT18	Northen Bypass – Entebbe (Old)	0.350187, 32.587344	0.046138, 32.453557	• Busega, Kibuye, Zzana, Kajjansi, Mpala, Entebbe Kitooro
JT19	Northen Bypass – Entebbe KEE	0.350187, 32.587344	0.164192, 32.543567	• Busega, Kajjansi, Mpala, Entebbe Kitooro
JT20	NBP Luwero – Nakasongora	0.842180,		• NBP, Kawempe, Matugga, Luweero,

DESCRIPTION OF JOURNEY TIME SURVEY ROUTES

Site Ref.	Description	Start Coordinates	End Coordinates	Intermediate Points
	– Kigumba - Karuma	32.493147	2.254905, 32.242658	Nakasongola, Kigumba, Karuma
JT21	NBP Matuga – Kapeka	0.465771, 32.521251	0.679214, 32.247670	• NBP, Matugga, Kapeeka, Luweero
JT22	Kapeka – Ngoma - Masindi	0.679214, 32.247670	1.680659, 31.727515	• Kapeeka, Ngoma, Masindi
JT23	NBP Kiboga - Hoima	0.928414, 31.756432	1.421313, 31.372358	• NBP, Wakiso, Busunju, Kiboga, Hoima
JT24	NBP Mityana – Mubende – Kyenjojo – Fort Portal	0.403931, 32.047801	0.659335, 30.292855	• NBP, Bujuko, Mityana, Mubende, Kyenjojo, Fort Portal
JT25	Kyenjojo – Hoima - Kigumba	0.611926, 30.643495	1.809860, 32.011770	• Kyenjojo, Kagadi, Hoima, Masindi, Kigumba
JT27	Kira – Kasangati – Matuga-Wakiso Nsangi	0.397350, 32.638625	0.282432, 32.451840	• Kira, Kasangati, Matugga, Wakiso, Buloba, Nsangi
JT28	Mubende – Semuto - Busunju	0.555027, 31.393196	0.565905, 32.204084	• Mityana, Busunju, Semuto, Kalule
JT29	NBP Mulago- Kitgum Jn - Kibuye	0.333766, 32.576993	0.293566, 32.572868	• NBP, Kubbiri, • Mulago Roundabout, Kitgum junction, Clock Tower, Kibuye
JT30	Kibuye Busega	0.293566, 32.572868	0.309657, 32.515191	• Kibuye, Nateete, Busega
JT31	Busega – Buwama - Masaka	0.309657, 32.515191	-0.327493, 31.751361	• Busega, Nsangi, Mpigi, Lukaya, Masaka
JT32	Masaka – Mutukula	-0.327493, 31.751361	-1.000252, 31.416566	• Masaka, Kalisizo, Mutukula
JT33	Masaka – Lyantode - Mbarara	-0.327493, 31.751361	-0.582672, 30.682528	• Masaka, Lyantonde, Mbarara
JT34	Mbarara - Kabale – Katuna	-0.582672, 30.682528	-1.433014, 30.015021	• Mbarara, Ntungamo, Kabale, Kabale, Kisoro, Bunagana
JT35	Mbarara – Ishaka – Kasese	-0.602408, 30.618351	0.172396, 30.085642	• Mbarara, Ishaka, Kichwamba, Kikorongo, Kikorongo, Kasese
JT36	Ishaka – Rukungiri - Kihihi	-0.545456, 30.139201	-0.750039, 29.700285	• Ishaka, Rukungiri, Kihihi

2.2.4 Location of Stated Preference Survey Sites

Figure 2-9 and Table 2-7 below present the locations where the Stated Preference (SP) survey was undertaken.

Figure 2-9: Location of journey time survey routes

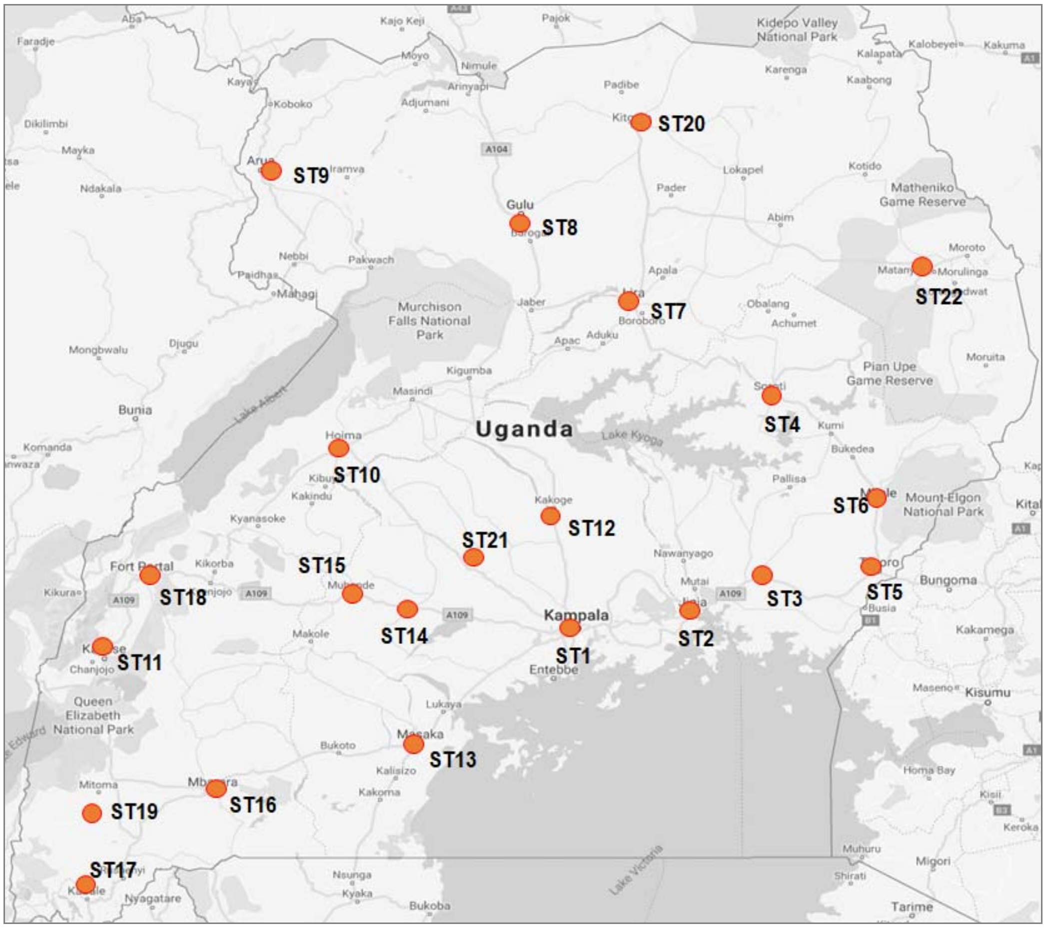


Table 2-7: Description of Stated Preference Survey sites

DESCRIPTION OF STATED PREFERENCE SURVEY SITES			
Site Ref.	**Site Name**	**Description**	**Start Coordinates**
SP1	Kampala	Taxi terminals, truck parking areas, shopping mall car parking areas, etc	0.315573, 32.563151
SP2	Jinja	Taxi terminals, truck parking areas, Jinja market, etc	0.427418, 33.210951
SP3	Iganga	Taxi terminals, truck parking areas, shopping areas, etc	0.613877, 33.479433
SP4	Soroti	Taxi terminals, truck parking areas, shopping areas, etc	1.715791, 33.610878
SP5	Tororo	Taxi terminals, truck parking areas, shopping areas, etc	0.691405, 34.186706

DESCRIPTION OF STATED PREFERENCE SURVEY SITES			
Site Ref.	**Site Name**	**Description**	**Start Coordinates**
SP6	Mbale	Taxi terminals, truck parking areas, shopping areas, etc	1.074244, 34.177307
SP7	Lira	Taxi terminals, truck parking areas, shopping areas, etc	2.254750, 32.896946
SP8	Gulu	Taxi terminals, truck parking areas, shopping areas, etc	2.772138, 32.299122
SP9	Arua	Taxi terminals, truck parking areas, shopping areas, etc	3.026798, 30.908611
SP10	Hoima	Taxi terminals, truck parking areas, shopping areas, etc	1.432452, 31.353082
SP11	Kasese	Taxi terminals, truck parking areas, shopping areas, etc	0.174955, 30.081822
SP12	Luwero	Taxi terminals, truck parking areas, shopping areas, etc	0.844825, 32.490566
SP13	Masaka	Taxi terminals, truck parking areas, shopping areas, etc	-0.341053, 31.737916
ST14	Mityana	Taxi terminals, truck parking areas, shopping areas, etc	0.399582, 32.047630
SP15	Mubende	Taxi terminals, truck parking areas, shopping areas, etc	0.560147, 31.389726
SP16	Mbarara	Taxi terminals, truck parking areas, shopping areas, etc	-0.615167, 30.652319
SP17	Kabale	Taxi terminals, truck parking areas, shopping areas, etc	-1.247297, 29.984444
SP18	Bushenyi	Taxi terminals, truck parking areas, shopping areas, etc	-0.540754, 30.190333
SP19	Rukungiri	Taxi terminals, truck parking areas, shopping areas, etc	-0.787624, 29.924806
SP20	Kitgum	Taxi terminals, truck parking areas, shopping areas, etc	3.294988, 32.881921
SP21	Wakiso	Entebbe, Nansana, Wakiso, Kira and Makindye Sabagabo divisions	0.384984, 32.479905

2.2.5 Location of Public Transport (Bus) Survey Sites

At the same locations where Stated Preference surveys were undertaken, public transport surveys were also undertaken. The only exception was Iganga.

The public transport survey focused on coach transport. As such, the survey was undertaken at bus parks and terminals in all locations where it was taken.

2.2.6 Accident Data

Accident data was collected from district and regional police stations across the country. The following locations provided the required information; Jinja, Soroti, Tororo, Mbale, Lira, Gulu, Hoima, Kasese, Luwero, Masaka, Mubende, Mbarara, Kabale, Bushenyi, Rukungiri, Kitgum, Iganga and Arua.

2.2.7 Locations for Ferry Surveys

Ferry surveys were undertaken at nine (9) locations as shown in Figure 2-10 and Table 2-8 below.

Figure 2-10: Location of ferry survey routes

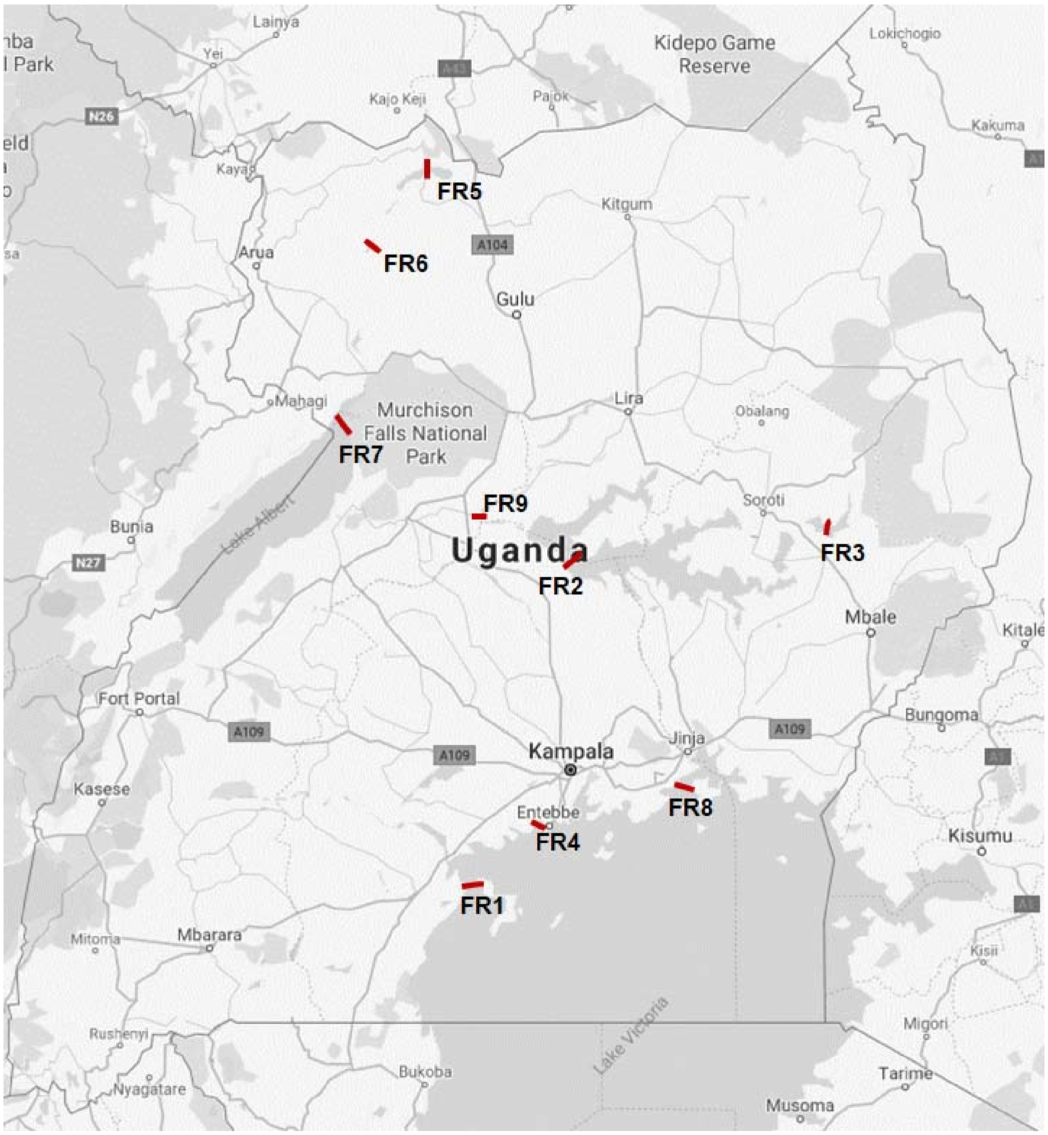


Table 2-8: Description and location of ferry survey routes

DESCRIPTION AND LOCATION OF FERRY SURVEY ROUTES					
Site Ref.	**Ferry Name**	**Survey Type**	**Description of Route**	**Start Coordinates**	**End Coordinates**
FR1	Kalangala	FR	Bukakata to Bugoma	-0.273115, 32.028387	-0.248300, 32.065592
FR2	Kyoga 1	FR	Zengebe to Namasale	1.475234, 32.529020	1.490234, 32.613617
FR3	Bisina	FR	Agule to Kikorio	1.623330, 33.922576	1.652045, 33.945973
FR4	Entebbe	FR	Nakiwogo to Buwaya	0.080005, 32.449259	0.089420, 32.439313
FR5	Laropi	FR	Laropi to Omi	3.551906, 31.812861	3.544139, 31.811156
FR6	Obongi	FR	Obongi to Sinyanya	3.236868, 31.557831	3.232850, 31.568090
FR7	Panyimur	FR	Wanseko - Panyimur	2.175454, 31.378363	2.294859, 31.347204
FR8	Kiyindi	FR	Kiyindi to Buvuma	0.278906, 33.147981	0.283373, 33.217005
FR9	Nakasingola	FR	Masindi Port to Kungu	1.694812, 32.092480	1.699416, 32.098755

2.2.8 Locations of Automatic Traffic Counts (ATC) Sites

ATC surveys were undertaken at seven (7) sites. These are identified and described in Figure 2-11 and Table 2-9 below.

Figure 2-11: Location of ATC survey sites

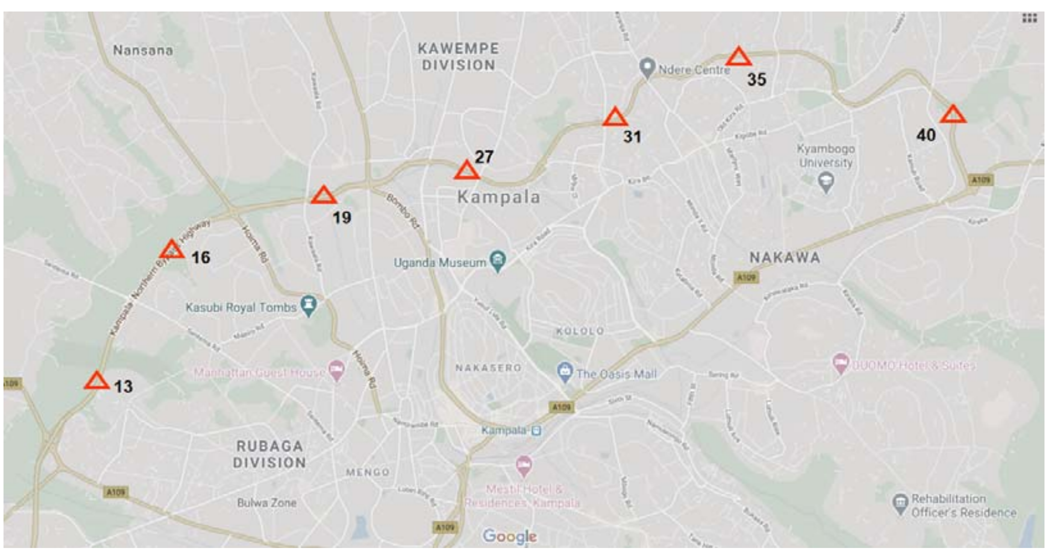


Table 2-9: Description and location of ATC survey sites

DESCRIPTION AND LOCATION OF ATC SURVEY SITES				
Site Ref.	**Highway / Link Name**	**Survey Type**	**Description of Site**	**Site Coordinates**
13	Kampala Northern Bypass	ATC	Between Mityana Road and Sentema Road junctions	0.322564, 32.522876
16	Kampala Northern Bypass	ATC	Between Sentema Road and Hoima Road junctions	0.339240, 32.532805
19	Kampala Northern Bypass	ATC	Between Hoima Road and Bombo Road junctions	0.347578, 32.556023
27	Kampala Northern Bypass	ATC	Between Gayaza Road and Kyebando Road junctions	0.350289, 32.580307
31	Kampala Northern Bypass	ATC	Between Kyebando Road and Kisaasi Road junctions	0.359111, 32.600155
35	Kampala Northern Bypass	ATC	Between Kisaasi Road and Nalya Road junctions	0.368873, 32.617906
40	Kampala Northern Bypass	ATC	Between Nalya Road and Bweyogerere junctions	0.367638, 32.632050

2.2.9 Locations of Axle Load Survey (ALS) Sites

ALS surveys were undertaken at ten (10) sites. These are identified and described in Figure 2-12 and Table 2-10 below.

Figure 2-12: Location of Axle Load survey

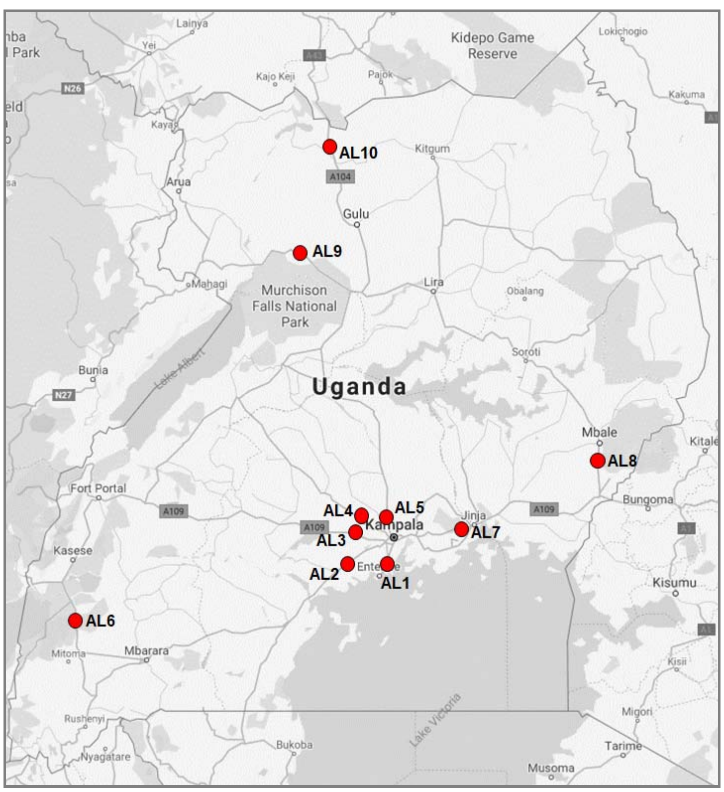


Table 2-10: Description and location of Axle Load survey sites

DESCRIPTION AND LOCATION OF ATC SURVEY SITES				
Site Ref.	**Highway / Link Name**	**Survey Type**	**Description of Site**	**Site Coordinates**
AL1	Kampala – Entebbe Highway	Axle	At the dual section of Kampala-Entebbe Road	0.061179, 32.472757
AL2	Kampala – Masaka Highway	Axle	Buwama	0.064800, 32.109264
AL3	Kampala – Mityana Highway	Axle	Bujuuko	0.333394, 32.370712
AL4	Kampala – Hoima Highway	Axle	Kakiri	0.419596, 32.402159
AL5	Kampala – Bombo Highway	Axle	Matugga	0.472375, 32.518501
AL6	Ishaka – Kasese Highway	Axle	Rubirizi town	-0.268438, 30.106536
AL7	Kampala – Jinja Highway	Axle	Nakibizzi former Weighbridge location	0.422428, 33.143343
AL8	Mbale – Tororo Highway	Axle	Busoba town	0.969742, 34.170843
AL9	Karuma – Arua Highway	Axle	Purongo Town	2.540505, 31.828369
AL10	Gulu – Nimule Highway	Axle	Abelokodi village	3.324474, 32.104841

2.3 Vehicle Classification

The proposal also identified the list of vehicle types to be considered. This list is in accordance with the vehicle types specified in the Manual for Road Investment Appraisal (UNRA, June 2015)[2]. Only motorized traffic was investigated in this assignment.

- Saloon car (sedan, saloon, estate, etc);
- Utility vehicle (pickup, four-wheel drive Station Wagon);
- Minibus or matatu (14-seat capacity);
- Medium bus or coaster (25-seat capacity);
- Large bus or coach (50 or higher seat capacity);
- Light truck (2 axle, gross weight of not more than 7 tonnes);
- Medium truck (2 axles, gross vehicle weight of between 7 and 15 tonnes);
- Heavy truck (3 or 4 axles, gross vehicle weight of up to 22 tonnes);

[2] Table 20, Page 76, Manual for Road Investment Appraisal

- Articulated truck (semi-trailer) (5 or more axles, gross vehicle weight of 22 - 56 tonnes);
- Motorcycles;
- Tractors – this vehicle type is not listed in the UNRA standard vehicle types. It was added to cater for agricultural vehicles (tractors) and construction equipment.

2.4 Survey Forms

The forms for use in the traffic survey exercise were designed carefully taking into consideration both the standard specifications in accordance with UNRA guidance and the specific needs of the proposed expressway master plan.

All survey forms are included in Appendix A.

3 MANUAL CLASSIFIED COUNTS (MCC) SURVEYS

3.1 Introduction and Methodology

MCC are typically undertaken by traffic enumerators standing by the roadside and recording each vehicle by category as it passes. They were undertaken at 172 locations countrywide. The specific locations of the survey sites are given by the location coordinates presented in Table 2-2.

Significant effort was made to ensure that errors in the data during collection are minimized. Errors in MCC data are usually due to unrecorded or unobserved traffic especially at road sections with significant volumes of traffic. To this end, the need for effective data collection was assessed on a site by site basis during an extensive pre-survey site inspection.

Errors can also result from poor site selection. To this end, the locations of the MCC surveys were carefully chosen to optimize the coverage of driver origins and destinations.

3.1.1 Survey Personnel and Training

The following categories of survey staff were used at all survey sites:

- Survey manager with overall responsibility for the survey programme;
- Site supervisors with overall responsibility for the operations of each of the survey sites;
- Two to four enumerators depending on the level of traffic on the road. Generally, survey sites in Cordon 1 required more enumerators. Quite frequently, one enumerator per direction was not adequate because of the high volume of traffic on the road;
- Two police officers per site for night counts.

Survey staff used for MCC surveys in this exercise were experienced, but were nonetheless trained prior to the survey. The training covered a range of issues including; details about the survey, the duties involved, roles and responsibilities, the associated health and safety requirements, vehicle recording procedure and others.

3.1.2 Survey Period

As described in Table 1-1, the MCC surveys were undertaken for a total of seven (7) days as follows:

- For 5 days, a 12-hour survey period (7am – 7pm) was used.
- For two days (one weekday and a Saturday), a 24-hour survey period (7am – 7am) was used.

Surveying was undertaken in both directions.

3.1.3 Survey Diary

For relatively small surveys (for a few survey stations) and a limited number of survey days, one of the key impacts is inclement weather. For this assignment, this was not considered a key issue. The survey was effectively collected for 7 days in a largely dry season (July to September).

The most significant issues that were experienced were as follows:

- For upcountry locations, the key issue was the ability of the limited local police resources to support the surveys or due to security reasons. At a few sites, in the Gulu area (Gulu and Atiak) and the Karuma area (Karuma-Pakwach road), the unavailability of police personnel to provide protection was experienced.

  For MCC surveys, where police personnel was unavailable, the survey stations were usually moved to nearby fuel stations or towns were available local security was hired to provide protection. But in all cases, care was taken to ensure that the collected data would be representative of the survey link.

- Another key issue was the sometimes-overwhelming traffic volumes, especially for some Cordon 1 survey sites. For all key routes into the City, at least two enumerators were required for each direction. In some cases, three were used to ensure that motorcycle counts (which frequently required a specific enumerator) could be taken with high level of accuracy. It is considered that survey sites near Kampala in the future should be undertaken using automatic means.

- There were also general issues with regard to vehicle classifications. There is now marked challenges in the differentiation between some vehicle types such as cars and utility vehicles, particularly station wagons. Vehicles models such as Toyota Kluger, Subaru Forester, Suzuki Vitara seem to fall between salon cars and station wagons. There also some issues with the differentiation between minibuses for example Toyota Noah and Toyota Hiace. Also, certain light trucks seem to be closer to pickups in shape and size. The Toyota Noah is widely used in many parts of Uganda as a minibus, whereas this is not the case in Greater Kampala.

3.2 Collected MCC Data

This section presents the analysis of the MCC data collected during the survey. The data is presented as site averages by cordon. The tables provide 12-hour average traffic volumes, 24-hour average daily traffic (ADT) and the implied Night Equivalent Factor (NEF). The NEF is the ratio of the 24-hour volume to the 12-hour volume, and is hence a measure of night traffic activity on a particular link or area. Higher NEF values imply high traffic movements and vice-versa.

Further, the Annual Average Daily Traffic (AADT) volume was estimated from the ADT volume and is included in the tables. The AADT is estimated from the AADT by factoring the ADT volume with the Seasonal Correction Factor (SCF). The SCF is a month by month factor that relates the ADT for a particular month and location to the AADT for that location.

The ADT at any given location on the road network in any given month is expected to vary from the ADT at the same location in another month due to seasonal impacts such as school holidays, harvest and market seasons, rainy or dry seasons, special events, and others. However, over an entire year, the average traffic is a specific number. The SCF therefore converts the ADT observed in a specific month to the AADT volume observed over a year.

3.2.1 ATC Data - Estimation of Seasonal Correction Factors

The UNRA collects automated traffic counts on a daily basis at 15 selected locations in all four regions of the country. This data was obtained by the Consultant and from it an estimation of the seasonal correction factors was undertaken. To obtain the SCF for each survey site, the ADT observed in a certain month was divided by the AADT over 12 months at that survey site. As such, for each survey site, SCF values were obtained for each month. From the monthly SCF values by site, monthly SCF values were estimated by region and for the country. Table 3-1 below presents the SCF values per site, region and national.

Table 3-1: Estimation of Seasonal Correction Factor (SCF)

	ATC Survey Site / Month	Region	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1	Kasangati - Kyaliwajala	Central	0.896	1.042	1.089	1.057	0.989	1.058	0.892	0.914	0.997	1.032	1.091	0.944
2	Mukono - Kalagi	Central	1.098	0.974	0.974	0.951	0.966	0.997	1.039	1.068	1.068	0.902	0.895	1.068
3	Njeru - Bukoloto	Central	0.921	0.930	1.085	1.006	0.899	0.999	0.989	0.964	1.028	1.016	1.050	1.113
4	Kampala - Mukono	Central	1.048	1.006	0.947	0.944	0.965	0.984	0.989	0.994	1.010	0.981	1.037	1.095
5	Buwama - Nyendo	Central	0.946	0.945	0.926	0.974	1.063	1.026	1.009	1.047	1.053	1.025	1.055	0.930
	SCF Central Region		***0.991***	***0.988***	***1.023***	***0.990***	***0.955***	***1.009***	***0.977***	***0.985***	***1.026***	***0.983***	***1.018***	***1.055***
6	Lyantonde - Ntusi	Western	1.012	1.095	0.875	0.962	0.969	1.015	0.949	1.130	1.141	0.826	0.950	1.076
7	Ishaka - Katunguru	Western	1.022	0.931	0.962	0.937	0.973	1.015	1.050	1.133	0.909	1.019	1.038	1.011
8	Kyenjojo - Fortportal	Western	1.088	1.081	1.022	1.002	1.186	0.952	1.049	1.027	0.583	0.990	0.975	1.045
9	Ibanda - Kamwenje Road	Western	1.137	1.183	0.937	1.066	1.078	0.974	0.682	0.981	1.019	0.956	1.021	0.965
	SCF Western Region		***1.065***	***1.073***	***0.949***	***0.992***	***1.052***	***0.989***	***0.932***	***1.068***	***0.913***	***0.948***	***0.996***	***1.024***
10	Tirinyi - Budaka - Kamonkoli	Eastern	1.052	0.995	1.105	0.936	0.979	0.935	0.990	1.022	0.976	0.971	1.022	1.017
11	Palisa - Kanyago Bridge	Eastern	1.107	0.839	0.769	1.558	1.171	0.963	0.862	1.088	0.905	0.970	0.862	0.907
12	Kumi - Brooks corner	Eastern	0.898	0.922	0.921	0.922	0.961	0.950	1.043	0.729	1.281	1.275	1.036	1.061
	SCF Eastern Region		***1.019***	***0.919***	***0.932***	***1.139***	***1.037***	***0.950***	***0.965***	***0.946***	***1.054***	***1.072***	***0.973***	***0.995***
13	Lira - Corner Kilak	Northern	0.985	1.099	1.019	0.901	0.996	0.914	0.949	1.059	1.001	0.949	0.931	1.198
14	Lamogi - Amuru Junction	Northern	0.981	1.012	1.061	1.055	1.015	0.870	0.965	0.986	1.058	1.080	0.926	0.991
15	Nebi - Olevu	Northern	0.949	0.977	1.009	0.930	1.042	1.049	0.892	0.988	0.965	1.069	1.047	1.082
	SCF Northern Region		***0.971***	***1.029***	***1.030***	***0.962***	***1.018***	***0.944***	***0.935***	***1.011***	***1.008***	***1.033***	***0.968***	***1.090***
	SCF - NATIONAL		***1.009***	***1.002***	***0.980***	***1.013***	***1.017***	***0.980***	***0.957***	***1.009***	***1.000***	***1.004***	***0.996***	***1.033***

Source: UNRA Automatic Traffic Counts Data, Consultants Analysis

The SCF values provides information on the traffic activity on a particular road link or region relative to the average traffic annual activity. As such, a SCF value less than 1.0 implies traffic volumes lower than the annual average and vice-versa. From Table 3-1 above, the month of September seems to result in ADT values that are equal to the national AADT. However, the picture is different within the individual regions.

It should be noted that seasons during any one year are expected to vary by region of the country. For example, the impact of school activity on traffic volumes may be significant in Kampala and the central region but unlikely to be so in Mbale. Similarly, the impact of rain seasons may be significant in Western Uganda but not so in Greater Kampala. Therefore, for regional specific studies or projects, regional SCF values are recommended for use.

The MCC survey exercise was undertaken between July and November 2019. Table 3-2 below identifies the survey dates for each station, the region in which the station is located and the appropriate SCF value. All ADT figures were divided by their respective SCF values to obtain the equivalent AADT.

Table 3-2: Seasonal Correction Factor (SCF) values used to convert ADT

SCF VALUES USED FOR CONVERSION OF ADT					
Ref	**Cordon**	**Stations**	**Region**	**Dates of Survey**	**SCF Value**
1	C1	1-25	Central	22nd July – 28th July 2019	0.977
2	C1	26-50	Central	29th July – 4th August 2019	0.981
3	C2	51-77	Central	23rd Sept – 29th Sept 2019	1.026
4	C3	78-92	Central	16th Oct – 22nd Oct 2019	0.983
5	C3	93-98	Eastern	23rd Oct – 28th Oct 2019	1.072
6	C4	99, 141	Central	23rd Oct – 28th Oct 2019	0.983
7	C4	100-112, 133-137	Western	23rd Oct – 28th Oct 2019	0.948
8	C4	118-119, 142, 143, 120, 144	Eastern	14th Aug – 27th Aug 2019	0.946
9	C5	113-116, 126-130, 138-140, 117, 121-125, 131-132,	Northern	31st July – 11th August 2019	0.935

3.2.2 Cordon 1 MCC Data

Table 3-3 below presents the collected MCC data for Cordon 1 survey sites. It presents the average 12-hour total traffic volumes, the average 24-hour daily traffic (ADT), the NEF and the AADT. Detailed station by station analysis results are included in Appendix B.

Table 3-3: MCC Survey Traffic Volumes for Cordon 1 (C1)

MCC SURVEY TRAFFIC VOLUMES – CORDON 1						
Site	**Description**	**Average 12-Hr**	**ADT (Average 24-Hr)**	**NEF**	**SCF**	**AADT**
01	Mobutu Road	26,962	38,339	1.422	0.977	39,241
01A	Salama Road	20,412	35,950	1.761	0.977	36,797
02	Entebbe Road	48,598	71,610	1.474	0.977	73,295
03	Kalinda Rd	29,590	38,500	1.301	0.977	39,406

04	Weraga Road	15,003	22,710	1.514	0.977	23,245
05	Kabusu Rd North	22,769	30,133	1.323	0.977	30,842
06	Kabusu Rd South	24,368	36,621	1.503	0.977	37,483
07	Masaka Rd Ndeeba	25,061	38,993	1.556	0.977	39,911
08	Wakaliga Rd North	25,690	40,410	1.573	0.977	41,362
09	Wakaliga Rd South	14,329	24,115	1.683	0.977	24,683
10	Masaka Rd Busega	33,908	53,416	1.575	0.977	54,674
11	Masaka Rd Kyengera	33,582	54,407	1.620	0.977	55,688
12	Mityana Rd	31,021	43,950	1.417	0.977	44,984
13	KNBP Mityana-Sentema	26,525	38,047	1.434	0.977	38,943
14	Sentema Rd North	13,279	22,734	1.712	0.977	23,269
15	Sentema Rd South	8,331	10,973	1.317	0.977	11,231
16	Northern Bypass	27,037	46,759	1.729	0.977	47,859
17	Hoima Rd South	22,335	33,307	1.491	0.977	34,091
18	Hoima Rd Nth	24,479	36,900	1.507	0.977	37,769
20	Kawaala Rd	7,229	12,462	1.724	0.977	12,755
21	Sir Apollo Kaggwa Road	25,412	41,929	1.650	0.977	42,916
22	Bombo Road North	32,687	47,182	1.443	0.977	48,293
23	Bombo Road North	26,186	38,520	1.471	0.977	39,427
24	Gayaza Rd	28,604	42,248	1.477	0.977	43,243
25	Tula Rd	21,520	31,763	1.476	0.977	32,511
26	Gayaza Rd	36,004	57,318	1.592	0.981	58,428
28	Kisingiri Street	19,993	29,829	1.492	0.981	30,407
29	Kyebando Ring	12,177	16,683	1.370	0.981	17,006
30	Kyebando-Kisalosalo	9,907	14,187	1.432	0.981	14,462
32	Bukoto Kisasi Rd	13,737	19,561	1.424	0.981	19,940
33	Kisasi-Ntinda Rd	17,947	26,848	1.496	0.981	27,368
34	Kisasi-Kyanja Rd	21,292	30,455	1.430	0.981	31,045
36	Mbogo Road	16,247	27,246	1.677	0.981	27,774
37	Naalya Road North	19,901	28,299	1.422	0.981	28,847
38	Naalya Road South	19,687	28,113	1.428	0.981	28,657
39	Namugongo Rd	21,038	26,592	1.264	0.981	27,107
40	KNBP Bweyogerere - Nalya	10,977	15,774	1.437	0.981	16,080
41	Namanve Ind Park	4,255	6,132	1.441	0.981	6,251
42	Kinawataka	19,942	32,346	1.622	0.981	32,972
43	Kampala-Jinja Highway	26,272	39,434	1.501	0.981	40,198
44	Nakawa-Jinja Rd	16,617	25,573	1.539	0.981	26,068
44A	Ntinda Road	16,651	23,794	1.429	0.981	24,255
45	New Port Bell Rd	22,995	30,422	1.323	0.981	31,011

46	Katalima Road	19,428	24,965	1.285	0.981	25,449
47	Lugogo Bypass	31,843	42,478	1.334	0.981	43,301
48	Old Port Bell Road	19,377	28,058	1.448	0.981	28,601
49	Access Road	59,732	79,682	1.334	0.981	81,225
50	Nsambya Rd	57,524	77,197	1.342	0.981	78,692

3.2.3 Cordon 2 MCC Data

Table 3-4 below presents the collected MCC data for Cordon 2 survey sites. It presents the average 12-hour total traffic volumes, the average 24-hour daily traffic (ADT), the NEF and the AADT. Detailed station by station analysis results are included in Appendix B.

Table 3-4: MCC Survey Traffic Volumes for Cordon 2 (C2)

MCC SURVEY TRAFFIC VOLUMES – CORDON 2						
Site	**Description**	**Average 12 -Hr Traffic**	**ADT (Average 24-Hr)**	**NEF**	**SCF**	**AADT**
51	Salama Road	8,994	14,600	1.623	1.026	14,230
52	Wavamunno Rd	10,225	16,348	1.599	1.026	15,934
53	Busabala Road	2,433	4,020	1.652	1.026	3,918
55	Lubugumu Road	1,348	1,864	1.383	1.026	1,817
56	Munyonyo Spur	7,503	10,713	1.428	1.026	10,442
57	Kigo Road	2,102	3,435	1.634	1.026	3,348
58	Entebbe Road	13,363	18,588	1.391	1.026	18,117
59	Kampala Entebbe Expressway	6,335	8,812	1.391	1.026	8,589
59A	Budo Road	4,570	7,330	1.604	1.026	7,144
60	Masaka Rd	11,100	16,632	1.498	1.026	16,211
61	Nsangi-Buloba Road	2,726	4,010	1.471	1.026	3,908
62	Mityana Road	6,429	9,218	1.434	1.026	8,984
63	Buloba-Bukasa Rd	2,375	3,494	1.471	1.026	3,406
64	Bukasa-Wakiso Road	2,787	3,907	1.402	1.026	3,808
65	Hoima Road	1,659	2,174	1.310	1.026	2,119
66	Wakiso-Matugga Rd	4,558	7,210	1.582	1.026	7,027
67	Bombo Road	16,232	22,330	1.376	1.026	21,764
68	Matugga-Kasangati Road	6,046	9,409	1.556	1.026	9,171
69	Gayaza Road	22,187	31,561	1.422	1.026	30,761
70	Kasangati-Kira Rd	3,760	5,550	1.476	1.026	5,409

MCC SURVEY TRAFFIC VOLUMES – CORDON 2						
Site	**Description**	**Average 12 -Hr Traffic**	**ADT (Average 24-Hr)**	**NEF**	**SCF**	**AADT**
71	Kira-Bulindo Rd	13,917	20,898	1.502	1.026	20,368
72	Kira-Kyaliwajjala Rd	14,622	21,193	1.449	1.026	20,656
73	Sonde Road	10,469	14,984	1.431	1.026	14,604
74	Kampala -Jinja Road	23,750	35,282	1.486	1.026	34,388
75	Mukono-Kalagi Rd	4,693	6,596	1.405	1.026	6,429
76	Kla-Jinja Highway	20,560	28,223	1.373	1.026	27,508
77	Katosi Rd	12,525	17,091	1.365	1.026	16,658

3.2.4 Cordon 3 MCC Data

Table 3-5 below presents the collected MCC data for Cordon 3 survey sites. It presents the average 12-hour total traffic volumes, the average 24-hour daily traffic (ADT), the NEF and the AADT. Detailed station by station analysis results are included in Appendix B.

Table 3-5: MCC Survey Traffic Volumes for Cordon 3 (C3)

MCC SURVEY TRAFFIC VOLUMES – CORDON 3						
Site	**Description**	**Average 12 -Hr Traffic**	**ADT (Average 24-Hr)**	**NEF**	**SCF**	**AADT**
78	Masaka - Mutukula	5,948	8,307	1.397	0.983	8,451
79	Masaka - Mbarara	3,850	5,695	1.479	0.983	5,793
80	Villa Rd	1,165	1,678	1.440	0.983	1,707
81	Sembabule-Kabulasoke	1,144	1,558	1.362	0.983	1,585
82	Kanoni-Maddu	1,558	2,023	1.298	0.983	2,058
83	Kanoni-Maddu	1,351	1,696	1.255	0.983	1,725
84	Mityana Road	6,573	8,714	1.326	0.983	8,865
84A	Mityana-Busunju	1,264	1,847	1.461	0.983	1,879
85	Hoima Rd	1,756	2,584	1.472	0.983	2,629
86	Semuto Rd	2,818	3,610	1.281	0.983	3,672
87	Kampala-Gulu Highway	10,752	13,258	1.233	0.983	13,487
88	Wobulenzi-Zirobwe	1,939	2,659	1.371	0.983	2,705
89	Galiraya-Kayunga	1,359	1,747	1.286	0.983	1,777
90	Kangulumira	4,308	5,462	1.268	0.983	5,557

MCC SURVEY TRAFFIC VOLUMES – CORDON 3						
Site	**Description**	**Average 12 -Hr Traffic**	**ADT (Average 24-Hr)**	**NEF**	**SCF**	**AADT**
91	Kampala-Jinja Highway	9,198	13,691	1.488	0.983	13,928
92	Nalubaale Rd	2,691	3,543	1.317	0.983	3,604
93	Jinja - Kamuli Rd	6,281	8,286	1.319	1.072	7,729
93A	Jinja - Musita	11,071	15,447	1.395	1.072	14,410
94	Jinja - Kaliro Rd	8,120	10,969	1.351	1.072	10,232
94A	Musita-Iganga	6,848	9,948	1.453	1.072	9,280
95	Tirinyi Road	3,071	4,118	1.341	1.072	3,841
96	Bugiri	3,843	6,057	1.576	1.072	5,650
97	Musita-Lumino	5,086	6,661	1.310	1.072	6,214
98	Busia Rd - Tororo	2,287	3,530	1.544	1.072	3,293
119	Tororo-Mbale	3,972	4,921	1.239	0.946	5,202
142	Busia Boarder	956	1,175	1.229	0.946	1,242
143	Malaba Boarder	3,187	4,077	1.279	0.946	4,310

3.2.5 Cordon 4 and 5 MCC Data

Table 3-6 below presents the collected MCC data for Cordons 4 and 5 survey sites. It presents the average 12-hour total traffic volumes, the average 24-hour daily traffic (ADT), the NEF and the AADT. Detailed station by station analysis results are included in Appendix B.

Table 3-6: MCC Survey Traffic Volumes for Cordon 4 and 5 (C4 and C5)

MCC SURVEY TRAFFIC VOLUMES – CORDON 4 AND 5						
Site	**Description**	**Average 12 -Hr Traffic**	**ADT (Average 24-Hr)**	**NEF**	**SCF**	**AADT**
Cordon 4 – Mid Country & Western Region						
99	Masaka-Mutukula	1,674	2,335	1.395	0.983	2,375
100	Mbarara-Ntungamo	3,571	4,917	1.377	0.948	5,187
100A	Mbarara-Kikagata	5,471	6,994	1.278	0.948	7,378
101	Mbarara-Bushenyi	8,095	9,675	1.195	0.948	10,206
102	Mbarara-Ibanda	1,977	2,565	1.297	0.948	2,706
103	Lyantonde-Kiruhura	1,315	2,144	1.630	0.948	2,262
104	Ishaka-Katunguru	5,647	7,357	1.303	0.948	7,761

MCC SURVEY TRAFFIC VOLUMES – CORDON 4 AND 5						
Site	**Description**	**Average 12 -Hr Traffic**	**ADT (Average 24-Hr)**	**NEF**	**SCF**	**AADT**
105	Bwera - Mpondwe	1,416	1,918	1.355	0.948	2,023
106	Kasese-Fort Portal	12,226	15,651	1.280	0.948	16,509
107	Fort Portal - Kyenjojo	868	1,209	1.393	0.948	1,275
108	Kyenjojo-Hoima	4,044	5,070	1.254	0.948	5,348
109	Kateera-Kyankwanzi	678	921	1.358	0.948	972
110	Ngoma-Kapeeka	1,279	1,734	1.356	0.948	1,829
111	Hoima-Masindi	2,438	3,008	1.234	0.948	3,173
111A	Bulisa-Hoima	567	721	1.272	0.948	761
112	Masindi-Kibangya	1,694	2,340	1.381	0.948	2,468
113	Kigumba-Karuma	1,516	2,399	1.582	0.935	2,565
118	Soroti-Kumi	1,617	2,343	1.449	0.946	2,477
119	Tororo - Mbale	1,604	2,325	1.450	0.946	2,458
120	Sironko Road	4,369	5,614	1.285	0.946	5,934
133	Mirama Hills Border	568	807	1.421	0.948	851
134	Katuna Border	996	1,407	1.413	0.948	1,484
135	Cyahafi Border	3,338	4,404	1.319	0.948	4,646
136	Bunagana Border	1,567	2,168	1.384	0.948	2,287
137	Mpondwe Border	6,156	7,580	1.231	0.948	7,996
141	Mutukula Border	543	854	1.573	0.983	869
144	Suam Border	1,031	1,299	1.260	0.946	1,373
Cordon 5 – Northern Region						
114	Nebbi-Arua	1,867	2,576	1.380	0.935	2,755
115	Karuma-Pakwach	834	1,357	1.627	0.935	1,451
116	Karuma -Gulu	6,614	10,245	1.549	0.935	10,957
117	Lira-Kitgum	916	1,415	1.545	0.935	1,513
121	Soroti-Moroto	636	857	1.347	0.935	917
122	Matany-Kilak	726	1,154	1.590	0.935	1,234
123	Moroto-Kotido	265	348	1.313	0.935	372
124	Abim - Kotido	651	1,070	1.643	0.935	1,144
125	Kotido - Kabong	146	260.5	1.784	0.935	279
126	Gulu - kitgum	916	1,415	1.545	0.935	1,513
127	Arua - Koboko	4,704	6,426	1.366	0.935	6,873
128	Koboko - Moyo	1,268	1,434	1.131	0.935	1,534

MCC SURVEY TRAFFIC VOLUMES – CORDON 4 AND 5						
Site	**Description**	**Average 12 -Hr Traffic**	**ADT (Average 24-Hr)**	**NEF**	**SCF**	**AADT**
129	Atiak - Nimule	485	746	1.538	0.935	798
130	Kitgum - Atiak	455	602	1.323	0.935	644
131	Kitgum - Madi Opei	1,803	2,248	1.247	0.935	2,404
132	Kabong - Apoka	226	257	1.137	0.935	275
138	Goli Border	1,484	1,897	1.278	0.935	2,029
139	Vurra Border	1,973	2,518	1.276	0.935	2,693
140	Nimule Border	1,169	1,212	1.037	0.935	1,296

3.3 Hourly Variation of Traffic

This section presents the analysis of the MCC data collected during the survey with respect to variation in traffic. The analysis is presented as average results by cordon.

3.3.1 MCC Data Analysis – Cordon 1

In this section, the MCC data collected at Cordon 1 is analysed. Figures 3-1 and 3-2 present the hourly variation of traffic and vehicle proportions for Cordon 1. Detailed station by station analysis results for this cordon are included in Appendix B.

Figure 3-1: Hourly variation of traffic for Cordon 1

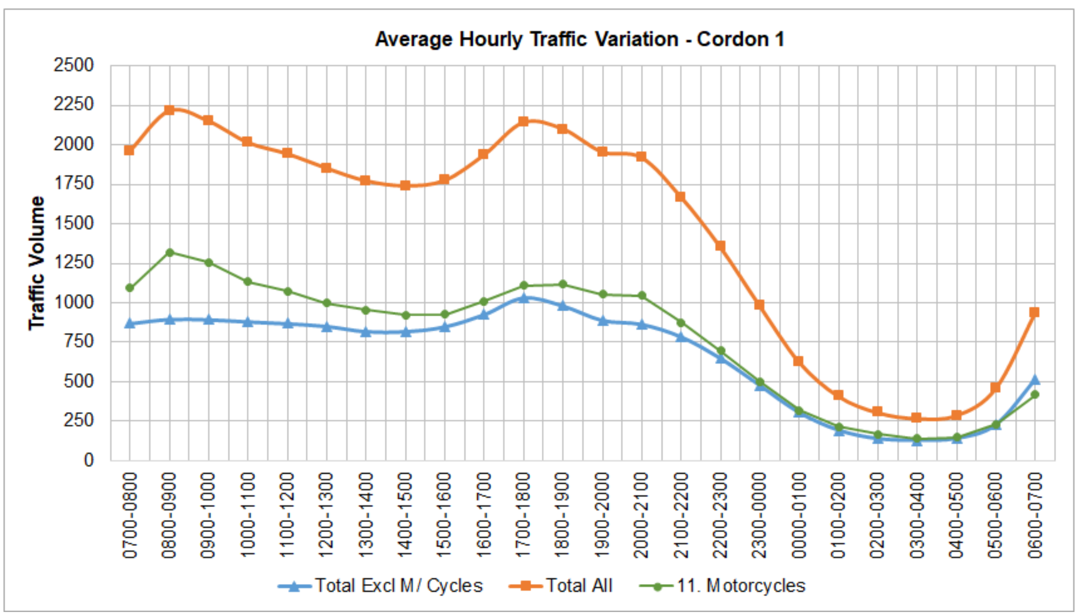
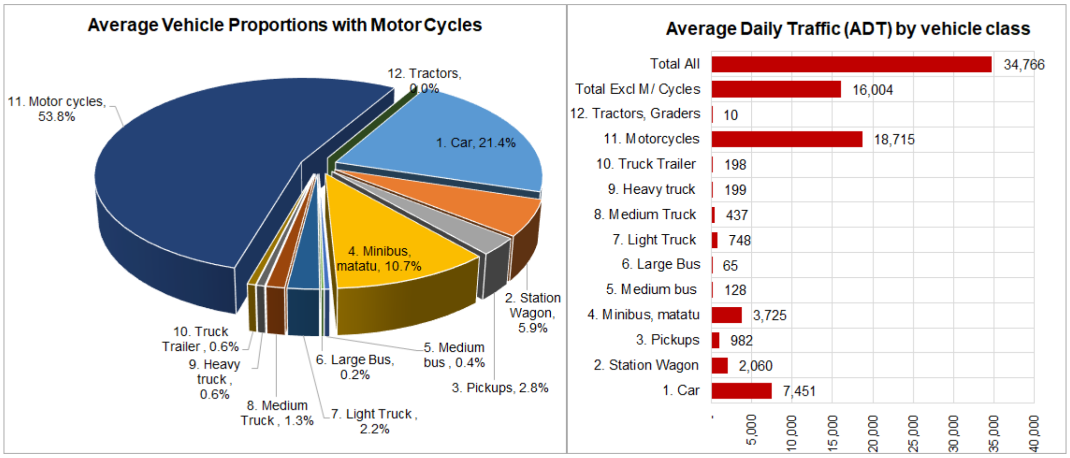


Figure 3-2: Average vehicle proportion for Cordon 1



The following observations are with regard to Figures 3-1 and 3-2 (Cordon 1):

- **Peak hours**: AM peak hours is observed to be 8-9am with a flow of 2,214 vehicles. The PM peak hour is observed as 5 – 6pm with a flow of 2,140 vehicles.

  It should be noted that traffic flow recorded from the surveys reflects vehicle movements past the survey point only and does not include queuing. The figures may hence not indicate how truly busy the link is. The 7 – 8am in the morning for cordon 1 is observed to be busier because of significant traffic jams and queuing. But MCC data is unlikely to show this.

- **Night activity**: there is strong night activity at Cordon 1 being an urban area. The least busy hour of the day is 3 – 4am in the morning with a significant flow of 269 vehicle per hour. Of this, 52% is motorcycle traffic, indicating higher night activity for this vulnerable class.

- **Vehicle proportions**: there is high motorcycle content of 53.8% (or 18,715 per day) followed by cars at 21.4% (7,451) and minibus at 10.7% (3,725).

- **Average ADT**: average ADT for all surveyed links across the cordon is 35,766 vehicles including motorcycles and 16,004 excluding motorcycles.

3.3.2 MCC Data Analysis – Cordon 2

This section presents the analysis of MCC data collected at Cordon 2. Figures 3-3 and 3-4 present the hourly variation of traffic and vehicle proportions for Cordon 2. Detailed station by station analysis results for this cordon are included in Appendix B.

Figure 3-3: Hourly variation of traffic for Cordon 2

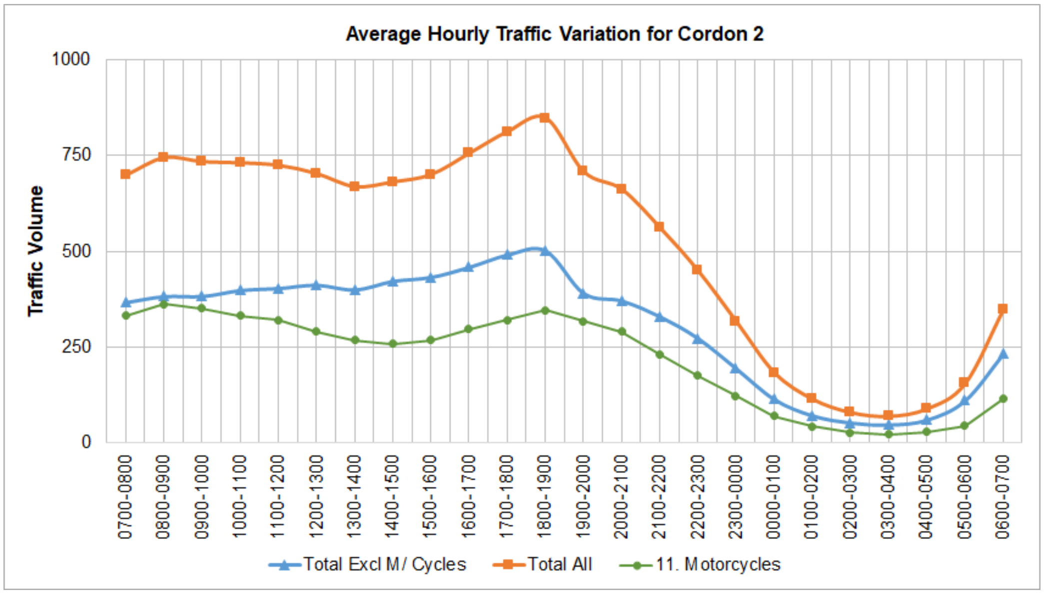
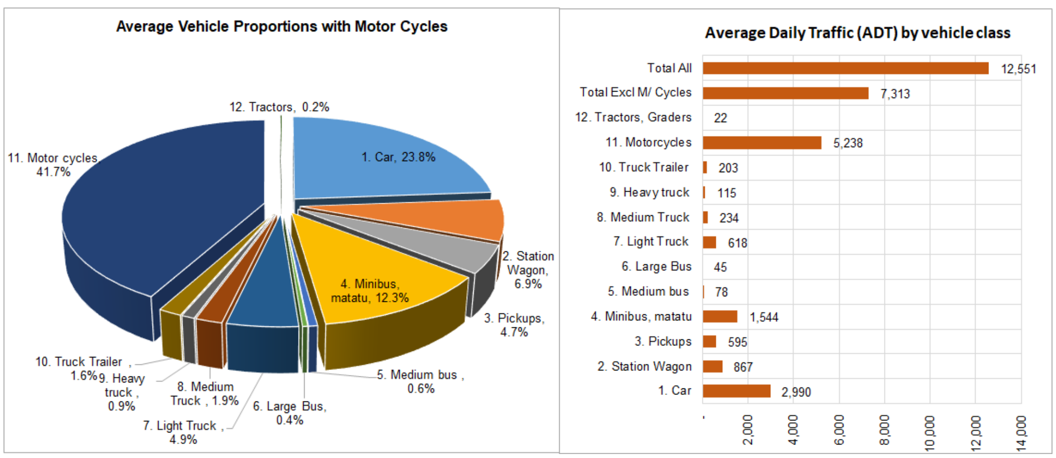


Figure 3-4: Average vehicle proportion for Cordon 2



The following observations are with regard to Figures 3-3 and 3-4 (Cordon 2):

- **Peak hours**: AM peak hours is observed to be 8-9am with a flow of 745 vehicles. The PM peak hour is observed as 6 – 7pm with a flow of 848 vehicles.
- **Night activity**: night activity is less strong than for Cordon 1. The least busy hour of the day is 3 – 4am in the morning with a flow of 70 vehicle per hour. Of this, 31% is motorcycle traffic.
- **Vehicle proportions**: there is high motorcycle content of 41.7% (5,238 vehicles per day) followed by cars at 23.8% (2,990) and minibus at 12.3% (1,544).
- **Average ADT**: average ADT for all surveyed links across the cordon is 12,551 vehicles including motorcycles and 7,313 excluding motorcycles.

3.3.3 MCC Data Analysis – Cordon 3

This section presents the analysis of MCC data collected at Cordon 3. Figures 3-5 and 3-6 present the hourly variation of traffic and vehicle proportions for Cordon 3. Detailed station by station analysis results for this cordon are included in Appendix B.

Figure 3-5: Hourly variation of traffic for Cordon 3

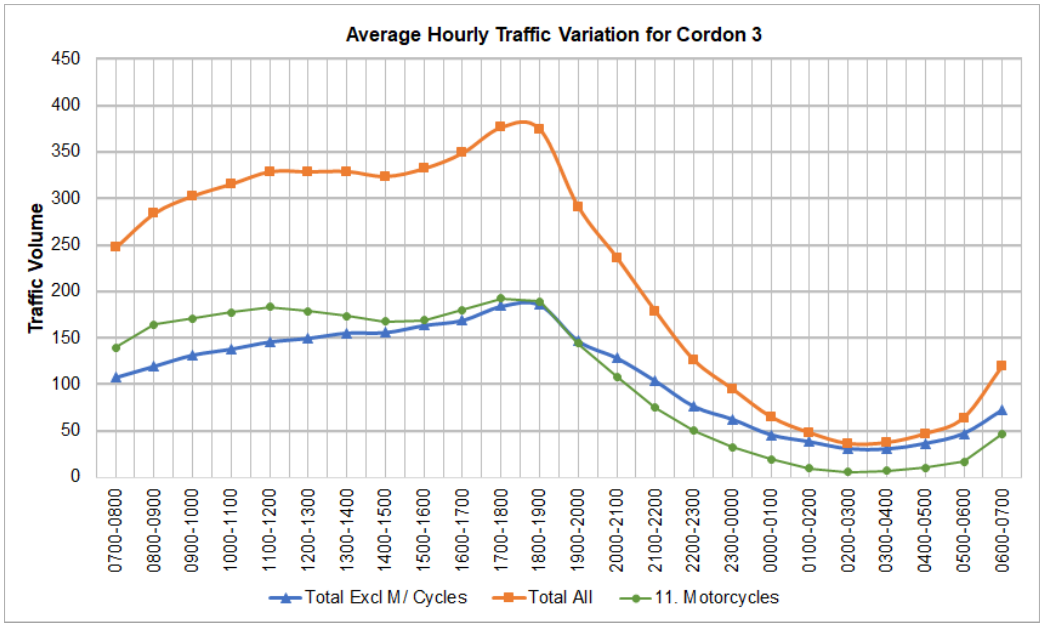
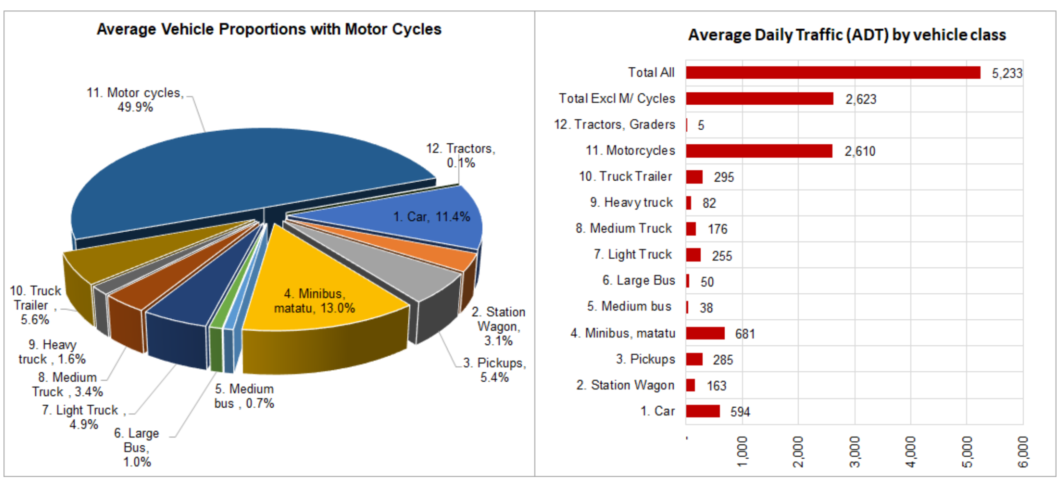


Figure 3-6: Average vehicle proportion for Cordon 3



The following observations are with regard to Figures 3-5 and 3-6 (Cordon 3):

- **Peak hours**: The AM peak hour is not well defined, with traffic volumes increase through the period. The PM peak hour is observed as 5 – 6pm with a flow of 376 vehicles.

- **Night activity**: night activity is not significant. The least busy hour of the day is 2 – 3am in the morning with a flow of 36 vehicle per hour. Of this, only 16.7% is motorcycle traffic.
- **Vehicle proportions**: there is high motorcycle content of 49.9% (2,610 vehicles per day) followed by minibus at 13.0% (681 vehicles) and cars at 11.4% (594 vehicles).
- **Average ADT**: average ADT for all surveyed links across the cordon is 5,233 vehicles including motorcycles and 2,623 excluding motorcycles.

3.3.4 MCC Data Analysis – Cordon 4

This section presents the analysis of MCC data collected at Cordon 4. Figures 3-7 and 3-8 present the hourly variation of traffic and vehicle proportions for Cordon 4. Detailed station by station analysis results for this cordon are included in Appendix B.

Figure 3-7: Hourly variation of traffic for Cordon 4

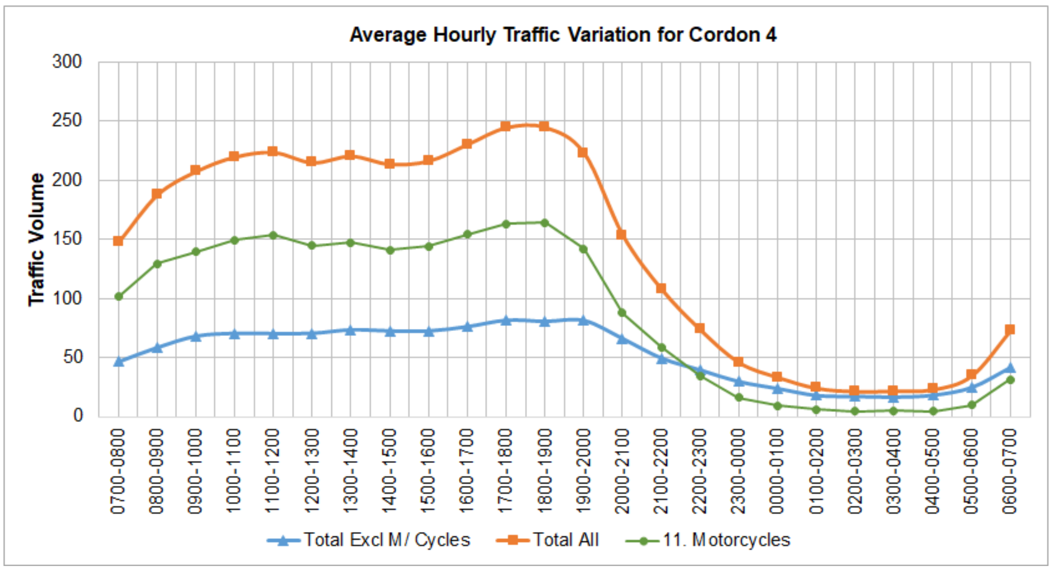
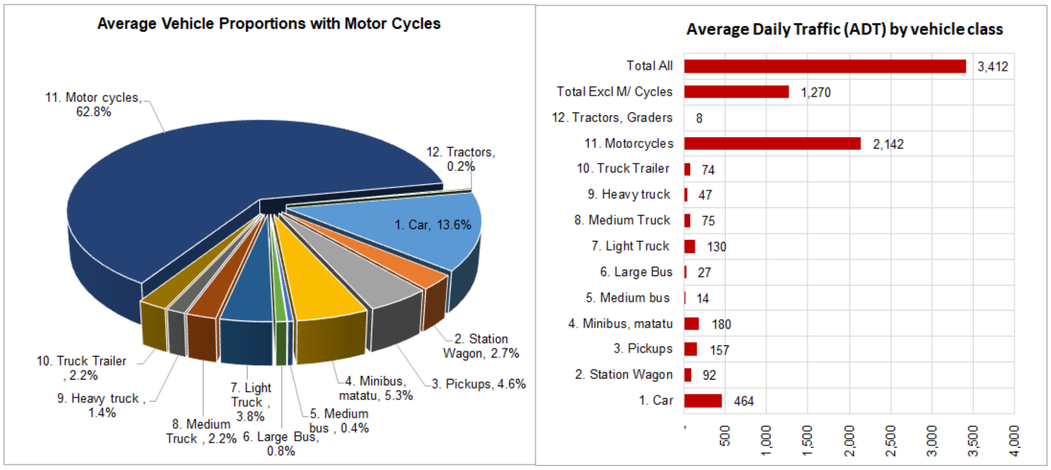


Figure 3-8: Average vehicle proportion for Cordon 4



The following observations are with regard to Figures 3-7 and 3-8 (Cordon 4):

- **Peak hours**: The AM peak hour is not well defined, with traffic volumes increase through the period. The PM peak hours are observed as 5 - 6pm and 6 – 7pm with a flow of 245 vehicles.
- **Night activity**: night activity is not significant. The least busy hour of the day is 2 – 3am in the morning with a flow of 22 vehicle per hour. Of this, only 18% is motorcycle traffic.
- **Vehicle proportions**: there is high motorcycle content of 62.8% (or 2,142 vehicles per day) followed by car at 13.6% (464) and minibus at 5.3% (180). Other notable proportions for this cordon are pickups at 4.6% and light trucks at 3.8%.
- **Average ADT**: average ADT for all surveyed links across the cordon is 3,412 vehicles including motorcycles and 1,270 excluding motorcycles.

3.3.5 MCC Data Analysis – Cordon 5

This section presents the analysis of MCC data collected at Cordon 5. Figures 3-9 and 3-10 present the hourly variation of traffic and vehicle proportions for Cordon 5. Detailed station by station analysis results for this cordon are included in Appendix B.

Figure 3-9: Hourly variation of traffic for Cordon 5

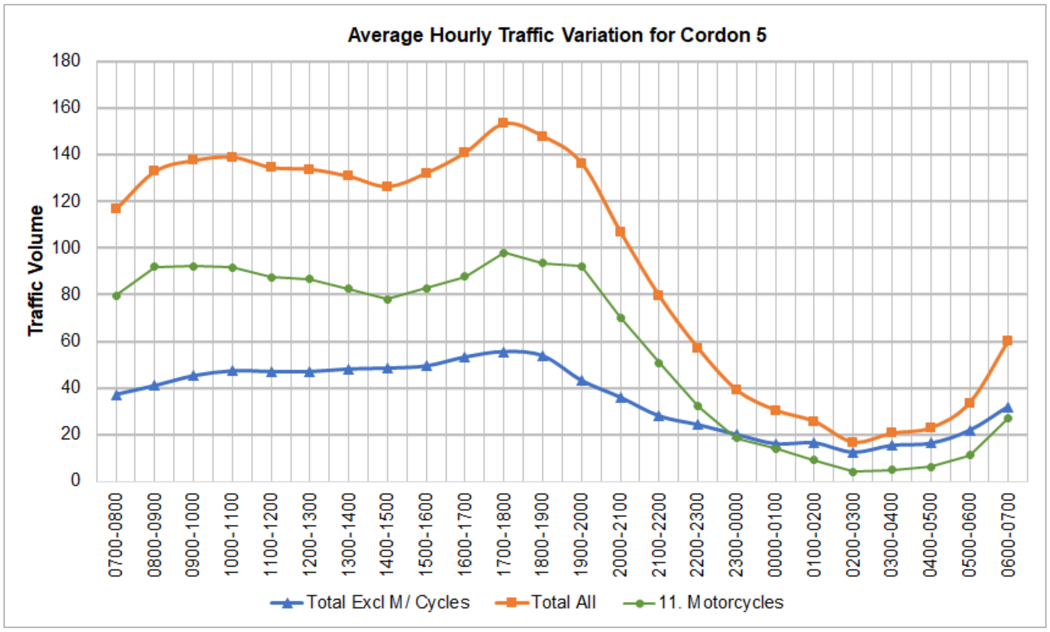
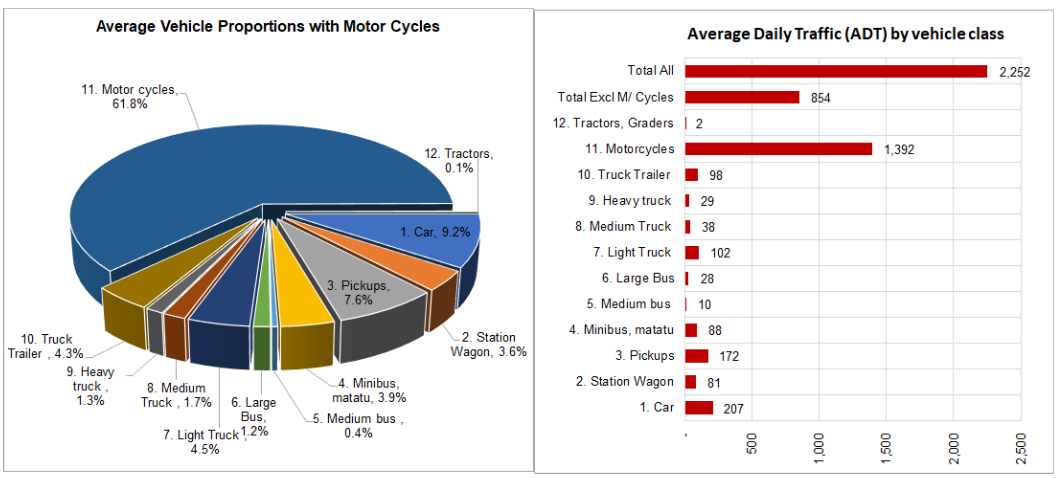


Figure 3-10: Average vehicle proportion for Cordon 5



The following observations are with regard to Figures 3-9 and 3-10 (Cordon 5):

- Peak **hours**: The AM peak hour is observed as 9 - 10am with a traffic flow of 137 vehicles. The PM peak hour is observed as 5 - 6pm with a flow of 153 vehicles.
- **Night activity**: night activity is significantly low as expected for this cordon. The least busy hour of the day is 2 – 3am in the morning with a flow of 17 vehicle per hour. Of this, 23.5% is motorcycle traffic.
- **Vehicle proportions**: there is high motorcycle content of 61.8% (1,490 vehicles per day) followed by car at 9.2% (207) and pickups at 7.6% (172). Other notable proportions for this cordon are truck/trailers at 4.3% and light trucks at 4.5%.

- **Average ADT**: average ADT for all surveyed links across the cordon is 2,252 vehicles including motorcycles and 854 excluding motorcycles.

3.3.6 Overall Observations from MCC Surveys and Data

Peak Travel Hours

The morning (AM) peak travel hour is not uniformly defined across all cordons. For the busy C1 and C2 cordons, the peak hour is 8-9am. For C3, C4 and C5, the AM peak hour is not marked and it was observed from the data that the inter peak (IP) is busier than the AM peak. The PM peak is generally defined for all cordons as 5-6pm.

Night Activity

As expected, this was observed to be significantly high for C1, being an urban area largely.

Proportion of Motor Cycles

Motor cycle content is high for all cordons. The average for all cordons is calculated as 54% - this represents the proportion of motor cycle trips countrywide.

Vehicle Classifications

The challenges experienced in the field with regard to vehicle classification were stated in the Survey Diary (Section 3.1.3). The current standard vehicle classification seems to have been overtaken by changes in vehicle makes and shapes. The Consultant recommends a review of the vehicle classification guidelines for better alignment between current vehicle models and the requirements for traffic analysis.

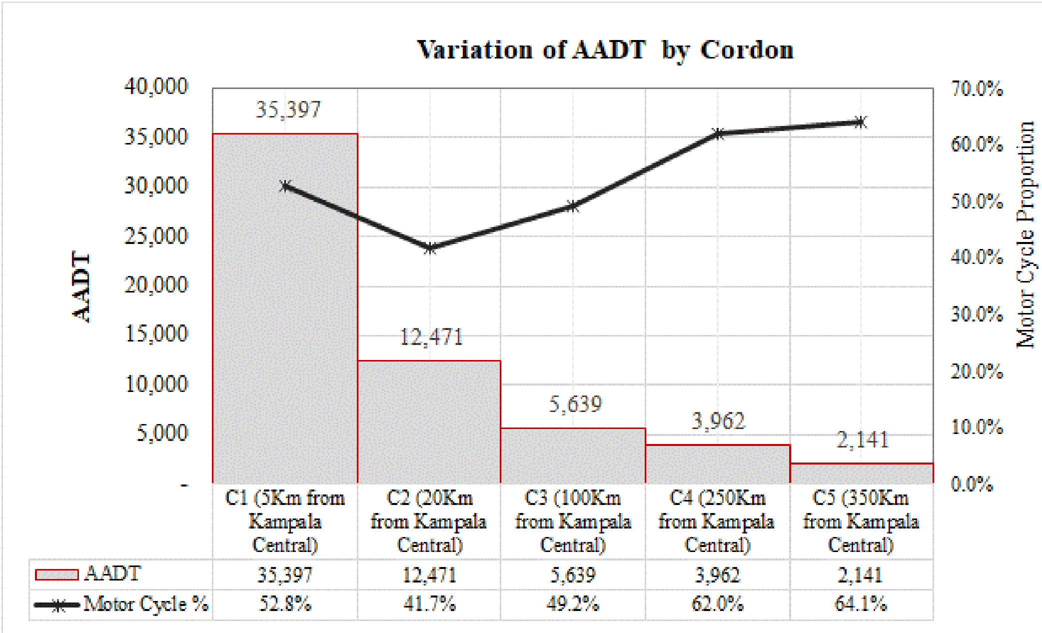
Adequacy of MCC Survey Method

MCC surveys provide vital information on traffic volumes and vehicle classification in a simple, flexible and cost-effective way. This survey type was largely adequate for this purpose throughout the country. However, as discussed in the survey diary, significant challenges and high costs of the survey were experienced in Cordon 1 – Kampala CBD. The Consultant believes that for Kampala CBD, as much as possible, automatic traffic surveys (using pneumatic tubes) should be employed. This will likely produce better quality data and reduce costs.

Annual Average Daily Traffic

The variation of AADT across the nation as one moves away from Kampala Central is shown in Figure 3-11 below. In estimating the AADT, the cordon ADT was divided by a Seasonal Correction Factor (SCF) of 0.9925 established in Section 3.2.1 of this report.

Figure 3-11: Variation of AADT by cordon



3.4 Vehicle Registrations in Uganda

3.4.1 New Vehicle Registration Data

As part of the survey exercise, vehicle registration data from the Uganda Revenue Authority was also analysed. The data which was requested through the Ministry of Works and Transport (MoWT) was provided for seven financial years from 2012/13 to 2018/19[3]. This data represented newly registered vehicles only. It does not represent the total number of vehicles in the country.

The data was received in Excel format and was organized by make and model. The data was then sorted into the 11 UNRA standard vehicle types, with a 12th type added for tractors and construction equipment. Figure 3-12 presents the form of the raw data. The complete set of raw data as provided by URA is included in Appendix B.

Table 3-7 presents the sorted vehicle registration data by vehicle type.

[3] According to URA, these are the years for which vehicle registration data is correctly recorded.

Figure 3-12: Format of raw vehicle registration data

FORM_NAME	CATEGORY_NAME	MAKE	MODEL	POWER	TYRE_SIZE	NET_WT	NUMBER_WHE	NUMBER_AXLE
First Time Registration - Motor Vehicle	Ordinary Vehicles	TOYOTA	LANDCRUISER PRADO TRJ150L	2700	265/65R17	2030	4	2
First Time Registration - Motor Vehicle	Ordinary Vehicles	MERCEDES	Benz, Saloon, C180	1794	195/65R15	1390	4	2
First Time Registration - Motor Vehicle	Ordinary Vehicles	TOYOTA	Harrier (Petrol)	2362	21570R16	1570	4	2
First Time Registration - Trailer	Duty Free/Tax Free	MAN	TRAILER TGA22- 400	0	385/65R225	13000	6	3
First Time Registration - Motor Vehicle	Ordinary Vehicles	TOYOTA	Corona Premio/Carina AT210	1587	175/70R14	1110	4	2
First Time Registration - Motor Vehicle	Ordinary Vehicles	TOYOTA	Harrier (Petrol)	2163	215/70R16	1570	4	2
First Time Registration - Motor Vehicle	Ordinary Vehicles	MITSUBISHI	Fuso Wing Truck	12020	275/70R22-5	11030	6	3
First Time Registration - Motor Vehicle	Ordinary Vehicles	TOYOTA	Mark 11 (Petrol)	1988	19565R15	1330	4	2
First Time Registration - Motor Cycle	Ordinary Vehicles	HONDA	NSR250R	250	110/150-70R17/1	150	2	2
First Time Registration - Motor Vehicle	Ordinary Vehicles	TOYOTA	Raum, model EXZ10	1496	175/65R14	1120	4	2
First Time Registration - Motor Vehicle	Ordinary Vehicles	NISSAN	Elgrand Van (Petrol)	3153	215/65R15	2100	4	2
First Time Registration - Motor Vehicle	Ordinary Vehicles	TOYOTA	Wish, ZNE10G	1794	19565R15	1300	4	2
First Time Registration - Motor Vehicle	Ordinary Vehicles	TOYOTA	Ipsum model SXM10	1998	195/65R14	1400	4	2

Table 3-7: New vehicle registrations in Uganda 2013-2019

VEHICLE REGISTRATION DATA 2013 - 2019													
Year	**Car**	**Station Wagon**	**Pickup**	**Mini bus**	**Med. Bus**	**Large Bus**	**Light Truck**	**Medium Truck**	**Hvy Truck**	**Truck/ Trailer**	**M/ Cycle**	**Tractor Grader**	**Total Vehicles**
2013	7,930	2,745	1,185	2,704	94	61	1,044	393	522	622	25,169	817	43,286
2014	7,892	3,165	1,243	3,800	98	45	1,217	509	958	468	14,234	766	34,395
2015	8,574	3,641	1,165	4,414	124	83	1,509	488	2,883	199	13,893	1,672	38,645
2016	6,626	3,072	1,272	2,978	146	43	2,460	780	818	551	14,897	1,222	34,865
2017	9,026	4,194	1,059	3,463	178	40	1,860	1,926	1,721	422	14,259	1,202	39,350
2018	13,628	5,294	3,216	5,792	215	71	4,551	1,705	728	350	36,297	1,469	73,316
2019	11,946	4,186	3,043	3,757	156	84	3,689	1,518	1,041	387	32,525	1,697	64,028
Avge	9,375	3,757	1,740	3,844	144	61	2,333	1,046	1,239	428	21,611	1,264	46,841
Grth	8.5%	8.7%	16.6%	5.7%	12.3%	4.1%	23.7%	28.0%	3.6%	-4.5%	9.5%	11.3%	9.7%

Table 3-7 also shows the average growth rates of vehicle registrations for each vehicle type. The medium truck vehicle type had the fastest growing vehicle registrations, followed by light trucks and pickups with 28%, 23.7% and 16.6%. Registrations for truck trailers declined over the period, with an average growth rate of -4.5%.

Motor cycles had the highest registrations but these grew at a lower average than the trucks and pickups of 9.5% over the period. The average motor cycle registrations over the seven-year period was 21,611 motorcycles. This is quite a high number when compared with other vehicle types.

Overall, vehicle registrations grew at 9.7% annually over the seven-year period, with an average total of 46,841 vehicles registered per year.

3.4.2 Total Number of Vehicles in the Country

It worth noting that the data presented in Table 3-7 is for vehicle registration per year and does not provide an indication of the total number vehicles in the country. To estimate the vehicle stock in the country, vehicle registrations data published in the

National Master Plan[4] were used. The NTMP reported the "Number of Registered Vehicles" in Uganda for the years 1997, 2002, and 2005 – 2009.

Assuming that this data was correct, the following were considered in this Report:

- To obtain 2012 data, an extrapolation of the NTMP data for the period 2005-2009. This would represent the number of registered vehicles in 2012.
- To obtain the cumulative number of registered vehicles for the period 2013-2019, the data in Table 3-7 above would be added to the 2012 data in sequence. That is, 2013 numbers would be obtained by adding the 2013 new registrations to the 2012 total vehicles.

The estimated total vehicles in the country obtained above was assumed not to consider scrappage i.e. the number of vehicles taken off the road per year. The correct rate of scrappage in Uganda in not known because vehicles censuses are not undertaken in the country. These would help to compare the vehicle stock in the country by year and hence enable the correct estimation of scrappage. Table 3-8 presents the estimated total number of vehicles without consideration for scrappage.

Table 3-8: Estimated vehicle population in Uganda (not adjusted for scrappage) 2005-2019

ESTIMATED VEHICLE POPULATION IN UGANDA (NOT ADJUSTED FOR SCRAPPAGE) 2005 - 2019													
Year	Car	Station Wagon	Pickup	Mini bus	Med. Bus	Large Bus	Light Truck	Medium Truck	Heavy Truck	Truck/ Trailer	M/ Cycle	Tractor Grader	Total Vehicles
2005	65,500	37,159	16,041	27,600	546	354	7,565	2,848	3,783	4,504	108,200	4,600	278,700
2006	70,700	37,159	16,041	32,000	546	354	8,293	3,122	4,147	4,938	134,000	4,800	316,100
2007	81,300	39,115	16,885	39,500	606	394	9,426	3,548	4,713	5,612	176,500	5,200	382,800
2008	90,900	40,721	17,579	49,200	728	472	11,530	4,340	5,765	6,865	236,500	5,900	470,500
2009	96,600	41,699	18,001	62,300	910	590	13,512	5,086	6,756	8,045	292,300	6,900	552,700
2010	101,179	40,780	17,604	72,585	985	639	14,852	5,591	7,426	8,843	356,235	7,261	633,980
2011	105,975	39,881	17,216	84,568	1,067	693	16,324	6,145	8,162	9,719	434,154	7,641	731,546
2012	110,998	39,002	16,837	98,529	1,156	750	17,942	6,754	8,971	10,683	529,117	8,041	848,781
2013	118,928	41,747	18,022	101,233	1,250	811	18,986	7,147	9,493	11,305	554,286	8,858	892,066
2014	134,750	47,657	20,450	107,737	1,442	917	21,247	8,049	10,973	12,394	593,689	10,441	969,747
2015	159,146	57,208	24,043	118,655	1,758	1,106	25,017	9,439	15,336	13,682	646,985	13,696	1,086,072
2016	190,168	69,831	28,908	132,551	2,220	1,338	31,247	11,609	20,517	15,522	715,178	18,173	1,237,263
2017	230,216	86,648	34,832	149,910	2,860	1,610	39,337	15,705	27,419	17,784	797,630	23,852	1,427,804
2018	283,892	108,759	43,972	173,061	3,715	1,953	51,978	21,506	35,049	20,396	916,379	31,000	1,691,661
2019	349,514	135,055	56,155	199,969	4,727	2,380	68,309	28,825	43,720	23,395	1,067,653	39,845	2,019,547

[4] National Transport Master Plan Including A Transport Master Plan For The Greater Kampala Metropolitan Area (NTMP/GKMA), Abridged Version (MoWT, August 2009)

3.4.3 Estimation of Scrappage Rates and Residual Vehicle Population

To overcome this challenge, significant literature has been reviewed with regard to used vehicle depreciation rates and scrappage. In particular, a study by Karl Storchmann[5], appeared to make a considerable amount of analysis to compared depreciation rates for OECD[6] and non-OECD countries (including a number of countries in sub-Saharan Africa). The paper estimated a depreciation rate of used vehicles in sub-Saharan Africa of 14.5% per year. This rate related to passenger vehicles.

The above value was considered reasonable by the Consultant for passenger and light goods vehicles (including motorcycles) as a scrappage rate. This assumes that a vehicle would be scrapped before or at the point of zero value. The percentage is also assumed to account for scrappage that occurs due to accidents – this is not considered to be a significant amount.

A lower rate of 10% was considered by the Consultant for heavier vehicles such as medium and large buses, and medium to large trucks. Larger vehicles from observation tend to operate on the road network at significantly higher vehicle age than lighter vehicles.

The above rates were combined with the estimates of total vehicle population in Table 3-8 to provide an estimate of the residual vehicle population by year for the period 2005-2019. The method of adjustment for scrappage was such that the existing stock was adjusted for scrappage prior to adding the newly registered vehicles. Table 3-9 presents this estimate.

Table 3-9: Estimated residual vehicle population in Uganda (adjusted for scrappage) 2005-2019

ESTIMATED RESIDUAL VEHICLE POPULATION IN UGANDA (ADJUSTED FOR SCRAPPAGE) 2005 - 2019													
Year	Car	Station Wagon	Pickup	Mini bus	Med. Bus	Large Bus	Light Truck	Medium Truck	Heavy Truck	Truck/ Trailer	M/ Cycle	Tractor Grader	Total Vehicles
2005	56,003	31,771	13,715	23,598	491	319	6,468	2,563	3,404	4,054	92,511	4,140	239,037
2006	60,449	31,771	13,715	27,360	491	319	7,091	2,810	3,732	4,444	114,570	4,320	271,071
2007	69,512	33,443	14,437	33,773	546	354	8,059	3,194	4,242	5,051	150,908	4,680	328,197
2008	77,720	34,816	15,030	42,066	655	425	9,858	3,906	5,188	6,178	202,208	5,310	403,361
2009	82,593	35,653	15,391	53,267	819	531	11,553	4,578	6,080	7,241	249,917	6,210	473,831
2010	86,508	34,867	15,052	62,060	887	576	12,698	5,032	6,683	7,959	304,581	6,535	543,436
2011	90,608	34,098	14,720	72,305	961	623	13,957	5,531	7,346	8,747	371,202	6,877	626,976
2012	94,903	33,347	14,395	84,242	1,041	675	15,341	6,079	8,074	9,615	452,395	7,237	727,344
2013	102,833	36,092	15,580	86,946	1,135	736	16,385	6,472	8,596	10,236	477,564	8,054	770,629
2014	117,505	41,603	17,837	93,058	1,317	836	18,494	7,334	10,024	11,264	513,317	9,555	842,146

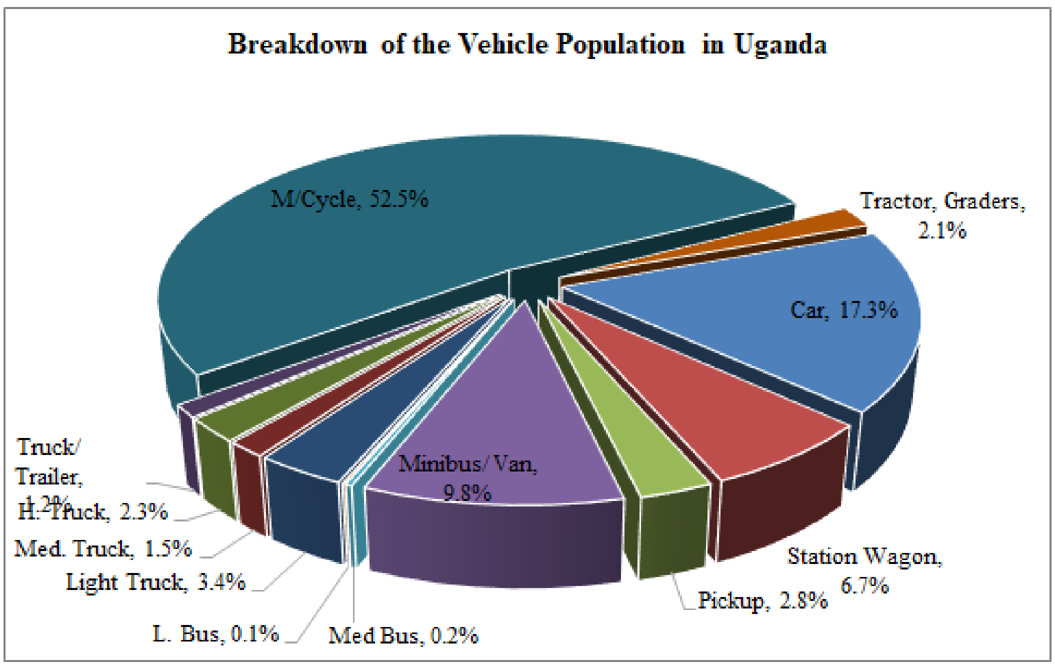
[5] On the depreciation of automobiles: An international comparison (Karl Storchmann, Department of Economics, Yale University) – Published in Transportation journal (issue 31, 2004, pages 371-408)

[6] OECD – Organisation for Economic Co-operation and Development

ESTIMATED RESIDUAL VEHICLE POPULATION IN UGANDA (ADJUSTED FOR SCRAPPAGE) 2005 - 2019													
Year	Car	Station Wagon	Pickup	Mini bus	Med. Bus	Large Bus	Light Truck	Medium Truck	Heavy Truck	Truck/ Trailer	M/ Cycle	Tractor Grader	Total Vehicles
2015	139,607	50,298	21,078	103,033	1,614	1,015	21,937	8,634	14,239	12,443	560,900	12,652	947,449
2016	167,092	61,536	25,422	115,346	2,045	1,228	27,620	10,665	18,984	14,154	621,365	16,804	1,082,258
2017	202,642	76,522	30,640	130,690	2,638	1,477	34,807	14,544	25,368	16,232	693,929	22,035	1,251,523
2018	250,511	96,195	38,921	151,324	3,429	1,792	46,275	19,936	32,307	18,618	800,722	28,615	1,488,645
2019	308,349	119,285	49,779	174,875	4,355	2,185	60,772	26,675	40,215	21,356	934,778	36,745	1,779,369
Proportion	**17.3%**	**6.7%**	**2.8%**	**9.8%**	**0.2%**	**0.1%**	**3.4%**	**1.5%**	**2.3%**	**1.2%**	**52.5%**	**2.1%**	
Growth, %	**11.1%**	**8.5%**	**8.0%**	**13.6%**	**15.1%**	**13.6%**	**14.7%**	**15.2%**	**17.1%**	**11.4%**	**15.6%**	**15.0%**	**13.6%**

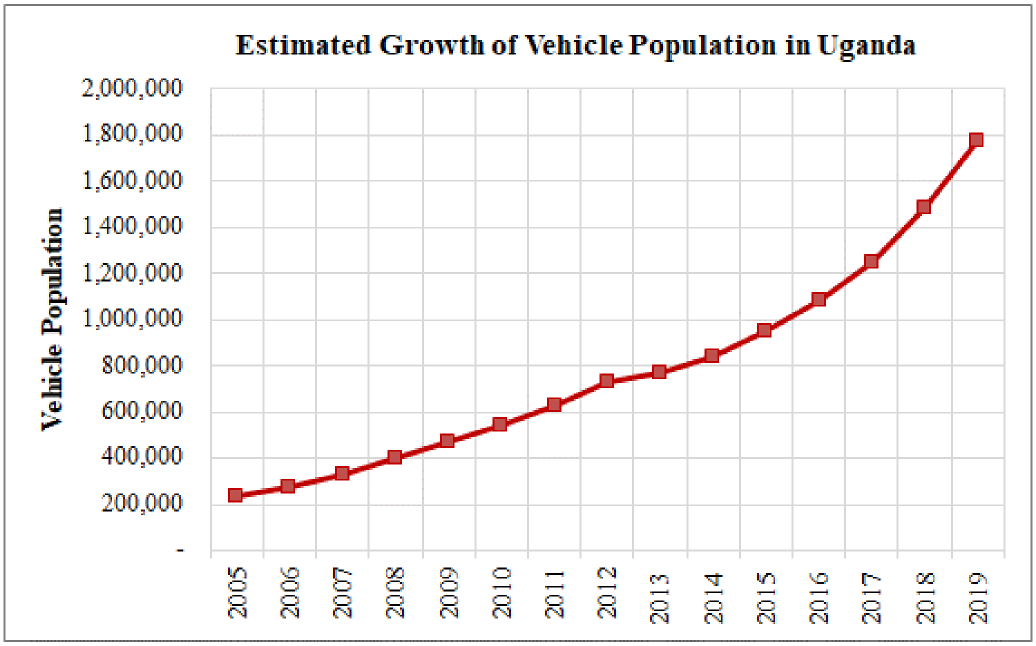
Table 3-9 shows that the estimated total vehicle population in Uganda was 1,844,078. The largest proportion of vehicles is motor cycles with 982,321 units or 52.5%. These are followed by car vehicles and minibus with 17.3% and 9.8% respectively. Large buses have the lowest proportion with 0.1%. Figure 3-13 confirms the breakdown of the vehicle populations.

Figure 3-13: Breakdown of the vehicle population in Uganda



With regard to growth, all vehicle types exhibit significant growth over the 15-year period 2005-2009. This is not surprising considering the significant growth the country has experienced. Overall, the stock of vehicles in country is estimated to have grown at 13.6% per year over the last 15 years. Figure 3-14 shows the growth trend of the vehicle stock in the country.

Figure 3-14: Growth of vehicle population in Uganda (2005-2019)



4 ORIGIN – DESTINATION SURVEYS

4.1 Introduction

OD surveys were undertaken at 132 locations countrywide. OD surveys are used to collect information on the origins and destinations on the road network.

The OD surveys were undertaken between 22th July 2019 and 5th November 2019. All vehicles classes were stopped for interview as part of the OD surveys except police, VIP and ambulance vehicles.

4.2 Methodology

4.2.1 Site Selection

The locations of the OD surveys were carefully chosen to optimize the coverage of driver origins and destinations.

The OD survey sites were also selected to maximize the safety of survey personnel and travelling public, and to minimize traffic disruptions and delays to traffic especially during the peak travel periods.

4.2.2 Survey Personnel and Training

The following categories of survey staff were used at all survey sites:

- Survey manager with overall responsibility for the survey programme;
- Site supervisors with overall responsibility for the operations of each of the survey sites;
- Four to ten interviewers depending on the level of traffic on the road and the required survey sample; and
- Two traffic officers per site.

Survey staff used for OD surveys in this exercise were experienced, but were nonetheless trained prior to the survey. The training covered a range of issues including; details about the survey, the duties involved, roles and responsibilities, the associated health and safety requirements, interview procedure and detailed instructions on how to complete the survey questionnaire.

4.2.3 Site Layouts, Signage and Notices

The site inspection undertaken as part of the survey planning enabled the determination of the actual locations for the survey sites. For safety reasons, all OD survey sites were chosen at areas of the road with adequate shoulder width and length to enable stopping traffic to safely pull off from the road.

Generally, a shoulder width that could accommodate about two truck trailers was considered. The length required for each site was estimated from the following relationship as recommended in the UK Design Manual for Roads and Bridges, Volume 5, Section 1, Part 4 (TA 11/09).

Length of site in metres = (4 + 8 x no. of interviewers). A minimum site length of 45m that would be adequate for 5 interviewers was generally considered.

Appropriate signage was also produced for each survey site. This objective of the signage was to enhance safety at the survey site by warning drivers to reduce speed as they approached the survey site.

4.2.4 Survey Direction and Period

As described in Table 1-1, the OD surveys were undertaken for a total of two (2) days. One (1) day for a 12-hour period (7am – 7pm) and one (1) day for a 24-hour period (7am – 7am). The survey periods sufficiently covered all travel periods.

Surveying was undertaken in both directions except where the OD survey site was located on a very busy urban road. These exception cases were used for 22 sites located on the busy cordon 1 (C1) line. For example, on the busy Bombo Road at Bwaise and other similar roads, it would be impossible to fit a survey site in both directions at the preferred short section between Bwaise and the Northern Bypass. A staggered arrangement was considered, but still in this short section, the risk of exacerbating the already high congestion was very high and Police support for the site would not be obtained. This was the consideration for all 22 sites.

The approach that was taken for the above cases was to interview in one direction on either side of the Northern Bypass. For example, sites 22 and 23, with interviews in a single direction on each site, effectively make up one dual direction survey site for Bombo Road on Cordon 1.

For an urban area were the largest proportion of trips are made daily, interview in a single direction is acceptable. OD data for the opposite direction, where required for model building and other analyses, can be obtained by transposing data from the direction that was surveyed.

4.2.5 Interviews

At the survey sites, enumerators positioned themselves along the survey site length ready to receive stopping traffic. A traffic police officer then waived traffic at random to stop for interview. Once all motorists had been interviewed and departed the officer waived another set of vehicles to stop. This cycle was repeated throughout the day.

4.2.6 Survey Diary

For relatively small surveys (for a few survey stations) and a limited number of survey days, one of the key impacts is inclement weather. For this assignment, this was not considered a key issue. The OD sample was effectively collected for 36 hours (1 day 24 hours, 1 day 12 hours) and in a largely dry season (July to September).

The most significant issue to the OD surveys frequently experienced was the unavailability of police personnel. The following issues were experienced:

- For Cordon 1 and Cordon 2 surveys, Police personnel was secured through the Greater Kampala Metropolitan Police office. The availability of these resources to support the surveys was frequently affected by activities in Greater Kampala such as political events, visiting international dignitaries, etc. The mitigation for this was to schedule the surveys in such as way that a relatively small number of police personnel would be required at any one time. This however led to significant delays to Cordon 1 and 2 surveys.
- For other cordons, the key issue was the ability of the limited local police resources to support the surveys while at the same time undertaking routine enforcement work. The mitigation for this was to request from the Officers in Charge (OCs) to avail ordinary police personnel to support the survey sites. However, even with this measure, some Police stations could not provide personnel for night surveys for example. This was experienced at Station 115.
- Further, for a limited number of stations, personnel could not be provided for night surveys due to insecurity in the regions. For example, the Police could not provide personnel for night surveys for survey sites 116, 124, 126, 129 and 130 (all in the northern region). No night OD surveys were undertaken at these stations.
- The issue of police personnel not being available for the whole survey period was also frequently encountered. For example, many police personnel argued that their working hours were 7am to 5pm. Quite often, Police personnel report at the survey sites between 8am and 9am, and left the sites at 6pm. This obviously shortens the survey period. In other cases, where police resources were limited (upcountry Police stations), the same resources supporting night surveys were also required to undertake duties the following morning at their stations. They would therefore leave the stations before end of the period, typically at 3am to 4am.

The mitigation for the above issues was to collect as much sample as possible with the availability the Police personnel could provide. In a limited few cases and on rural roads, the survey team, helped by appropriate signage, was successful in stopping vehicles for interview especially for morning periods prior to the arrival of police personnel.

It worth stating that even with the above issues, a representative OD sample was obtained generally. For those northern region survey sites where Police personnel could not be deployed for night surveys, surveys on adjacent stations or for survey sites along the same roads up or downstream were considered adequate. For example, for the key survey site 116 which had no night OD surveys, sites 113 and 118 provide adequate usable night data.

4.3 Sampling

The number of interviews (sample size) required to obtain a representative sample of vehicles using the project road is determined chiefly by the current volume of traffic on the road. Once current volume of traffic can be estimated (prior to commencement of surveys), the size of the required sample size can then be estimated.

There is no specific guidance from the Ministry of Works and Transport (MoWT), UNRA or internationally for the required sample rate. The TA 11/09 of the UK DMRB[7] states that "When data is being collected for a large multi-zoned modal, it is impossible to calculate required sample rates for each OD site as the origins and destinations are not known. Once a survey site has been established it is best practice to collect as much data as reasonably practical".

However, traffic engineering experience worldwide indicates that a minimum sample rate of 10% is sufficient for average traffic volume roads. The sample rate may be adjusted for lower or higher volume roads.

For reference, the Consultant has considered the following guidelines reasonable. These have also been used for another countrywide traffic survey commissioned by the MoWT.

The minimum OD sample rates were set as follows:

- 20% for survey links with ADT less than 1,000 vehicles per day;
- 10% for ADT of 1,000 to 4,999;
- 5% for ADT 5,000 to 10,000;
- 2% for ADT greater than 10,000;

During the surveys, vehicles were selected at random in order to have a representative sample with at least the minimum sample rate for all vehicle types. In order to do this, vehicles were directed at interview bays as soon as the bays were free. However, in a few cases, traffic officers seemed more inclined to stop certain vehicle type and not others. For example, significantly less large trucks, truck-trailers and coaches were stopped than minibuses and other small vehicles. To mitigate this, the survey supervisors were asked to be vigilant and encourage the officers to select the traffic at random.

Another aspect that was experienced especially in the higher trafficked areas such as Cordons 1 and 2, was the difficulty of stopping motorcycles (the majority being Boda Boda cyclists). This class of vehicles was less cooperative and the majority did not readily stop when flagged down by traffic police personnel. This resulted in significantly low samples of motorcycles especially in Cordon 1.

[7] UK Design Manual for Roads and Bridges (DMRB), TA 11/09, Annex 8

The above observation notwithstanding, even in less urban areas, with lower traffic levels, sample rates for motorcycles were generally low because trips at any survey site are predominantly made by the same riders i.e. one rider makes multiple trips past the site. Once he has stopped for survey once, he is less cooperative when stopped subsequently.

4.4 Collected OD Data

4.4.1 Cordon 1 OD Sample Rates

Tables 4-1 to 4-4 below compares the collected and expected OD sample for various survey site and cordon lines. Detailed data including vehicle by vehicle and directional disaggregation is included in Appendix C.

Table 4-1: Comparison of collected and required OD sample for Cordon 1 (C1)

COMPARISON OF COLLECTED AND REQUIRED OD SAMPLES – CORDON 1					
Site	**Description**	**Interviewed Sample**	**Average Daily Traffic (ADT)**	**Collected Sample %**	**Required Sample, %**
01	Mobutu Road	1,398	38,339	3.6%	2%
01A	Salama Road	1,643	35,950	4.6%	2%
02	Entebbe Road	1,475	71,465	2.1%	2%
03	Kalinda Rd	668	18,500	3.6%	2%
04	Weraga Road	502	10,615	4.7%	2%
05	Kabusu Rd	860	9,150	9.4%	5%
06	Wankulukuku Rd	284	13,618	2.1%	2%
07	Masaka Rd Ndeeba	1,306	38,993	3.3%	2%
08	Wakaliga Rd North	336	19,200	1.8%	2%
09	Wakaliga Rd South	374	12,757	2.9%	2%
10	Masaka Rd Busega	1,238	53,416	2.3%	2%
11	Masaka Rd Kyengera	1,255	54,407	2.3%	2%
12	Mityana Rd	1,200	43,950	2.7%	2%
13	Mityana-Sentema	726	38,047	1.9%	2%
14	Sentema Rd South	604	11,233	5.4%	2%
15	Sentema Rd North	574	5,385	10.7%	5%
16	Northern Bypass	ATC Only	46,759	-	-
17	Hoima Rd South	938	18,378	5.1%	2%
18	Hoima Rd Nth	1,514	17,936	8.4%	2%
20	Kawaala Rd	448	12,462	3.6%	2%
21	Sir Apollo Kaggwa Road	620	41,929	1.5%	2%
22	Bombo Road South	900	25,668	3.5%	2%
23	Bombo Road North	628	21,501	2.9%	2%
24	Gayaza Rd South	1,105	21,469	5.1%	2%
25	Tula Rd	1,219	21,519	5.7%	2%

COMPARISON OF COLLECTED AND REQUIRED OD SAMPLES – CORDON 1					
Site	**Description**	**Interviewed Sample**	**Average Daily Traffic (ADT)**	**Collected Sample %**	**Required Sample, %**
26	Gayaza Rd North	1,224	15,299	8.0%	2%
28	Kisingiri Street	1,260	19,988	6.3%	2%
29	Kyebando Ring Rd	400	5,788	6.9%	5%
30	Kyebando-Kisalosalo	398	4,977	8.0%	5%
31	Bukoto Kisasi Rd	678	13,740	4.9%	2%
33	Kisasi-Ntinda Rd	661	12,333	5.4%	2%
34	Kisasi-Kyanja Rd	399	14,623	2.7%	2%
36	Mbogo Road	728	27,246	2.7%	2%
37	Naalya Road West	489	14,588	3.4%	2%
38	Naalya Road East	692	13,936	5.0%	2%
39	Namugongo Rd	696	26,592	2.6%	2%
40	Bweyogerere - Nalya	572	15,774	3.6%	2%
41	Namanve Ind Park	MCC Only	6,132	-	-
42	Kinawataka Road	644	32,346	2.0%	2%
43	Kla-Jinja Hway	1,957	63,727	3.1%	2%
44	Kinawataka - Spear Road	798	11,469	7.0%	2%
44A	Ntinda Road	591	11,493	5.1%	2%
45	New Port Bell Rd	905	30,422	3.0%	2%
46	Katalima Road	896	24,965	3.6%	2%
47	Lugogo Bypass	775	42,478	1.8%	2%
48	Old Port Bell Road	630	28,058	2.2%	2%
49	Access Road	1,511	79,682	1.9%	2%
50	Nsambya Rd	1,205	57,503	2.1%	2%

The following observations are with regard to Table 4-1 (OD samples for C1):

- OD surveys were conducted at 49 links for C1;
- Of the 49 survey sites, higher than the minimum sample rate was achieved at 44 sites (90%).
- Five stations did not achieve the minimum sample rates. On average these achieved 1.8%. These sites collected less than adequate motorcycle samples. Nonetheless, the collected sample totals are considered adequate.
- For Stations 3, 4, 5, 6 , 8, 9, 14, 15, 17, 18, 22, 23, 24, 26, 29, 30, 33, 34, 37, 38, 44, 44A (22 stations), OD surveys were undertaken in one direction (see Section 4.2.4) because of lack of adequate survey space. As such, the ADT has been used to calculate sample rates are the one direction (as appropriate) totals.
- Overall, nearly 40,000 interviews were undertaken in Cordon 1. This is a significant sample that is adequate for modelling and other analyses.

4.4.2 Cordon 2 Sample Rates

Table 4-2: Comparison of collected and required OD sample for Cordon 2 (C2)

COMPARISON OF COLLECTED AND REQUIRED OD SAMPLES – CORDON 2					
Site	**Description**	**Interviewed Sample**	**Average Daily Traffic (ADT)**	**Collected Sample %**	**Required Sample, %**
51	Salama Road	487	14,600	3.3%	2%
52	Wavamunno Rd	783	16,004	4.9%	2%
53	Busabala Road	240	2,054	11.7%	10%
55	Lubugumu Road	441	1,864	23.7%	10%
56	Munyonyo Spur	768	10,713	7.2%	2%
57	Kigo Road	228	3,435	6.6%	10%
58	Entebbe Road	587	18,588	3.2%	2%
59	Kampala Entebbe Expressway	833	8,812	9.5%	5%
59A	Budo Road	554	7,330	7.6%	5%
60	Masaka Rd	739	16,632	4.4%	2%
61	Nsangi-Buloba Road	*Road was closed during survey period due to heavy rains*			
62	Mityana Road	411	9,218	4.5%	5%
63	Buloba-Bukasa Rd	599	3,494	17.1%	10%
64	Bukasa-Wakiso Road	459	3,907	11.7%	10%
65	Hoima Road	877	2,174	40.3%	10%
66	Wakiso-Matugga Rd	768	7,210	10.7%	5%
67	Bombo Road	862	22,330	3.9%	2%
68	Matugga-Kasangati Road	485	9,409	5.2%	5%
69	Gayaza Road	1123	31,561	3.6%	2%
70	Kasangati-Kira Rd	717	16,632	4.3%	2%
71	Kira-Bulindo Rd	634	20,898	3.0%	2%
72	Kira-Kyaliwajjala Rd	883	21,193	4.2%	2%
73	Sonde Road	529	14,984	3.5%	2%
74	Kampala -Jinja Road	852	35,282	2.4%	2%
75	Mukono-Kalagi Rd	630	6,596	9.6%	5%
76	Kla-Jinja Highway	923	28,223	3.3%	2%
77	Katosi Rd	798	17,091	4.7%	2%

The following observations are with regard to Table 4-2 (OD samples for C2):

- OD surveys were conducted at 26 links for C2;
- Of the 26 survey sites, 24 (92.3%) achieved higher than the set minimum sample rate. Only site 57 achieved less due to smaller motorcycle sample; and
- Overall, 17,210 interviews were undertaken in Cordon 2. This is considered a significant and adequate sample.

4.4.3 Cordon 3 Sample Rates

Table 4-3: Comparison of collected and required OD sample for Cordon 3 (C3)

COMPARISON OF COLLECTED AND REQUIRED OD SAMPLES – CORDON 3					
Site	**Description**	**Interviewed Sample**	**Average Daily Traffic (ADT)**	**Collected Sample %**	**Required Sample, %**
78	Masaka - Mutukula	848	8,307	10.2%	5%
79	Masaka - Mbarara	748	5,695	13.1%	5%
80	Villa Rd	436	1,678	26.0%	10%
81	Sembabule-Kabulasoke	392	1,558	25.2%	10%
82	Kanoni-Maddu	396	2,023	19.6%	10%
83	Kanoni-Maddu	274	1,696	16.2%	10%
84	Mityana Road	619	8,714	7.1%	5%
84 A	Mityana-Busunju	292	1,847	15.8%	10%
85	Hoima Rd	1,235	2,432	50.8%	10%
86	Semuto Rd	721	3,610	20.0%	10%
87	Kampala-Gulu Highway	510	14,806	3.4%	2%
88	Wobulenzi-Zirobwe	271	2,659	10.2%	10%
89	Galiraya-Kayunga	378	917	41.2%	20%
90	Kangulumira	1,661	5,462	30.4%	10%
91	Kampala-Jinja Highway	1,008	13,691	7.4%	2%
92	Nalubaale Rd	1,258	3,543	35.5%	10%
93	Jinja - Kamuli Rd	907	8,286	10.9%	5%
93 A	Jinja - Musita	*MCC survey only, no OD*			
94	Jinja - Kaliro Rd	282	10,969	2.6%	2%
94A	Musita-Iganga	*MCC survey only, no OD*			
95	Tirinyi Road	1,819	4,118	44.2%	10%
96	Bugiri	616	6,057	10.2%	5%
97	Musita-Lumino	471	6,661	7.1%	5%
98	Busia Rd - Tororo	*MCC survey only, no OD*			
119	Tororo-Mbale	1,971	4,921	40.1%	10%
142	Busia Boarder	649	1,175	55.2%	10%
143	Malaba Boarder	870	4,077	21.3%	10%

The following observations are with regard to Table 4-3 (OD samples for C3):

- OD surveys were conducted at 27 links for C3. On three links (Jinja – Iganga, Musita-Iganga and Busia Rd - Tororo) only MCC surveys were undertaken in accordance to the survey plan;
- Average ADT for all links was 5,013 trips – moderately trafficked;

- All surveyed links had OD samples greater than the set minimum; and
- Overall, 18,632 interviews were conducted at Cordon 3.

4.4.4 Cordon 4 and Cordon 5 Sample Rates

Table 4-4: Comparison of collected and required OD sample for Cordon 4 and 5 (C4 and C5)

COMPARISON OF COLLECTED AND REQUIRED OD SAMPLES – CORDON 4 AND 5					
Site	**Description**	**Interviewed Sample**	**Average Daily Traffic (ADT)**	**Collected Sample %**	**Required Sample, %**
Cordon 4 – Mid Country & Western Region					
99	Masaka-Mutukula	493	2,335	21.1%	10%
100	Mbarara-Ntungamo	318	4,917	6.5%	10%
100 A	Mbarara-Kikagata	361	6,994	5.2%	5%
101	Mbarara-Bushenyi	390	9,675	4.0%	5%
102	Mbarara-Ibanda	341	2,565	13.3%	10%
103	Lyantonde-Kiruhura	272	2,144	12.7%	10%
104	Ishaka-Katunguru	677	7,357	9.2%	5%
105	Bwera - Mpondwe	401	1,918	20.9%	10%
106	Kasese-Fort Portal	611	15,651	3.9%	2%
107	Fort Portal - Kyenjojo	350	1,209	28.9%	10%
108	Kyenjojo-Hoima	303	5,070	6.0%	5%
109	Kateera-Kyankwanzi	*MCC survey only, no OD*			
110	Ngoma-Kapeeka	*MCC survey only, no OD*			
111	Hoima-Masindi	296	4,726	6.3%	10%
111A	Bulisa-Hoima	294	721	40.8%	20%
112	Masindi-Kibangya	649	2,340	27.7%	10%
113	Kigumba-Karuma	972	2,399	40.5%	10%
118	Soroti-Kumi	881	2,343	37.6%	10%
120	Sironko Road	628	5,614	11.2%	5%
133	Mirama Hills Border	287	807	35.6%	10%
134	Katuna Border	392	1,407	27.9%	10%
135	Cyahafi Border	324	4,404	7.4%	5%
136	Bunagana Border	243	2,168	11.2%	10%
137	Mpondwe Border	544	7,580	7.2%	5%
144	Suam Border	275	1,299	21.2%	10%
Cordon 5 – Northern Region					
114	Nebbi-Arua	341	2,576	13.2%	10%
115	Karuma-Pakwach	625	1,357	46.1%	10%
116	Karuma -Gulu	629	11,576	5.4%	2%
117	Lira-Kitgum	335	1,415	23.7%	10%

COMPARISON OF COLLECTED AND REQUIRED OD SAMPLES – CORDON 4 AND 5					
Site	**Description**	**Interviewed Sample**	**Average Daily Traffic (ADT)**	**Collected Sample %**	**Required Sample, %**
121	Soroti-Moroto	146	857	17.0%	20%
122	Matany-Kilak	212	1,154	18.4%	10%
123	Moroto-Kotido	76	348	21.8%	20%
124	Abim - Kotido	131	1,181	11.1%	10%
125	Kotido - Kabong	*MCC survey only, no OD*			
126	Gulu - kitgum	391	1,415	27.6%	10%
127	Arua - Koboko	*MCC survey only, no OD*			
128	Koboko - Moyo	800	1,434	55.8%	10%
129	Atiak - Nimule	392	746	52.5%	20%
130	Kitgum - Atiak	*MCC survey only, no OD*			
131	Kitgum - Madi Opei	*MCC survey only, no OD*			
132	Kabong - Apoka	*MCC survey only, no OD*			
138	Goli Border	278	1,897	14.7%	10%
139	Vurra Border	521	2,518	20.7%	10%
140	Nimule Border	388	1,212	32.0%	10%

The following observations are with regard to Table 4-4 (OD samples for C4 and C5):

- OD surveys were conducted at 23 links for C4 and 15 links for C5;
- Average ADT for all links in C4 was 4,290 trips and for C5, average ADT was 2,019;
- For both C4 and C5, 87% of all surveyed links had a collected sample greater than the required sample. Survey sites which collected less than the minimum sample, collected less motorcycle interviews; and
- Overall, 15,179 interviews were collected.

4.5 Sample Rates by Vehicle Type

4.5.1 Analysis Results

Figures 4-1 to 4-5 present a summary of the average sample rates for each vehicle type captured in the OD data for each Cordon. Analyses for each survey site are provided in Appendix C.

Figure 4-1: Sample rates by vehicle type for Cordon 1

Vehicle Types	OD Sample	ADT	OD Sample, %
Car	329	5462	6.0%
Station Wagon	84	1537	5.5%
Pickups	46	704	6.5%
Minibus, matatu	140	2711	5.2%
Medium bus	11	88	12.2%
Large Bus	5	41	11.7%
Light Truck	79	519	15.2%
Medium Truck	54	299	18.0%
Heavy truck	17	117	14.8%
Truck Trailer	8	126	6.5%
Motor cycles	95	13817	0.7%
Tractors	0	5	7.3%
Average	**868**	**25427**	

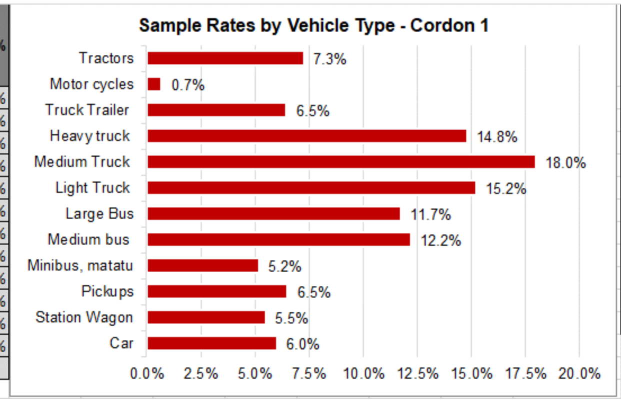


Figure 4-2: Sample rates by vehicle type for Cordon 2

Vehicle Types	OD Sample	ADT	OD Sample, %
Car	259	3,230	8.0%
Station Wagon	82	992	8.2%
Pickups	39	602	6.5%
Minibus, matatu	86	1,674	5.1%
Medium bus	9	85	10.9%
Large Bus	9	49	18.8%
Light Truck	72	660	10.9%
Medium Truck	48	244	19.7%
Heavy truck	14	121	11.8%
Truck Trailer	9	212	4.2%
Motor cycles	35	5,510	0.6%
Tractors	0	5	4.0%
Total	**662**	**13,385**	

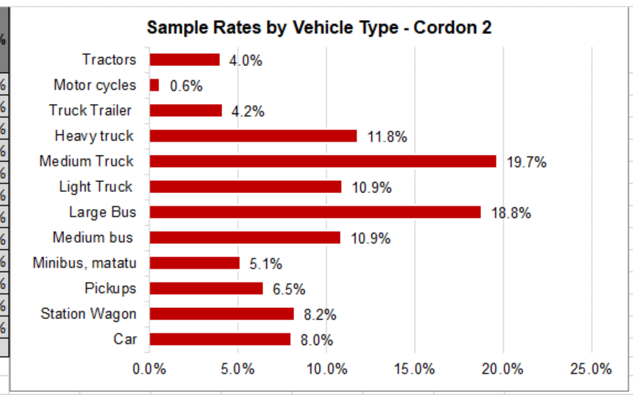


Figure 4-3: Sample rates by vehicle type for Cordon 3

Vehicle Types	OD Sample	ADT	OD Sample, %
Car	173	553	31.4%
Station Wagon	67	159	42.2%
Pickups	86	262	32.8%
Minibus, matatu	69	582	11.9%
Medium bus	14	36	37.8%
Large Bus	16	47	33.6%
Light Truck	71	252	28.1%
Medium Truck	77	151	51.1%
Heavy truck	31	75	41.0%
Truck Trailer	30	181	16.8%
Motor cycles	65	2698	2.4%
Tractors	0	4	11.4%
Total	**700**	**5,001**	

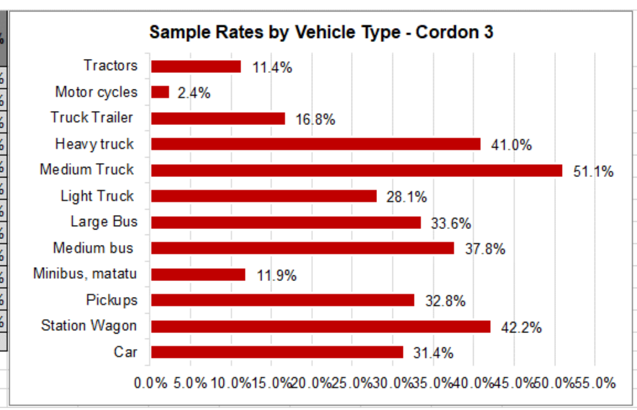


Figure 4-4: Sample rates by vehicle type for Cordon 4

Vehicle Types	OD Sample	ADT	OD Sample, %
Car	124	666	18.6%
Station Wagon	61	154	39.4%
Pickups	55	230	23.9%
Minibus, matatu	54	277	19.3%
Medium bus	6	11	53.3%
Large Bus	12	30	39.5%
Light Truck	55	179	30.8%
Medium Truck	32	87	36.4%
Heavy truck	20	58	34.0%
Truck Trailer	10	83	11.5%
Motor cycles	31	2,804	1.1%
Tractors	1	3	39.3%
Total	**459**	**4,582**	

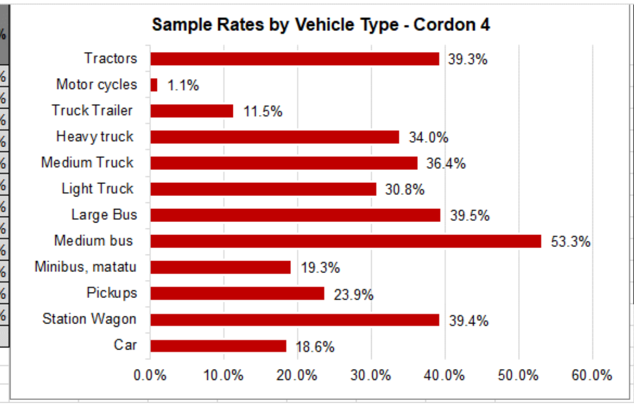
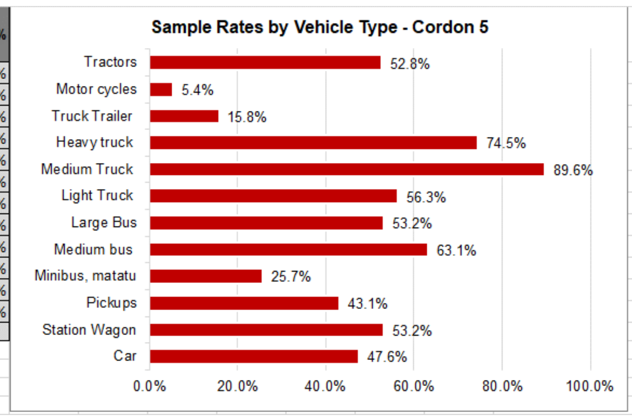


Figure 4-5: Sample rates by vehicle type for Cordon 5

Vehicle Types	OD Sample	ADT	OD Sample, %
Car	95	200	47.6%
Station Wagon	52	98	53.2%
Pickups	79	184	43.1%
Minibus, matatu	25	97	25.7%
Medium bus	6	9	63.1%
Large Bus	14	27	53.2%
Light Truck	64	113	56.3%
Medium Truck	37	41	89.6%
Heavy truck	29	39	74.5%
Truck Trailer	19	119	15.8%
Motor cycles	67	1239	5.4%
Tractors	1	3	52.8%
Total	**489**	**2,169**	



4.5.2 Observations

- Main objective for the above analysis is to ascertain the coverage of the OD surveys with regard to the various vehicle types. It is important to ensure that surveys covered all vehicle types at all or most survey stations.
- Sample rates for each vehicle type are generally low for Cordons 1 and 2 due to the high traffic volumes. Figures 4-1 and 4-2 show that the sample sizes are large enough, but the high average ADT values for the two cordons translate into lower sample rates. Sample rates increase with decreasing ADT in Cordons 3, 4 and 5.
- Sample rates for motorcycles were generally low, especially in Cordons 1 and 2 due to challenges in stopping them for survey.
- Overall conclusions:
    - all vehicle types were represented in the sample OD surveys at all cordon lines.
    - The spread of vehicle proportions was observed to be better at sites with lower traffic volumes (cordons 3, 4 and 5).

- Stopping motorcycles for interview was difficult overall and this explains the low proportions at all cordon lines. They easily pass without stopping even when stopped and their up and down trip making nature means that a rider who has been interviewed once would not be willing to stop again.

4.6 Journey Purpose

4.6.1 Analysis Results

Figures 4-6 to 4-10 present a summary of the average journey purposes across the five cordon lines. Analysis for each survey site is included in Appendix C.

Figure 4-6: Journey purpose analysis for Cordon 1

	Home	Work	Study	Commercial	Tourism	Leisure/Social	Hospital	Others
Home	3.8%	16.3%	1.1%	1.3%	0.1%	0.9%	0.9%	2.8%
Work	14.9%	33.1%	0.7%	2.0%	0.1%	0.4%	0.2%	2.5%
Study	1.2%	0.4%	0.4%	0.1%	0.1%	0.0%	0.0%	0.1%
Commercial	1.1%	1.0%	0.1%	4.1%	0.1%	0.0%	0.0%	0.5%
Tourism	0.1%	0.1%	0.0%	0.0%	0.1%	0.0%	0.0%	0.0%
Leisure/Social	0.5%	0.2%	0.0%	0.1%	0.0%	0.3%	0.0%	0.1%
Hospital	0.9%	0.2%	0.0%	0.0%	0.0%	0.0%	0.2%	0.1%
Others	1.9%	1.2%	0.1%	0.3%	0.0%	0.1%	0.0%	2.8%

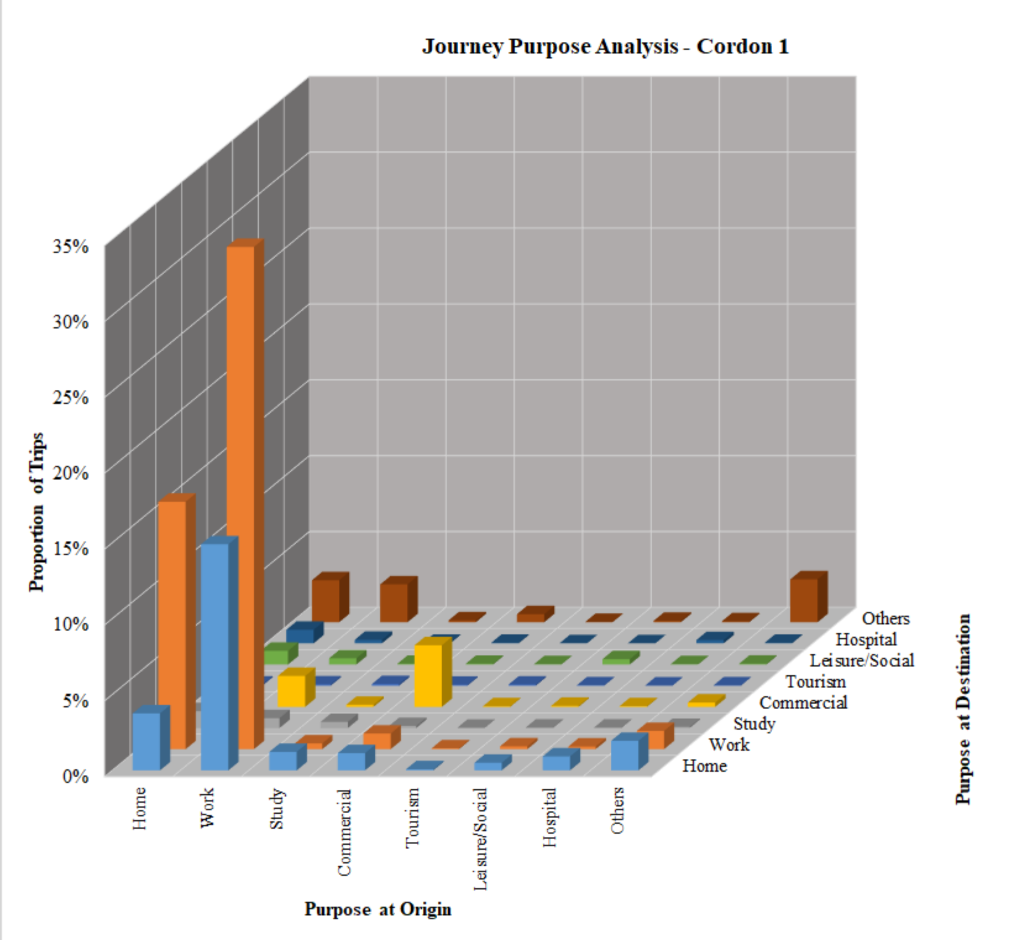


Figure 4-7: Journey purpose analysis for Cordon 2

	Home	Work	Study	Commercial	Tourism	Leisure/Social	Hospital	Others
Home	5.6%	17.1%	1.5%	1.7%	0.1%	2.7%	1.2%	5.3%
Work	16.1%	26.2%	0.3%	1.6%	0.1%	0.4%	0.1%	1.6%
Study	1.5%	0.2%	0.4%	0.1%	0.0%	0.0%	0.0%	0.1%
Commercial	1.4%	0.7%	0.0%	3.8%	0.0%	0.1%	0.0%	0.3%
Tourism	0.1%	0.0%	0.0%	0.0%	0.1%	0.0%	0.0%	0.0%
Leisure/Social	1.2%	0.2%	0.0%	0.0%	0.0%	0.3%	0.0%	0.1%
Hospital	0.7%	0.1%	0.0%	0.0%	0.0%	0.0%	0.1%	0.0%
Others	2.9%	1.1%	0.1%	0.2%	0.1%	0.1%	0.0%	2.0%

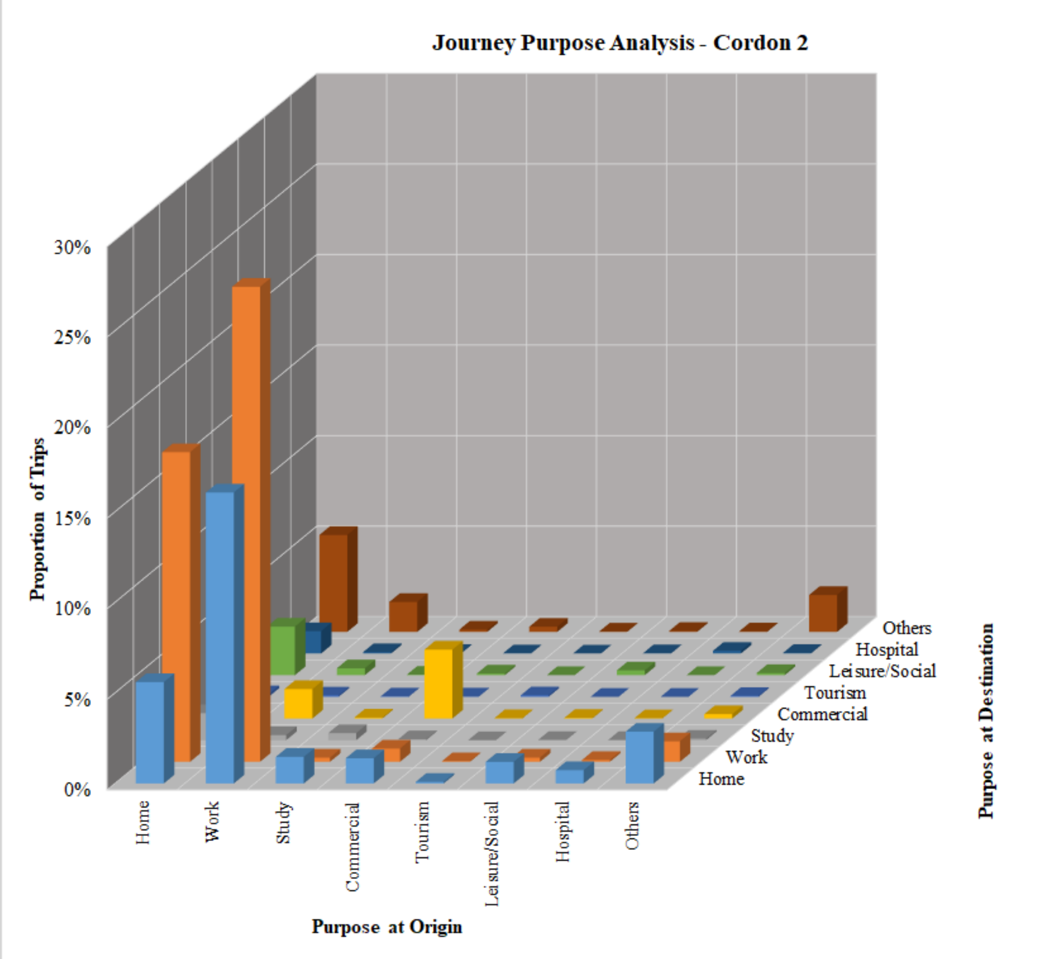


Figure 4-8: Journey purpose analysis for Cordon 3

	Home	Work	Study	Commercial	Tourism	Leisure/Social	Hospital	Others
Home	4.3%	13.7%	0.5%	2.8%	0.2%	2.8%	1.0%	3.1%
Work	10.6%	40.6%	0.2%	2.4%	0.1%	0.5%	0.1%	1.0%
Study	0.5%	0.1%	0.2%	0.0%	0.0%	0.0%	0.0%	0.1%
Commercial	1.5%	1.4%	0.1%	6.0%	0.0%	0.1%	0.0%	0.2%
Tourism	0.1%	0.0%	0.0%	0.0%	0.1%	0.0%	0.0%	0.0%
Leisure/Social	0.8%	0.1%	0.0%	0.1%	0.0%	0.2%	0.0%	0.0%
Hospital	0.4%	0.1%	0.0%	0.0%	0.0%	0.0%	0.1%	0.0%
Others	1.3%	0.4%	0.0%	0.4%	0.0%	0.1%	0.0%	1.4%

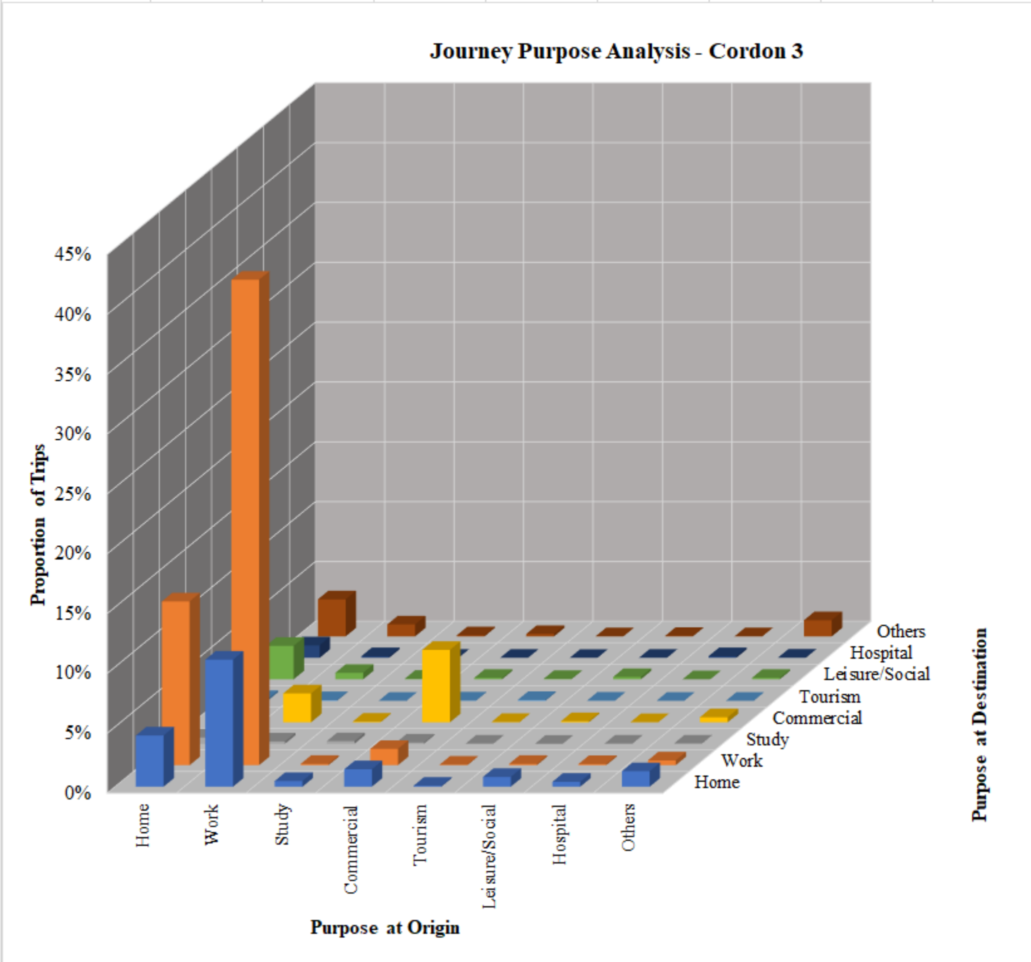


Figure 4-9: Journey purpose analysis for Cordon 4

	Home	Work	Study	Commercial	Tourism	Leisure/Social	Hospital	Others
Home	6.5%	16.3%	1.2%	2.3%	0.4%	1.2%	0.6%	2.4%
Work	10.4%	34.4%	0.7%	2.7%	0.2%	0.4%	0.2%	1.0%
Study	1.0%	0.5%	0.7%	0.1%	0.1%	0.1%	0.0%	0.0%
Commercial	2.0%	1.2%	0.2%	6.5%	0.1%	0.0%	0.0%	0.2%
Tourism	0.3%	0.2%	0.0%	0.0%	0.2%	0.0%	0.0%	0.0%
Leisure/Social	0.8%	0.1%	0.0%	0.1%	0.0%	0.2%	0.0%	0.0%
Hospital	0.6%	0.1%	0.0%	0.0%	0.0%	0.0%	0.1%	0.0%
Others	1.1%	0.7%	0.1%	0.1%	0.1%	0.0%	0.0%	1.3%

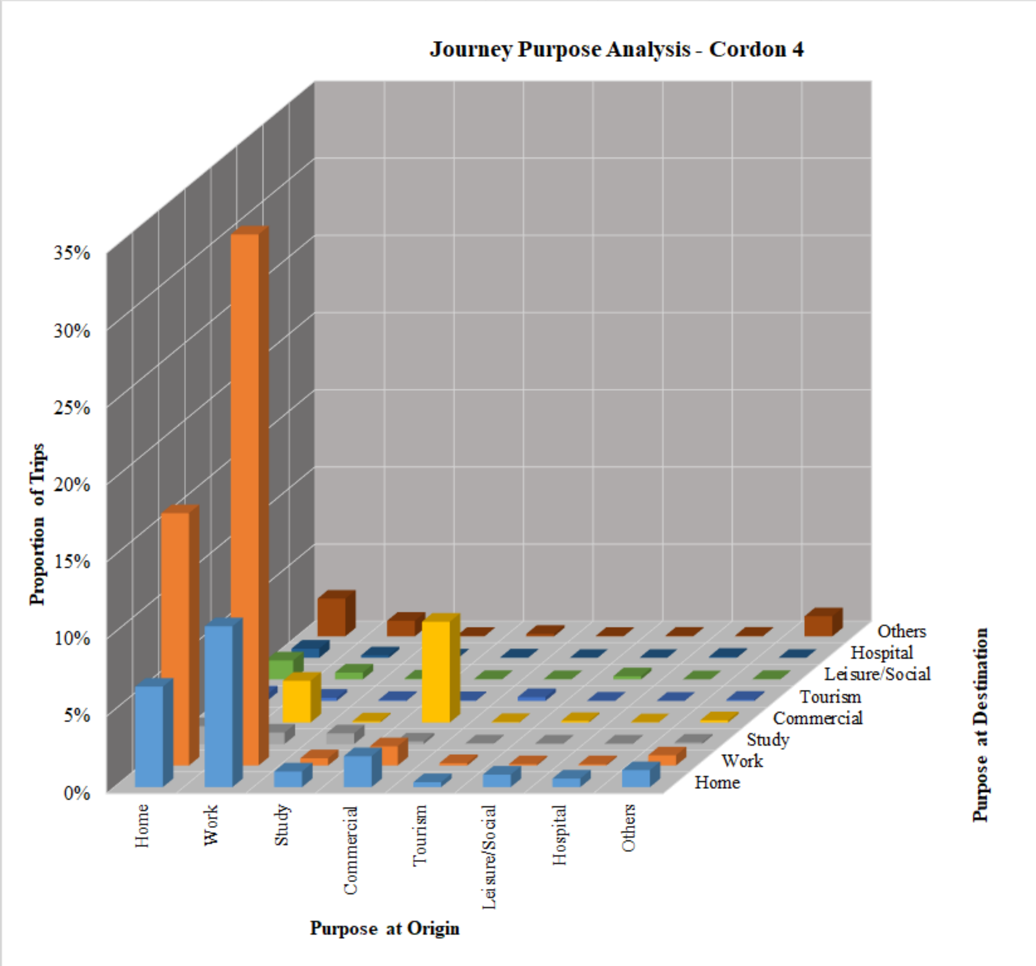
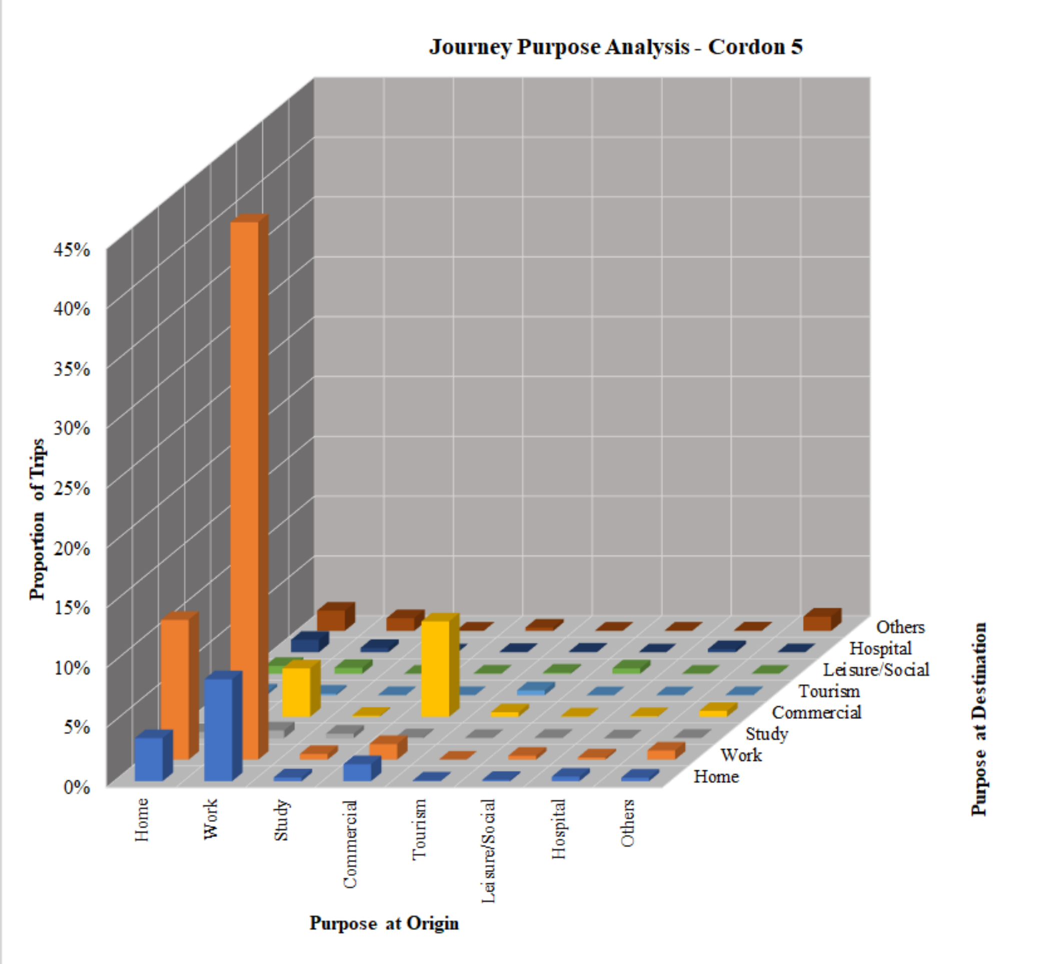


Figure 4-10: Journey purpose analysis for Cordon 5

	Home	Work	Study	Commercial	Tourism	Leisure/Social	Hospital	Others
Home	3.6%	11.7%	0.5%	2.2%	0.2%	0.7%	1.0%	1.7%
Work	8.5%	44.9%	0.6%	4.0%	0.1%	0.5%	0.4%	1.1%
Study	0.3%	0.5%	0.4%	0.1%	0.0%	0.0%	0.0%	0.0%
Commercial	1.4%	1.3%	0.1%	8.0%	0.0%	0.0%	0.0%	0.3%
Tourism	0.1%	0.0%	0.0%	0.4%	0.4%	0.1%	0.1%	0.0%
Leisure/Social	0.1%	0.3%	0.0%	0.0%	0.0%	0.4%	0.0%	0.0%
Hospital	0.4%	0.2%	0.0%	0.1%	0.0%	0.0%	0.3%	0.0%
Others	0.3%	0.8%	0.0%	0.5%	0.0%	0.0%	0.0%	1.2%



4.6.2 Observations

- The main objective of this analysis is to identify the various journey purposes observed in the OD surveys. The identification of the different journey purposes and their proportions helps to estimate the right influences for travel across the network.

- The dominant journey purposes are home-work, work-home and work-work. Home-home and commercial-commercial were also seen to be significant. These five journey purposes together account for 70% to 80% of all trips at all cordons.
- Tourism related trips account for 1.7% and 1.5% of all trips at Cordons 4 and 5 respectively. At cordons 1, 2 and 3, the average share for tourism related trips is 0.6%.
- Trips related to "Other" i.e. those that fall outside the seven main journey purposes are highest in Cordons 1 and 2, accounting for 12.5% and 14% respectively. For Cordons 3, 4 and 5, the average share is 7%.

4.7 Average Trip Length

4.7.1 Analysis Results

Figures 4-11 to 4-15 present a summary of the average trip length across the five cordon lines. Analysis for each survey site is included in Appendix C.

Figure 4-11: Average trip length for Cordon 1

Trip Length Range (Km)	Cumm-ulative	No of Vehicles	Percentage
0-1	42	42	7.3%
1-2.5	87	45	7.9%
2.5-5.0	217	130	22.6%
5.0-10.0	361	144	25.0%
10.0-20.0	464	103	17.9%
20.0-50.0	525	61	10.6%
50.0-100.0	544	20	3.4%
100.0-250.0	564	20	3.4%
Over 250	575	11	2.0%

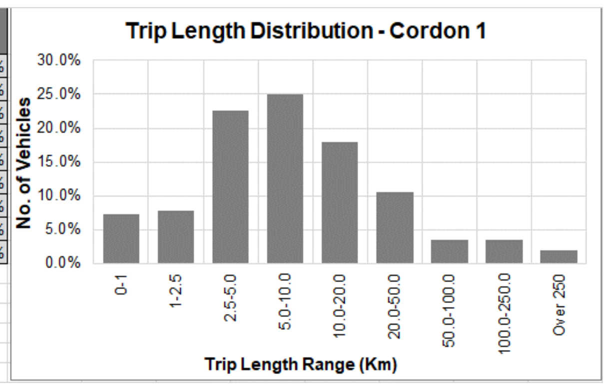


Figure 4-12: Average trip length for Cordon 2

Trip Length Range (Km)	Cumm-ulative	No of Vehicles	Percentage
0-1	12	12	4.2%
1-2.5	32	20	6.8%
2.5-5.0	75	43	14.7%
5.0-10.0	121	46	15.8%
10.0-20.0	172	51	17.3%
20.0-50.0	229	57	19.6%
50.0-100.0	257	28	9.5%
100.0-250.0	277	20	6.7%
Over 250	293	16	5.4%

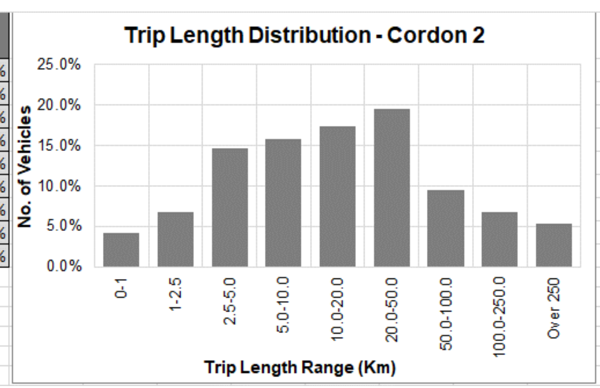


Figure 4-13: Average trip length for Cordon 3

Trip Length Range (Km)	Cumm-ulative	No of Vehicles	Percentage
0-1	3	3	2.0%
1-2.5	11	7	4.4%
2.5-5.0	22	12	7.0%
5.0-10.0	36	14	8.2%
10.0-20.0	52	16	9.7%
20.0-50.0	83	32	19.3%
50.0-100.0	114	30	18.3%
100.0-250.0	152	38	23.1%
Over 250	165	13	8.0%

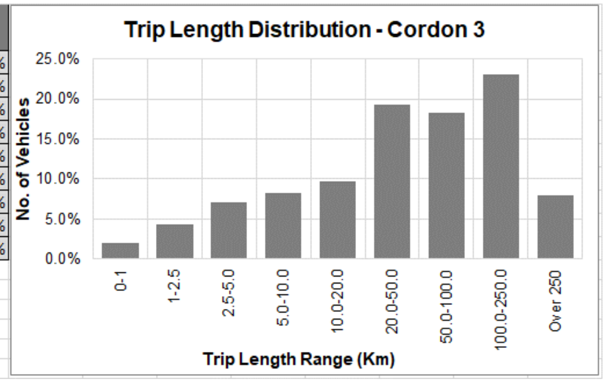


Figure 4-14: Average trip length for Cordon 4

Trip Length Range (Km)	Cumm-ulative	No of Vehicles	Percentage
0-1	15	15	4.5%
1-2.5	37	22	6.6%
2.5-5.0	99	62	18.5%
5.0-10.0	157	59	17.5%
10.0-20.0	201	44	13.1%
20.0-50.0	252	51	15.3%
50.0-100.0	283	31	9.2%
100.0-250.0	319	37	11.0%
Over 250	334	15	4.4%

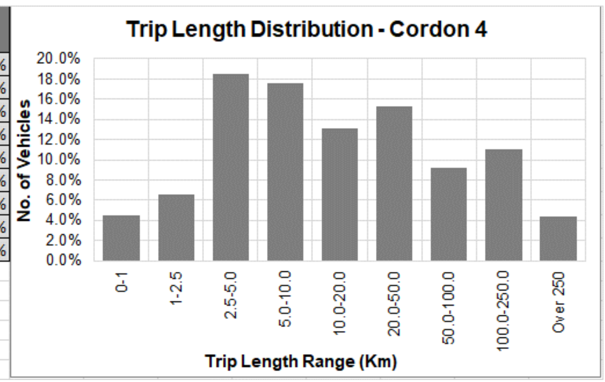
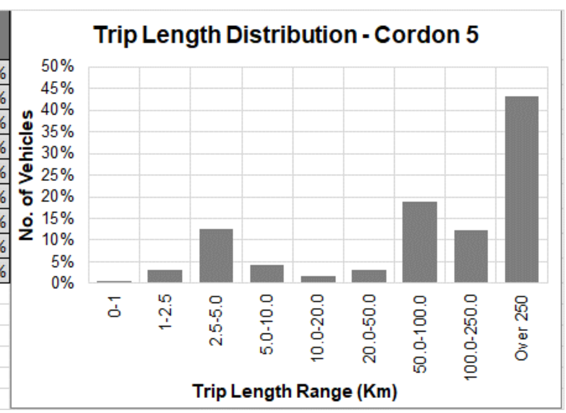


Figure 4-15: Average trip length for Cordon 5

Trip Length Range (Km)	Cumm-ulative	No of Vehicles	Percentage
0-1	2	2	0.7%
1-2.5	11	9	3.2%
2.5-5.0	47	36	12.6%
5.0-10.0	59	12	4.2%
10.0-20.0	64	5	1.8%
20.0-50.0	73	9	3.2%
50.0-100.0	127	54	18.9%
100.0-250.0	162	35	12.3%
Over 250	285	123	43.2%



4.7.2 Observations

- For purposes of analysis of the distribution of trips by length of journey, the trip lengths have been classified as follows; short trips (0 – 20Km), medium trips (21 – 100Km) and long trips (over 100Km). Short trips are typically the distance travelled by an average commuter to Kampala City from the boundaries of the GKMA. The boundaries of the GKMA are 10 – 20Km from central Kampala.

- For C1, Figure 4-11 shows that the majority of the trips are short length trips (80.7%). Medium trips account for 14% of all trips and long trips are 5.4%. This pattern is expected for such an urban area.
- For C2, short trips reduce significantly compared to C1, accounting for 58.8%, medium and long trips take 29.1% and 12.1% respectively.
- At C3, the distribution of the trips is more even with short, medium and long trips accounting for 31.3%, 37.5% and 31.1% respectively.
- Cordon C4 provides an unexpected distribution. Short, medium and long trips account for 60.1%, 12.2% and 15.4% respectively. It would be expected that C4 follows similar trends for C2 and C3 i.e. short trips reduce and long trips dominate as one moves away from Kampala. Further analysis shows that the Western region (which forms the bulk of Cordon 4), 83% of all trips sampled start and end within the region as opposed to C3 where only 36% start and end within the boundaries of C3. The presence of major commercial centres such as Mbarara, Kasese, Hoima, Fort Portal and others seems to influence the local / regional nature of trips in C4. ***This nature of trip making in this area would likely influence expressway use.***
- For C5, the distribution is 22.5%, 22.1% and 55.4% respectively for short, medium and long trips. Comparing C5 and C4 distributions, the absence of strong commercial centres in the Northern region seems to influence trip making in C5 and would influence expressway use.

4.8 Average Journey Time

4.8.1 Analysis Results

Figures 4-16 to 4-20 present a summary of the average journey time for all vehicles across the five cordon lines. Analysis for each survey site is included in Appendix C.

Figure 4-16: Average journey time for Cordon 1

Journey Time Range (Hrs)	Cummu-lative	No of Vehicles	No. of Vehicles, %
0-0.25	129	129	16.9%
0.25-0.50	337	208	27.2%
0.50-0.75	418	81	10.6%
0.75-1.00	592	173	22.7%
1.00-1.50	629	38	4.9%
1.5-2.0	691	62	8.0%
2.0-3.0	724	33	4.3%
3.0-5.0	747	23	3.0%
5.0-7.5	757	10	1.3%
7.5-10	763	6	0.7%
Over 10	765	2	0.3%

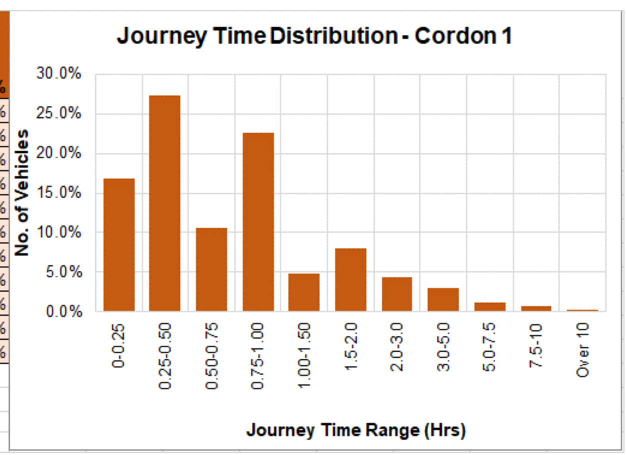


Figure 4-17: Average journey time for Cordon 2

Journey Time Range (Hrs)	Cummu-lative	No of Vehicles	No. of Vehicles, %
0-0.25	93	93	15.7%
0.25-0.50	213	120	20.4%
0.50-0.75	264	52	8.8%
0.75-1.00	390	125	21.3%
1.00-1.50	425	35	6.0%
1.5-2.0	489	64	10.9%
2.0-3.0	530	41	7.0%
3.0-5.0	563	32	5.5%
5.0-7.5	577	14	2.4%
7.5-10	585	9	1.5%
Over 10	589	3	0.6%

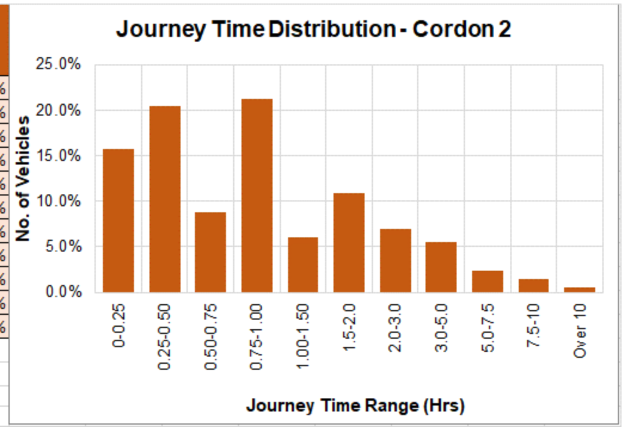


Figure 4-18: Average journey time for Cordon 3

Journey Time Range (Hrs)	Cummu-lative	No of Vehicles	No. of Vehicles, %
0-0.25	49	49	8.9%
0.25-0.50	118	69	12.7%
0.50-0.75	146	29	5.2%
0.75-1.00	236	90	16.5%
1.00-1.50	273	37	6.9%
1.5-2.0	349	76	14.0%
2.0-3.0	424	74	13.7%
3.0-5.0	496	72	13.2%
5.0-7.5	524	28	5.2%
7.5-10	538	13	2.5%
Over 10	544	6	1.1%

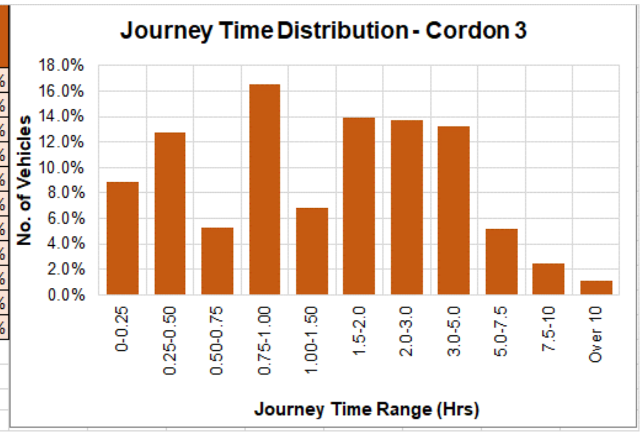


Figure 4-19: Average journey time for Cordon 4

Journey Time Range (Hrs)	Cummu-lative	No of Vehicles	No. of Vehicles, %
0-0.25	74	74	13.8%
0.25-0.50	179	105	19.8%
0.50-0.75	216	37	7.0%
0.75-1.00	310	94	17.7%
1.00-1.50	334	24	4.4%
1.5-2.0	378	44	8.3%
2.0-3.0	422	44	8.3%
3.0-5.0	472	50	9.3%
5.0-7.5	504	33	6.2%
7.5-10	524	19	3.7%
Over 10	532	8	1.5%

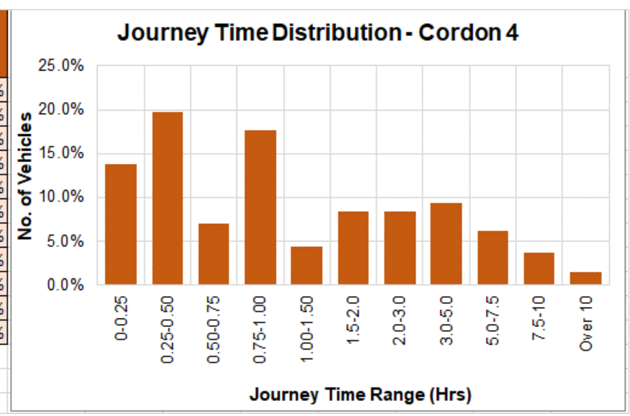
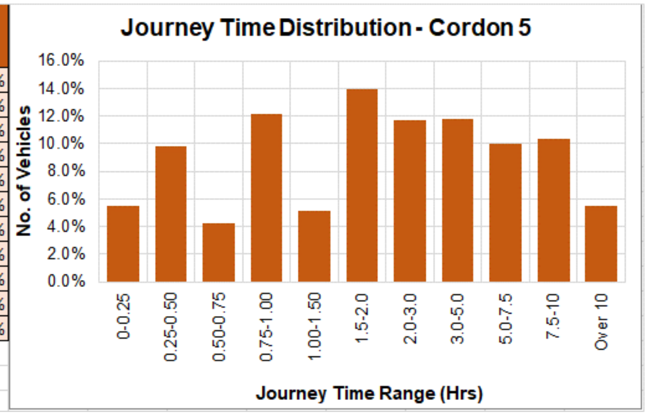


Figure 4-20: Average journey time for Cordon 5

Journey Time Range (Hrs)	Cummu-lative	No of Vehicles	No. of Vehicles, %
0-0.25	17	17	5.5%
0.25-0.50	47	30	9.8%
0.50-0.75	60	13	4.2%
0.75-1.00	98	38	12.1%
1.00-1.50	114	16	5.2%
1.5-2.0	157	43	13.9%
2.0-3.0	193	36	11.7%
3.0-5.0	229	37	11.8%
5.0-7.5	260	31	10.0%
7.5-10	292	32	10.3%
Over 10	309	17	5.5%



4.8.2 Observations

- For purposes of analysis of journey time data, trips have been classified as follows; short trips (up to 1 hour), medium trips (1 – 3 hours) and long trips (over 3 hours). Similar to the analogy of trip length, the average journey time (from data) for trips to Kampala central from the GKMA is one hour. Classifications for medium and long journey trips are based on this.
- For C1, short journeys account for 77.1%, medium and long journeys are 17.3% and 5.3% respectively. This is broadly in line with the trip length distribution.
- For C2, short journeys are 66.2%, medium 23.9% and long 9.9%. Short trips reduce and medium and long trips increase.
- For C3, short trips account for 43.4%, medium 34.5% and long 22.1%.
- The pattern at C4 is similar to the unusual pattern observed in the trip length distribution i.e. short, medium and long trips accounting for 58.3%, 21.1% and 20.6% respectively. Again, this pattern is expected to influence expressway use in the Western region, at least in the short term.
- For C5, short, medium and long trips account for 31.6%, 30.8% and 37.6% respectively. The trip length distribution shows that 55.4% are long distance trips, yet with regard to journey time, only 37.6% are categorised as long trips. This implies that long distance trips captured in C5, do not necessarily take long journey times (i.e. journeys longer than 3 hours). This is expected.

4.9 Trip Frequency

4.9.1 Analysis Results

Figures 4-21 to 4-25 present a summary of the analysis of average trip frequencies for all vehicles across the five cordon lines. Analysis for each survey site is included in Appendix C.

Figure 4-21: Average trip frequency for Cordon 1

Trip Frequency	No of Vehicles	No. of Vehicles, %
More than 2 time	231	27.1%
Daily (Up and Do	366	42.8%
2/3 times a week	94	11.0%
Once a week	60	7.0%
Once a month	28	3.3%
Once in a while	75	8.8%
Total	855	

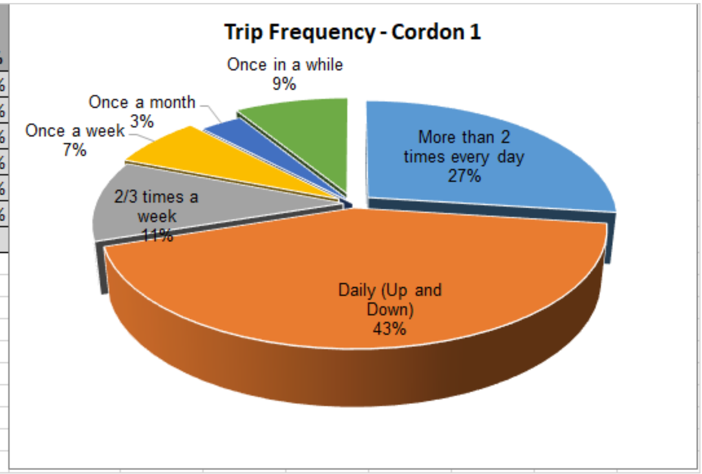


Figure 4-22: Average trip frequency for Cordon 2

Trip Frequency	No of Vehicles	No. of Vehicles, %
More than 2 time	159	24.7%
Daily (Up and D	249	38.8%
2/3 times a wee	93	14.4%
Once a week	50	7.8%
Once a month	19	2.9%
Once in a while	73	11.4%
	643	

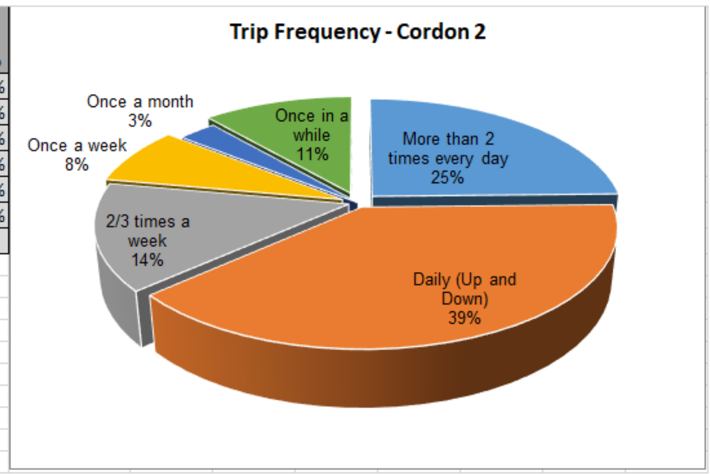


Figure 4-23: Average trip frequency for Cordon 3

Trip Frequency	No of Vehicles	No. of Vehicles, %
More than 2 tim	135	20.6%
Daily (Up and D	207	31.4%
2/3 times a wee	166	25.2%
Once a week	80	12.2%
Once a month	28	4.3%
Once in a while	42	6.4%
Total	658	

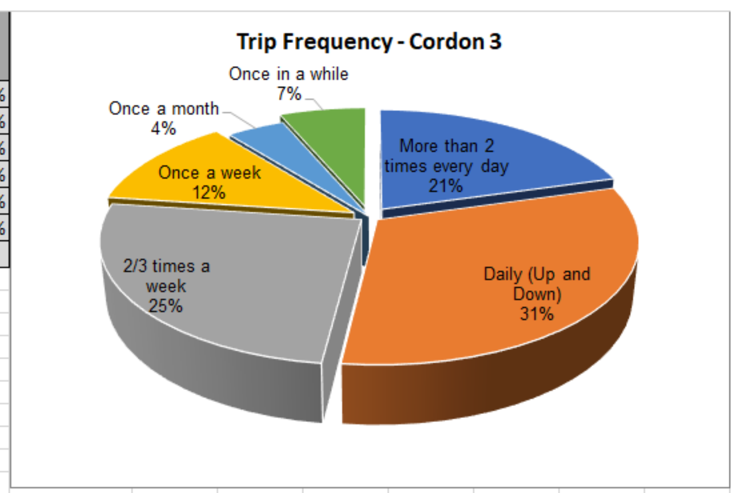


Figure 4-24: Average trip frequency for Cordon 4

Trip Frequency	No of Vehicles	No. of Vehicles, %
More than 2 time	152	26.3%
Daily (Up and Do	195	34.0%
2/3 times a week	104	18.1%
Once a week	70	12.2%
Once a month	20	3.5%
Once in a while	34	5.9%
	575	

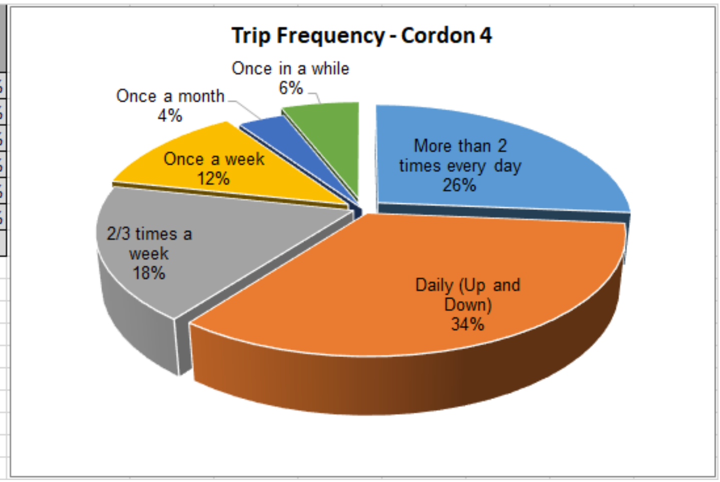
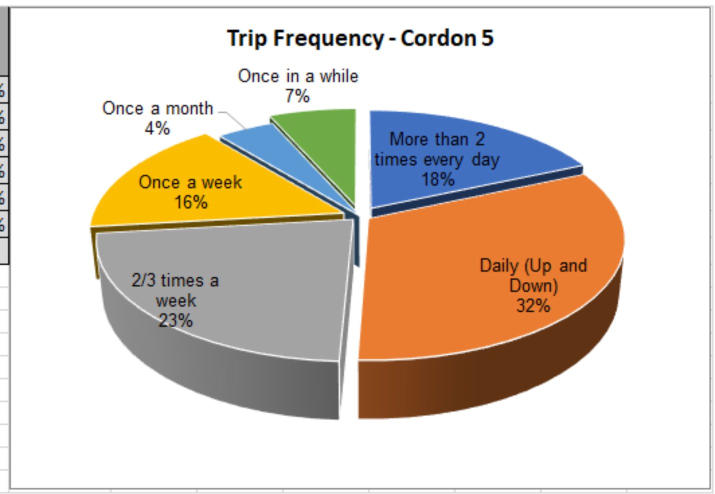


Figure 4-25: Average trip frequency for Cordon 5

Trip Frequency	No of Vehicles	No. of Vehicles, %
More than 2 time	80	18.4%
Daily (Up and Do	141	32.3%
2/3 times a week	98	22.5%
Once a week	70	16.0%
Once a month	19	4.3%
Once in a while	29	6.6%
	437	



4.9.2 Observations

- For C1 (Figure 4-21), the majority of trips run daily (70%), with 43% being “up and down” and 27% running “twice daily”. Weekly trips account for only 18% and the remainder being those that are made occasionally. The high proportion of daily trips is not surprising and is likely heavily influenced by the high proportion of motorcycles and minibuses.
- At C2, the daily trips are somewhat lower at 64% with “up and down” trips accounting for 39% and “twice daily” trips, 25%. Weekly trips account for 22%. The increase in proportions is considered small, mainly because C2 also located in the GKMA is significantly urban in nature and the use of motorcycle and minibuses is significant.
- For C3, the daily trips reduce further to 52% and weekly trips increase to 37%. Occasional trips at C3 account for 10.7%.

- At C4, daily trips are 60% with up and down trips accounting for 34% and "twice daily" trips 26%. Weekly trips account for 30%. The pattern continues to strengthen the observations with trip length and journey time distributions for C4, that the trips observed at C4 are more local and regional compared to C5 for example.
- At C5, daily trips account for 40% and weekly trips 39%. Occasional (once a month and once in a while) trips account for 11%. This is broadly in line with the trip length and journey time distributions.
- Overall, there seems to be a direct correlation between trip length (distance and journey time) and trip frequency. Trips with lower distance and journey time characteristics seem to be associated with high trip frequency as observed at C1 and longer trips associated with lower trip frequencies.

4.10 Future Use of Expressways

As part of the OD surveys, motorists were asked whether they would use the proposed expressways to quicken their journeys. More data in this regard was also collected and has been analyzed as part of the Stated Preference Survey exercise.

The key responses analysed are two-fold; first, whether the respondent would use the available expressways to quicken their journey, and secondly, how much they would be willing to pay to use the expressway.

The questionnaire (included in Appendix A) that was used indicated the average journey time saving with expressway use to help appreciation by respondent. Also, the questionnaire indicated the potential toll charge – this was included to try and get the respondent to measure their responses accordingly. It was considered that this was the best way to ensure sensible results given the lack of experience of tolling in the country currently.

4.10.1 Analysis Results

Figures 4-26 to 4-30 present a summary of the outcomes for all vehicles across the five cordon lines. Analysis for each survey site is included in Appendix C.

Figure 4-26: Potential use of expressways and willingness to pay toll charges for Cordon 1

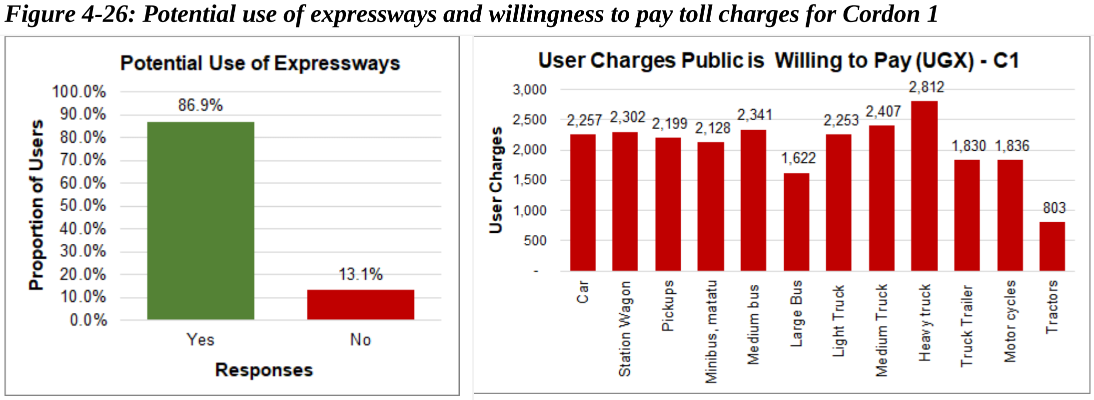


Figure 4-27: Potential use of expressways and willingness to pay toll charges for Cordon 2

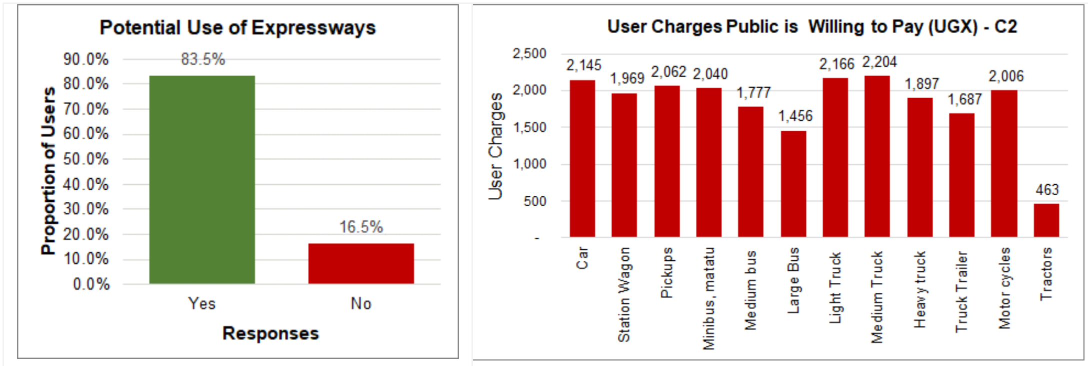


Figure 4-28: Potential use of expressways and willingness to pay toll charges for Cordon 3

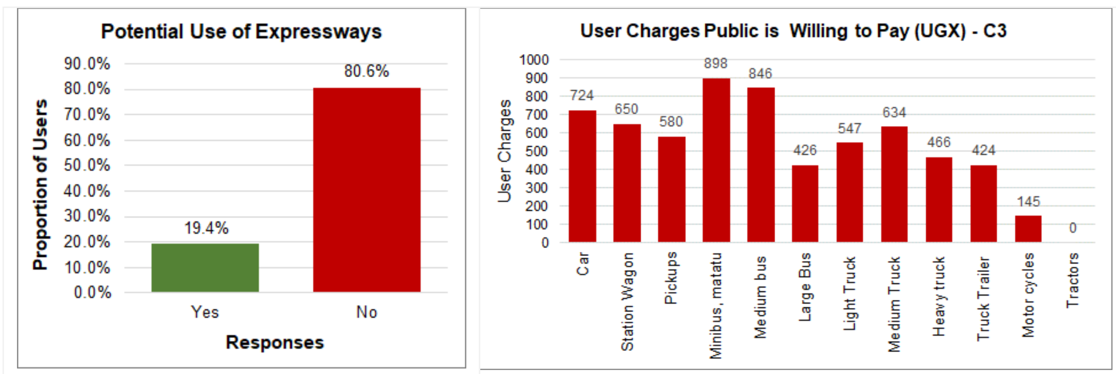


Figure 4-29: Potential use of expressways and willingness to pay toll charges for Cordon 4

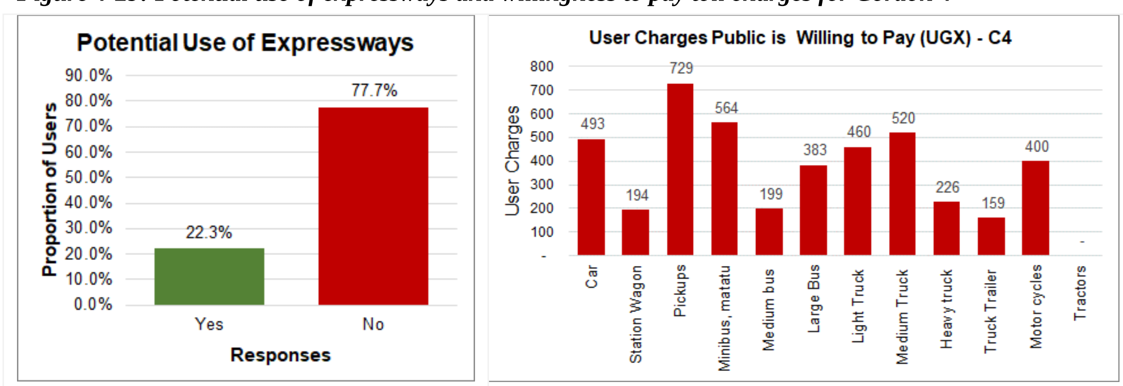
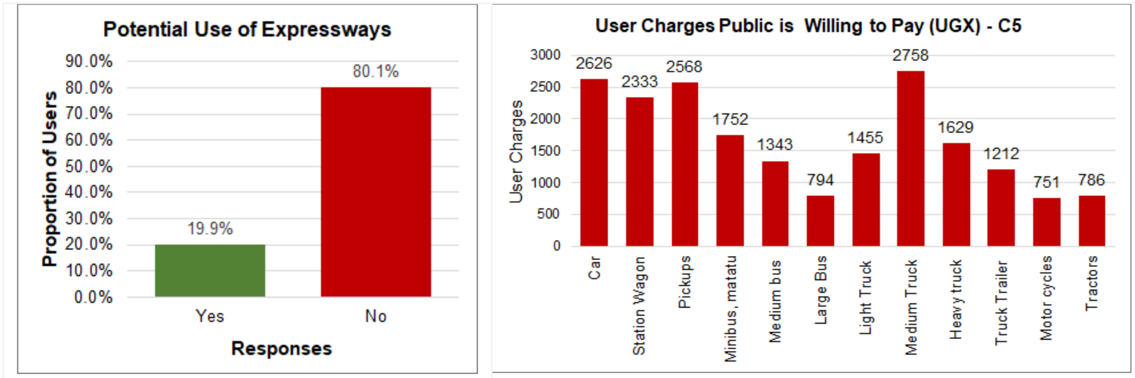


Figure 4-30: Potential use of expressways and willingness to pay toll charges for Cordon 5



4.10.2 Observations

- From Figure 4-26 (C1), 87% of all respondents indicated that they would use an available expressway to ease their journey.

  Those that indicated use of expressways also indicated that they would be willing to pay toll charges for the use of the expressway. Figure 3-30 shows that the charges they are willing to pay are relatively low; cars, station wagons, pickups, mini and medium buses paying UGX2,250 to UGX2,500. Willingness to pay for larger vehicles does not match the expectation that they would pay more than smaller vehicles. This is likely to be the drivers' view and not the company (vehicle owner's) view.
- For C2, willingness to use the expressway drops to 83.5% and toll charges that would be paid reduce to an average of UGX2,000 for the first five vehicle types. Charges for larger vehicles also reduce.
- At C3, the willingness to use expressways reverses, with only 19.4% responding in the affirmative and 80.6% negative. The main reason is likely to be that drivers at this point on the road network do not see a reason for using the expressways because of the less congestion. For those who would use the expressway network, the toll charges they would be willing to pay are also lower; on average UGX750 for the first five vehicle types.
- At C4, responses to the use of the expressway network are similar to C3. Toll charges that would be paid are also low; on average UGX500 for the first five vehicle types.`
- At C5, the potential use of the expressway network is 20%. However, for those willing to use the expressways, the toll charges that they are willing to pay are higher; on average UGX2,500 for car, station wagon and pickups. The analysis of the data also indicates evidence of willingness to pay higher tolls for trucks. The data also shows a recognition that longer distances (from C5 to Kampala) would incur higher tolls and does not support a "once for all" payment.
- In conclusion, willingness to pay data provides some indication of the likely acceptable toll charges especially for smaller vehicles.

- For larger vehicles, data from extended surveys covering vehicle owners (such as from Stated Preference exercises) would likely provide better results.
- Generally, drivers on the more congested roads (Cordons 1 and 2) showed an appreciation of the use of the expressway, with the large majority showing interest.
- Willingness to pay was high for Cordons 1 and 2. However, what would be paid was generally on the low side. With currently no operating tolled expressway, there is likely to be a generally low appreciation of expressway use in the country. This would have influenced the responses.
- Further detailed studies in this area are required, especially when the Kampala Entebbe Expressway starts toll operations. Data from this operation can be used to help streamline future expressway operations.

4.11 Nationwide OD Matrix

The ToR required the creation of a nationwide OD matrix by vehicle type. From the OD survey data, nationwide OD matrices have been created for each identified vehicle class and for all vehicles.

4.11.1 Vehicle Classes

The identified specific vehicle classes are as follows:

- Class 1 – Cars, utility vehicles and pickups
- Class 2 – Buses (minibuses, medium buses, large buses)
- Class 3 – Motorcycles
- Class 4 – Small trucks
- Class 5 – Medium trucks
- Class 6 – Heavy trucks
- Class 7 – Tractors and construction plant (graders, wheel loaders, etc)

4.11.2 Zone Structure

In order to create the OD matrices, the country was demarcated into zones covering all major domestic locations such as cities and significant towns, sub counties and districts. The zone system within the country was based on the countrywide parish and sub country structure.

In Kampala, the zone structure was based on parishes. That is, the various parishes in Kampala were the zones. Outside Kampala, the structure was based on sub counties. For districts close to Kampala, such Wakiso, Mukono, Mpigi and most of the central region where there is significant trip generation, each sub county was allocated a zone number. For more remote districts, sub counties were aggregated to form zones.

A total of 1,450 zones were identified. The zone structure including the aggregations is included in Appendix C.

4.11.3 OD Matrices

The created OD nationwide OD matrices are included in Appendix C for each of the identified vehicle classes and for all vehicles.

5 JUNCTION TURNING COUNTS (JTC) SURVEYS

5.1 Introduction and Methodology

Similar to MCC, JTC surveys are undertaken by traffic enumerators standing by the survey junction and recording each vehicle by category as it turns. They were undertaken at 5 locations within Kampala. The specific locations of the survey sites are given by the location coordinates presented in Table 2-12. The junctions were:

- Kitgum Junction;
- Clock Tower Junction;
- Kibuye Junction;
- Mulago Junction; and
- Nateete Junction.

All five survey junctions are significantly busy junctions. Again, similar to MCC surveys, errors in JTC data are usually due to unrecorded or unobserved traffic especially at road sections with significant volumes of traffic. The most critical vehicle type at all the junctions is the motorcycle. In order to ensure more robust data, turns that comprised of significant motorcycle movements were allocated a specific enumerator to collect motorcycles data.

5.1.1 Survey Period

As described in Table 1-1, the JTC surveys were undertaken for a total of three (3) days; two weekdays and one Saturday. On each day, the survey was undertaken for a 12-hour period between 7am and 7pm.

5.1.2 Survey Diary

No significant adverse weather was experienced on all survey days. There were no unusual traffic conditions that could affect the data quality at any of the junctions.

5.2 Results and Analysis

Tables 5-1 to 5-5 summarise all JTC data collected during the surveys. The tables present average daily traffic (ADT) for all turning movements at each of the junctions. It should be noted that for roundabout junctions (Mulago and Nateete), typical turning counts cannot be undertaken. For these junctions, only entries and exits on each arm are recorded.

Full survey results are included in Appendix D.

Table 5-1: Summary JTC data for Kitgum Junction (ADT)

SUMMARY JTC DATA FOR KITGUM JUNCTION						
		Access Road	Kampala Road	Yusuf Lule Road	Jinja Road	**TOTAL**
From Arm	Access Road	-	3,603	9,382	13,555	**26,540**
	Kampala Road	3,912	-	1,033	16,535	**21,479**
	Yusuf Lule Road	6,516	585	-	18,945	**26,046**
	Jinja Road	7,312	22,854	25,169	-	**55,334**
	TOTAL	**17,740**	**27,041**	**35,583**	**49,034**	

Table 5-2: Summary JTC data for Clock Tower Junction (ADT)

SUMMARY JTC DATA FOR CLOCK TOWER JUNCTION						
		Shoprite	Nsambya Rd	Queensway	Nakivubo Blue	**TOTAL**
From Arm	Shoprite	-	34,611	45,584	-	**80,195**
	Nsambya Rd	28,549	-	18,914	-	**47,463**
	Queensway	-	-	-	-	**-**
	Nakivubo Blue	64,561	4,749	-	-	**69,310**
	TOTAL	**93,111**	**39,360**	**64,498**	**-**	

Table 5-3: Summary JTC data for Kibuye Junction (ADT)

SUMMARY JTC DATA FOR KIBUYE JUNCTION								
		Kibuye RBT	Entebbe	Ndeeba	Katwe	Queens Way	Makindye	**TOTAL**
From Arm	Kibuye RBT	-	49,696	15,804	50,331	-	27,620	**143,451**
	Entebbe	44,619						**44,619**
	Ndeeba	17,941						**17,941**
	Katwe	25,187						**25,187**
	Queens Way	38,140						**38,140**
	Makindye	31,911						**31,911**
	TOTAL	**157,798**	**49,696**	**15,804**	**50,331**	**-**	**27,620**	

Table 5-4: Summary JTC data for Mulago Junction (ADT)

SUMMARY JTC DATA FOR MULAGO JUNCTION							
		Mulago Jn	Ku Bbiri	Kamwokya	Wandegeya	Yusuf Lule	TOTAL
From Arm	Mulago Jn	-	19,974	24,755	33,016	13,014	**90,759**
	Ku Bbiri	23,018					**23,018**
	Kamwokya	25,763					**25,763**
	Wandegeya	33,133					**33,133**
	Yusuf Lule	19,089					**19,089**

SUMMARY JTC DATA FOR MULAGO JUNCTION							
		Mulago Jn	Ku Bbiri	Kamwokya	Wandegeya	Yusuf Lule	TOTAL
	TOTAL	**101,002**	**19,974**	**24,755**	**33,016**	**13,014**	

Table 5-5: Summary JTC data for Nateete Junction (ADT)

SUMMARY JTC DATA FOR NATEETE JUNCTION						
		Nakawuka	Busega	Wakaliga	Ndeeba	**TOTAL**
From Arm	Nakawuka	-	1,790	4,114	515	**6,418**
	Busega	1,270	-	16,425	17,087	**34,781**
	Wakaliga	3,527	8,315	-	1,634	**13,476**
	Ndeeba	4,028	14,795	956	-	**19,779**
	TOTAL	**8,825**	**24,900**	**21,494**	**19,236**	

6 JOURNEY TIME AND GIS DATA

6.1 Journey Time Surveys

6.1.1 General Approach

Journey time survey data is important at traffic modelling stage to enable the correct calibration of the base traffic model to the existing journey times. Journey time surveys were undertaken over a period of 30 days in August/September 2019. The routes along which the survey was undertaken are presented in Figure 2-11 and Table 2-14. A total of 36 routes were surveyed.

The Moving Observer Method (also referred to as the Floating Vehicle Method) - a widely used method for undertaking this type of survey was employed. In the Moving Observer Method, a survey vehicle containing one driver and an enumerator is driven along the road link under survey, ensuring that it travels at the average speed of the road link on which it is travelling. To avoid spurious data, the driver of the survey vehicle in congested environments, should ignore vehicles moving at significantly faster or slower speeds than other vehicles. The enumerator records the time taken to move from one point to another within the road link.

However, to minimize the time required to survey such long routes, the use of GPS was also employed. Currently, satellite data from Google can be used to estimate journey time and distance. However, GPS data can suffer from inaccuracies due to poor receivers, characteristics of the surrounding landscape, positioning of the satellites as the time of measurement, traffic congestion, and others. As such, GPS data was only used to validate field journey time measurements.

The survey was undertaken in both directions on the selected routes. The road links were subdivided into shorter timing segments to ensure data accuracy. As the vehicle travelled along the link, the enumerator noted the start and finish clock time pertaining to each timing segments. The travel times for each segment were then computed at the end of the survey.

6.1.2 Survey Period

The approach for the survey was as follows:

- Each route and each timing segment was surveyed in each of the three travel periods; AM, IP and PM.
- In each of the periods, three measurements were undertaken i.e. in the AM, three measurements were undertaken, and the same was done for the IP and PM.

6.1.3 Survey Diary

No significant adverse weather was experienced on all survey days. There were no unusual traffic conditions that could affect the data quality at any of the junctions.

6.2 Journey Time Surveys - Results and Analysis

Tables 6-1 to 6-3 below summarise the results of the analysis of the journey time survey data. Detailed analysis is included in Appendix E.

Table 6-1: Journey time analysis results for routes JT1 to JT15

					From Kampala Average Journey Time (min)		Towards Kampala Average Journey Time (min)		From Kampala Average Journey Time/Km (min/Km)		Towards Kampala Average Journey Time/Km (min/Km)	
				Distance (Km)	AM	PM	AM	PM	AM	PM	AM	PM
JT1	Kampala - Jinja	Spear Junctio	Bweyogerere	5.6	16.7	21.0	17.0	15.3	2.98	3.75	3.04	2.74
		Bweyogerere	Mukono	11.0	24.7	37.7	40.3	40.7	2.24	3.42	3.67	3.70
		Mukono	Lugazi	24.0	33.3	36.0	34.7	40.0	1.39	1.50	1.44	1.67
		Lugazi	Jinja Java Hou	37.0	39.0	46.0	36.0	41.0	1.05	1.24	0.97	1.11
		Spear Junctio	Jinja Java Hou	77.6	113.7	140.7	128.0	137.0	1.46	1.81	1.65	1.77
JT2	Jinja - Tororo - Mbale	Jinja Java Hou	Iganga	40.0	45.0	48.3	45.0	48.3	1.13	1.21	1.13	1.21
		Iganga	Bugiri	34.0	41.0	42.0	42.0	41.0	1.21	1.24	1.24	1.21
		Bugiri	Busia	45.0	47.0	47.0	47.0	47.0	1.04	1.04	1.04	1.04
		Busia	Tororo	48.0	52.0	52.0	52.0	52.0	1.08	1.08	1.08	1.08
		Tororo	Mbale	51.0	61.0	61.0	60.0	61.0	1.20	1.20	1.18	1.20
		Jinja Java Hou	Mbale	218.0	246.0	250.3	246.0	249.3	1.13	1.15	1.13	1.14
JT3	Jinja - Kamuli - Bukungu	Jinja	Buwenge	28.0	33.7	33.0	36.0	34.0	1.20	1.18	1.29	1.21
		Buwenge	Kamuli	36.0	36.7	37.0	36.7	38.0	1.02	1.03	1.02	1.06
		Kamuli	Bukungu	68.0	82.7	83.0	82.3	83.0	1.22	1.22	1.21	1.22
		Jinja	Bukungu	132.0	153.0	153.0	155.0	155.0	1.16	1.16	1.17	1.17
JT4	Mukono - Nkokonjeru - Jin	Mukono	Kisoga	15.0	22.3	22.0	21.7	23.0	1.49	1.47	1.44	1.53
		Kisoga	Nkokonjeru	15.0	18.3	18.0	18.7	18.0	1.22	1.20	1.24	1.20
		Nkokonjeru	Buikwe	24.0	33.7	34.0	34.7	34.0	1.40	1.42	1.44	1.42
		Buikwe	New Bridge	28.0	40.3	41.0	41.0	40.0	1.44	1.46	1.46	1.43
		mukono	New Bridge	82.0	114.7	115.0	116.0	115.0	1.40	1.40	1.41	1.40
JT5	Mukono - Kayunga - Jinja	Mukono	Kalagi	19.0	23.7	24.0	25.7	24.0	1.25	1.26	1.35	1.26
		Kalagi	Kayunga	32.0	32.7	33.0	35.7	32.0	1.02	1.03	1.11	1.00
		Kayunga	New Bridge	46.0	44.0	45.0	45.3	43.7	0.96	0.98	0.99	0.95
		Mukono	Jinja	97.0	100.3	102.0	106.7	99.7	1.03	1.05	1.10	1.03
JT6	Iganga - Budaka - Mbale	Iganga	Namutumba	38.0	44.7	45.0	45.7	45.3	1.18	1.18	1.20	1.19
		Namutumba	Budaka	42.0	38.3	38.7	38.0	38.0	0.91	0.92	0.90	0.90
		Budaka	Mbale	28.0	28.7	29.0	28.0	28.0	1.02	1.04	1.00	1.00
		Iganga	Mbale	108.0	111.7	112.7	111.7	111.3	1.03	1.04	1.03	1.03
JT7	Iganga - Pallisa - Kumi	Iganga	Kaliro	35.0	38.7	38.0	40.3	40.0	1.10	1.09	1.15	1.14
		Kaliro	Pallisa	44	58.3	58.0	57.7	58.0	1.33	1.32	1.31	1.32
		Pallisa	Kumi	48	57.3	57.0	57.3	57.7	1.19	1.19	1.19	1.20
		Iganga	Kumi	127.0	154.3	153.0	155.3	155.7	1.22	1.20	1.22	1.23
JT8	Tororo - Mbale - Kumi - Sor	Tororo	Mbale	51	60.0	59.0	58.0	58.7	1.18	1.16	1.14	1.15
		Mbale	Kumi	55	49.0	48.7	48.0	48.7	0.89	0.88	0.87	0.88
		Kumi	Soroti	50	42.0	41.0	42.0	41.0	0.84	0.82	0.84	0.82
		Tororo	Soroti	156.0	151.0	148.7	148.0	148.3	0.97	0.95	0.95	0.95
JT9	Soroti - Katakwi - Moroto	Soroti	Katakwi	52	47.0	40.4	47.0	46.7	0.90	0.78	0.90	0.90
		Katakwi	moroto	118	102.0	102.0	99.0	100.0	0.86	0.86	0.84	0.85
		Soroti	moroto	170.0	149.0	142.4	146.0	146.7	0.88	0.84	0.86	0.86
JT10	Soroti-Lira-Karuma	Soroti	Dokolo	67	57.0	56.7	61.0	60.3	0.85	0.85	0.91	0.90
		Dokolo	Lira	58	60.0	60.7	60.3	59.0	1.03	1.05	1.04	1.02
		Lira	Karuma	78	75.0	74.7	75.0	74.7	0.96	0.96	0.96	0.96
		Soroti	Karuma	203.0	192.0	192.0	196.3	194.0	0.95	0.95	0.97	0.96
JT11	Lira -Kalak- Kitgum	Lira	Kalak	116	126.0	126.3	126.0	128.3	1.09	1.09	1.09	1.11
		kalak	kitgum	123	112.0	110.0	113.3	113.3	0.91	0.89	0.92	0.92
		Lira	kitgum	239.0	238.0	236.3	239.3	241.7	1.00	0.99	1.00	1.01
JT12	Kilak - Abim - Moroto	Kilak	Abim	123	225.7	223.7	234.0	234.0	1.83	1.82	1.90	1.90
		Abim	moroto	198	140.7	140.7	140.3	138.7	0.71	0.71	0.71	0.70
		Kilak	moroto	321.0	366.3	364.3	374.3	372.7	1.14	1.13	1.17	1.16
JT13	Karuma - Pakwach - Arua	Karuma	Pakwach	108	85.0	85.0	84.0	84.0	0.79	0.79	0.78	0.78
		pakwach	Nebbi	52	45.0	46.0	42.0	42.0	0.87	0.88	0.81	0.81
		Nebbi	Arua	77	76.7	75.0	73.7	74.0	1.00	0.97	0.96	0.96
		Karuma	Arua	237.0	206.7	206.0	199.7	200.0	0.87	0.87	0.84	0.84
JT14	Karuma - Gulu - Atiak - Nim	Karuma	Gulu	75	64.3	61.7	63.7	44.0	0.86	0.82	0.85	0.59
		Gulu	Atiak	69	62.0	60.3	60.0	75.3	0.90	0.87	0.87	1.09
		Atiak	Elegu	52	42.0	40.7	42.7	62.0	0.81	0.78	0.82	1.19
		Karuma	Atiak-Nimule	196.0	168.3	162.7	166.3	181.3	0.86	0.83	0.85	0.93
JT15	Moroto - Karenga - Kitgum	Moroto	Kotido	107	114.3	113.0	115.3	114.7	1.07	1.06	1.08	1.07
		Kotido	Kaabong	69	90.0	90.0	90.0	89.3	1.30	1.30	1.30	1.29
		Kaabong	Karenga	72	85.0	86.0	87.3	88.0	1.18	1.19	1.21	1.22
		Karenga	Kitgum	114	138.0	137.3	137.0	133.7	1.21	1.20	1.20	1.17
		Moroto	kitgum	362.0	427.3	426.3	429.7	425.7	1.18	1.18	1.19	1.18

Table 6-2: Journey time analysis results for routes JT16 to JT28

					From Kampala		Towards Kampala		From Kampala		Towards Kampala	
					Average Journey Time (min)		Average Journey Time (min)		Average Journey Time/Km (min/Km)		Average Journey Time/Km (min/Km)	
				Distance (Km)	AM	PM	AM	PM	AM	PM	AM	PM
JT16	Gulu - Kitgum	Gulu	Kitgum	104.0	92.7	91.3	95.7	94.3	0.89	0.88	0.92	0.91
		Gulu	Kitgum	104.0	92.7	91.3	95.7	94.3	0.89	0.88	0.92	0.91
JT17	Arua - Koboko - Moyo	Arua	Koboko	55.0	48.0	48.0	47.3	88.3	0.87	0.87	0.86	1.61
		Koboko	Moyo	103.0	130.0	129.0	129.3	104.7	1.26	1.25	1.26	1.02
		Moyo	Adjumani	41.0	74.7	75.3	75.0	119.3	1.82	1.84	1.83	2.91
		Adjumani	Atiak	47.0	56.7	58.3	57.0	78.7	1.21	1.24	1.21	1.67
		Arua	Atiak	246.0	309.3	310.7	308.7	391.0	1.26	1.26	1.25	1.59
JT18	Northern Bypass - Entebbe (Old)	Busega	Kibuye	7.2	23.0	25.0	19.3	29.0	3.19	3.47	2.69	4.03
		Kibuye	Zzana	5.1	10.3	13.0	25.3	12.0	2.03	2.55	4.97	2.35
		Zzana	Kajjansi	5.8	11.7	11.0	13.7	12.0	2.01	1.90	2.36	2.07
		Kajjansi	Mpala	14.0	21.7	23.0	22.0	28.0	1.55	1.64	1.57	2.00
		Mpala	Entebbe Kitooro	7.9	13.7	13.0	15.0	17.0	1.73	1.65	1.90	2.15
		Busega	Entebbe Kitooro	40.0	80.3	85.0	95.3	98.0	2.01	2.13	2.38	2.45
JT19		Busega	Kajjansi	12.0	12.7	11.0	12.7	11.0	1.06	0.92	1.06	0.92
	Northern Bypass - Entebbe (KEE)	Kajjansi	Mpala	15.0	19.0	12.3	12.7	15.0	1.27	0.82	0.84	1.00
		Mpala	Entebbe Kitooro	7.9	12.0	9.3	15.0	10.0	1.52	1.18	1.90	1.27
		Busega	Entebbe Kitooro	34.9	43.7	32.7	40.3	36.0	1.25	0.94	1.16	1.03
JT20	NBP - Luwero - Nakasongola - Kigumba - Karuma	KNBP	Kawempe	3.4	13.3	15.3	15.0	15.7	3.92	4.51	4.41	4.61
		Kawempe	Matugga	10.0	20.0	21.7	21.3	25.0	2.00	2.17	2.13	2.50
		Matugga	Luweero	45.0	50.0	50.7	52.0	52.0	1.11	1.13	1.16	1.16
		Luweero	Nakasongola	47.0	36.0	36.0	36.0	36.0	0.77	0.77	0.77	0.77
		Nakasongola	Kigumba	93.0	62.7	63.0	64.0	64.3	0.67	0.68	0.69	0.69
		Kigumba	Karuma	59.0	48.7	48.7	50.0	50.0	0.82	0.82	0.85	0.85
		KNBP	Karuma	257.4	230.7	235.3	238.3	243.0	0.90	0.91	0.93	0.94
JT21	NBP - Matugga - Kapeeka	KNBP	Matugga	13.0	32.0	35.7	36.0	36.7	2.46	2.74	2.77	2.82
		Matugga	Kapeeka	44.0	50.7	51.3	49.7	51.3	1.15	1.17	1.13	1.17
		Kapeeka	Luweero	41.0	65.0	65.0	65.0	65.0	1.59	1.59	1.59	1.59
		KNBP	Luweero	98.0	147.7	152.0	150.7	153.0	1.51	1.55	1.54	1.56
JT22	Kapeeka - Ngoma - Masindi	Kapeeka	Ngoma	70.0	86.0	86.0	86.0	86.0	1.23	1.23	1.23	1.23
		Ngoma	Masindi	80.0	93.0	93.0	92.0	92.0	1.16	1.16	1.15	1.15
		Kapeeka	Masindi	150.0	179.0	179.0	178.0	178.0	1.19	1.19	1.19	1.19
JT23	NBP - Kiboga - Hoima	KNBP	Wakiso	10.0	18.3	29.0	19.3	31.0	1.83	2.90	1.93	3.10
		Wakiso	Busunju	38.0	38.0	40.0	41.3	42.0	1.00	1.05	1.09	1.11
		Busunju	Kiboga	69.0	66.0	66.0	66.0	66.0	0.96	0.96	0.96	0.96
		Kiboga	Hoima	79.0	70.0	70.0	71.0	71.0	0.89	0.89	0.90	0.90
		KNBP	Hoima	196.0	192.3	205.0	197.7	210.0	0.98	1.05	1.01	1.07
JT24	NBP - Mityana - Mubende - F/Portal	KNBP	Bujuko	18.0	26.7	37.3	36.0	36.0	1.48	2.07	2.00	2.00
		Bujuko	Mityana	41.0	39.0	38.0	39.3	39.0	0.95	0.93	0.96	0.95
		Mityana	Mubende	80.0	67.0	67.0	69.0	69.0	0.84	0.84	0.86	0.86
		Mubende	Kyenjojo	96.0	74.0	74.0	76.0	76.0	0.77	0.77	0.79	0.79
		Kyenjojo	Fortportal	50.0	47.0	47.0	47.7	48.0	0.94	0.94	0.95	0.96
		KNBP	Fortportal	285.0	253.7	263.3	268.0	268.0	0.89	0.92	0.94	0.94
JT25	Kyenjojo - Hoima - Kigumba	Kyenjojo	Kagadi	54.0	78.0	78.0	76.0	76.0	1.44	1.44	1.41	1.41
		Kagadi	Hoima	92.0	89.0	89.0	86.0	86.0	0.97	0.97	0.93	0.93
		Hoima	Masindi	57.0	78.0	78.0	76.0	76.0	1.37	1.37	1.33	1.33
		Masindi	Kigumba	42.0	51.0	51.0	51.3	51.3	1.21	1.21	1.22	1.22
		Kyenjojo	Kigumba	245.0	296.0	296.0	289.3	289.3	1.21	1.21	1.18	1.18
JT27	Kira - Kasangati - Matugga	Kira	Kasangati	6.6	15.3	18.0	15.7	18.0	2.32	2.73	2.37	2.73
		Kasangati	Matugga	9.6	22.0	22.7	21.7	22.0	2.29	2.36	2.26	2.29
		Matugga	Wakiso	10.0	20.0	19.3	20.3	19.0	2.00	1.93	2.03	1.90
		Wakiso	Buloba	10.0	17.7	18.0	18.0	18.7	1.77	1.80	1.80	1.87
		Buloba	Nsangi	7.4	14.0	15.7	14.3	14.7	1.89	2.12	1.94	1.98
		Kira	Nsangi	43.6	89.0	93.7	90.0	92.3	2.04	2.15	2.06	2.12
JT28	Mityana - Semuto - Busunju	Mityana	Busunju	29.0	38.0	38.0	37.7	37.7	1.31	1.31	1.30	1.30
		Busunju	Semuto	17.0	30.0	30.0	31.0	31.0	1.76	1.76	1.82	1.82
		Semuto	Kalule	27.0	45.0	45.0	44.0	44.0	1.67	1.67	1.63	1.63
		Mityana	Kalule	73.0	113.0	113.0	112.7	112.7	1.55	1.55	1.54	1.54

Table 6-3: Journey time analysis results for routes JT29 to JT36

					From Kampala		Towards Kampala		From Kampala		Towards Kampala	
					Average Journey Time (min)		Average Journey Time (min)		Average Journey Time/Km (min/Km)		Average Journey Time/Km (min/Km)	
				Distance (Km)	AM	PM	AM	PM	AM	PM	AM	PM
JT29	NBP - Mulago - Kitgum JN - Kibuye	KNBP	Kubbiri	1.5	5.0	5.3	4.7	5.0	3.33	3.56	3.11	3.33
		Kubbiri	Mulago RB	1.1	6.0	3.7	3.3	3.3	5.45	3.33	3.03	3.03
		Mulago RB	Kitgum JN	2.7	8.3	10.3	6.7	9.0	3.09	3.83	2.47	3.33
		Kitgum JN	Clock Tower	2.1	6.7	8.3	7.3	9.7	3.17	3.97	3.49	4.60
		Clock Tower	Kibuye	2.7	6.3	10.7	8.7	9.0	2.35	3.95	3.21	3.33
		KNBP	Kibuye	10.1	32.3	38.3	30.7	36.0	3.20	3.80	3.04	3.56
JT30	Kibuye - Busega	Kibuye	Nateete	5.2	12.7	15.0	15.0	23.3	2.44	2.88	2.88	4.49
		Nateete	Busega	1.9	5.7	8.7	12.7	10.7	2.98	4.56	6.67	5.61
		Kibuye	Busega	7.1	18.3	23.7	27.7	34.0	2.58	3.33	3.90	4.79
JT31	Busega - Buwama - Masaka	Busega	Nsangi	10.0	22.0	29.0	25.0	34.0	2.20	2.90	2.50	3.40
		Nsangi	Mpigi	17.0	16.7	18.7	17.3	19.7	0.98	1.10	1.02	1.16
		Mpigi	Lukaya	67.0	66.0	66.3	63.3	69.0	0.99	0.99	0.95	1.03
		Lukaya	Masaka	26.0	28.3	32.3	29.0	31.3	1.09	1.24	1.12	1.21
		Busega	Masaka	120.0	133.0	146.3	134.7	154.0	1.11	1.22	1.12	1.28
JT32	Masaka - Mutukula	Masaka	Kalisizo	30.0	33.7	33.0	34.0	35.0	1.12	1.10	1.13	1.17
		Kalisizo	Mutukula	61.0	61.0	61.0	45.9	46.4	1.00	1.00	0.75	0.76
		Masaka	Mutukula	91.0	94.7	94.0	79.9	81.4	1.04	1.03	0.88	0.89
JT33	Masaka - Lyantonde - Mbarara	Masaka	Lyantonde	74.0	68.7	69.0	72.0	72.0	0.93	0.93	0.97	0.97
		Lyantonde	Mbarara	67.0	64.3	64.0	64.3	65.0	0.96	0.96	0.96	0.97
		Masaka	Mbarara	141.0	133.0	133.0	136.3	137.0	0.94	0.94	0.97	0.97
JT34	Mbarara - Kabale - Katuna	Mbarara	Ntungamo	64.0	68.7	68.7	66.7	66.3	1.07	1.07	1.04	1.04
		Ntungamo	Kabale	78.0	73.3	73.0	69.0	69.0	0.94	0.94	0.88	0.88
		Kabale	Katuna	25.0	28.0	28.0	28.0	28.0	1.12	1.12	1.12	1.12
		Kabale	Kisoro	75.0	101.0	101.0	101.0	101.0	1.35	1.35	1.35	1.35
		Kisoro	Bunagana	12.0	23.0	22.7	23.0	33.7	1.92	1.89	1.92	2.81
		Mbarara	Bunagana	254.0	294.0	293.3	287.7	298.0	1.16	1.15	1.13	1.17
JT35	Mbarara - Ishaka - Kasese	Mbarara	Ishaka	63.0	72.0	72.0	63.0	71.3	1.14	1.14	1.00	1.13
		Ishaka	Kichwamba	45.0	55.0	55.0	45.0	58.3	1.22	1.22	1.00	1.30
		Kichwamba	Kikorongo Jn	29.0	29.0	28.7	29.0	29.0	1.00	0.99	1.00	1.00
		Kikorongo Jn	Kaseese	22.0	22.0	20.0	22.0	19.0	1.00	0.91	1.00	0.86
		Kikorongo Jn	Mpondwe	38.0	38.0	36.0	38.0	35.3	1.00	0.95	1.00	0.93
		Mbarara	Kaseese	197.0	216.0	211.7	197.0	213.0	1.10	1.07	1.00	1.08
JT36	Ishaka - Kihihi - Rukungiri	Ishaka	Rukungirir	71.0	74.0	74.0	71.0	74.0	1.04	1.04	1.00	1.04
		Rukungiri	Ntungamo	71.0	53.0	53.0	51.0	53.0	0.75	0.75	0.72	0.75
		Ishka	Rukungirir	142.0	127.0	127.0	122.0	127.0	0.89	0.89	0.86	0.89

Tables 6-1, 6-2 and 6-3 provide an understanding of the journey time for each of the selected 36 routes. For each of the route, the tables also show the journey time for each of the individual sections on the route. From this segmentation, it is possible to understand the estimated time it takes to travel on each link.

However, absolute journey time values are less informative in trying to understand where actual delays occur. A long section will always result in longer journey time than a short section. Tables 6-1, 6-2 and 6-3 also present the journey time per Km values. These values attempt to measure the journey time for each section regardless of length. A higher journey time per Km means more delay on that link than on the other link.

To understand he journey time analysis, six key arterial routes were selected for a closer analysis. These were:

- JT1 Kampala – Jinja highway;
- JT18 Kampala Northern Bypass (Busega) to Entebbe (Old Entebbe Road)
- JT20 Kampala Northern Bypass – Luwero – Karuma
- JT23 Kampala Northern Bypass – Kiboga – Hoima
- JT24 Kampala Northern Bypass – Mityana – Mubende – Fort Portal
- JT30 Busega – Buwama – Masaka

6.2.1 Journey Time Analysis for JT1 Kampala – Jinja Highway

Figures 6-1 and 6-2 present the journey time analysis for Kampala – Jinja Highway for each of the directions; Kampala – Jinja and Jinja - Kampala. Figure 6-2 presents the journey time per Km for the route.

Figure 6-1: Average journey time by period for JT1 Kampala - Jinja

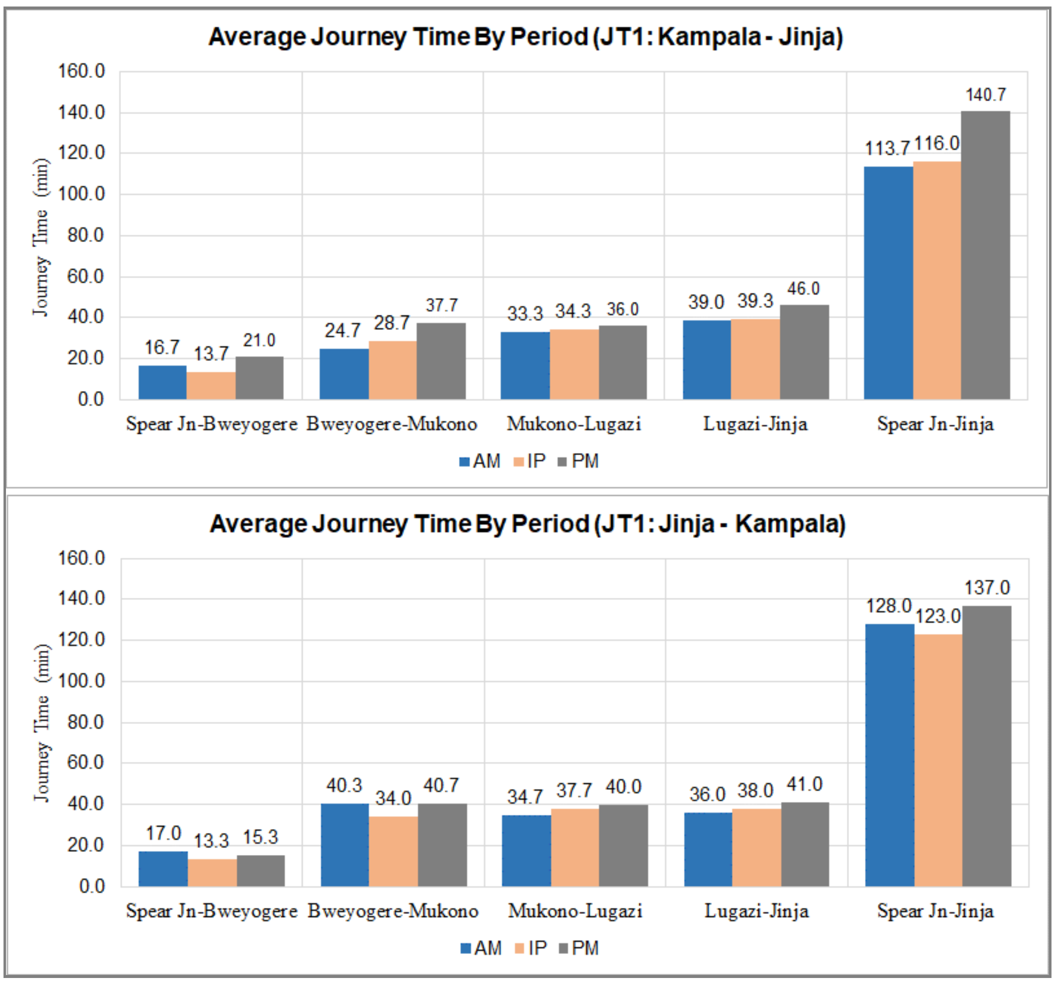
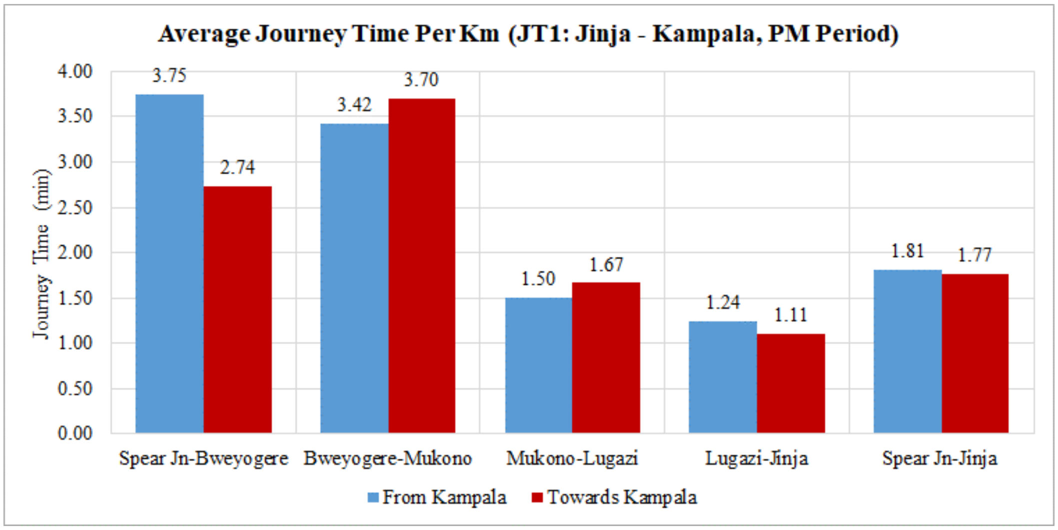


Figure 6-2: Average journey time per Km for PM for JT1 Kampala - Jinja



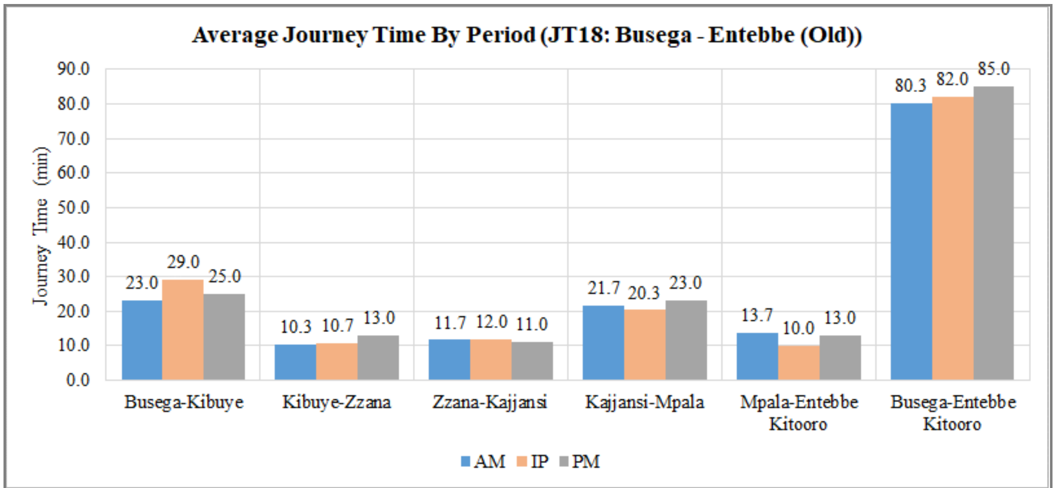
The following observations are with regard to Figures 6-1 and 6-2:

- Journey time: the total journey time for the route is 126 minutes or 2.1 hours on average. The stretch between Spear Junction and Mukono alone accounts for over 40 minutes on average.
- There is significant variability in journey time between the peak periods and between directions, especially in the Kampala – Jinja direction. For example, travel between Spear Junction and Bweyogerere varies between 17 minutes and 21 minutes, Bweyogerere to Mukono varying between 25 minutes and 38 minutes, and Lugazi to Jinja between 39 minutes and 46 minutes. The overall journey from Kampala to Jinja varies between 114 minutes and 141 minutes. This variation is significant.
- For both directions, the journey time in each direction is longest in the PM generally. In the PM, the total journey time in the Kampala – Jinja direction is longest with 140.7 minutes in journey time. This implies an average speed of 30 Kph for the entire journey.
- From Figure 6-2, it is observed that in the busier PM peak, the section between Spear Junction and Mukono incurs the highest journey time per Km in both directions; 3.75 min/Km towards Kampala and 3.70 min/Km towards Jinja. This implies and average speed of 16 Kph in this section. In the PM peak, it takes 70 minutes to travel the 20 Km section between Spear Junction and Mukono.

6.2.2 Journey Time Analysis for JT18 Kampala Northern Bypass (Busega) to Entebbe (Old Entebbe Road)

Figures 6-3 and 6-4 present the journey time analysis for Kampala Northern Bypass (Busega) to Entebbe (using the old Entebbe Road) for each of the directions; Busega - Entebbe and Entebbe - Busega. Figure 6-4 presents the journey time per Km for the route.

Figure 6-3: Average journey time by period for JT18 Busega – Entebbe (Old Route)



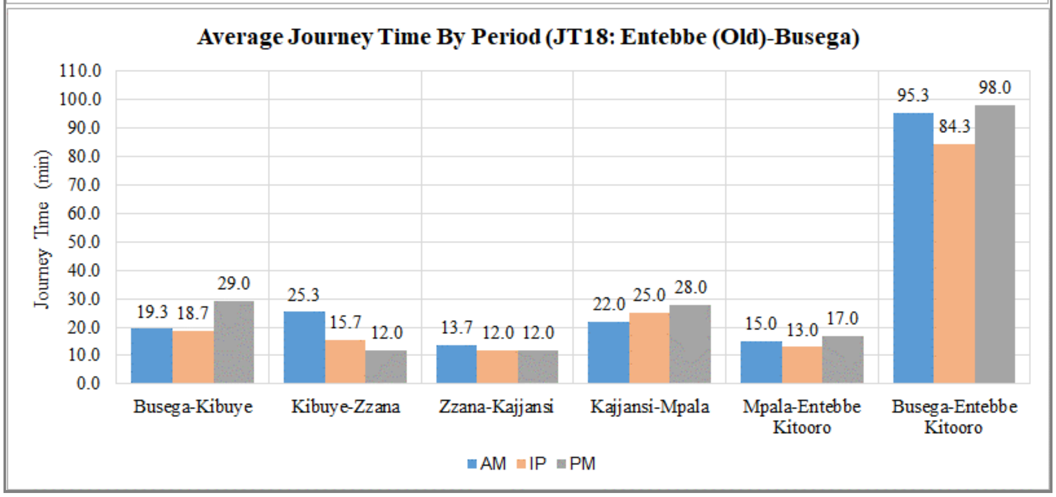
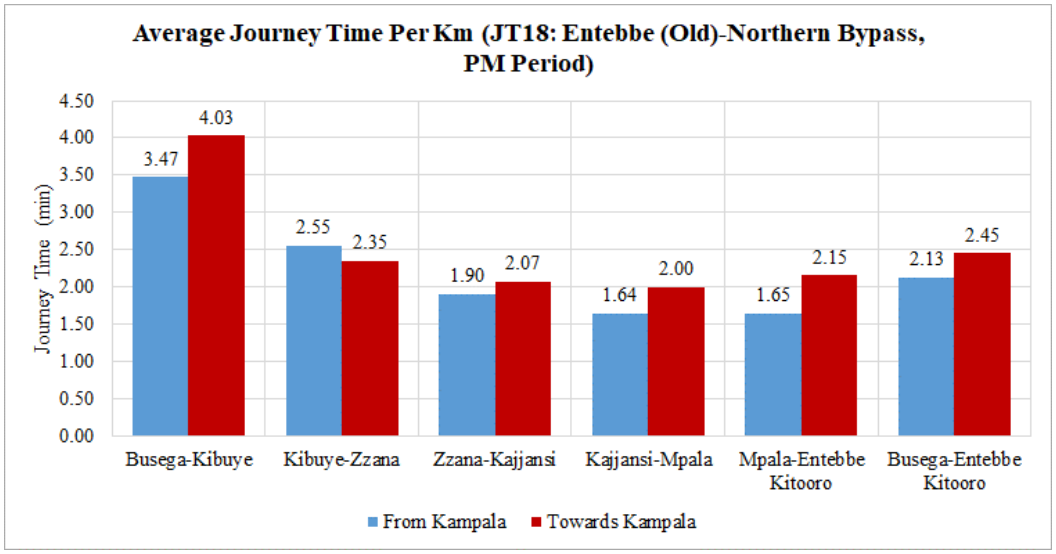


Figure 6-4: Average journey time per Km for PM for JT18 Busega – Entebbe (Old Route)



The following observations are with regard to Figures 6-3 and 6-4:

- Journey time: The total journey time for the route is 87.5 minutes (or 1.46 hours) on average. Busega – Kibuye and Kajjansi-Impala sections account for two-thirds of the journey time but are only 50% of the total distance.
- There is noticeable variability of journey time by section, by peak period and by direction. This is best demonstrated in Figure 6-4 for journey time rates between directions.
- The PM peak is the busier peak period of the three travel periods for both directions.
- Figure 6-4 confirms the Busega – Kibuye to be the slowest section especially in the Entebbe – Kampala direction (4.03 min/Km or 14.9 Kph). However, the entire route is generally slow, the quickest section on the route in the Entebbe – Kampala direction being Mpala – Kajjansi with 2 min/Km or 30 Kph.

6.2.3 Journey Time Analysis for JT20 Kampala Northern Bypass – Luwero – Karuma

Figures 6-5 and 6-6 present the journey time analysis for Kampala Northern Bypass to Karuma via Luwero for each of the two directions. Figure 6-6 presents the journey time per Km for the route.

Figure 6-5: Average journey time by period for JT20 Northern Bypass – Luwero - Karuma

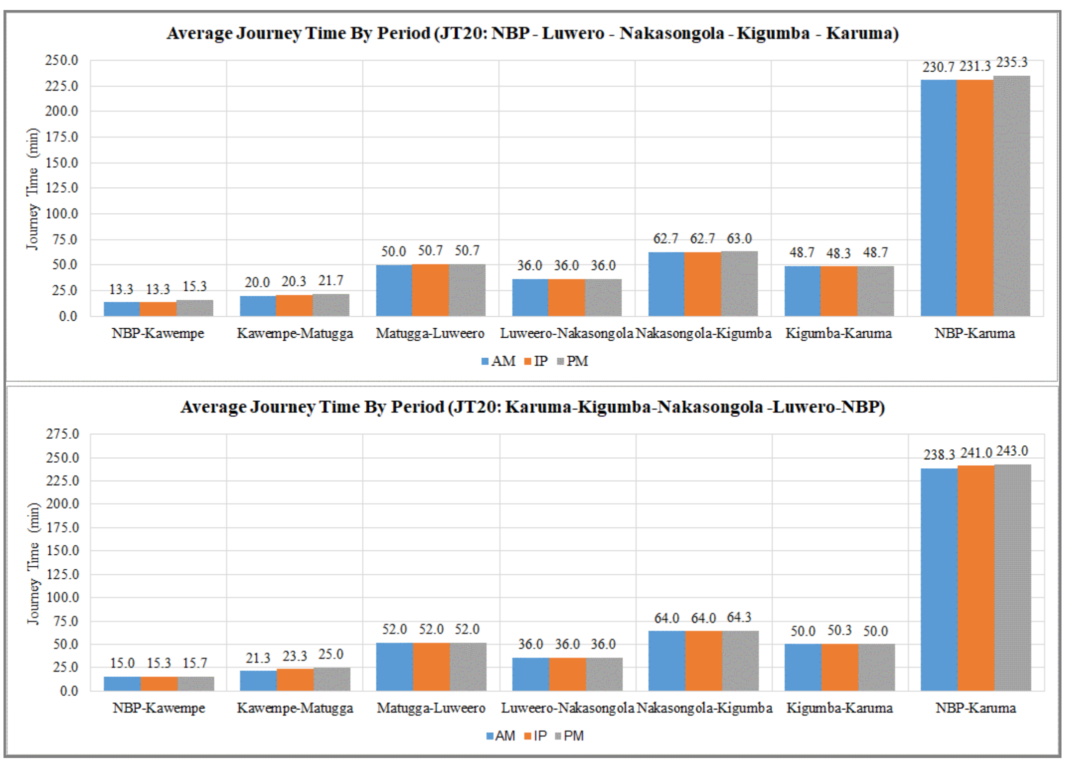
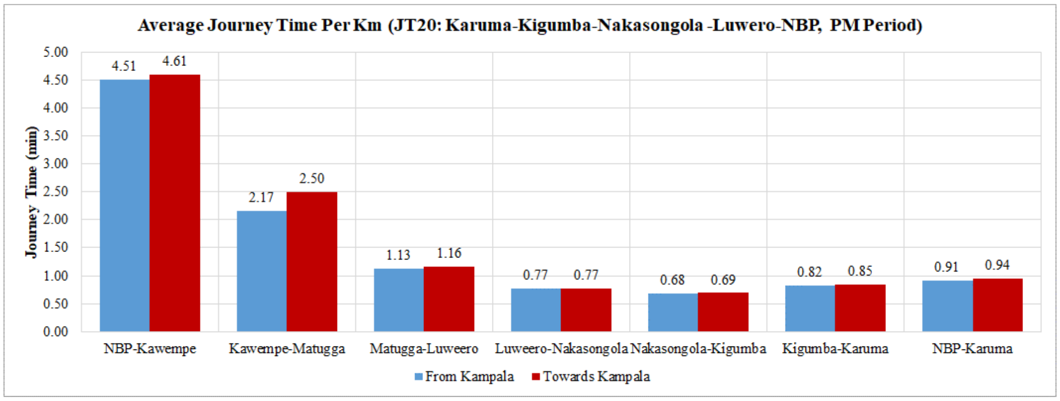


Figure 6-6: Average journey time per Km for PM for JT20 Northern Bypass – Luwero - Karuma



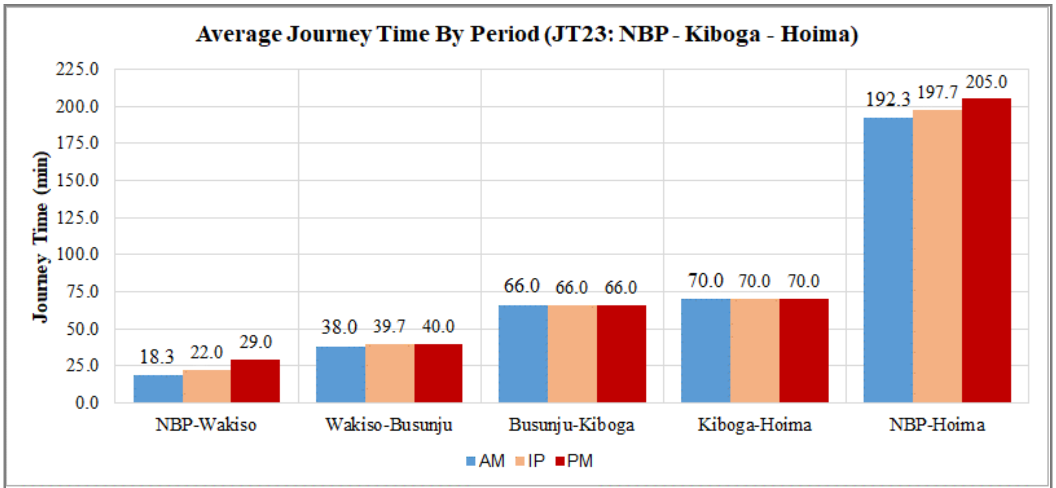
The following observations are with regard to Figures 6-5 and 6-6:

- Journey time: the total journey time for the route (257.4 Km) is 237 minutes or 3.95 hours on average. Nearly 50% of this is spent between Matugga - Luwero (50 min) and Nakasongola – Kigumba (63 min).
- There is little variability in journey time between Matugga and Karuma due to the semi-rural or rural nature of the route.
- Figure 6-6 shows that the slowest sections of the route as expected is the section between Northern Bypass and Kawempe with 4.6 min/Km and Kawempe – Matugga with 2.5 min/Km. This implies average speeds of 13 Kph and 24 Kph. The quickest section is the Nakasongola – Kigumba section with 0.68 min/Km or 88.2 Kph.

6.2.4 Journey Time Analysis for JT23 Kampala Northern Bypass – Kiboga – Hoima

Figures 6-7 and 6-8 present the journey time analysis for Kampala Northern Bypass to Hoima (Hoima Road) for each of the two directions. Figure 6-8 presents the journey time per Km for the route.

Figure 6-7: Average journey time by period for JT23 Northern Bypass – Kiboga - Hoima



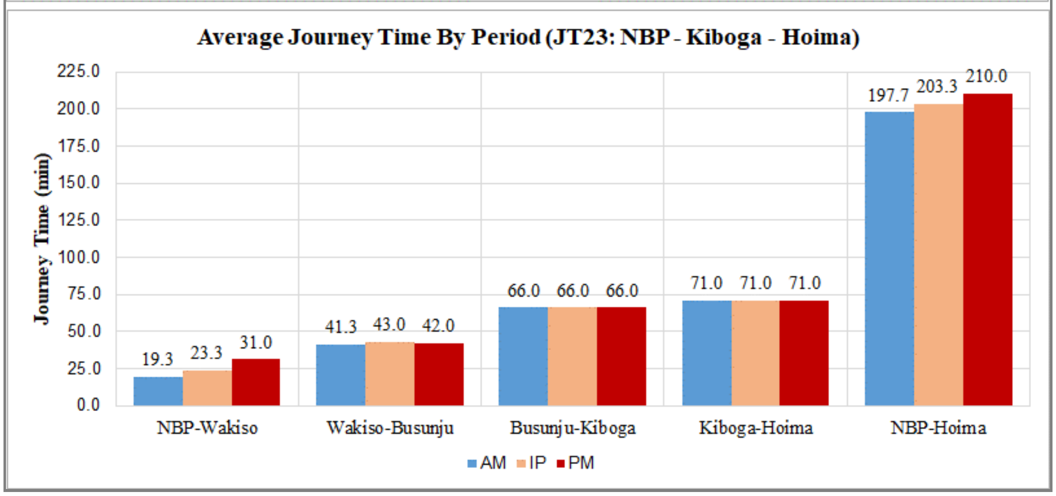
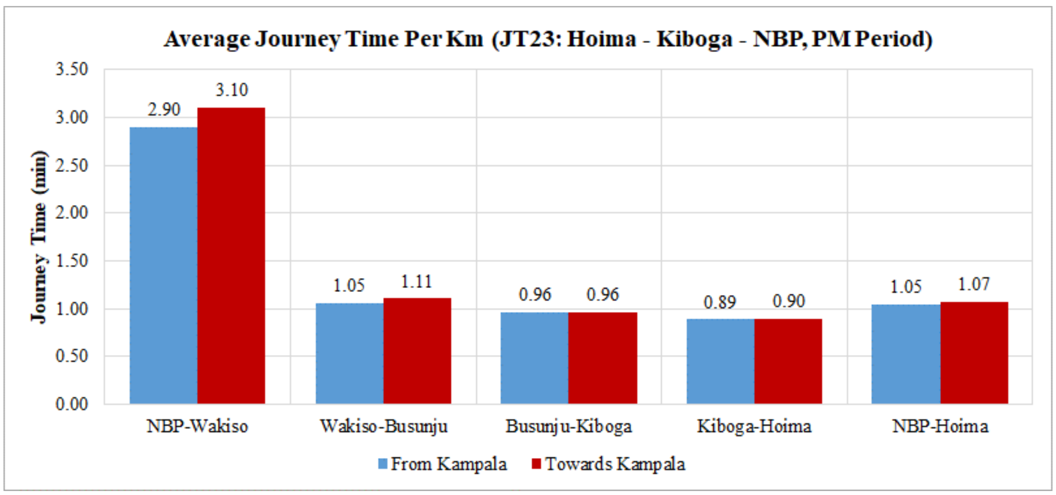


Figure 6-8: Average journey time per Km for PM for JT23 Northern Bypass – Kiboga - Hoima



The following observations are with regard to Figures 6-7 and 6-8:

- Journey time: the total journey time for the route (196 Km) is on average 200 min or 3.35 hours. More that half of this is spent in the Busunju – Kiboga (66 min) and Kiboga – Hoima (70 min). These are also the longest sections on the route accounting for 75% of its length.
- Journey time variability between pea periods and directions is only noticeable in the Northern Bypass – Wakiso section. The PM peak accounts for the longest journey time in this section.
- Figure 6-8 confirms that Northern Bypass – Wakiso is the slowest section with 3 min/Km on average or 20 Kph.
- The remainder of the route travels well at an average of 60 Kph.

6.2.5 Journey Time Analysis for JT24 Kampala Northern Bypass – Mubende – Fort Portal

Figures 6-9 and 6-10 present the journey time analysis for Kampala Northern Bypass to Fort Portal (Mityana Road) for each of the two directions. Figure 6-10 presents the journey time per Km for the route.

Figure 6-9: Average journey time by period for JT24 Northern Bypass – Mubende – Fort Portal

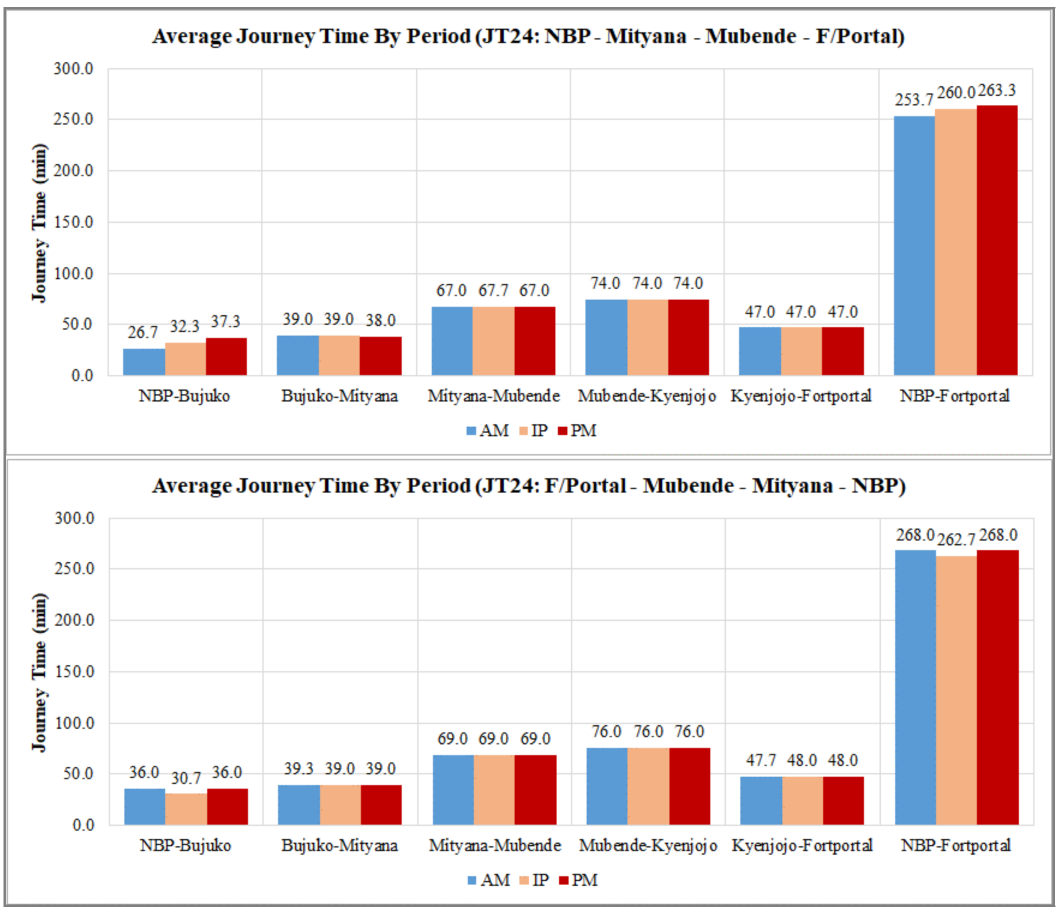
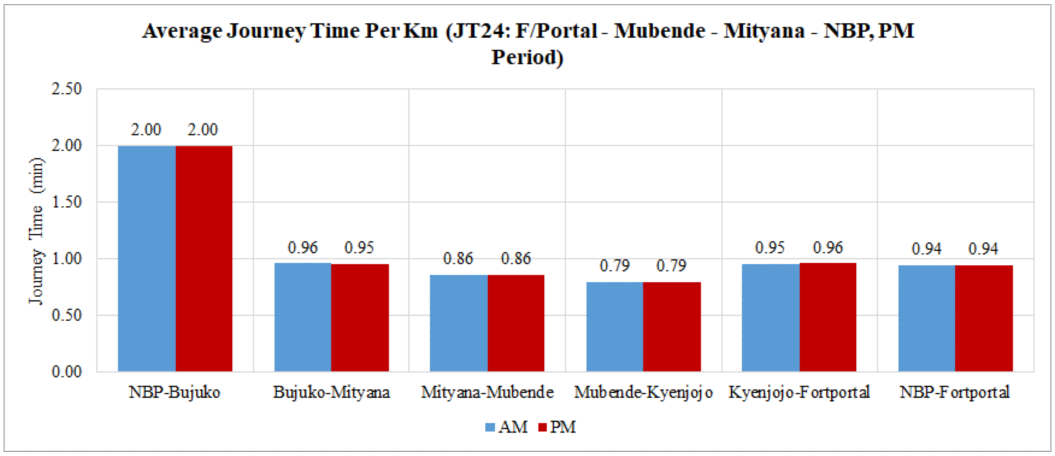


Figure 6-10: Average journey time per Km for PM for JT24 Northern Bypass – Mubende – Fort Portal



The following observations are with regard to Figures 6-9 and 6-10:

- Journey time: the total journey time for the entire route (285 Km) is about 263 minutes or 4.4 hours. More than 50% of the journey time is incurred in the section Mityana – Kyenjojo whch is the longest section (176 Km) accounting for 72% of the total length.
- Journey time variability only noticeable in the Northern Bypass – Bujuuko section, the most urbanised section on the route.
- The PM is the busiest peak period as with other routes.
- Figure 6-10 confirms that the Northern Bypass – Bujuuko section is indeed the slowest section with journey time per Km of 2 min/Km or 30 Kph. All other sections travel well at slightly over 60 Kph.

6.2.6 Journey Time Analysis for JT31 Busega – Buwama - Masaka

Figures 6-11 and 6-12 present the journey time analysis for Busega to Masaka (Masaka Road) for each of the two directions. Figure 6-12 presents the journey time per Km for the route.

Figure 6-11: Average journey time by period for JT31 Busega – Buwama - Masaka

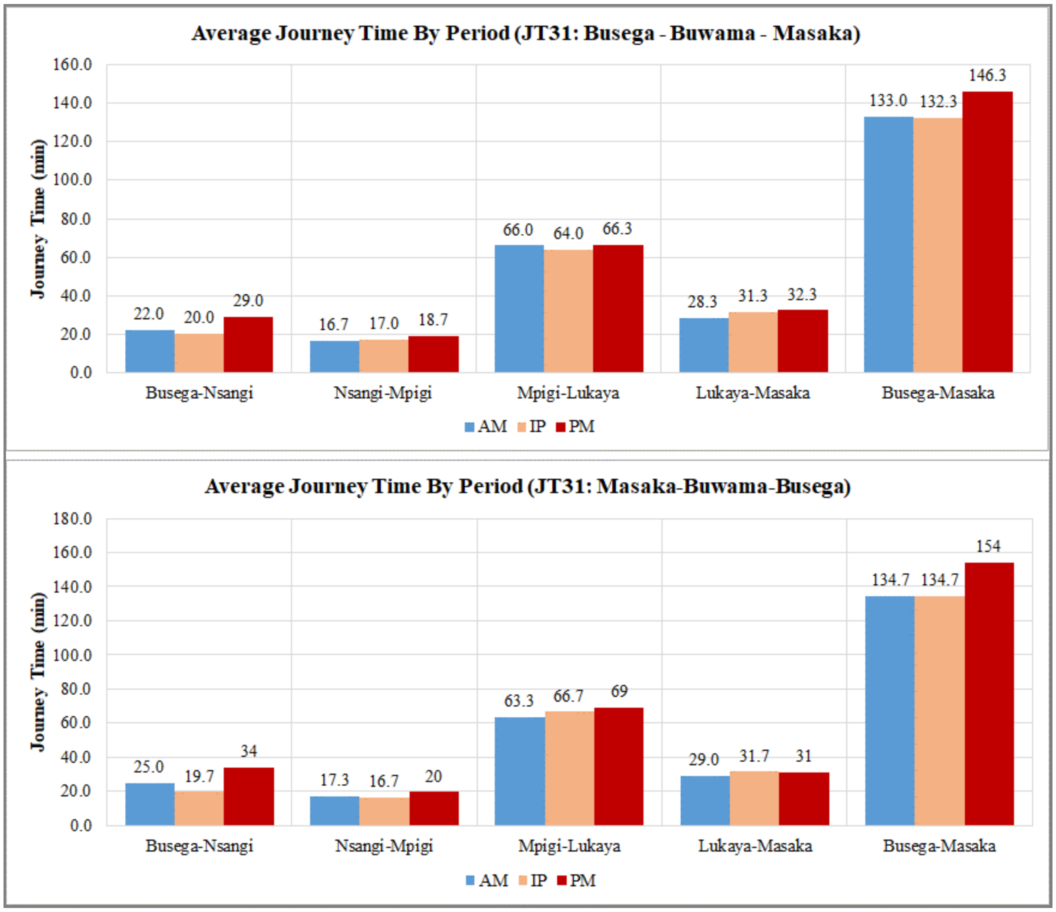
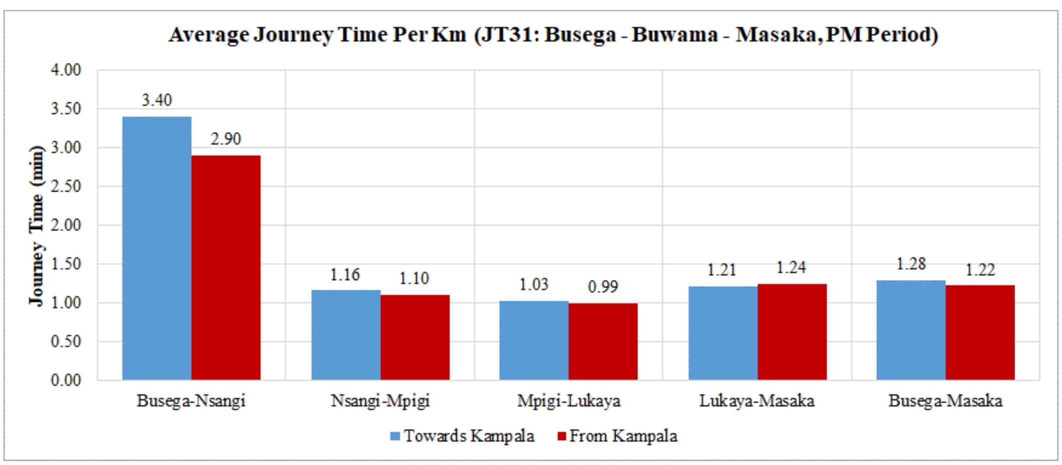


Figure 6-12: Average journey time per Km for PM for JT31 Busega – Buwama - Masaka



The following observations are with regard to Figures 6-11 and 6-12:

- Journey time: the total journey time for the entire route (120 km) is 139 min (or 2.31 hours) on average. This implies an average speed for the route of 51.9 Kph.

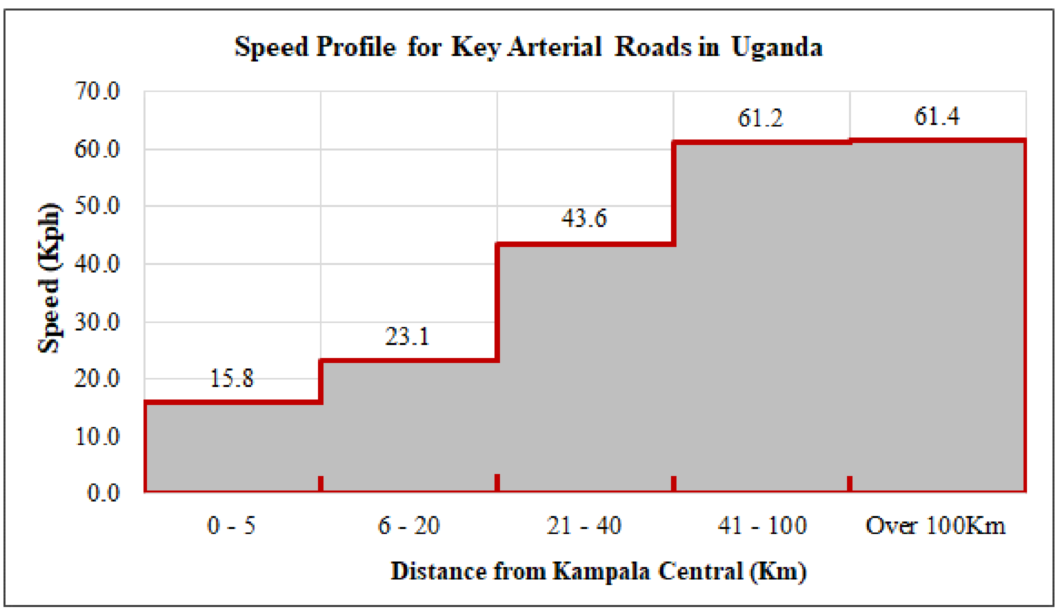
The section between Mpigi and Lukaya (67 Km) accounts for the longest journey time (65 minutes).

- There is high variability of journey time between peaks and directions at all sections of the route. Generally the PM peak is the busiest peak.
- Figure 6-12 shows that the urban section Busega – Nsangi is the slowest section on average especially in the Masaka – Kampala direction with 3.4 min/Km. This implies a speed of 17.6 Kph on average.
- The quickest section is Mpigi – Lukaya with a journey time rate of about 1 min/Km or 60 Kph travel speed.

6.2.7 Overall Journey Time Analysis

- Journey time along the arterial routes is significantly influenced by the slow-moving urban areas of the GKMA.
- There is greater variability of journey time and hence unreliability of travel between peak periods and directions in the urban areas than other sections of the routes.
- The direction to Kampala is clearly the busiest direction of the routes.
- From the data gathered, the general speed profile along the arterial routes has been constructed and is presented in Figure 6-13 below.

Figure 6-13: Average speed profile for arterial routes in Uganda



From Figure 6-13 above, the following observations are made:

- Within 5Km of Kampala City centre, the speed on the arterial roads is an average of just 16 Kph.

- Between 5Km and the boundary of the GKMA (approximately 20 Km from Kampala central), the speed increases slightly to 23 Kph.
- For the section 21 – 40 Km, the speed increases significantly to an average of 43.1 Kph. Beyond this distance, it was observed that the roads travel at just over 60 Kph on average and no higher. This is expected due to a number of factors. Key of these however is the significant presence of traffic calming measures especially near towns and the significant roadside activity in the bigger urban areas along the routes.

6.3 GIS Data

In line with the ToR with respect to road network GIS data, the Consultant conducted field and desktop (online GIS data) survey to update the road inventories (number of lanes, speed, pavement conditions). The GIS database provided by UNRA was accordingly updated.

In the future, when the expressway network is finally decided, the updated GIS database will be submitted by overlaying the future expressway network.

Figure 6-14: Statement on the Methodology of GIS Database

■ **Statement on the Methodology of GIS Database**

Step 1.

The Consultant will review the values of attributes with TransCAD programs on the existing Geographic Information System (GIS) Data in *.shp, *.shx, and *.dbf which will be provided by UNRA after the Contract signing.

Step 2.

All the values of attributes of national road network such as length of roads, number(s) of lanes, maximum/minimum speed, pavement conditions, etc. being collected through site surveys will be appropriately updated on the current road conditions after carefully comparing with the values of GIS data which were provided by the Client, UNRA.

Step 3.

The potential expressway network which will be selected through this assignment will be overlaid with the current national road network for the development of GIS data.

Step 4.

All the values of attributes of roads newly created by Step 1 to Step 3 will be converted in types of e-files such as *.shp, *.shx and *.dbf for the purpose of the users as a final deliverables of GIS data.

The following was collected and updated:

- Road link number;
- Length of road link (in kilometres);
- Number of lanes(each links)
- Travel speed(each links)
- UNRA Station in charge;
- Road class of the link;
- Road link name;
- Road surface type – sealed or unsealed;
- Region of country where link is located; and

The format of the collected and updated data is shown in Figure 6-14 below. All of the collected and updated GIS data is provided in Appendix E2.

Figure 6-155: Format of collected and updated GIS data

objectid	Road	Length	Number of lanes	speed	Station	Class	Linkid	Link_name	Surface	Region
650	C596	3.66	1	40	Hoima	C	C596_Link01	Kiziranfumbi-Kaiso	Unsealed	West
691	C592	4.16	1	50	Hoima	C	C592_Link02	Buseruka-Kabaale	Bituminous	West
42	A005	12.4	1	80	Fortportal	A	A005_Link09	Kyegegwa - Kakabala	Bituminous	West
277	B202	7.02	1	50	Hoima	B	B202_Link03	Kagadi - Mabale - Ka	Unsealed	West
709	C591	13.52	1	40	Hoima	C	C591_Link01	Kabwoya - Kituti	Unsealed	West
419	C553	29.04	1	40	Hoima	C	C553_Link01	Kibaale-Kyebando	Unsealed	West
560	C566	27.21	1	40	Hoima	C	C566_Link02	Bugwara-Kikwaya	Unsealed	West
619	C599	23.23	1	40	Hoima	C	C599_Link01	Kaiso-Tonya	Unsealed	West
540	C550	2.22	1	40	Hoima	C	C550_Link02	Kibale-Kagadi	Unsealed	West
604	C552	15.53	1	40	Fortportal	C	C552_Link02	Kabagole-Kyegegwa	Unsealed	West
471	C541	10.11	1	40	Fortportal	C	C541_Link01	Karugutu - N-roko	Unsealed	West
474	C540	0.2	1	50	Fortportal	C	C540_Link02	Karugutu-Bundibugyo	Bituminous	West
433	C545	12.51	1	40	Fortportal	C	C545_Link01	Busalu - Butoogo	Unsealed	West
729	C558	19.71	1	40	Fortportal	C	C558_Link01	Kakara - Rwebisengo	Unsealed	West
642	C565	30.21	1	40	Hoima	C	C565_Link01	Kyenzige-Rugashali-M	Unsealed	West
435	C544	1.3	1	40	Fortportal	C	C544_Link01	Bubandi - Kahuka -Ma	Unsealed	West
435	C544	5.7	1	40	Fortportal	C	C544_Link01	Bubandi - Kahuka -Ma	Unsealed	West
443	C543	4.76	1	40	Fortportal	C	C543_Link01	Bubandi - Bundibugyo	Unsealed	West
435	C544	0	1	40	Fortportal	C	C544_Link01	Bubandi - Kahuka -Ma	Unsealed	West

7 STATED PREFERENCE SURVEYS

7.1 Introduction

These surveys offer a great advantage for overcoming the problem of the "new option" such as a proposed tolled expressway system. Other transport scenarios where SP is useful are those when new transport modes such as high-speed rail, buses, etc are to be introduced. An analyst seeks to forecast the use of this new option, and requires to understand the public's potential perception of the transport system.

Stated Preference (SP) surveys present hypothetical situations to the respondents, who are then asked to choose based on the given attributes for each alternative, without necessarily experiencing them in real situations.

As such, SP surveys are not usually carried out on the road like other surveys. Instead, key locations such as markets, bus and lorry parks, business and commercial parks and other such locations are used. The approach is to undertake interviews in which the respondents are asked to provide answers to hypothetical questions.

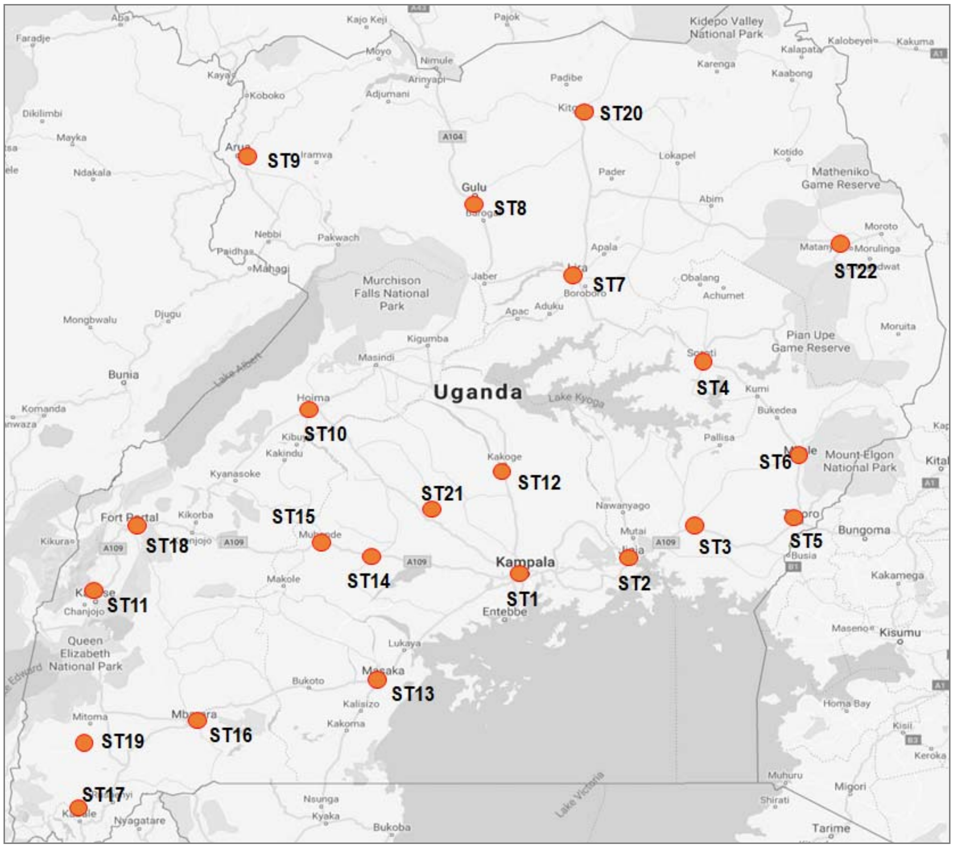
The proposed locations for undertaking stated preference surveys are presented in Table 2-16. The surveys were undertaken in 21 key towns and cities in Uganda. Within these locations, the surveys were undertaken at key terminals, car parking areas, markets and other such locations.

7.2 Methodology

7.2.1 Site Selection

The locations of the SP surveys were carefully chosen through stratification to maximise coverage of the entire country. Figure 7-1 below shows the location of the towns/cities where SP surveys were undertaken.

Figure 7-1: SP Survey Locations



7.2.2 Survey Personnel and Training

The following categories of survey staff were used at all survey locations:

- Survey manager with overall responsibility for the survey programme;
- Location supervisors with overall responsibility for the operations of each of the survey locations; and
- Four to six interviewers per location.

Survey staff used for SP surveys in this exercise were experienced socioeconomic survey staff, but were nonetheless trained prior to the survey. The training covered a range of issues including; details about the survey, the duties involved, roles and responsibilities, the associated health and safety requirements, interview procedure and detailed instructions on how to complete the survey questionnaire.

7.2.3 Survey Period

As described in Table 1-1, the SP surveys were undertaken for a total of three days at every location. The survey period was 7am to 7pm everyday survey day.

7.2.4 Willingness to Pay (WTP) Surveys

These are useful in providing data on unit costs of travel (value of time and unit vehicle operating costs) and the toll rates that the public would be willing pay in case a new road is to be tolled. Questions regarding the public's willingness to pay for the use of the proposed expressway system were part of the SP survey questionnaire.

7.2.5 Sampling

Two relatively simple sampling methods were considered – simple random sampling and stratified random sampling. Simple random sampling involves taking random sample from the entire population. In this case, surveys would be taken randomly throughout Uganda.

On the other hand, the method of stratified random sampling involves taking samples of the population data subdivided into smaller groups known as strata. Stratification is the process of dividing members of the population into homogeneous subgroups before sampling. In this case, the country would first be divided into representative regions or towns and then random samples taken from these towns. Stratified random sampling is also referred to as proportional random sampling or quota random sampling.

From the above definitions, it is clear than stratified sampling would likely produce a better sample of the population (data). Simple random sampling risks disproportionately sampling one area of the country than the other, resulting sampling bias or error. Stratified sampling improves the precision of the sample by reducing sampling error.

To use stratified sampling, it is necessary that some prior information about the population is available before sampling takes place. This information is used to subdivide the population. For this survey, information available on key commercial centres in the country, areas that have been accorded city status, regional demarcations of the country, etc were all considered before the subdivisions were made. Based on this information, 21 locations were chosen to be representative of vehicular activity in the country. These were discussed with UNRA at inception and confirmed. They are presented in Figure 7-1 above.

An overall sample size of about 5,000 valid interviews was required, based on experience and the objectives of the survey i.e. to analyse travel behavior that would influence expressway use. So that target number of valid interviews per survey location was approximately 230.

7.3 SP Data Analysis Results

7.3.1 Summary Totals

A significant amount of data was collected as part of the SP surveys. Table 7-1 below summarises the number of valid interviews from each survey location.

Table 7-1: Number of SP interviews

NUMBER OF VALID INTERVIEWS FROM SP SURVEYS			
Site	**Description**	**Region**	**No. of Interviews**
SP1A	Arua Park	Central	74
SP1B	Kisenyi	Central	182
SP1C	New Taxi Park	Central	133
SP1D	Old Taxi Park	Central	176
SP2	Jinja	Eastern	401
SP3	Iganga	Eastern	293
SP4	Soroti	Eastern	146
SP5	Tororo	Eastern	173
SP6	Mbale	Eastern	173
SP7	Lira	Northern	112
SP8	Gulu	Northern	178
SP9	Arua	Northern	417
SP10	Hoima	Western	347
SP11	Kasese	Western	250
SP12	Luwero	Central	312
SP13	Masaka	Central	266
SP14	Mityana	Central	270
SP15	Mubende	Central	304
SP16	Mbarara	Western	212
SP17	Kabale	Western	269
SP18	Bushenyi	Western	257
SP19	Rukungiri	Western	216
SP20	Kitgum	Northern	106
SP21A	Entebbe	Central	241
SP21B	Wakiso	Central	362
TOTALS			**5,870**

Table 7-1 shows that a total of 5,870 interviews were undertaken and data collected as part of the SP surveys.

The distribution of the SP survey locations was also influenced by the need to investigation regional variations. For the main four regions of the country, a representative number of locations were surveyed. The analysis was therefore undertaken on a regional basis. However, for this report analysis results for the entire country is presented. Detailed region by region data is included in Appendix F.

The amount of data collected is significant and has a number of features as shown on the survey tool (Appendix A). All the data collected cannot feasibly be analysed. The aspects selected for analysis were based on what was considered relevant for the expressway master plan formulation. The following was analysed:

- Gender and age of respondents;
- Journey purpose by gender for a typical/representative journey;
- Number of respondents by typical journey distance;
- Number of respondents by typical journey time;
- Toll charges respondents were willing to pay by typical journey distance;
- Toll charges respondents were willing to pay by monthly income of the respondent; and
- Toll charges respondents were willing to pay by vehicle type operated.

7.3.2 Summary Results – All Regions

The analysis results are summarized below in a number of tables and figures.

- Figure 7-2 below shows that the interviews were conducted predominantly on males. This is because, the interviews targeted vehicle drivers and female drivers were significantly less than males. Female interviewees accounted for only 27.4% of all interviews.
- Further, Figure 7-2 below shows that the dominant age ranges interviewed were 30-39 (50.6%) and 40-49 (24.6%).

Figure 7-2: Gender and age of respondents

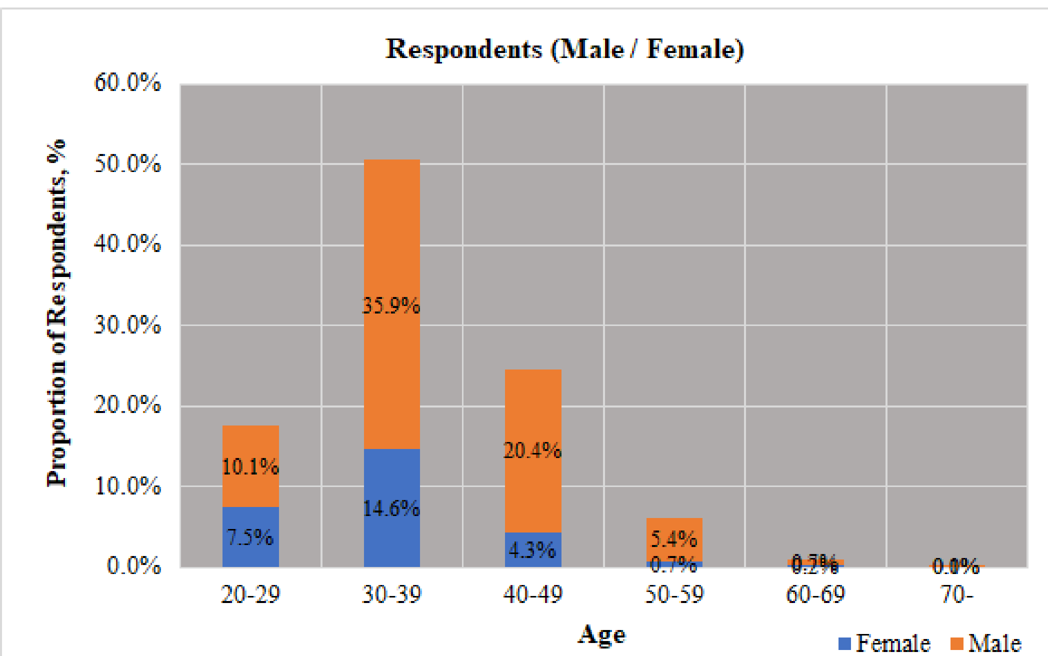
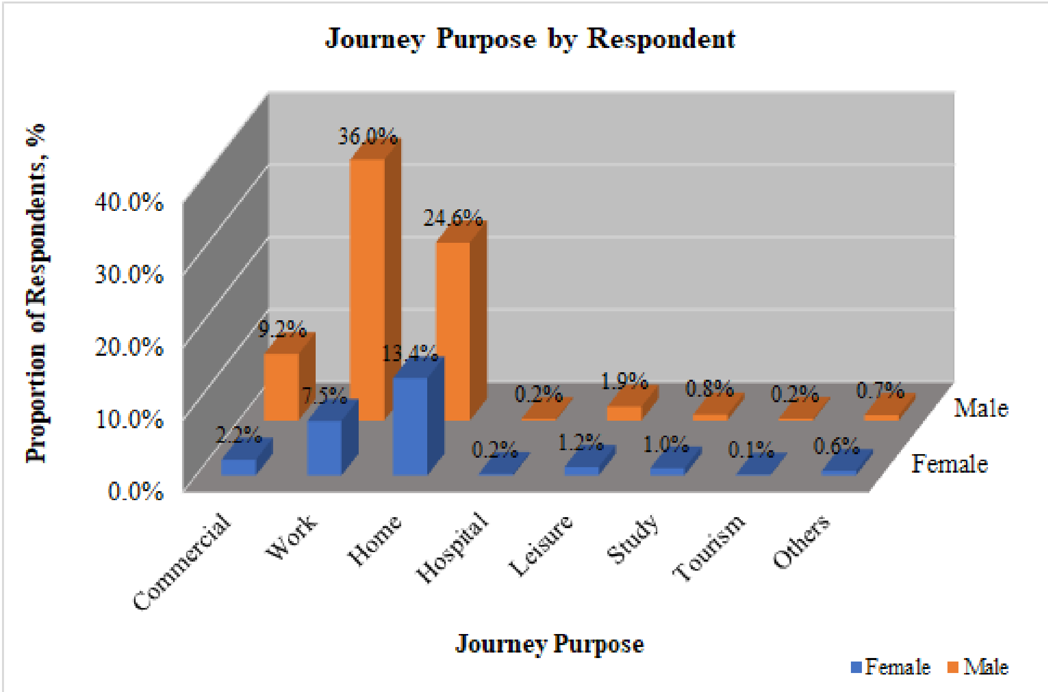
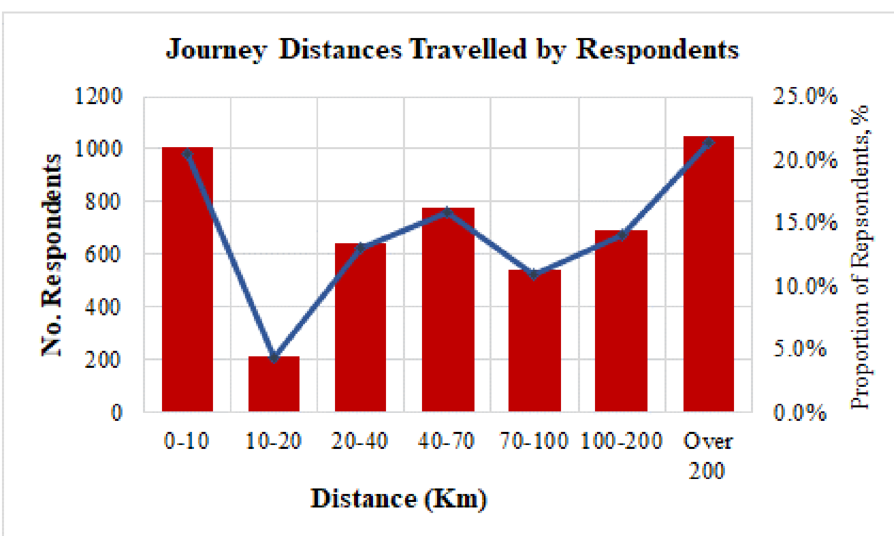


Figure 7-3: Journey purpose by gender or respondents – Eastern Region



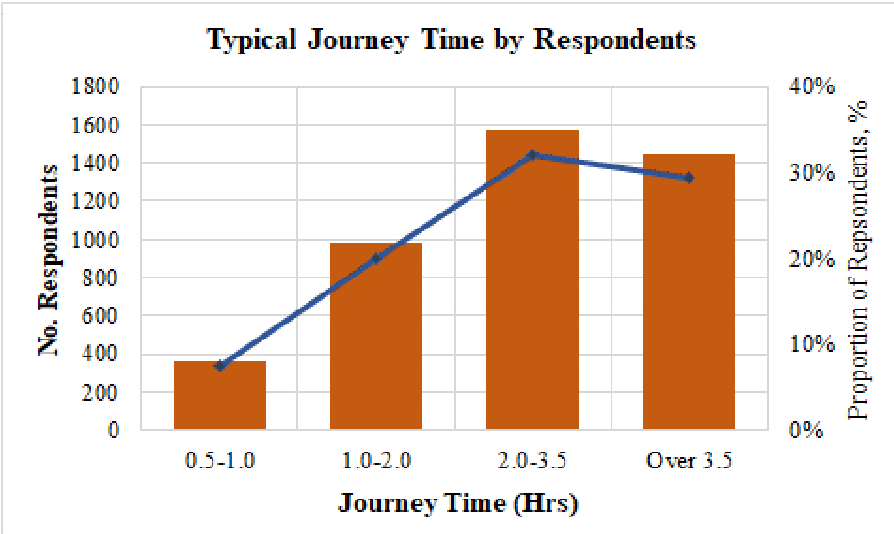
- Figure 7-3 above shows that the dominant number of trips work related i.e. the origin purpose was work (43.5%). The second highest trip purpose were home related i.e. home to work, home to school, etc. These accounted for 38% of all trips.
- Most male respondents were making work-based trips and most female interviewees made home-based trips.

Figure 7-4: Journey distance for a typical journey



- Figure 7-4 above shows that the majority of respondents' typical journeys were either very short or very long. Short journeys (0 – 10Km), accounted for 20.5% and very long journeys (over 200Km), accounted for 21.4%. Middle distance trips (typically 40-100 km) accounted for 26.8% of all trips.

Figure 7-5: Journey time for a typical journey



- Figure 7-5 above shows that the majority of respondents' typical journey times were 2 hours and above. Those that travelled for 2 – 3.5 hours accounted for 32% and those travelling for over 3.5 hours were 29.4%.

Table 7-2: Willingness to pay toll charges by typical journey distance

	Potential Expressway Toll Charge (UGX)						
Journey Distance (Km)	0-1,000	1,000-2,000	2,000-3,000	3,000-5,000	5,000-10,000	Over 10,000	Totals
0-10	8.3%	5.1%	2.7%	1.7%	1.6%	1.1%	**20.5%**
10-20	4.8%	3.3%	2.4%	1.5%	1.5%	0.4%	**14.0%**
20-40	2.0%	1.3%	0.4%	0.2%	0.2%	0.1%	**4.3%**
40-70	5.1%	4.0%	1.7%	0.9%	1.1%	0.2%	**13.0%**
70-100	6.5%	4.3%	2.4%	1.1%	1.3%	0.3%	**15.8%**
100-150	3.8%	2.8%	2.1%	1.0%	1.0%	0.3%	**11.0%**
Over 150	6.0%	5.1%	3.8%	2.2%	3.0%	1.2%	**21.4%**
Totals	**36.5%**	**25.9%**	**15.5%**	**8.6%**	**9.9%**	**3.6%**	

Figure 7-6: Willingness to pay toll charges by typical journey distance

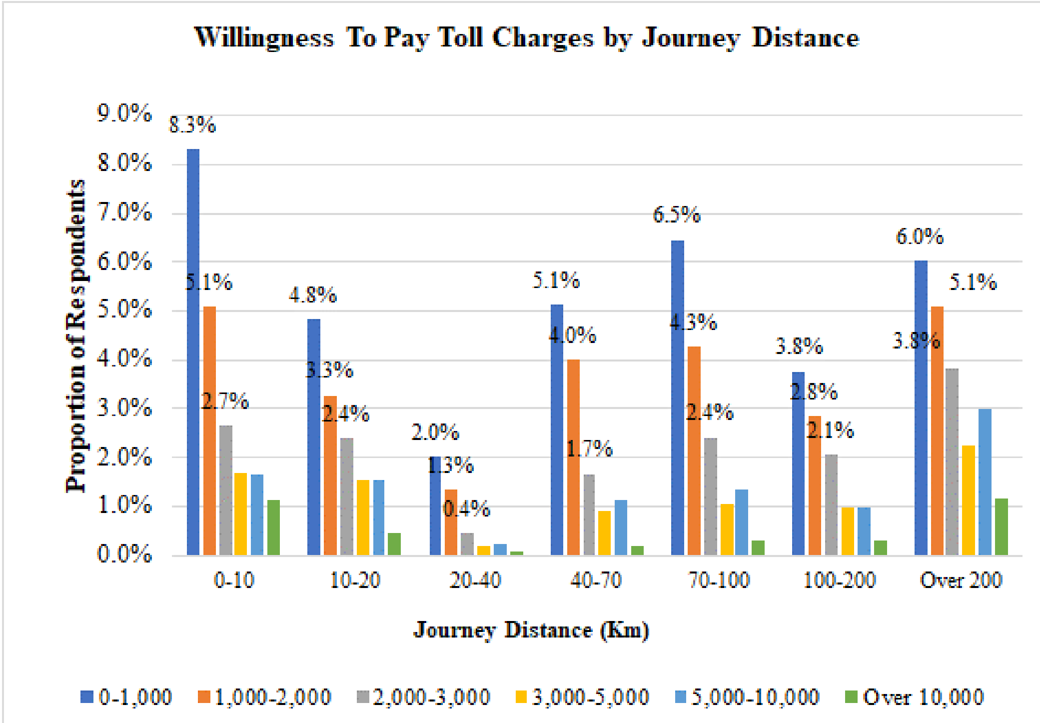


Table 7-2 and Figure 7-6 present the critical issue of willingness to pay. The toll charges that respondents were willing to pay for the use of expressways (toll per use) were compared to the typical journey distance. The charges are for each single journey regardless of distance. It would reasonably be expected that higher travel distance travelers would be willing to pay higher toll charges.

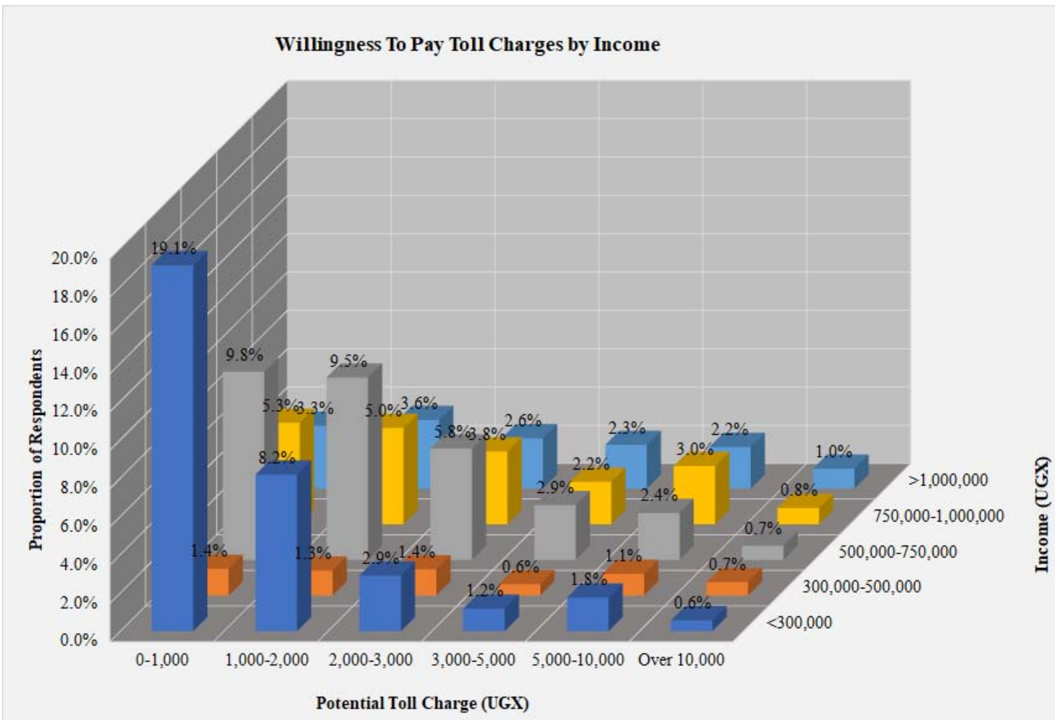
Table 7-2 and Figure 7-6 indicate the following:

- The largest majority of respondents would only be willing to pay up to UGX1,000. These account for 36.5%. Those willing to pay UGX1,000 – UGX2,000 were 25.9% of all respondents, UGX2,000-UGX3,000 were 15.5% and 22.1% were willing to pay higher than UGX3,000.
- For every range of journey distance, the data shows that the largest majority of respondents would pay up to UGX1,000. The proportion of these respondents does not clearly reduce with increase in journey distance.
- The data shows indications that long-distance travelers would be willing to pay higher charges. From Figure 7-6, the grey, yellow and light blue bars (representing toll charges between UGX2,000 and UGX5,000) clearly increase in length with increase in travel distance. For journey distance of over 200 Km, it is clear that the number of respondents willing to pay higher charges is significantly higher.

Table 7-3: Willingness to pay toll charges by income of respondent

	Potential Expressway Toll Charge (UGX)						
Income (UGX)	**0-1,000**	**1,000-2,000**	**2,000-3,000**	**3,000-5,000**	**5,000-10,000**	**Over 10,000**	**Totals**
<300,000	19.1%	8.2%	2.9%	1.2%	1.8%	0.6%	33.7%
300,000-500,000	1.4%	1.3%	1.4%	0.6%	1.1%	0.7%	6.5%
500,000-750,000	9.8%	9.5%	5.8%	2.9%	2.4%	0.7%	31.2%
750,000-1,000,000	5.3%	5.0%	3.8%	2.2%	3.0%	0.8%	20.2%
>1,000,000	3.3%	3.6%	2.6%	2.3%	2.2%	1.0%	15.0%
Totals	**38.9%**	**27.6%**	**16.5%**	**9.1%**	**10.5%**	**3.9%**	

Figure 7-7: Willingness to pay toll charges by income of respondent



The toll charges that respondents were willing to pay for the use of expressways were compared to the stated income of the respondents. It would reasonably be expected that higher income earners would be willing to pay higher toll charges.

Table 7-3 and Figure 7-7 indicate the following:

- The largest majority of respondents would be willing to pay up to UGX1,000. These account for 38.9%. Those willing to pay UGX1,000 – UGX2,000 were 27.6% of all respondents, UGX2,000-UGX3,000 were 16.5% and 23.5% were willing to pay higher than UGX3,000.

- The data shows indications that higher income respondents would be willing to pay higher charges. From Figure 7-7, it is clear that the proportion of those willing to pay UGX3,000 and higher increases with increasing income.

Table 7-4: Willingness to pay toll charges by vehicle type

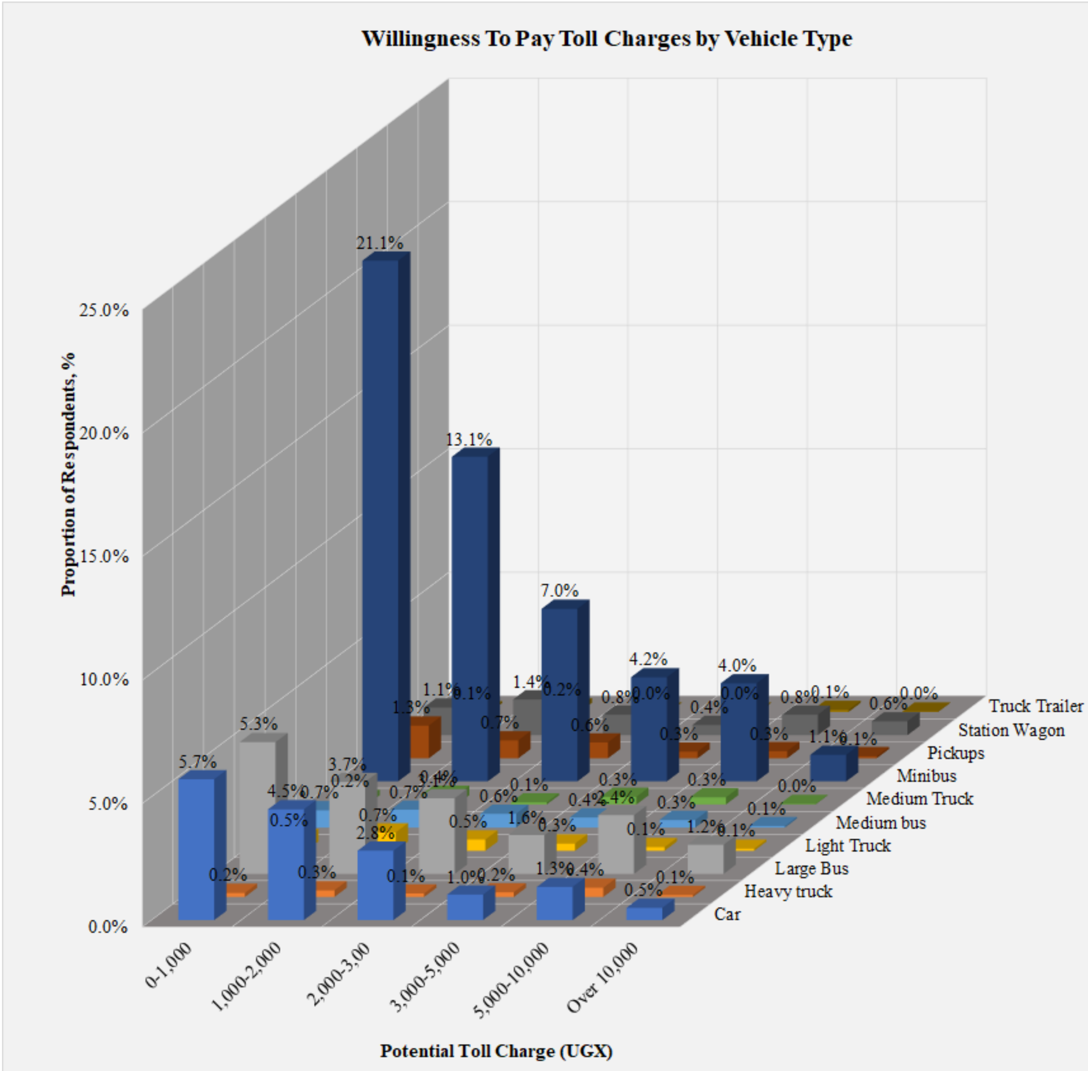
	Potential Expressway Toll Charge (UGX)						
Vehicle Type	0-1,000	1,000-2,000	2,000-3,000	3,000-5,000	5,000-10,000	Over 10,000	Totals
Car	5.7%	4.5%	2.8%	1.0%	1.3%	0.5%	**15.9%**
Heavy truck	0.2%	0.3%	0.1%	0.2%	0.4%	0.1%	**1.2%**
Large Bus	5.3%	3.7%	3.1%	1.6%	2.4%	1.2%	**17.2%**
Light Truck	0.5%	0.7%	0.5%	0.3%	0.1%	0.1%	**2.2%**
Medium bus	0.7%	0.7%	0.6%	0.4%	0.3%	0.1%	**2.8%**
Medium Truck	0.2%	0.4%	0.1%	0.3%	0.3%	0.0%	**1.3%**
Minibus	21.1%	13.1%	7.0%	4.2%	4.0%	1.1%	**50.5%**
Pickups	1.3%	0.7%	0.6%	0.3%	0.3%	0.1%	**3.3%**
Station Wagon	1.1%	1.4%	0.8%	0.4%	0.8%	0.6%	**5.2%**
Truck Trailer	0.1%	0.2%	0.0%	0.0%	0.1%	0.0%	**0.4%**
Totals	**36.2%**	**25.8%**	**15.6%**	**8.7%**	**10.0%**	**3.7%**	**100.0%**

The toll charges that respondents were willing to pay for the use of expressways were vehicle type of the respondents. It would reasonably be expected that larger vehicles would be willing to pay higher toll charges.

Table 7-4 and Figure 7-8 indicate the following:

- The largest majority of vehicles would be willing to pay up to UGX1,000. These account for 36.2%. Those willing to pay UGX1,000 – UGX2,000 were 25.8% of all respondents, UGX2,000-UGX3,000 were 15.6% and 22.4% were willing to pay higher than UGX3,000.
- The data shows that a significant number of respondents were minibus drivers, although all other vehicle types are well represented in the large database.
- The data shows that for cars, minibuses, station wagons and large buses a significant number of respondents were willing to pay higher toll charges. The proportion of respondents who were willing to pay UGX3,000 and higher for cars, minibuses, station wagons and large buses were 2.9%, 9.3%, 1.8% and 5.1% respectively of all respondents. Overall, a total of 22.4% of all vehicle types were willing to pay UGX3,000 or higher.

Figure 7-8: Willingness to pay toll charges by vehicle type



8 PUBLIC TRANSPORT (PT) SURVEYS

8.1 Introduction

Public transport surveys undertaken as part of the survey exercise focused solely on large buses. The main objective was to determine the large bus demand for key routes within the country.

Origin-destination surveys were not undertaken as part of this exercise. Only bus counts were undertaken. It was considered that OD surveys undertaken at roadside would provide data that could be required if needed.

The locations for undertaking PT surveys were the same as those where SP surveys were undertaken, with the exception of Iganga. Iganga did not possess a designated bus parking area. These were presented in Table 2-16. The surveys were undertaken in 20 key towns and cities in Uganda. Within these locations, the surveys were undertaken at formal bus terminals and key bus stops.

8.2 Methodology

8.2.1 Site Selection

The locations of the PT surveys were carefully chosen to maximise coverage of the entire country. The locations shown in Figure 7-1 where SP was undertaken are also the locations where PT surveys were undertaken generally.

Two key pieces of information were collected as part of the PT surveys:

i. the number of buses passing in either direction through the bus terminal. Those arriving at the terminal (to the town) and those leaving the terminal (from the town);

ii. The number of boarding and alighting passengers for each stopping bus. For most towns, boarding and alighting of passengers is a small activity before the bus departs to travel farther on. At other towns that serve as end terminals for bus services, the activity takes a significant amount of time and the number of boarding or alighting passengers is significant.

Kampala, having a number of key bus terminals was considered for detailed surveys. The major bus terminals were also surveyed, a total of 11 bus terminals including; Global Coaches, Bismarkan Coaches, Jaguar, Easy Coaches, KK Coaches, YY Coaches, Trinity, Swift Coach, Post Bus, Modern Coaches and Link Buses.

8.2.2 Survey Personnel and Training

The following categories of survey staff were used at all survey locations:

- Survey manager with overall responsibility for the survey programme;
- Location supervisors with overall responsibility for the operations of each of the survey locations; and
- Two enumerators interviewers per location.

Survey staff used for PT surveys in this exercise were experienced socioeconomic survey staff, but were nonetheless trained prior to the survey. The training covered a range of issues including; details about the survey, the duties involved, roles and responsibilities, the associated health and safety requirements, interview procedure and detailed instructions on how to complete the survey questionnaire.

8.2.3 Survey Period

As described in Table 1-1, the PT surveys were undertaken for one day at every location. The survey period was 7am to 7pm everyday survey day.

8.3 PT Data Analysis Results

8.3.1 Summary Analysis - Kampala

Table 8-1 presents the PT survey summaries for Kampala. PT surveys in Kampala were undertaken at ten locations. Coach companies in Kampala largely operate at their own terminals. The largest coach operators were identified to be eleven (11). The PT surveys were therefore undertaken at the terminals of these identified operators. The identified operators were:

- Global Coaches;
- Bismarkan Coaches;
- Jaguar Bus Services;
- Easy Coaches;
- KK Bus Services;
- YY Coaches;
- Trinity Bus Services
- Swift Coaches;
- Posta Bus Services;
- Modern Coast Coaches; and
- Link Bus Services.

The collected data for Kampala is hence presented by operator. Detailed data is included in Appendix G.

Table 8-1: PT survey results for Kampala

GLOBAL COACHES		BUS JOURNEY					PASSENGER DETAILS	
STATION	DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
KAMPALA	FROM KAMPALA	KAMPALA	MBARARA	48	270	4.9	65	20,000
KAMPALA	TO KAMPALA	MBARARA	KAMPALA	43	270	4.9	66	20,000

BISMARKAN COACHES		BUS JOURNEY					PASSENGER DETAILS	
STATION	DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
KAMPALA	FROM KAMPALA	KAMPALA	BUSHENYI	6	331	6.0	57	25,000
KAMPALA	FROM KAMPALA	KAMPALA	ISHAKA	3	328	5.5	43	25,000
KAMPALA	FROM KAMPALA	KAMPALA	KATENDE-ISHAKA	1	328	5.5	44	25,000
KAMPALA	FROM KAMPALA	KAMPALA	KISORO	5	467	9.0	66	40,000
KAMPALA	FROM KAMPALA	KAMPALA	MBARARA	1	270	4.9	33	20,000
KAMPALA	FROM KAMPALA	KAMPALA	RUBIRIZI	1	365	6.3	67	25,000
KAMPALA	TO KAMPALA	BUSHENYI	KAMPALA	7	331	6.0	37	25,000
KAMPALA	TO KAMPALA	KISORO	KAMPALA	5	467	9.0	62	35,000
KAMPALA	TO KAMPALA	MBARARA	KAMPALA	1	270	4.9	55	20,000
KAMPALA	TO KAMPALA	RUBIRIZI	KAMPALA	1	365	6.3	40	21,500

JAGUAR BUS SERVICES		BUS JOURNEY					PASSENGER DETAILS	
STATION	DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
KAMPALA	FROM KAMPALA	KAMPALA	DRC	1	570	12.0	64	50,000
KAMPALA	FROM KAMPALA	KAMPALA	KAMEMBE	1	744	13.8	55	60,000
KAMPALA	FROM KAMPALA	KAMPALA	KANYALU	1	617	11.0	46	60,000
KAMPALA	FROM KAMPALA	KAMPALA	KIGALI	2	510	8.6	48	50,000
KAMPALA	FROM KAMPALA	KAMPALA	GOMA	1	570	12.0	48	50,000
KAMPALA	FROM KAMPALA	KAMPALA	KISORO	5	467	8.5	57	40,000
KAMPALA	FROM KAMPALA	KAMPALA	KIVU/KISENYI	1	694	13.3	48	50,000
KAMPALA	FROM KAMPALA	KAMPALA	LUBAVU	1	570	12.0	48	50,000
KAMPALA	TO KAMPALA	GISENYI	KAMPALA	1	669	12.3	46	40,000
KAMPALA	TO KAMPALA	GOMA	KAMPALA	1	570	12.0	49	46,500
KAMPALA	TO KAMPALA	KABALE	KAMPALA	1	410	6.6	50	30,000
KAMPALA	TO KAMPALA	KIGALI	KAMPALA	2	510	8.6	49	50,000
KAMPALA	TO KAMPALA	KANYALU	KAMPALA	1	617	11.0	50	60,000
KAMPALA	TO KAMPALA	KATUNA	KAMPALA	1	430	7.0	45	40,000
KAMPALA	TO KAMPALA	KISENYI	KAMPALA	1	669	12.3	42	50,000
KAMPALA	TO KAMPALA	KISORO	KAMPALA	4	467	8.5	55	40,000
KAMPALA	TO KAMPALA	NARUKOKO	KAMPALA	1	520	11.0	45	40,000

EASY COACHES		BUS JOURNEY					PASSENGER DETAILS	
STATION	DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
KAMPALA	FROM KAMPALA	KAMPALA	NAIROBI	3	664	12.8	30	70,000
KAMPALA	TO KAMPALA	NAIROBI	KAMPALA	3	664	12.8	33	73,000
KAMPALA	TO KAMPALA	KISUMU	KAMPALA	1	316	6.8	32	42,000
KAMPALA	TO KAMPALA	MAKUENI	KAMPALA	1	846	15.5	34	76,000

KK BUS SERVICES		BUS JOURNEY					PASSENGER DETAILS	
STATION	DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
KAMPALA	FROM KAMPALA	KAMPALA	ARUA	4	495	8.4	46	43,000
KAMPALA	FROM KAMPALA	KAMPALA	KOBOKO	2	549	8.0	48	40,000
KAMPALA	FROM KAMPALA	KAMPALA	ZOMBO	2	458	7.0	45	30,000
KAMPALA	TO KAMPALA	ARUA	KAMPALA	3	495	8.4	48	40,000
KAMPALA	TO KAMPALA	KOBOKO	KAMPALA	2	549	8.0	53	40,000
KAMPALA	TO KAMPALA	ZOMBO	KAMPALA	2	458	7.0	47	27,571

YY COACHES		BUS JOURNEY					PASSENGER DETAILS	
STATION	DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
KAMPALA	FROM KAMPALA	KAMPALA	LIRA	2	337	5.0	61	25,000
KAMPALA	FROM KAMPALA	KAMPALA	MBALE	12	226	4.5	61	20,000
KAMPALA	FROM KAMPALA	KAMPALA	SOROTI	9	327	6.0	61	25,000
KAMPALA	TO KAMPALA	APAC	KAMPALA	1	268	4.5	67	20,000
KAMPALA	TO KAMPALA	LIRA	KAMPALA	2	337	5.0	67	20,000
KAMPALA	TO KAMPALA	MBALE	KAMPALA	13	226	4.5	63	19,077
KAMPALA	TO KAMPALA	SOROTI	KAMPALA	7	327	6.0	62	25,000

TRINITY BUS SERVICES		BUS JOURNEY					PASSENGER DETAILS	
STATION	DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
KAMPALA	FROM KAMPALA	KAMPALA	BUKAVU-CONGO	1	570	9.7	50	58,667
KAMPALA	FROM KAMPALA	KAMPALA	GISENYI	1	669	7.0	51	40,000
KAMPALA	FROM KAMPALA	KAMPALA	GISORO	1	467	7.0	49	40,000
KAMPALA	FROM KAMPALA	KAMPALA	GOMA	1	570	12.3	51	50,000
KAMPALA	FROM KAMPALA	KAMPALA	JUBA	2	635	10.3	52	100,000
KAMPALA	FROM KAMPALA	KAMPALA	KIGALI	2	510	8.6	44	50,000
KAMPALA	FROM KAMPALA	KAMPALA	KISORO	1	467	8.6	58	50,000
KAMPALA	TO KAMPALA	GISENYI	KAMPALA	1	669	12.3	48	45,000
KAMPALA	TO KAMPALA	GISORO	KAMPALA	1	467	8.5	67	40,000
KAMPALA	TO KAMPALA	GOMA	KAMPALA	1	570	12.0	51	50,000
KAMPALA	TO KAMPALA	JUBA	KAMPALA	1	635	10.3	51	90,000
KAMPALA	TO KAMPALA	KIGALI	KAMPALA	1	510	8.6	52	50,000
KAMPALA	TO KAMPALA	KYEGEGWA	KAMPALA	1	192	3.3	44	15,000
KAMPALA	TO KAMPALA	LUKAYA	KAMPALA	1	105	2.3	67	10,000
KAMPALA	TO KAMPALA	NAIROBI	KAMPALA	1	664	12.8	35	70,000

SWIFT COACHES		BUS JOURNEY					PASSENGER DETAILS	
STATION	DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
KAMPALA	FROM KAMPALA	KAMPALA	BUSHENYI	4	331	6.0	63	25,000
KAMPALA	FROM KAMPALA	KAMPALA	ISHAKA	2	328	5.4	65	25,000
KAMPALA	FROM KAMPALA	KAMPALA	MBARARA	1	270	5.0	53	20,000
KAMPALA	FROM KAMPALA	KAMPALA	RUBIRIZI	1	365	6.3	65	25,000
KAMPALA	TO KAMPALA	BUSHENYI	KAMPALA	2	331	6.0	37	23,000

POSTA BUS SERVICES		BUS JOURNEY					PASSENGER DETAILS	
STATION	DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
KAMPALA	FROM KAMPALA	KAMPALA	GULU	1	334	5.3	66	25,000
KAMPALA	FROM KAMPALA	KAMPALA	KITGUM	2	435	6.5	65	30,000
KAMPALA	TO KAMPALA	GULU	KAMPALA	1	334	5.3	52	20,000
KAMPALA	TO KAMPALA	KITGUM	KAMPALA	2	435	6.5	64	27,000

MODERN COAST COACHES		BUS JOURNEY					PASSENGER DETAILS	
STATION	DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
KAMPALA	FROM KAMPALA	KAMPALA	NAIROBI	4	664	13.0	40	90,000
KAMPALA	TO KAMPALA	KIGALI	KAMPALA	2	510	8.6	34	51,000
KAMPALA	TO KAMPALA	MOMBASA	KAMPALA	1	1150	20.0	45	115,000
KAMPALA	TO KAMPALA	NAIROBI	KAMPALA	4	664	13.0	37	80,000

LINK BUS SERVICES		BUS JOURNEY					PASSENGER DETAILS	
STATION	DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
KAMPALA	FROM KAMPALA	KAMPALA	BUNDIBUGYO	3	379	6.3	68	25,000
KAMPALA	FROM KAMPALA	KAMPALA	BWERA	1	424	7.3	66	35,000
KAMPALA	FROM KAMPALA	KAMPALA	FORT PORTAL	1	295	5.0	67	22,500
KAMPALA	FROM KAMPALA	KAMPALA	HOIMA	10	199	3.8	66	15,000
KAMPALA	FROM KAMPALA	KAMPALA	KASESE	17	370	7.0	66	35,000
KAMPALA	FROM KAMPALA	KAMPALA	MASINDI	6	211	3.3	66	15,000
KAMPALA	FROM KAMPALA	KAMPALA	NYAHUKA	1	391	6.8	64	25,000
KAMPALA	TO KAMPALA	BUNDIBUGYO	KAMPALA	3	379	6.3	65	23,000
KAMPALA	TO KAMPALA	BWERA	KAMPALA	1	424	7.3	55	35,000
KAMPALA	TO KAMPALA	FORT PORTAL	KAMPALA	1	295	5.0	66	20,000
KAMPALA	TO KAMPALA	HOIMA	KAMPALA	7	199	3.8	60	15,000
KAMPALA	TO KAMPALA	KAIIGO	KAMPALA	1	215	3.4	35	15,000
KAMPALA	TO KAMPALA	KASESE	KAMPALA	14	370	7.0	60	30,000
KAMPALA	TO KAMPALA	MASINDI	KAMPALA	5	211	3.3	53	12,000

8.3.2 Summary Analysis - Countrywide

Table 8-2 presents the PT survey summary for all countrywide locations.

Table 8-2: PT survey results for Jinja

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	MBALE	8	226	4.0	59	25,000
FROM KAMPALA	KAMPALA	SOROTI	14	327	6.1	61	25,000
FROM KAMPALA	KAMPALA	MBALE	8	226	4.0	59	25,000
FROM KAMPALA	KAMPALA	TORORO	1	211	4.5	45	25,000
FROM KAMPALA	KAMPALA	NAIROBI	2	667	13.9	44	78,900
FROM KAMPALA	KAMPALA	KOTIDO	2	527	9.0	44	51,250
FROM KAMPALA	KAMPALA	LIRA	4	337	6.0	47	30,000
FROM KAMPALA	KAMPALA	MOROTO	2	495	9.5	50	41,000
FROM KAMPALA	KAMPALA	KAABONG	1	586	10.0	40	55,000
TO KAMPALA	LIRA	KAMPALA	3	337	6.0	59	27,000
TO KAMPALA	MBALE	KAMPALA	4	226	4.0	60	18,000
TO KAMPALA	MOROTO	KAMPALA	2	495	8.5	60	40,000
TO KAMPALA	NAIROBI	KAMPALA	1	667	13.2	64	70,000
TO KAMPALA	SOROTI	KAMPALA	5	327	6.0	65	20,000

Table 8-3: PT survey results for Soroti

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	SOROTI	6	327	6.0	46	25,000
TO KAMPALA	SOROTI	BUSIA	1	201	3.5	41	15,000
TO KAMPALA	SOROTI	GULU	1	224	4.0	57	20,000
TO KAMPALA	SOROTI	KAMPALA	11	327	6.5	63	25,000

Table 8-4: PT survey results for Tororo

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	TORORO	12	210	4.8	14	17,552
TO KAMPALA	TORORO	KAMPALA	2	210	4.8	14	18,000
TO KAMPALA	BUSIA	KAMPALA	1	194	4.3	14	20,000
TO KAMPALA	LIRA	KAMPALA	1	337	5.3	46	25,000
TO TORORO	BUGIRI	TORORO	1	59	1.3	14	6,000
TO TORORO	BUSIA	TORORO	6	52	1.3	14	5,036
TO TORORO	BUTALEJA	TORORO	1	52	1.3	14	5,000
TO TORORO	IGANGA	TORORO	2	93	1.8	14	10,000
TO TORORO	JINJA	TORORO	11	131	2.3	14	13,000
TO TORORO	MBALE	TORORO	18	51	1.9	14	5,112
FROM TORORO	TORORO	BUSIA	1	52	0.5	14	6,000
FROM TORORO	TORORO	JINJA	3	131	2.3	14	13,000
FROM TORORO	MBALE	BUSIA	1	98	1.8	14	6,000
FROM TORORO	MBALE	JINJA	2	145	2.5	14	15,000
TO MBALE	IGANGA	MBALE	1	108	1.7	14	8,000
FROM LIRA	LIRA	BUSIA	1	324	5.3	57	25,000

Table 8-5: PT survey results for Mbale

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	MBALE	19	226	4.5	60	25,000
TO KAMPALA	MBALE	KAMPALA	14	226	4.5	66	20,000
FROM MBALE	MBALE	GULU	5	326	5.2	60	25,000
FROM MBALE	MBALE	BUSIA	1	98	1.8	60	6,000
FROM MBALE	MBALE	LIRA	3	227	3.3	60	18,000
FROM MBALE	MBALE	KILYANDONGO	1	341	5.0	60	20,000
FROM MBALE	MBALE	MALABA	4	63	1.4	62	5,000
FROM MBALE	MBALE	MOROTO	4	271	4.0	67	20,000
FROM MBALE	MBALE	NAMAYINGO	1	121	2.5	50	10,000
FROM MBALE	MBALE	SOROTI	1	103	1.5	50	8,000
FROM MBALE	MBALE	SUDAN	3	430	7.0	50	35,000
TO MBALE	ARUA	MBALE	1	539	8.5	67	35,000
TO MBALE	BUSIA	MBALE	2	98	1.8	67	6,000
TO MBALE	GULU	MBALE	2	326	5.2	67	25,000
TO MBALE	IGANGA	MBALE	1	108	1.7	28	8,000
TO MBALE	MOROTO	MBALE	1	271	4.0	50	20,000
TO MBALE	NEBBI	MBALE	1	431	6.5	58	30,000
TO MBALE	NIMULI	MBALE	1	430	7.0	50	35,000
TO MBALE	SOROTI	MBALE	4	103	1.5	59	8,000
TO MBALE	TORORO	MBALE	2	51	1.1	48	4,000
TO MBALE	KITGUM	MBALE	2	349	5.5	67	20,000
TO MBALE	LIRA	MBALE	4	227	3.3	62	18,000
TO MBALE	KUMI	MBALE	1	56	1.0	67	5,000

Table 8-6: PT survey results for Lira

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	ABIM	1	458	7.5	48	35,000
FROM KAMPALA	KAMPALA	APAC	3	268	4.5	44	20,000
FROM KAMPALA	KAMPALA	GULU	1	334	5.0	40	25,000
FROM KAMPALA	KAMPALA	LIRA	8	337	5.0	49	25,000
TO KAMPALA	LIRA	KAMPALA	21	337	5.0	49	25,000
TO KAMPALA	PADER	KAMPALA	1	408	6.5	46	30,000
TO KAMPALA	ABIM	KAMPALA	1	458	7.5	38	35,000
TO KAMPALA	APAC	KAMPALA	9	268	4.3	43	20,000
FROM LIRA	LIRA	APAC	1	61	1.0	41	6,000
FROM LIRA	LIRA	BUSIA	1	324	5.5	64	25,000
FROM LIRA	LIRA	GULU	1	101	2.0	39	10,000
TO LIRA	AGAGO	LIRA	1	114	2.3	29	15,000
TO LIRA	DOKOLO	LIRA	1	47	1.0	29	8,000
FROM BUSIA	BUSIA	GULU	1	424	7.0	42	30,000
FROM GULU	GULU	BUSIA	2	424	6.5	44	27,000

Table 8-7: PT survey results for Gulu

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	ADJUMANI	1	443	6.7	67	35,000
FROM KAMPALA	KAMPALA	AMURU	1	352	5.3	67	30,000
FROM KAMPALA	KAMPALA	GULU	6	334	5.0	60	25,000
FROM KAMPALA	KAMPALA	KAABONG	1	586	10.0	50	45,000
FROM KAMPALA	KAMPALA	KITGUM	5	435	6.5	53	30,000
FROM KAMPALA	KAMPALA	NIMULE	1	437	6.5	30	30,000
TO KAMPALA	ADJUMANI	KAMPALA	1	443	6.7	40	30,000
TO KAMPALA	AMURU	KAMPALA	3	334	5.3	40	25,000
TO KAMPALA	GULU	KAMPALA	14	334	5.0	58	25,000
TO KAMPALA	KITGUM	KAMPALA	6	435	6.5	32	30,000
TO KAMPALA	MOYO	KAMPALA	2	485	8.0	44	40,000
TO KAMPALA	NIMULE	KAMPALA	5	437	6.5	48	30,000
TO KAMPALA	PADER	KAMPALA	1	408	6.6	36	30,000
FROM GULU	GULU	KITGUM	1	104	1.5	67	15,000
FROM GULU	GULU	SOROTI	3	224	3.9	62	20,000
FROM GULU	GULU	KAABONG	1	276	5.3	67	25,000
TO GULU	LIRA	GULU	1	101	2.0	42	12,000
TO GULU	BUSIA	GULU	2	500	7.5	40	38,000
TO GULU	NIMULE	GULU	1	108	1.9	30	10,000
TO GULU	PADER	GULU	1	118	2.3	36	10,000

Table 8-8: PT survey results for Arua

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	ARUA	9	495	8.0	61	41,667
TO KAMPALA	ARUA	KAMPALA	9	495	7.5	57	38,111
FROM ARUA	ARUA	ADJUMANI	5	195	4.0	55	20,000
FROM ARUA	ARUA	HOIMA	3	271	5.5	52	30,000
TO ARUA	ADJUMANI	ARUA	3	195	4.0	55	20,000
TO ARUA	HOIMA	ARUA	1	271	5.5	67	30,000
TO ARUA	LIRA	ARUA	1	314	4.5	40	35,000
TO ARUA	KOBOKO	ARUA	1	55	1.0	40	7,000
TO ARUA	MASINDI	ARUA	1	240	5.0	50	25,000

Table 8-9: PT survey results for Hoima

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	HOIMA	2	199	3.5	57	15,000
FROM KAMPALA	KAMPALA	KIKUUBE	3	213	4.0	62	22,000
TO KAMPALA	HOIMA	KAMPALA	5	199	3.5	59	15,000
TO KAMPALA	KIKUUBE	KAMPALA	1	213	4.0	53	22,000
FROM HOIMA	HOIMA	ARUA	1	271	5.5	28	30,000
TO HOIMA	KIKUUBE	HOIMA	1	23	0.5	57	5,000

Table 8-10: PT survey results for Kasese

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	KASESE	2	369	7.0	58	35,000
TO KAMPALA	KASESE	KAMPALA	16	369	7.0	39	30,000

Table 8-11: PT survey results for Luwero

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	LUWERO	13	64	1.5	14	6,000
TO KAMPALA	LUWERO	KAMPALA	11	64	1.5	14	5,000
FROM LUWERO	LUWERO	ARUA	1	449	6.5	14	35,000
FROM LUWERO	LUWERO	BUIKWE	1	96	2.3	14	10,000
FROM LUWERO	LUWERO	KIRYANDONGO	1	176	2.5	14	15,000
FROM LUWERO	LUWERO	NAKASONGOLA	2	69	1.0	14	5,000
FROM LUWERO	LUWERO	NEBBI	2	341	4.7	14	30,000
FROM LUWERO	LUWERO	PAKWACH	1	321	4.4	14	30,000
FROM LUWERO	LUWERO	WAKISO	3	58	1.5	14	5,000
TO LUWERO	ARUA	LUWERO	2	449	6.5	14	35,000
TO LUWERO	GULU	LUWERO	1	288	4.0	14	25,000
TO LUWERO	KIRYANDONGO	LUWERO	5	176	2.5	14	15,000
TO LUWERO	KITGUM	LUWERO	1	389	5.3	14	30,000
TO LUWERO	KOBOKO	LUWERO	1	503	7.3	14	45,000
TO LUWERO	KUMI	LUWERO	1	260	5.0	14	25,000
TO LUWERO	LIRA	LUWERO	1	291	4.3	14	28,000
TO LUWERO	MASAKA	LUWERO	1	189	4.0	14	15,000
TO LUWERO	MBALE	LUWERO	1	241	4.3	14	20,000
TO LUWERO	NAKASONGOLA	LUWERO	2	69	1.0	14	5,000
TO LUWERO	NEBBI	LUWERO	1	341	4.4	14	30,000
TO LUWERO	PAIDHA	LUWERO	1	321	4.4	14	35,000
TO LUWERO	PAKWACH	LUWERO	2	321	4.4	12	30,000
TO LUWERO	SOROTI	LUWERO	1	343	5.4	14	30,000
TO LUWERO	WAKISO	LUWERO	5	58	1.5	14	5,000
TO LUWERO	YUMBE	LUWERO	1	479	7.5	14	40,000

Table 8-12: PT survey results for Masaka

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	RUKUNGIRI	7	382	6.3	63	24,000
FROM KAMPALA	KAMPALA	BUSHENYI	1	331	5.5	64	24,000
FROM KAMPALA	KAMPALA	IBANDA	4	310	5.2	55	21,000
FROM KAMPALA	KAMPALA	ISHAKA	12	328	5.3	61	20,000
FROM KAMPALA	KAMPALA	KABALE	6	410	6.8	55	35,000
FROM KAMPALA	KAMPALA	KATETE	1	418	7.5	64	35,000
FROM KAMPALA	KAMPALA	KIGALI	1	510	8.5	64	50,000
FROM KAMPALA	KAMPALA	MBARARA	8	270	4.8	59	20,000
FROM KAMPALA	KAMPALA	NTUNGAMO	10	332	5.5	56	25,000
FROM KAMPALA	KAMPALA	LYANTODE	1	203	3.8	64	20,000
TO KAMPALA	BUSHENYI	KAMPALA	6	331	5.5	59	21,000
TO KAMPALA	IBANDA	KAMPALA	5	310	5.2	54	21,000
TO KAMPALA	KABALE	KAMPALA	4	410	6.8	64	35,000
TO KAMPALA	MBARARA	KAMPALA	10	270	4.8	56	20,000
TO KAMPALA	KIGALI	KAMPALA	1	510	8.5	67	50,000
TO KAMPALA	NTUNGAMO	KAMPALA	4	332	5.5	64	21,000
TO KAMPALA	RUKUNGIRI	KAMPALA	2	382	6.3	64	21,000

Table 8-13: PT survey results for Mityana

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	MITYANA	24	70	1.8	14	7,000
TO KAMPALA	MITYANA	KAMPALA	42	70	1.8	12	7,000
FROM MITYANA	MITYANA	NAMUTAMBA	2	19	0.8	6	3,000
TO MITYANA	NAMUTAMBA	MITYANA	1	19	0.8	8	5,000

Table 8-14: PT survey results for Mubende

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	KABAROLE	7	302	5.0	51	30,000
FROM KAMPALA	KAMPALA	KASESE	3	369	6.3	58	35,000
TO KAMPALA	KABAROLE	KAMPALA	12	302	5.0	51	30,000

Table 8-15: PT survey results for Mbarara

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	BUSHENYI	2	331	6.0	36	25,000
FROM KAMPALA	KAMPALA	IBANDA	1	310	7.0	65	25,000
FROM KAMPALA	KAMPALA	KISORO	6	467	9.0	51	40,000
FROM KAMPALA	KAMPALA	MBARARA	13	270	5.0	56	20,000
FROM KAMPALA	KAMPALA	RUKUNGIRI	2	382	7.0	64	24,500
TO KAMPALA	BUSHENYI	KAMPALA	10	268	5.9	48	25,000
TO KAMPALA	KABALE	KAMPALA	1	410	7.0	41	30,000
TO KAMPALA	MBARARA	KAMPALA	33	270	5.0	51	20,000
TO KAMPALA	RUKUNGIRI	KAMPALA	2	382	8.5	36	25,000
FROM MBARARA	MBARARA	KISORO	1	200	5.0	65	20,000

Table 8-16: PT survey results for Kabale

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	ISHAKA	1	331	5.5	64	25,000
FROM KAMPALA	KAMPALA	KABALE	3	410	6.5	49	30,000
FROM KAMPALA	KAMPALA	KISORO	3	467	9.0	42	40,000
FROM KAMPALA	KAMPALA	KYANIKA	1	470	8.0	64	40,000
TO KAMPALA	KABALE	KAMPALA	5	410	6.6	54	30,000
TO KAMPALA	KISORO	KAMPALA	2	467	9.0	47	40,000

Table 8-17: PT survey results for Bushenyi

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	BUSHENYI	2	331	5.5	41	25,000
TO KAMPALA	BUSHENYI	KAMPALA	4	331	5.5	42	20,000
TO KAMPALA	RUBIRIZI	KAMPALA	3	365	7.0	47	23,333
TO BUSHENYI	KASESE	BUSHENYI	2	96	1.9	47	8,000
TO BUSHENYI	MBARARA	BUSHENYI	15	64	1.3	45	7,000
FROM BUSHENY	BUSHENYI	KASESE	2	96	1.9	44	8,000
FROM BUSHENY	BUSHENYI	MBARARA	12	64	1.3	56	7,000
FROM MBARARA	MBARARA	KASESE	3	155	2.5	39	12,000
TO MBARARA	KASESE	MBARARA	7	155	2.5	46	11,286

Table 8-18: PT survey results for Rukungiri

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	RUKUNGIRI	4	331	5.5	26	25,000
TO KAMPALA	RUKUNGIRI	KAMPALA	5	331	5.5	16	25,000
TO RUKUNGIRI	KABALE	RUKUNGIRI	1	127	2.5	3	15,000
TO RUKUNGIRI	NTUNGAMO	RUKUNGIRI	2	50	1.0	6	6,000
FROM RUKUNGIF	RUKUNGIRI	MBARARA	2	115	2.3	5	12,500

Table 8-19: PT survey results for Kitgum

BUS JOURNEY						PASSENGER DETAILS	
DIRECTION	ORIGIN	DESTINATION	BUS TRIPS / DAY	JOURNEY DISTANCE (KM)	JOURNEY TIME/ TRIP (HRS)	EST. PAX/ BUS	AVERAGE PAX FARE (UGX)
FROM KAMPALA	KAMPALA	KITGUM	4	435	6.5	39	30,000
TO KAMPALA	KAABONG	KAMPALA	1	586	9.5	29	45,000
TO KAMPALA	KITGUM	KAMPALA	14	435	6.5	46	30,000
TO KAMPALA	LAMWO	KAMPALA	2	454	7.5	28	35,000
FROM KITGUM	KITGUM	KAABONG	1	185	4.0	24	15,000

9 ACCIDENT DATA COLLECTION

9.1 Introduction

As part of the traffic survey exercise, it was required to collect accident data for all regions in the country. The actual locations from which data was to be collected were established at survey proposal stage.

Accident data was collected from district and regional police stations across the country. The following locations provided the required information; Jinja, Soroti, Tororo, Mbale, Lira, Gulu, Hoima, Kasese, Luwero, Masaka, Mubende, Mbarara, Kabale, Bushenyi, Rukungiri, Kitgum, Iganga and Arua.

9.2 Methodology

The data was requested from the District or divisional police station by means of a formal authority letter from Police Headquarters in Kampala.

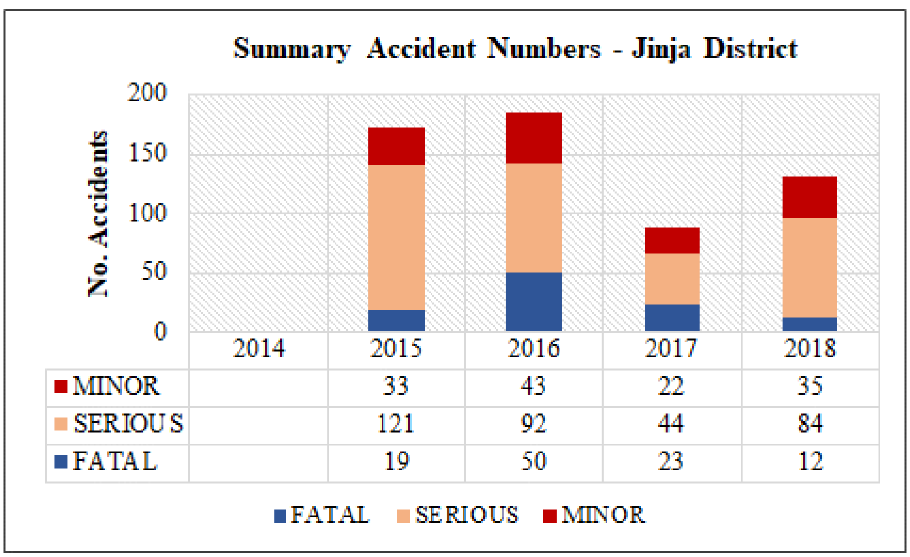
Accident data was requested for at least the previous five years. This was to be annual data classified into fatal, serious and minor accidents with the Police Station's jurisdiction.

9.3 Accident Data Analysis Results

9.3.1 Summary Results – Individual Stations

Figures 9-1 to 9-18 present the accident data analysis for all 18 stations on a station by station basis. Detailed raw data including monthly variations for selected stations is included in Appendix H.

Figure 9-1: Accident analysis – Jinja District

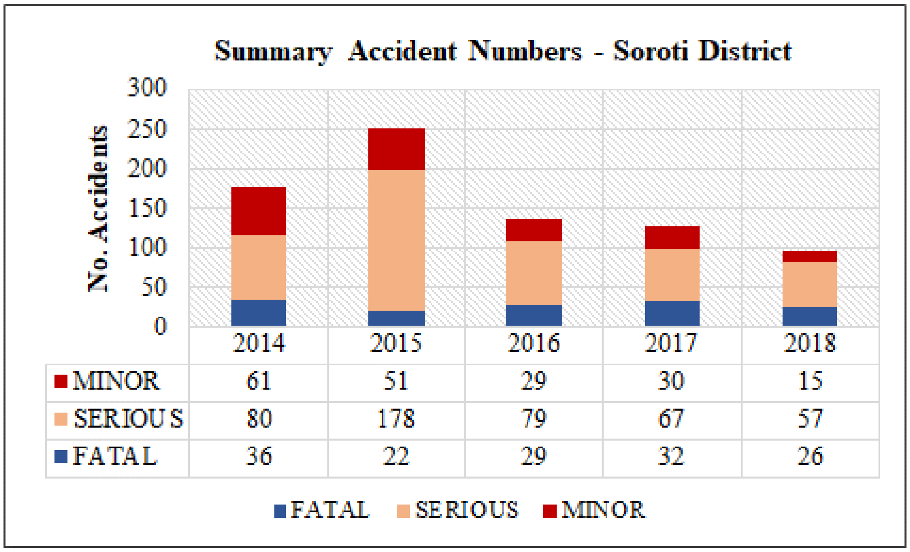


The following observations are made from the figure above (Jinja District):

- Data was only availed to the Consultant for the previous four years

- There is a reduction in total accidents mainly driven by the significant reduction in serious and fatal accidents.

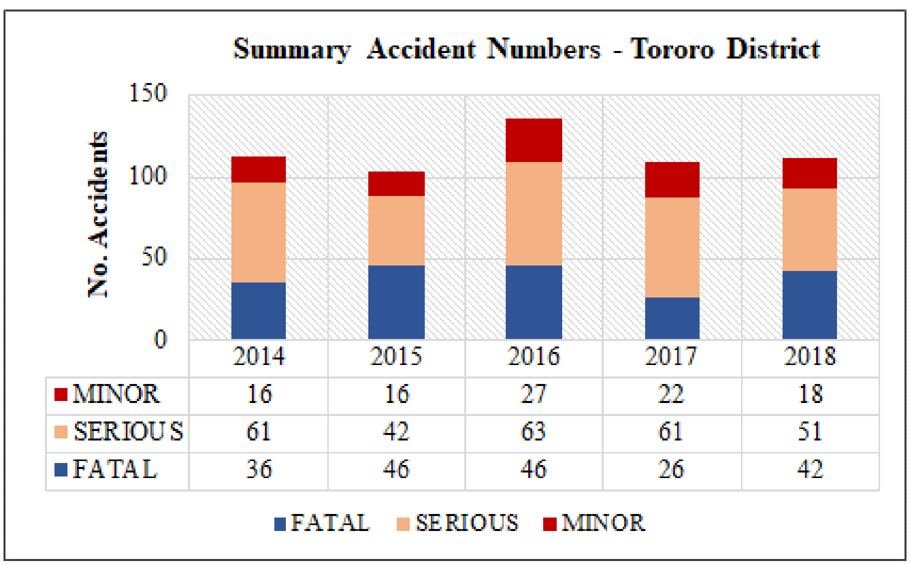
Figure 9-2: Accident analysis – Soroti District



The following observations are made from the figure above (Soroti District):

- There is a significant reduction in total accidents mainly influenced by significant reduction in serious accidents.
- There is an observed doubling in recorded accidents in 2015 and a following reduction by about 60%.

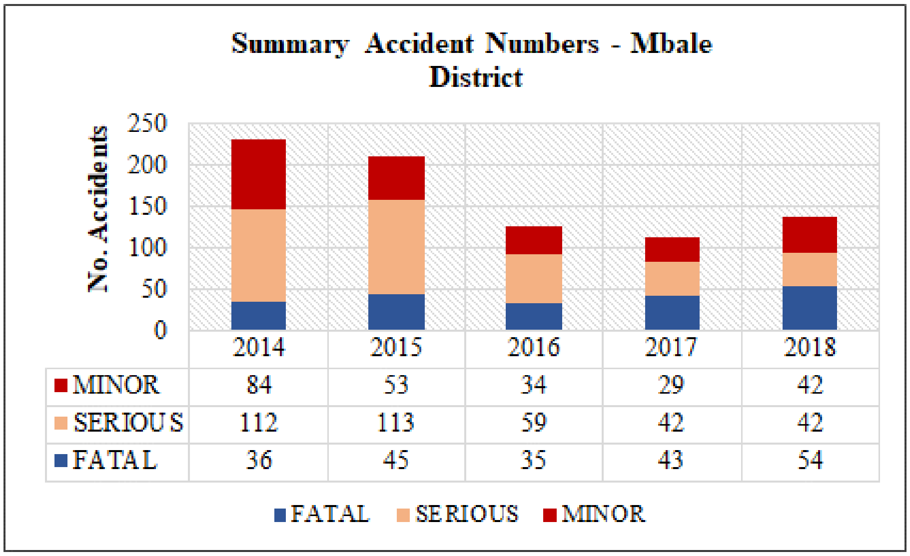
Figure 9-3: Accident analysis – Tororo District



The following observations are made from the figure above (Tororo District):

- There seems no reduction in total accident numbers. There is however some reduction in fatal accidents.

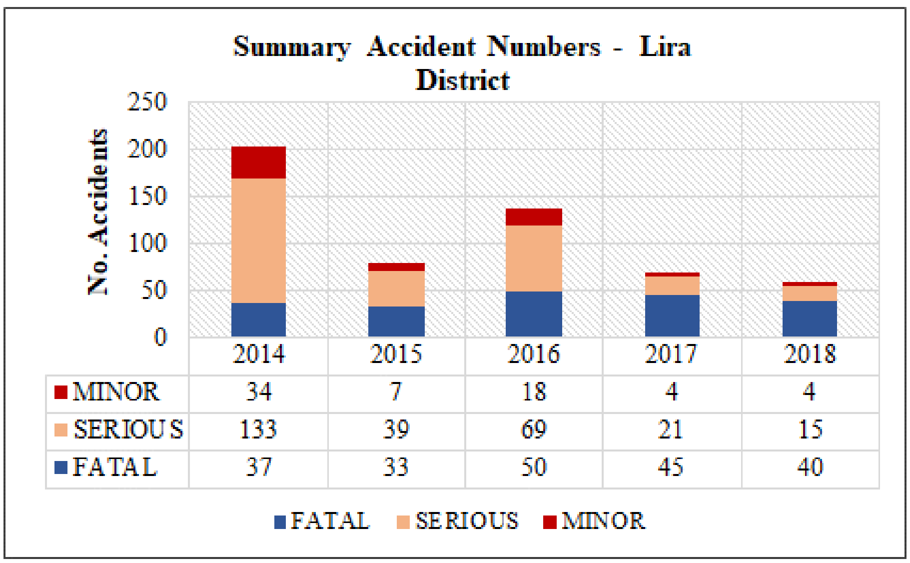
Figure 9-4: Accident analysis – Mbale District



The following observations are made from the figure above (Mbale District):

- There is an observable reduction in total accident numbers influenced by a significant reduction in serious accidents. Recorded serious accidents reduced by 63% between 2014-15 and 2018. There is seems no reduction in fatal accidents.

Figure 9-5: Accident analysis – Lira District

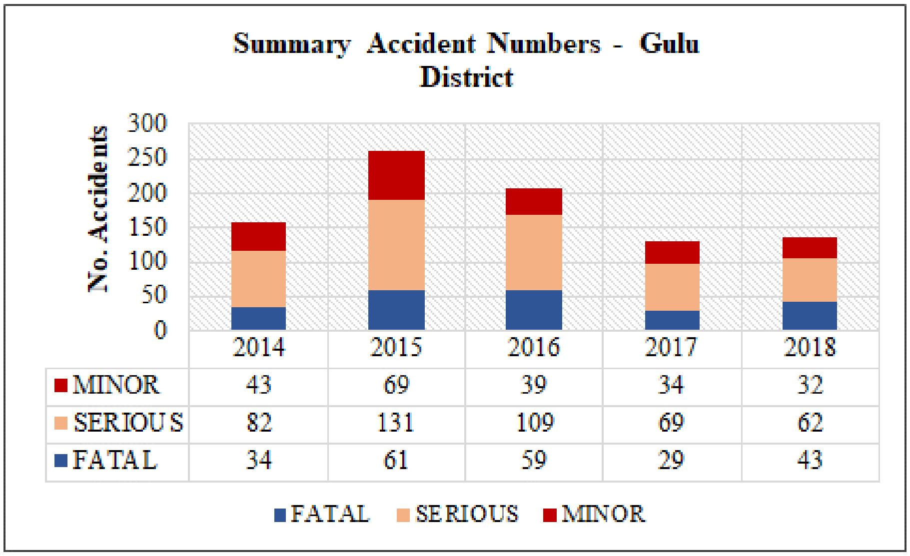


The following observations are made from the figure above (Lira District):

- There is a significant reduction in total accidents mainly influenced by significant reduction in serious accidents. Recorded serious accidents reduced by 89%.

- There seems to be a reduction in minor accidents. However, minor accidents are not comprehensively recorded because they are not inspected adequately.

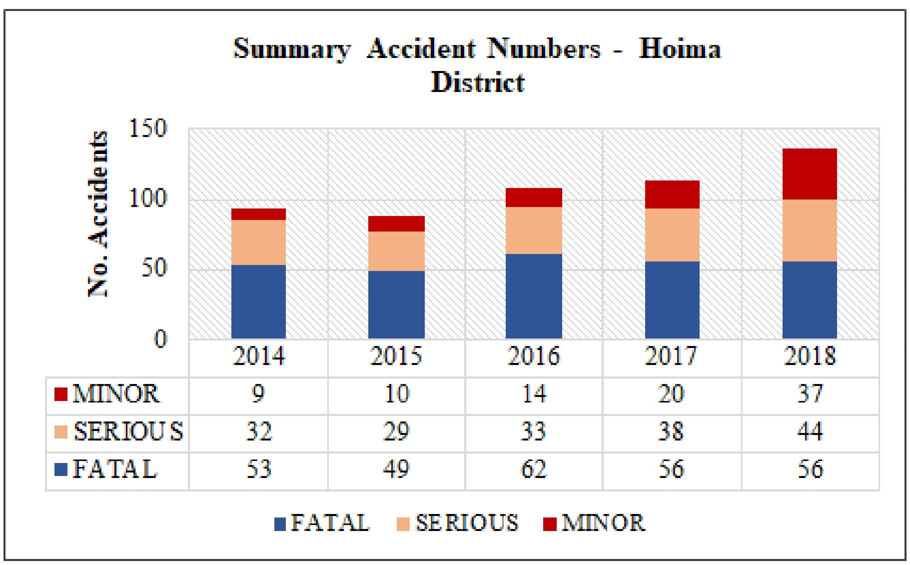
Figure 9-6: Accident analysis – Gulu District



The following observations are made from the figure above (Gulu District):

- There is an observable reduction in total accident numbers influenced by a significant reduction in serious accidents. Recorded serious accidents reduced by 42% between 2014-15 and 2018. There is also a reduction in fatal accidents.

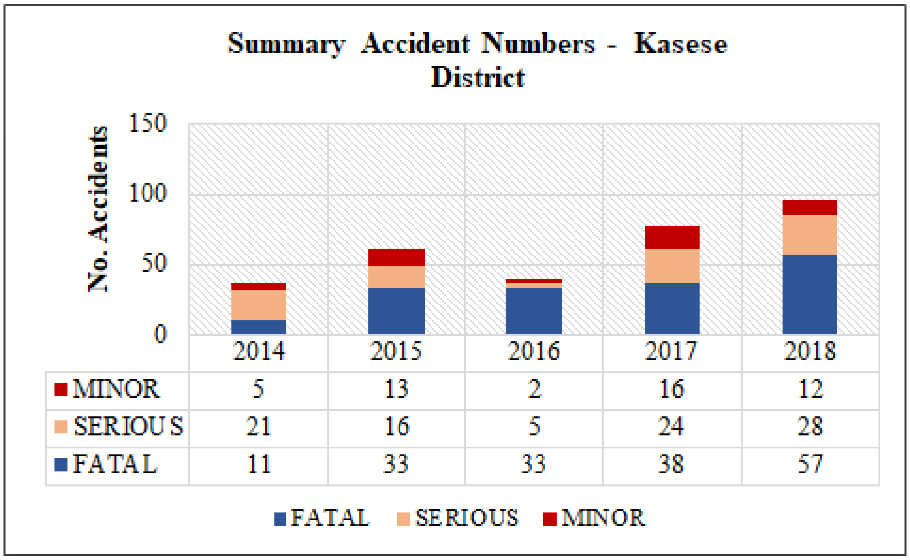
Figure 9-7: Accident analysis – Hoima District



The following observations are made from the figure above (Hoima District):

- There is an observable increase in total accident numbers influenced by a slight increase in serious accidents between 2015 and 2018. There is also an increase in recorded minor accidents.

Figure 9-8: Accident analysis – Kasese District

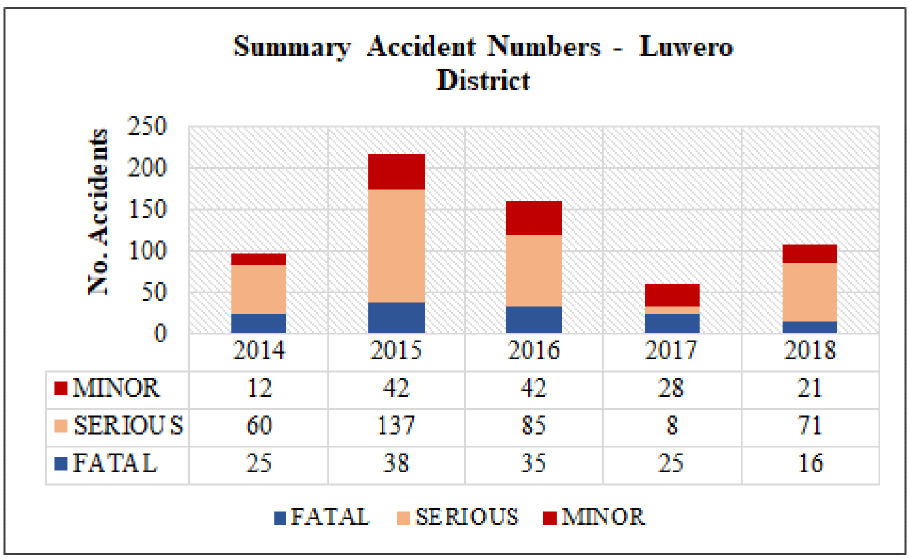


	2014	2015	2016	2017	2018
MINOR	5	13	2	16	12
SERIOUS	21	16	5	24	28
FATAL	11	33	33	38	57

The following observations are made from the figure above (Kasese District):

- There is an observable increase in total accident numbers influenced by a 418% increase in fatal accidents between 2014 and 2018. There is also an increase in recorded minor accidents.

Figure 9-9: Accident analysis – Luwero District

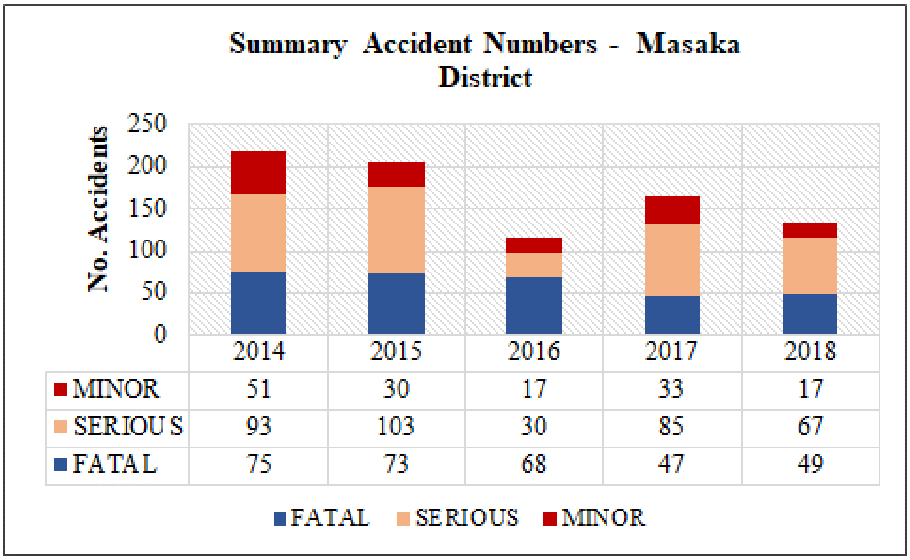


	2014	2015	2016	2017	2018
MINOR	12	42	42	28	21
SERIOUS	60	137	85	8	71
FATAL	25	38	35	25	16

The following observations are made from the figure above (Luwero District):

- There is an observable reduction in total accident numbers influenced by a significant reduction in serious accidents. Recorded serious accidents reduced by 60% between 2014-15 and 2017-18. There is also a reduction in fatal accidents.

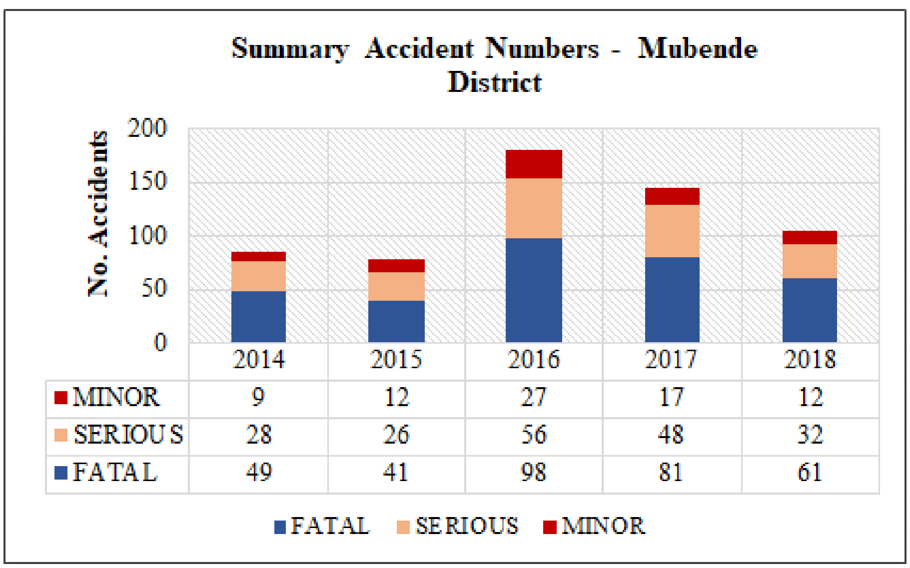
Figure 9-10: Accident analysis – Masaka District



The following observations are made from the figure above (Masaka District):

- There is an observable reduction in total accident numbers influenced by a reduction in serious and fatal accidents. Recorded serious accidents reduced by 22% between 2014-15 and 2017-18 and fatal accidents reduced by 35% over the same period.

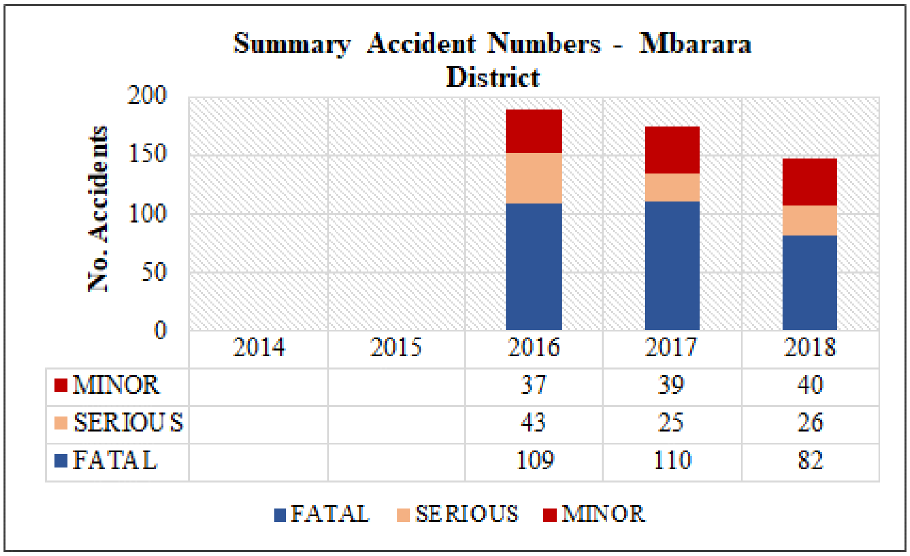
Figure 9-11: Accident analysis – Mubende District



The following observations are made from the figure above (Mubende District):

- There is an observable increase in total accident numbers in the period 2017-18 compared to 2014-15, influenced by increases in serious and fatal accidents. The increase is influenced by the significant increase in recorded accidents in 2016. However, from 2016, there is a declining trend in all accidents.

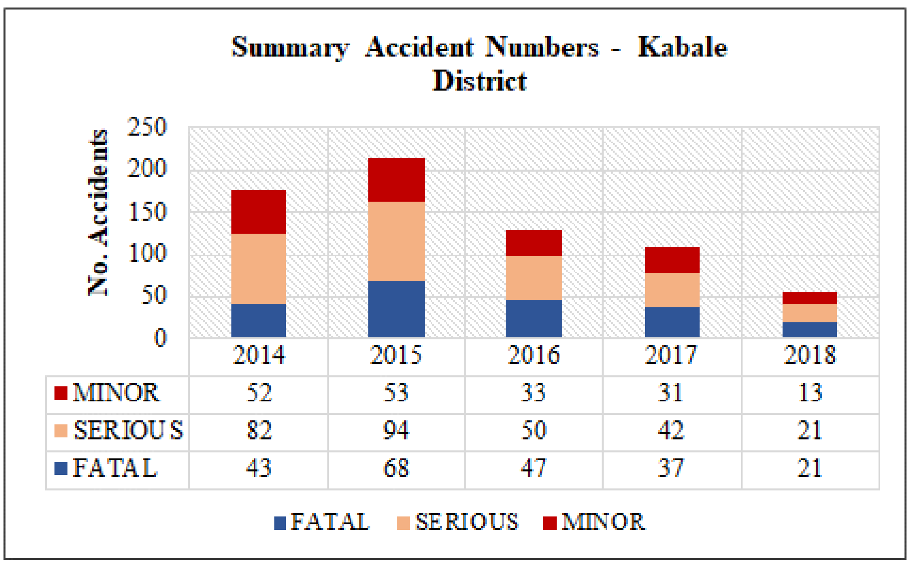
Figure 9-12: Accident analysis – Mbarara District



The following observations are made from the figure above (Mbarara District):

- Only data for the three years 2016, 2017 and 2018 was availed.
- There is a declining trend in all accident numbers. Fatal accidents are significantly high even though they are on a declining trend, likely driven by the busy Kampala – Mbarara – Kasese corridor.

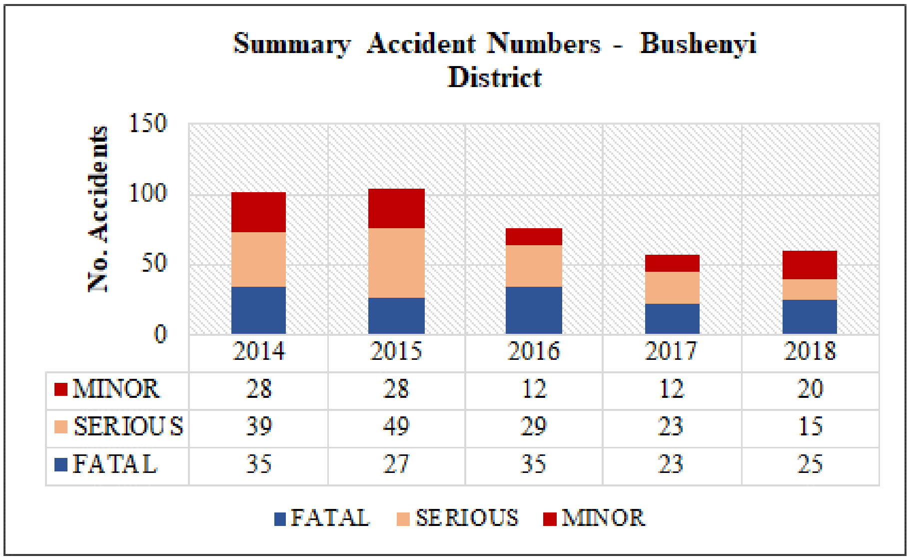
Figure 9-13: Accident analysis – Kabale District



The following observations are made from the figure above (Kabale District):

- There is an observable reduction in total accident numbers influenced by a significant reduction in serious and fatal accidents. Recorded serious accidents reduced by 76% between 2014-15 and 2018. Fatal accidents reduced by 62% over the same period.

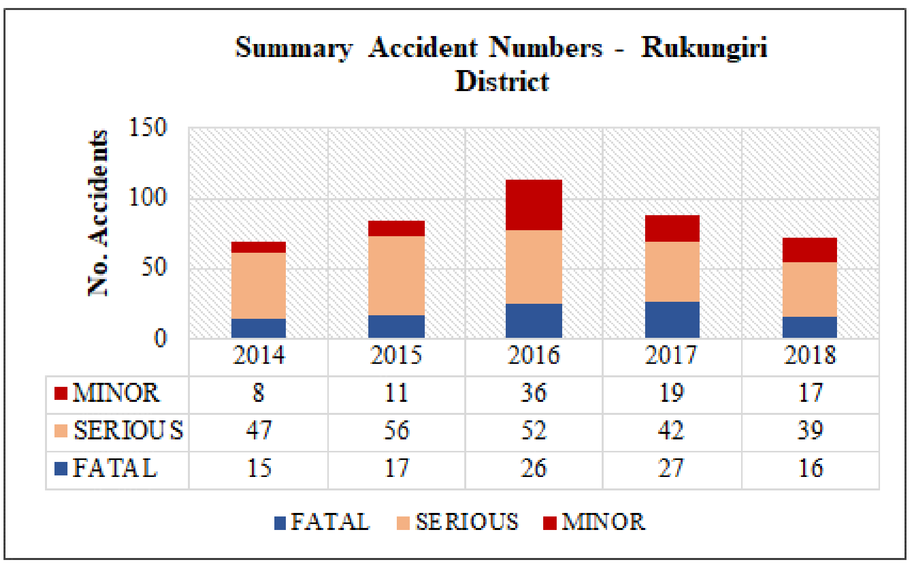
Figure 9-14: Accident analysis – Bushenyi District



The following observations are made from the figure above (Bushenyi District):

- Similar to Kabale, there is an observable reduction in total accident numbers influenced by a significant reduction in serious accidents. Recorded serious accidents reduced by 67% between 2014-15 and 2018. Fatal accidents reduced by 19% over the same period.

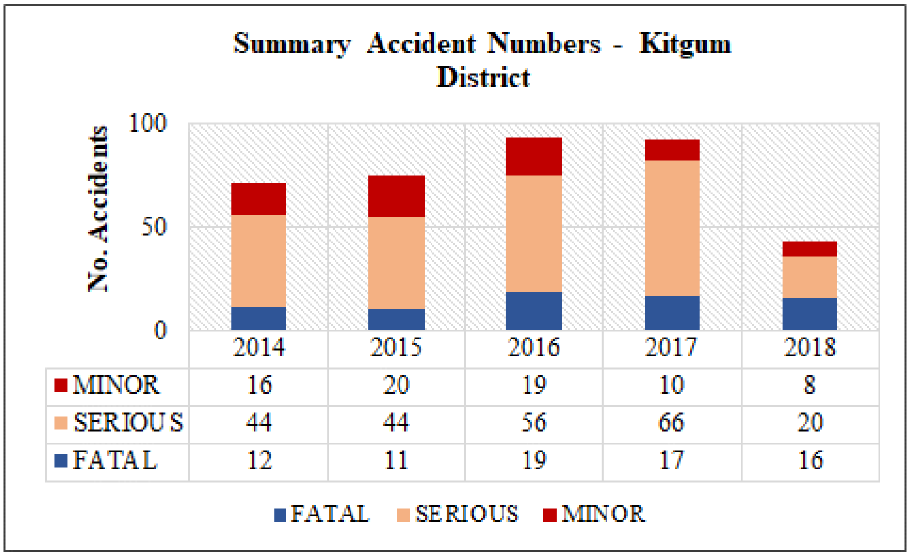
Figure 9-15: Accident analysis – Rukungiri District



The following observations are made from the figure above (Rukungiri District):

- Recorded accidents in Rukungiri increased between 2014 and 2016 but have since been on a clear declining trend. The reduction is mainly influenced by a reduction in serious accidents between 2016 and 2018.

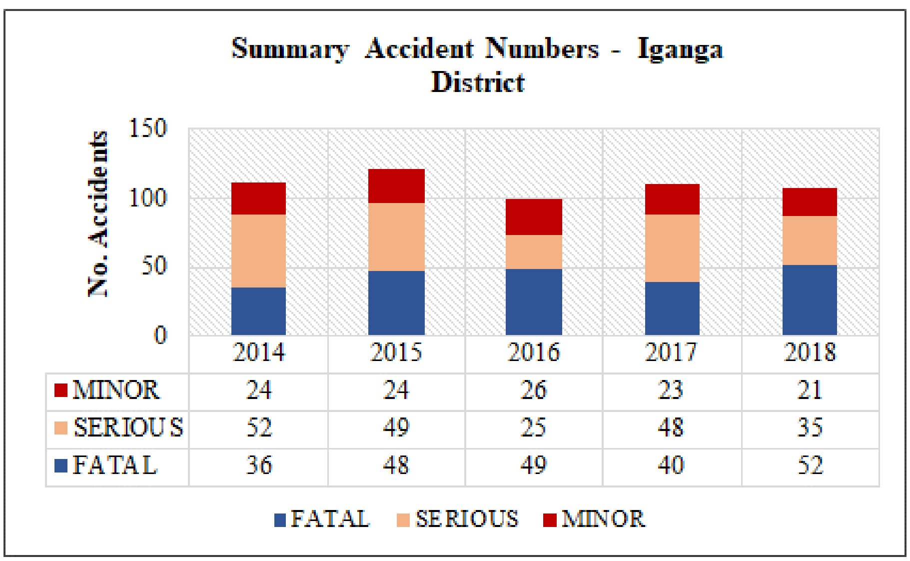
Figure 9-16: Accident analysis – Kitgum District



The following observations are made from the figure above (Kitgum District):

- Total accidents in 2018 are significantly lower than for the period 2014-17. There seems to have been an increasing trend between 2017-18. The 2018 record seems a sudden reversal of the 2017-18 trend. It may be an actual decrease in accidents or more likely an issue with accident recording.

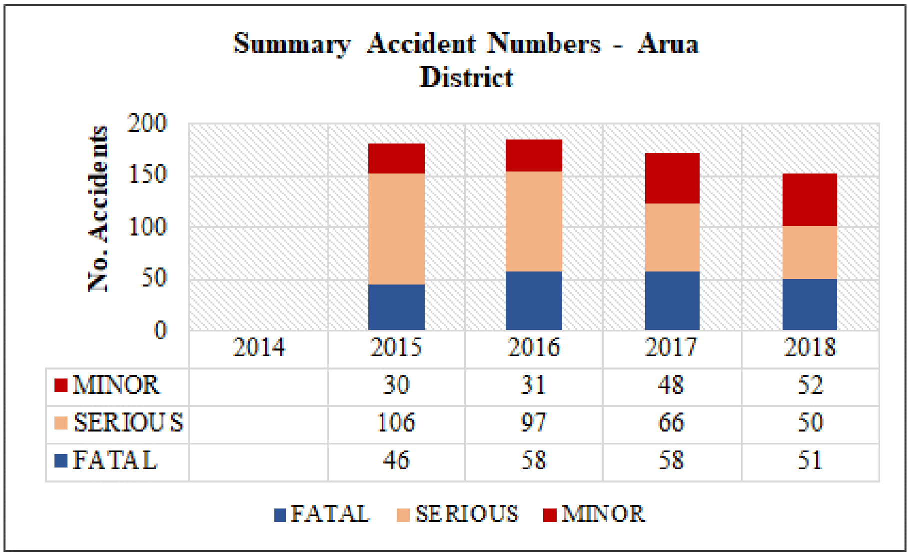
Figure 9-17: Accident analysis – Iganga District



The following observations are made from the figure above (Iganga District):

- There seems no reduction in total accident numbers. There is no significant decline or increase variation in either serious or fatal accidents over the period 2014-2015.

Figure 9-18: Accident analysis – Arua District



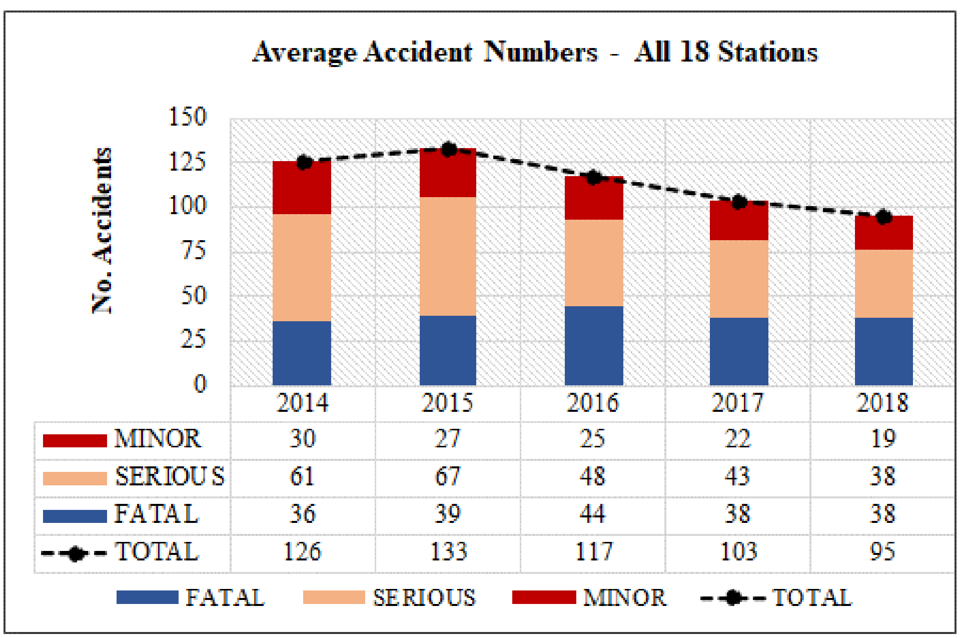
The following observations are made from the figure above (Arua District):

- Only data for the three years 2015, 2016, 2017 and 2018 was availed.
- There is a small decline in total accident numbers. Recorded serious accidents are clearly lower in 2018 than in 2015 but fatal accidents seem slightly higher.

9.3.2 Summary Results – Countrywide

Figures 9-19 presents the accident data analysis for all 18 stations combined. This analysis was used as a proxy to obtain a representative national accident trend.

Figure 9-19: Accident analysis – All stations combined



It is recognized that accident data was not collected from every district Police Station in Uganda and therefore cannot be fully representative of national statistics. In developing this national analysis, the average accident numbers at all 18 stations was taken.

Figure 9-19 indicates the following:

- There is a decreasing trend in recorded accidents.
- The decrease is observable in all three accident types but is most significant with serious accidents. Recorded serious accidents indicate a reduction of 62% over the period 2014 – 2018. Fatal accidents also indicate a 25% reduction over the 5-year period.

The reasons for the general decrease in accidents may vary from one region or district to another, especially with regard to the level of urbanization. However, at national level, the reduction in accidents seems to be the result of increase traffic police present on major highways. Nationally, the largest cause of accidents is reckless and dangerous driving, which accounts for nearly 80% of all accidents[8].

[8] Statistical Abstract 2019, Table 2.6.13

10 FERRY SURVEYS

10.1 Introduction

As part of the traffic survey exercise, traffic surveys on ferries was also conducted at UNRA ferry crossings. The objective of this survey was to determine the following:

- The schedule of the ferry services;
- The length and journey time of ferry services;
- The number of passengers boarding and alighting;
- The estimated freight volume alighting and boarding;
- The volume and vehicle types of motorized and non-motorized traffic carried by the ferries.

The last objective was considered most important. In the planning of the future expressway network, it is considered that some or all of the critical ferry crossings may be converted into road crossings. It is therefore important to understand the current and future traffic volumes carried by the ferries.

The locations of the ferry surveys were presented in Figure 2-17 and Table 2-18.

10.2 Methodology

In order to effectively undertake the survey, enumerators were stationed at one end of the ferry crossing. From this point, the amount of goods alighting and boarding, passengers, vehicular and non-vehicular traffic could be accurately recorded.

The ferry data was collected by means of a specific survey form. The format was well designed to enable the recording of the data specified above for every ferry service that arrived at the survey point.

10.3 Historic Ferry Data

As part of the survey process, historic data on the UNRA ferry routes was obtained. Table 10-1 presents an analysis of the data. The data provided was for the months of July, August and September 2019. The figures shown in the table are three month averages for traffic that crosses on the ferries.

Table 10-1: Analysis of historic ferry data (three-month ADT)

TRAFFIC CROSSING AT FERRY ROUTES (ADT)							
Ferry Name	**Route**	**Buses**	**Trucks**	**Cars/ Pickups**	**Motor-cycles**	**Bicycles**	**TOTAL**
Kyoga Ferries	Zengebe-Namasale	0	22	42	35	46	**146**
Bisina Ferry	Agule-Kokorio	0	8	9	143	40	**200**
Nakiwogo Ferry	Nakiwogo-Buwaya	0	63	40	115	101	**319**

Laropi Ferry	Omi-Laropi	14	23	79	119	19	**254**
Obongi Ferry	Obongi-Sinyanya	1	30	102	112	65	**310**
Albert Nile	Wanseko-Panyimur	0	11	3	18	5	**37**
Kiyindi Ferry	Kiyindi-Buvuma	0	13	16	50	10	**89**
Masindi Port	Masindi-Kungu	2	32	29	168	58	**290**

10.4 Ferry Survey Data Analysis Results

The analysis was undertaken on a ferry by ferry basis. Nine ferries in total were surveyed. The presentation of results in this section is therefore made on a ferry by ferry basis. Detailed data is included in Appendix I.

Figures 10-1 to 10-9 present the analysis results for the ferry services.

Figure 10-1: Analysis results for Bukakata-Kalangala ferry

	Bugoma-Bukakata	Bukakata-Bugoma	Both Directions
Cars	33	24	57
4WD	20	9	30
Pick-ups	9	6	16
matatu	2	1	3
Medium bus	0	0	0
Large Bus	2	0	2
Light Truck	5	0	5
Medium Truck	12	6	18
Heavy Truck	6	3	8
Truck/Trailer	0	0	0
Motorcycle	101	51	152
Tractors	0	0	0
Bicycle	7	4	11
Total	**197**	**104**	**302**

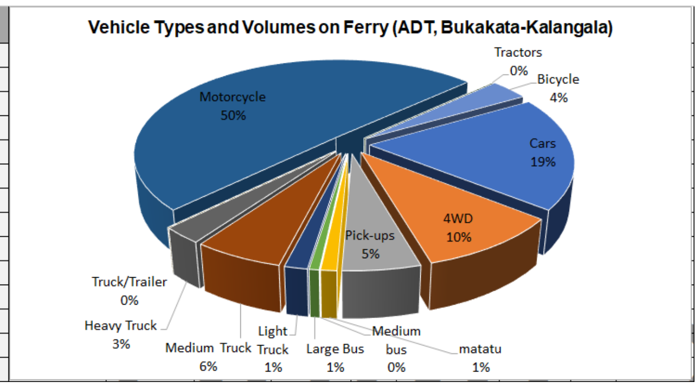


Figure 10-2: Analysis results for Zengebe-Namasale (Amolatar) ferry

	Zengebe-Namasale	Namasale-Zengebe	Both Directions
Cars	4	10	14
4WD	4	7	11
Pick-ups	1	11	12
matatu	0	0	0
Medium bus	0	0	0
Large Bus	0	0	0
Light Truck	1	3	4
Medium Truck	5	10	15
Heavy Truck	0	0	0
Truck/Trailer	0	0	0
Motorcycle	14	26	40
Tractors	0	0	0
Bicycle	16	20	36
Total	**44**	**87**	**131**

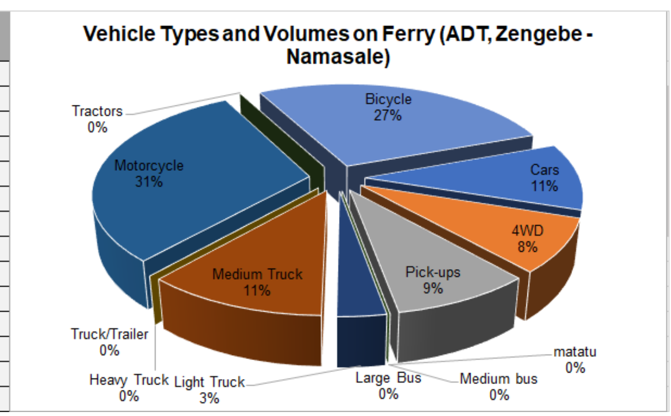


Figure 10-3: Analysis results for Agule-Kokorio (Lake Bisina) ferry

	Agule-Kokorio	Kokorio-Agule	Both Directions
Cars	4	2	5
4WD	2	1	3
Pick-ups	2	2	4
matatu	0	0	0
Medium bus	0	0	0
Large Bus	0	0	0
Light Truck	0	0	0
Medium Truck	4	7	11
Heavy Truck	0	0	0
Truck/Trailer	0	0	0
Motorcycle	45	48	92
Tractors	0	0	0
Bicycle	34	25	59
Total	**90**	**84**	**174**

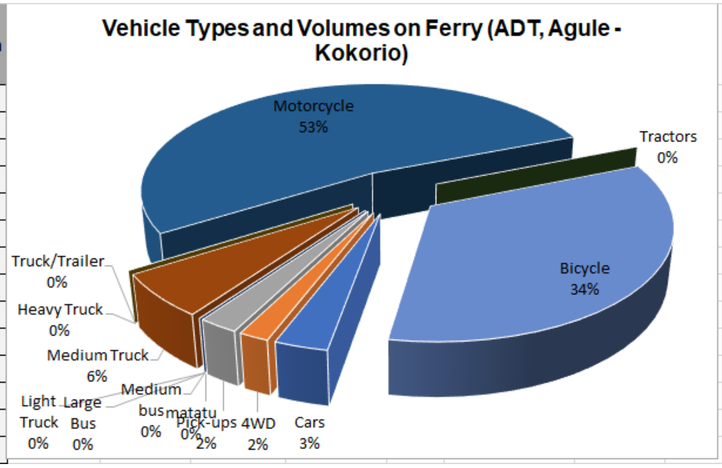


Figure 10-4: Analysis results for Nakiwogo-Buwaya ferry

	Buwaya-Nakiwogo	Nakiwogo-Buwaya	Both Directions
Cars	22	29	51
4WD	7	8	15
Pick-ups	1	0	1
matatu	1	0	1
Medium bus	0	0	0
Large Bus	8	0	8
Light Truck	4	8	11
Medium Truck	3	15	18
Heavy Truck	6	5	11
Truck/Trailer	0	0	0
Motorcycle	63	84	147
Tractors	0	0	0
Bicycle	10	38	48
Total	**123**	**185**	**308**

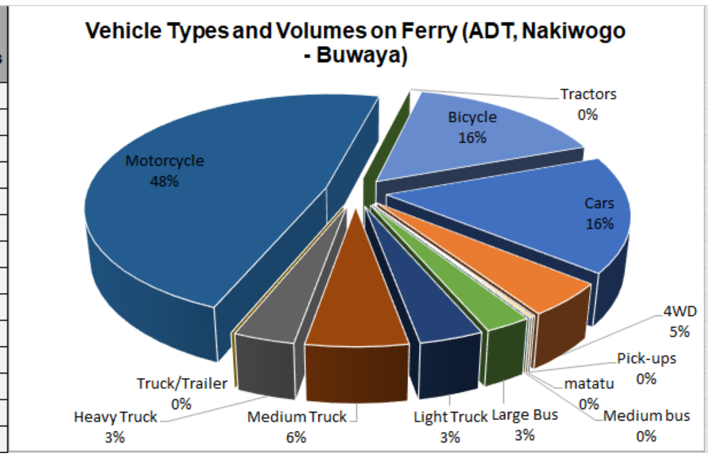


Figure 10-5: Analysis results for Omi-Laropi ferry

	Laropi-Omi	Omi-Laropi	Both Directions
Cars	7	8	15
4WD	37	33	70
Pick-ups	40	3	43
matatu	0	0	0
Medium bus	7	8	15
Large Bus	0	0	0
Light Truck	4	0	4
Medium Truck	7	4	11
Heavy Truck	5	4	9
Truck/Trailer	0	0	0
Motorcycle	41	40	81
Tractors	0	0	0
Bicycle	8	6	14
Total	**155**	**107**	**261**

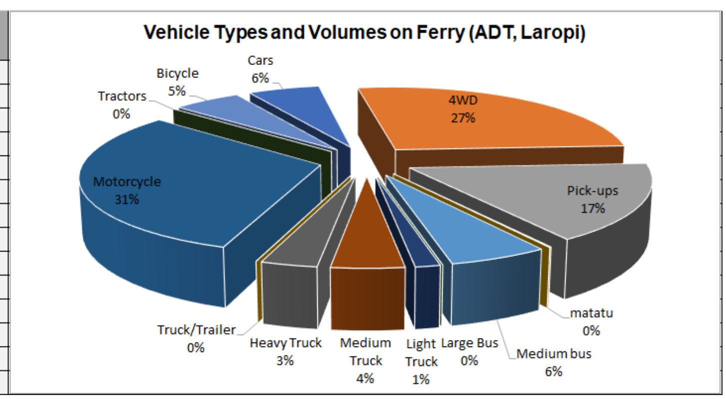


Figure 10-6: Analysis results for Obongi-Sinyanya ferry

	Obongi-Sinyanya	Sinyanya-Obongi	Both Directions
Cars	45	33	78
4WD	7	0	7
Pick-ups	1	1	2
matatu	0	2	2
Medium bus	0	0	0
Large Bus	0	0	0
Light Truck	0	1	1
Medium Truck	3	8	11
Heavy Truck	5	0	5
Truck/Trailer	0	0	0
Motorcycle	46	0	46
Tractors	0	0	0
Bicycle	34	0	34
Total	**141**	**44**	**185**

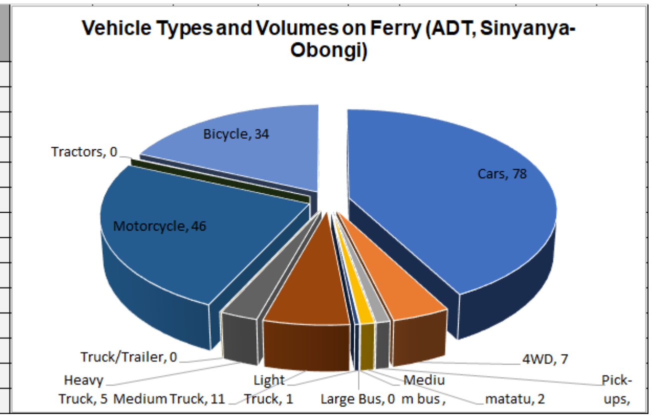


Figure 10-7: Analysis results for Wanseko-Panyimur ferry

	Panyimur-Wanseko	Wanseko-Panyimur	Both Directions
Cars	0	0	1
4WD	2	1	3
Pick-ups	1	0	2
matatu	0	0	0
Medium bus	0	0	0
Large Bus	0	0	0
Light Truck	0	1	1
Medium Truck	6	6	12
Heavy Truck	1	0	1
Truck/Trailer	0	0	0
Motorcycle	10	9	19
Tractors	0	0	0
Bicycle	3	3	6
Total	**23**	**21**	**44**

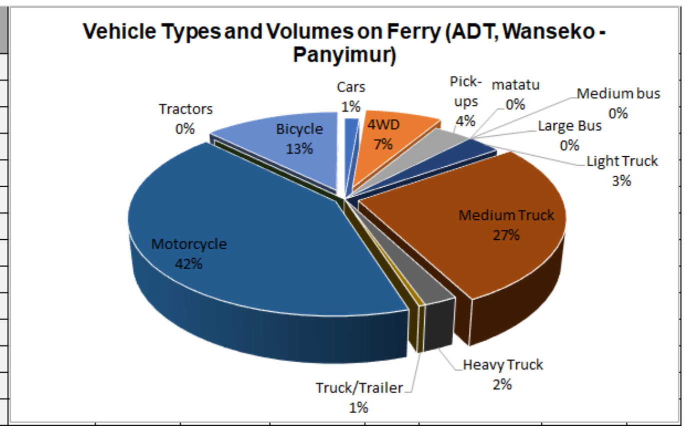


Figure 10-8: Analysis results for Kiyindi-Buvuma ferry

	Buvuma-Kiyindi	Kiyindi-Buvuma	Both Directions
Cars	17	19	36
4WD	5	9	13
Pick-ups	5	4	9
matatu	2	1	3
Medium bus	0	0	0
Large Bus	0	0	0
Light Truck	3	2	4
Medium Truck	7	6	13
Heavy Truck	1	2	3
Truck/Trailer	0	0	0
Motorcycle	28	37	66
Tractors	0	0	0
Bicycle	2	8	10
Total	**70**	**88**	**158**

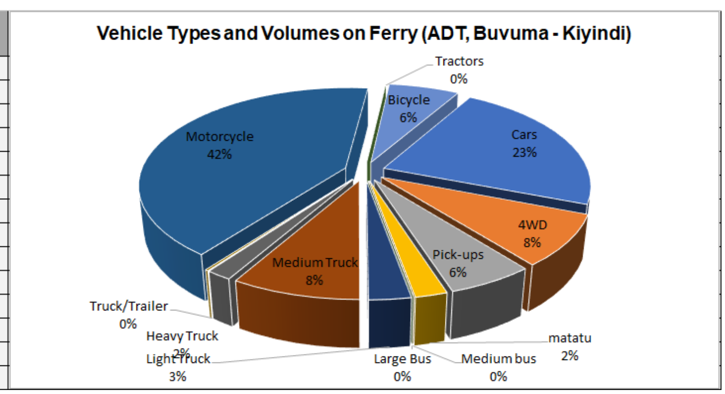
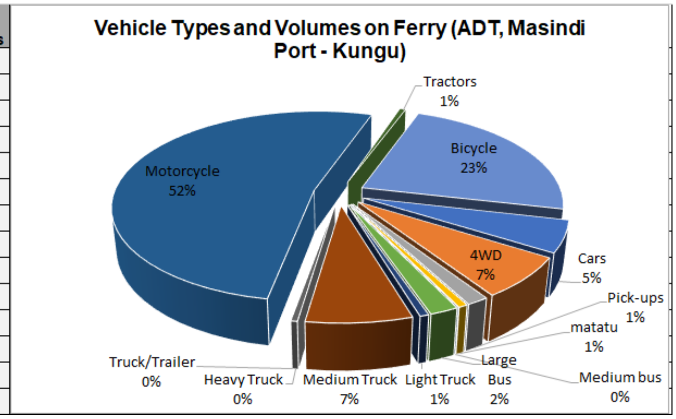


Figure 10-9: Analysis results for Masindi-Kungu ferry

	Kungu-Masindi	Masindi-Kungu	Both Directions
Cars	11	7	18
4WD	12	10	23
Pick-ups	3	2	5
matatu	1	1	2
Medium bus	0	0	0
Large Bus	6	0	6
Light Truck	1	1	2
Medium Truck	18	6	24
Heavy Truck	1	1	1
Truck/Trailer	0	0	0
Motorcycle	117	57	175
Tractors	1	1	2
Bicycle	55	22	77
Total	**226**	**108**	**334**



11 AUTOMATIC TRAFFIC COUNT (ATC) SURVEYS

11.1 Introduction

Roadside automated traffic counters are frequently used to count and classify traffic on heavily trafficked roads. The method has a number of advantages over manual counts, key of which is the convenience of not having to deal with large numbers of enumerators.

A comparison of the automated and manual counts is presented in Table 11-1 below.

Table 11-1: Advantages of ATC surveys

Advantages of ATC Surveys
• Eliminates the need to handle large numbers of enumerators and associated issues of training, mobilisation, late coming, etc
• Cheaper and convenient for counts spanning days and weeks
• More suitable for surveys at highly trafficked road sections
• Saves time as more survey stations can be done at the same time
• Produces better quality data by eliminating enumerators errors due to inexperience, inadequate supervision, tiredness, inclement weather, etc
• With regard to counting, the ATC method also produces more accurate data because it is able to record 24-hour counts on each day as opposed the normal manual method of taking 24-hour counts on only two days.
• Suitable for extended survey periods and regular assessment of traffic flows across the wider road network
• RSU can also measure vehicle speed
Advantages of MCC Surveys
• May be more accurate for low volume road for surveys spanning a few days using well trained enumerators
• Maintains human contact with road incidents that may impact on the quality of the data.
• Cheaper for smaller, shorter surveys

11.2 Objective of the Survey

The Kampala Northern Bypass is a critical highway for Kampala and the nation. The road was constructed to relieve traffic congestion within the city center, allowing cross-country traffic to bypass the city's downtown area. Since its opening, the Northern Bypass has become a highly trafficked, high-speed highway that facilitates the effective distribution of traffic to and from key arterial routes located north of Kampala.

The Kampala Northern Bypass is considered a key route in the proposed EDMP. An investigation of the section by section traffic volumes on the highway was considered key. For that reason, ATC surveys were planned with the objective of determining the ADT on the highway more accurately. This was the main objective of this survey.

11.3 Methodology

11.3.1 Method of Collection

The method involves installation of a roadside unit (RSU) from which two pneumatic tubes are laid across the road. As traffic moves across the two tubes, the RSU records the number of vehicles axles and their spacing for each tube. This information is then used by specialist software to determine the type and number of vehicles moving across the count station. The software utilises inbuilt or custom vehicle profiles that are representative of all vehicle types using on the road network.

ATC surveys were undertaken at 7 survey stations along the Kampala Northern Bypass highway as indicated in Table 2-20.

11.3.2 Site Selection

RSUs were installed at all 7 stations. The stations were carefully chosen along the relevant sections of the highway in accordance with RSU manufacturer's recommendations. The following considerations were made in choosing the ATC survey sites:

- Sites where selected at points where most traffic is travelling at a constant speed. Sites where vehicles are accelerating or decelerating due to bends, steep inclines, traffic signals, or intersections where avoided as much as possible.
- Effort was made to avoid sites where vehicles stop over the tubes although this could not be totally achieved with the existing traffic jam of the highway.
- Sites were chosen to ensure that most traffic would cross perpendicular to the tubes.
- Sites were also chosen to minimise single tube hits due to excessive swerving or lane changing.
- Sites with a suitable securing point for the RSU, such as a post or tree were also given priority.

Good installation of the RSUs is key because it is the main determinant of the quality of data. Installation requires trained or experienced personnel but is largely simple. The following need to be observed when making each installation:

- The pneumatic tubes need to be installed at exactly one-metre spacing for optimal results;
- The length of both tubes should be equal particularly the length from the kerb to the RSU;

- The tension in each tube should be similar; and
- The tube should be pressed against the road surface for its entire length.

11.4 ATC Equipment

11.4.1 Vehicle Classification System

The survey employed the MC5600 roadside unit. This unit and other traffic survey technology are supplied by MetroCount Limited, a renowned traffic survey equipment manufacturer.

Using two pneumatic tubes, the MC5600 stores every axle and the length between axles. Subsequent to the survey, the user determines the method of classifying the recorded data. Each axle combination combined with the length between axles implies a certain vehicle type. All this information is stored in the RSU at the time of survey.

The interpretation of the stored data i.e. what type of vehicle is represented by a particular axle combination depends on the local vehicle classification system. For example, the UNRA classification system comprises of 10 vehicle types (Section 2.3). When the utility vehicle class is further subdivided into pure utility vehicles (4-wheel drives) and pickups, the vehicle classes are 11.

11.4.2 Calibration of the RSU

This process of calibrating the RSU to local traffic conditions was as follows.

The ATC equipment is supplied with windows-based “Traffic Executive” software. This software is used to read the recoded data. Within this software is eight (8) inbuilt vehicle classification systems that broadly represent various classification systems around the world. Some of these systems are newer versions of others. As such, four classifications were considered for calibration purposes. None of these systems would be expected to precisely match the UNRA system. This is because vehicle types and classes used in various countries differ.

The calibration approach sought to ascertain which classification system results in similar traffic volumes as those resulting from a controlled manual count. Once, a classification system is confirmed, the traffic volumes recorded by the RSU can then be disaggregated further into local classes in accordance with the vehicle type splits from the manual counts.

Control Manual Counts and Comparison With ATC Counts

The control manual counts were conducted for a period of two hours at three stations along the Northern Bypass; Kyebando – Kisaasi, Kisaasi – Naalya and Naalya – Bweyogerere. The period of the manual count was limited to 2 hours at each location on one day during the 7-day ATC at each of the three stations. All control MCC surveys were undertaken on 3rd February 2020.

It was intended to use one hour (the test hour) to compare with the ATC count (see Table 11-1). The period of the manual count was limited to ensure that it would be controlled as much as possible to avoid the common sources of error in MCC counts such as fatigue, loss of concentration, tiredness, etc.

Table 11-2 presents a comparison of the manual counts with the ATC counts from the selected classification systems. The classification system traffic volume that results in the smallest difference with the MCC count would be preferred. The vehicle classification systems selected from the RSU software are:

- AustRoads94 – this is an Australian vehicle classification scheme that was developed in 1994. It consisted of 12 vehicle classes. Details on this scheme are included in Appendix J.
- ARX - ARX is a modification of AustRoads94. It removes class 12, moves all other classes up by one, and inserts a motorcycle class as Class 1.
- Scheme F2 - Scheme F2 is USA classification system. It is an implementation of the Federal Highway Administration (FHWA)'s visual classification scheme as an axle-based classification scheme. Details are included in Appendix J.
- TNZ 1999 – TNZ 1999 is a scheme developed by Transit New Zealand. It has 14 classes. Details are included in Appendix J.

Table 11-2: Comparison of control MCC count with ATC count

COMPARISON OF CONTROL MCC COUNT WITH ATC COUNT								
Section of KNBP	**Test Survey Period**	**Test Hour**	**Test Hour MCC**	**Test Hour ATC count**				**Min Difference, %**
				Aust-Roads94	**ARX**	**Scheme F2**	**TNZ 1999**	
Kyebando - Kisaasi	16.45-18.45	17.00-18.00	2,017	2,056	2,056	1,983	2,062	1.7%
Kisaasi - Naalya	14.30-16.30	15.00-16.00	1,906	1,956	1,956	1,911	1,964	0.5%
Bweyogerere - Naalya	12.15-14.15	13.00-14.00	946	1,047	1,047	1,026	1,071	1.1%

From Table 11-2 above, the US Scheme F2 seems to match the control MCC better than the others. It consistently results in a closer traffic volume to MCC volume. As it was decided to extract the ATC data from the RSU using Scheme F2.

11.5 ATC Data Analysis Results

The analysis was undertaken on site by site basis. Figures 10-2 to 10-8 present the summary analysis of the ATC data. In the analysis, a plot of ADT for each of the week is plotted to show daily variation of the data.

In additional the traffic volume profile for the Kampala Northern Bypass highway is plotted. The section between Bwaise and Gayaza Road was not surveyed. For purposes of this analysis, the volume for Bwaise and Gayaza Road is assumed to be equal to the volume in the Hoima Road to Bwaise section.

Detailed data is included in Appendix J.

Figure 11-1: ADT variation for KNBP Mityana Rd to Sentema Rd

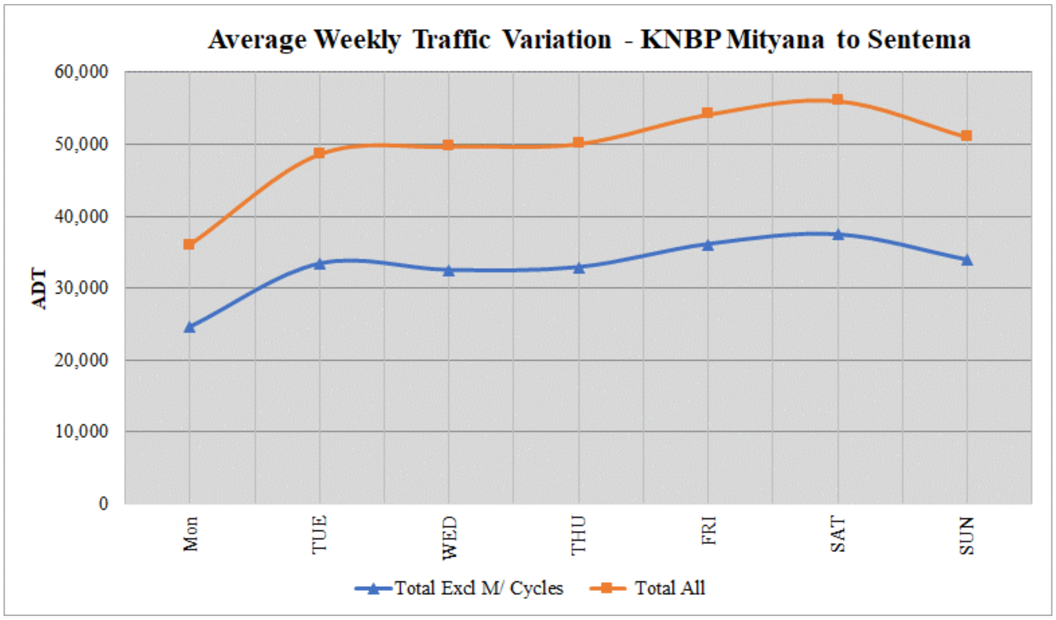


Figure 11-2: ADT variation for KNBP Sentema Rd to Hoima Rd

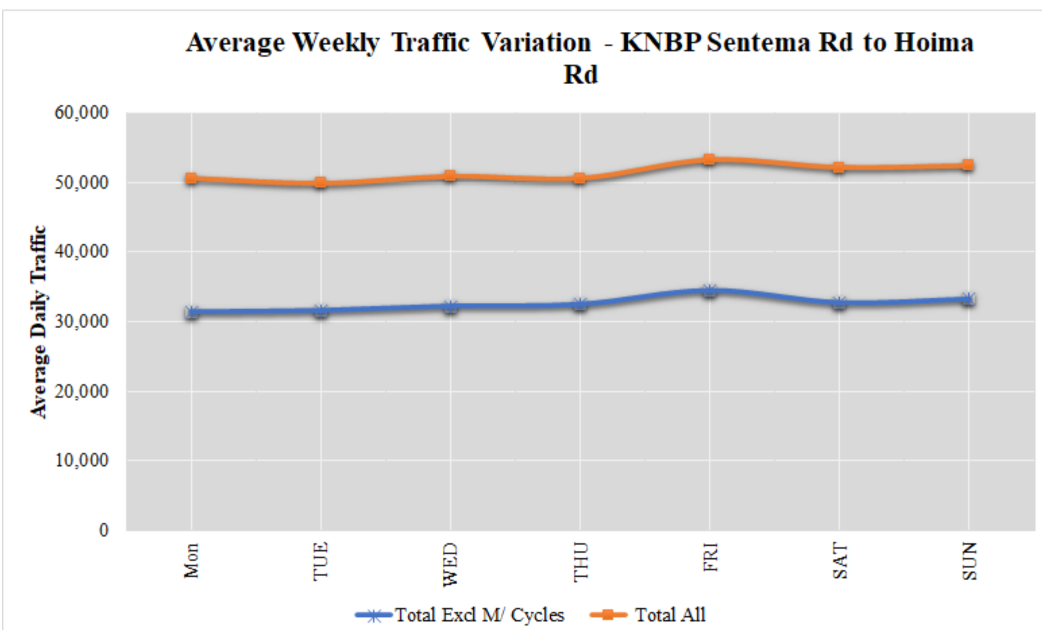


Figure 11-3: ADT variation for KNBP Hoima Rd to Bwaise Junction

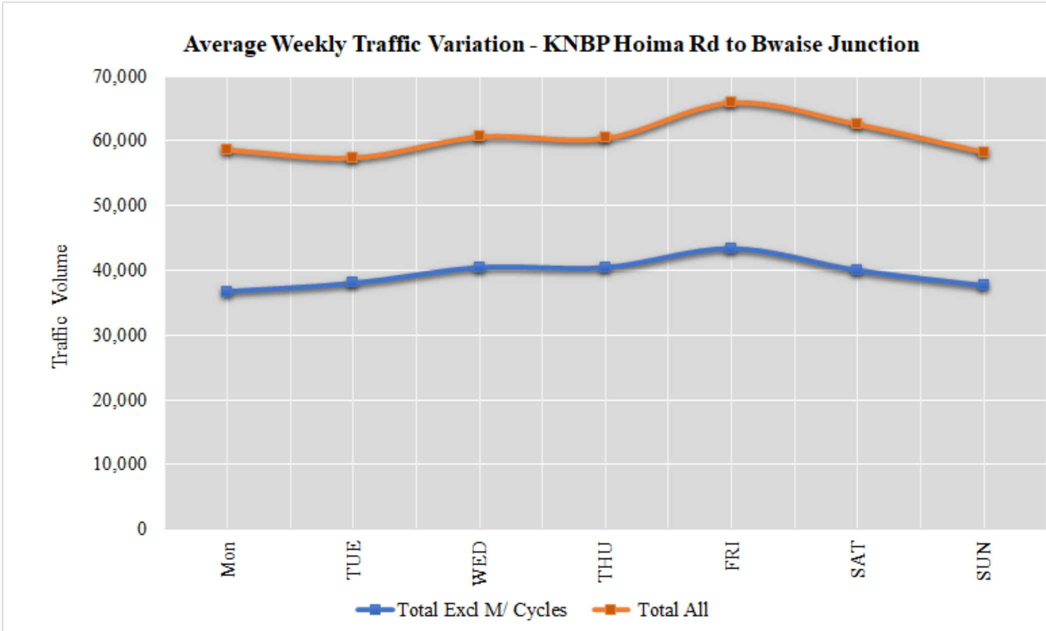


Figure 11-4: ADT variation for KNBP Gayaza Rd to Kyebando Rd

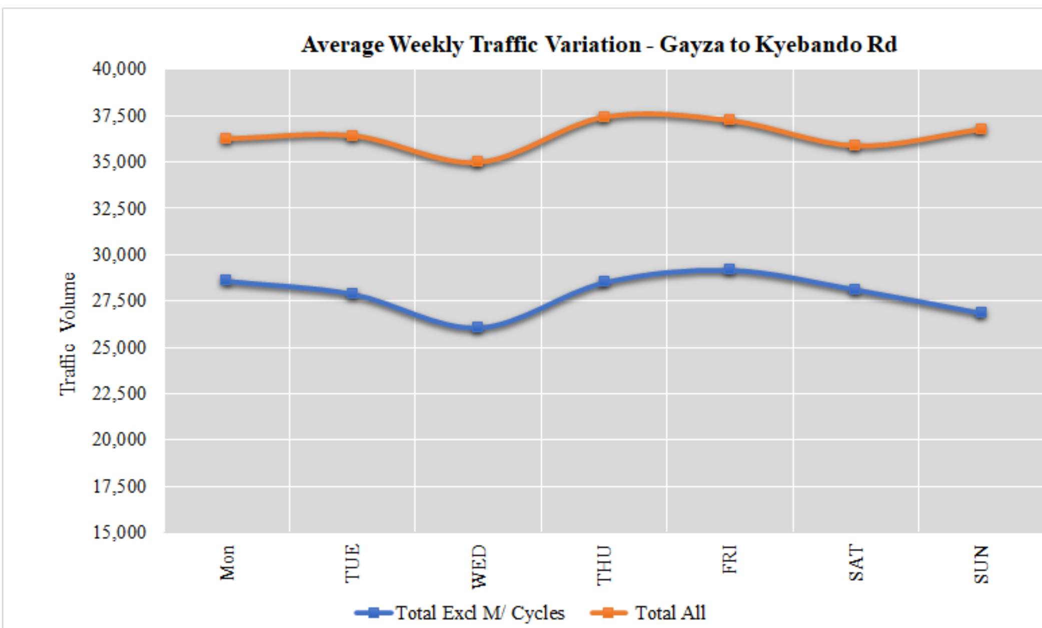


Figure 11-5: ADT variation for KNBP Kyebando Rd to Kisaasi Rd

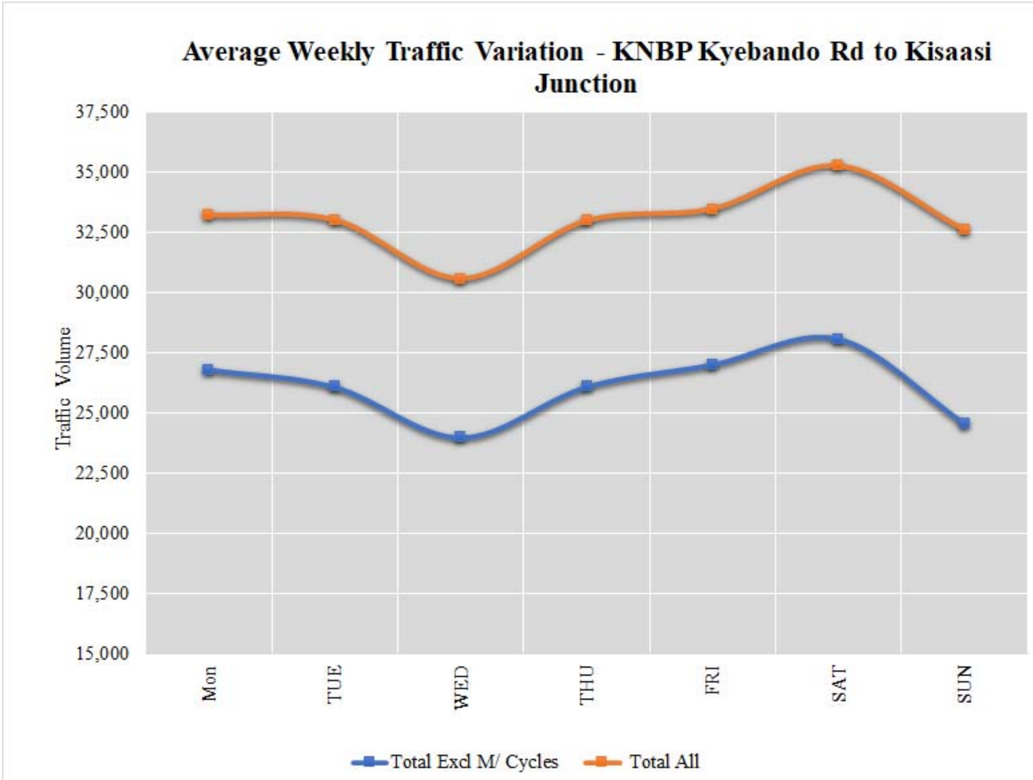


Figure 11-6: ADT variation for KNBP Kisaasi Rd to Nalya Junction

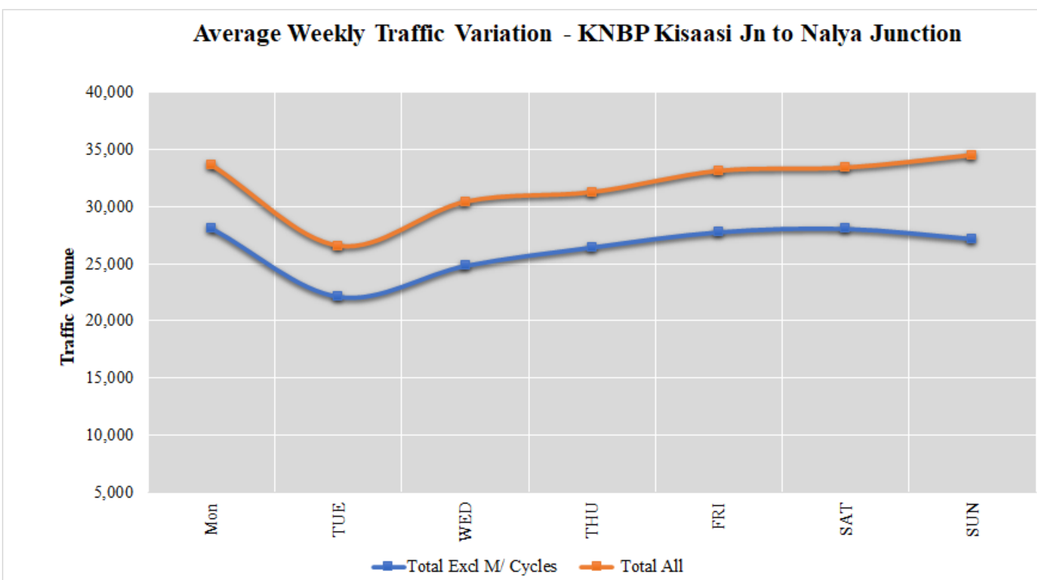


Figure 11-7: ADT variation for KNBP Nalya Junction to Bweyogerere

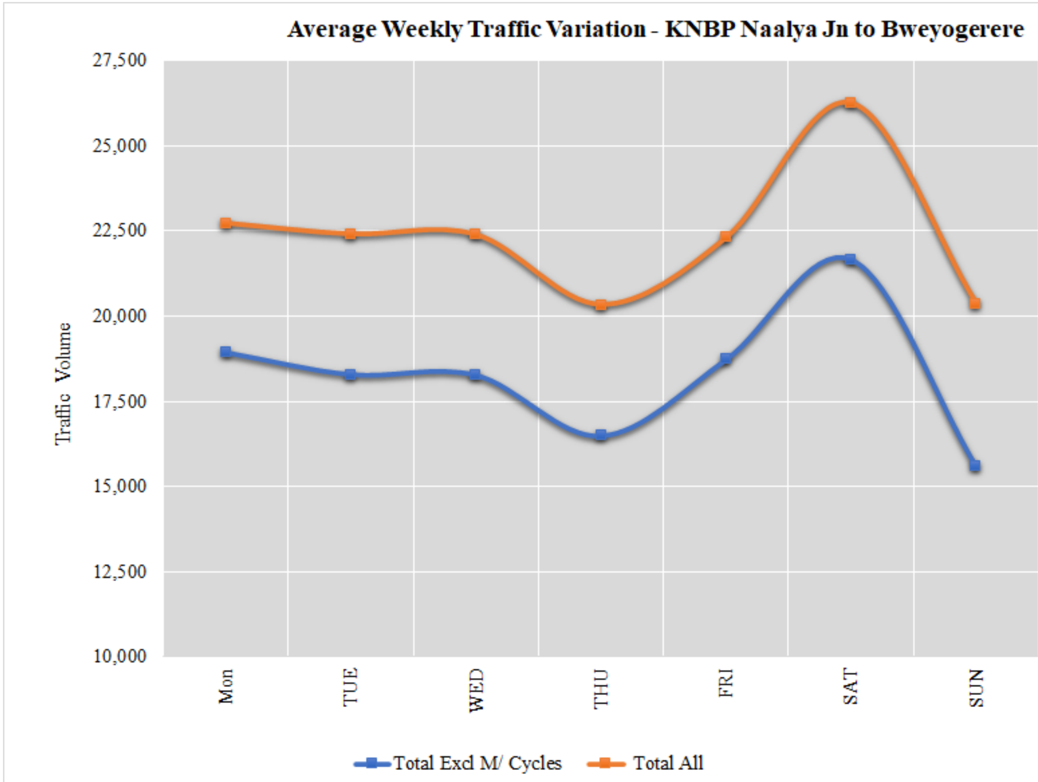
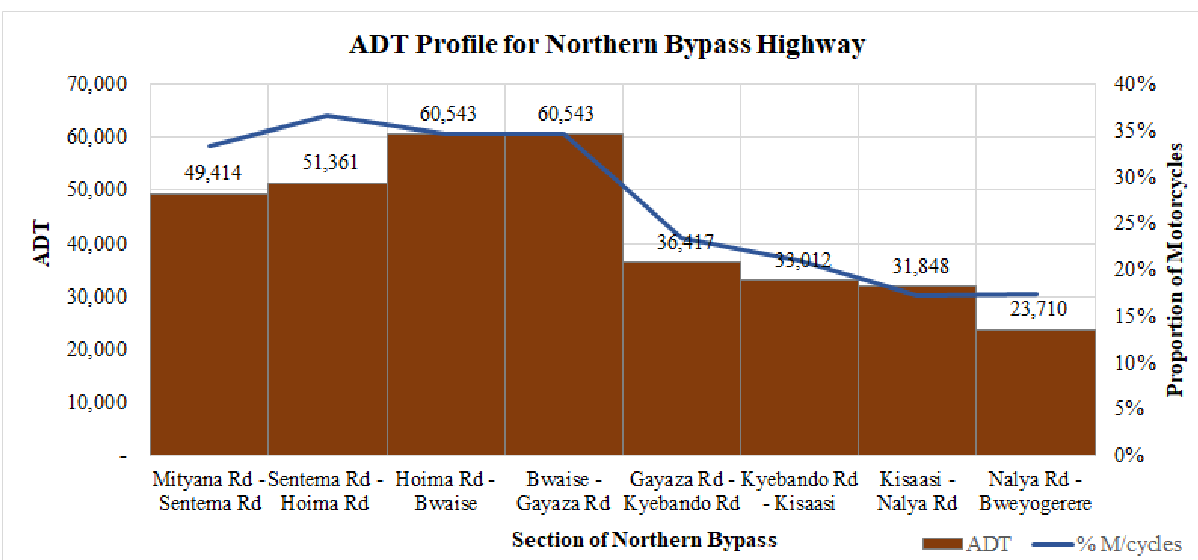


Figure 11-8: ADT profile for Northern Bypass highway



12 AXLE LOAD SURVEYS

12.1 Introduction

The main objectives of this survey were;

- to determine the level of loading of trucks on key highways; and
- to identify the type of loads carried on the selected highways

12.1.1 Location of Survey Sites

Axle load surveys were undertaken at 10 locations as described in Table 12-1 below.

Table 12-1: Description and location of axle load surveys

DESCRIPTION AND LOCATION OF ATC SURVEY SITES				
Site Ref.	**Highway / Link Name**	**Survey Type**	**Description of Site**	**Site Coordinates**
AL1	Kampala – Entebbe Highway	Axle	At the dual section of Kampala-Entebbe Road	0.061179, 32.472757
AL2	Kampala – Masaka Highway	Axle	Buwama	0.064800, 32.109264
AL3	Kampala – Mityana Highway	Axle	Bujuuko	0.333394, 32.370712
AL4	Kampala – Hoima Highway	Axle	Kakiri	0.419596, 32.402159
AL5	Kampala – Bombo Highway	Axle	Matugga	0.472375, 32.518501
AL6	Ishaka – Kasese Highway	Axle	Rubirizi town	-0.268438, 30.106536
AL7	Kampala – Jinja Highway	Axle	Nakibizzi former Weighbridge location	0.422428, 33.143343
AL8	Mbale – Tororo Highway	Axle	Busoba town	0.969742, 34.170843
AL9	Karuma – Arua Highway	Axle	Purongo Town	2.540505, 31.828369
AL10	Gulu – Nimule Highway	Axle	Abelokodi village	3.324474, 32.104841

12.2 Methodology

12.2.1 Survey Equipment

The surveys were undertaken using one Haenni Model 103 mobile weighbridge. The Haenni weighbridges are light weight, easy to use, accurate and generally have a number of features that have made them widely used mobile weighbridge.

The equipment was under valid calibration. It is a requirement that such equipment is calibrated annually by the Uganda National Bureau of Standards (UNBS) and must possess a valid calibration certificate. A copy of the calibration certificate for the equipment, valid at the time of survey was furnished to UNRA prior to commencement of surveys and is also included in Appendix K.

12.2.2 Survey Team

The survey team generally consisted of three members comprising of one supervisor and two scale operators. The scale operators also acted as the enumerators, recording scale readings, load types and other data

In addition, each site also consisted of two traffic police personnel to direct truck traffic to pull over and to ensure that traffic flowed safely around the survey site. The axle load measurements were undertaken on a lay-by adjacent to the main carriageway and therefore traffic generally flowed around the survey site adequately.

12.2.3 Survey Period

Axle Load Surveys were undertaken for three consecutive days at each station. The survey period was 12-hours (7am – 7pm) on each day.

The trucks were weighed one axle at a time and weights recorded on a pre-designed survey form attached in Appendix A. Each day's completed form was verified in the field and then dispatched to the office for data entry, logical checks and general data cleaning.

12.3 Vehicle Classifications and Axle Load Configurations

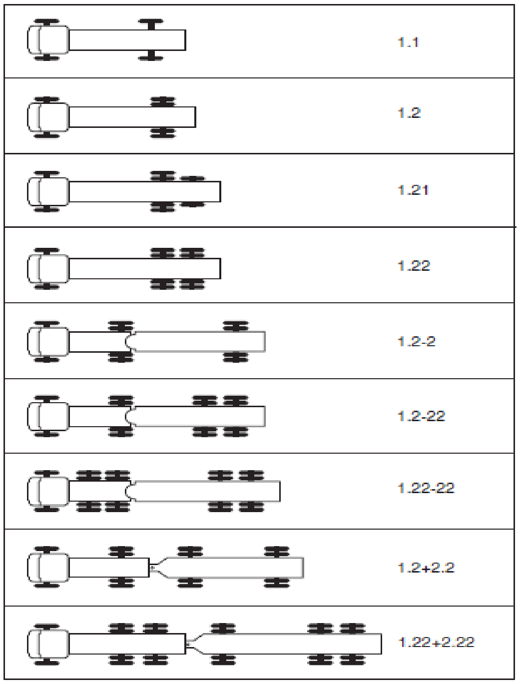
Axle load surveys were conducted on truck vehicles only. The truck vehicle types according to the vehicle classification stated in Section 2.3 are four:

- Vehicle type 7 - Light truck (2 axle, gross weight of not more than 7 tonnes);
- Vehicle type 8 - Medium truck (2 axles, gross vehicle weight of between 7 and 15 tonnes);
- Vehicle type 9 - Heavy truck (3 or 4 axles, gross vehicle weight of up to 22 tonnes);
- Vehicle type 10 - Articulated truck (semi-trailer) (5 or more axles, gross vehicle weight of 22 - 56 tonnes);

Large buses were not considered for survey. Although load-carrying, from experience in load data analyses, buses do not particularly carry very bulky goods that are of significant influence on pavement deterioration. Also, the selected key routes had significantly high truck volumes. It is the truck loadings that are considered to be the main source of loading for the expressway network.

In order to record axle data accurately and to ensure that it is meaningful for further use, it is important to record the axle load configurations. Axle load configuration is the sequencing of tyre arrangements of any truck from front to back. Weights are taken for each configuration. The following guidance was used.

Figure 12-1: Truck configuration guide



- Light and medium trucks generally have two axles i.e. 1:1 and 1:2. As such, for axle load record purposes, these two were merged as classified as one truck type "Light/Medium Trucks".
- Heavy trucks typically comprise of three or four axles. However, the most common heavy trucks such as the Fuso 10-tyre truck comprises of three axles. Because semi-trailer and truck trailer trucks often comprise of four axles and above, for simplicity of analysis, heavy trucks were taken to comprise of only three axles i.e. 1:21, 1:2-2 and 1:22
- Semi-trailers and truck trailers were considered to comprise of four axles and above.

12.4 Axle Weight Limits

To enable a robust analysis of the axle load data, it is important to appreciate the weight limits imposed by UNRA on trucks carrying goods. Any truck is considered overloaded if the weight on any one axle exceeds the load limit for that axle by more than 5%. Figure 12-2 below sets out the axle weight limits for the various truck types and axles obtained from UNRA[9].

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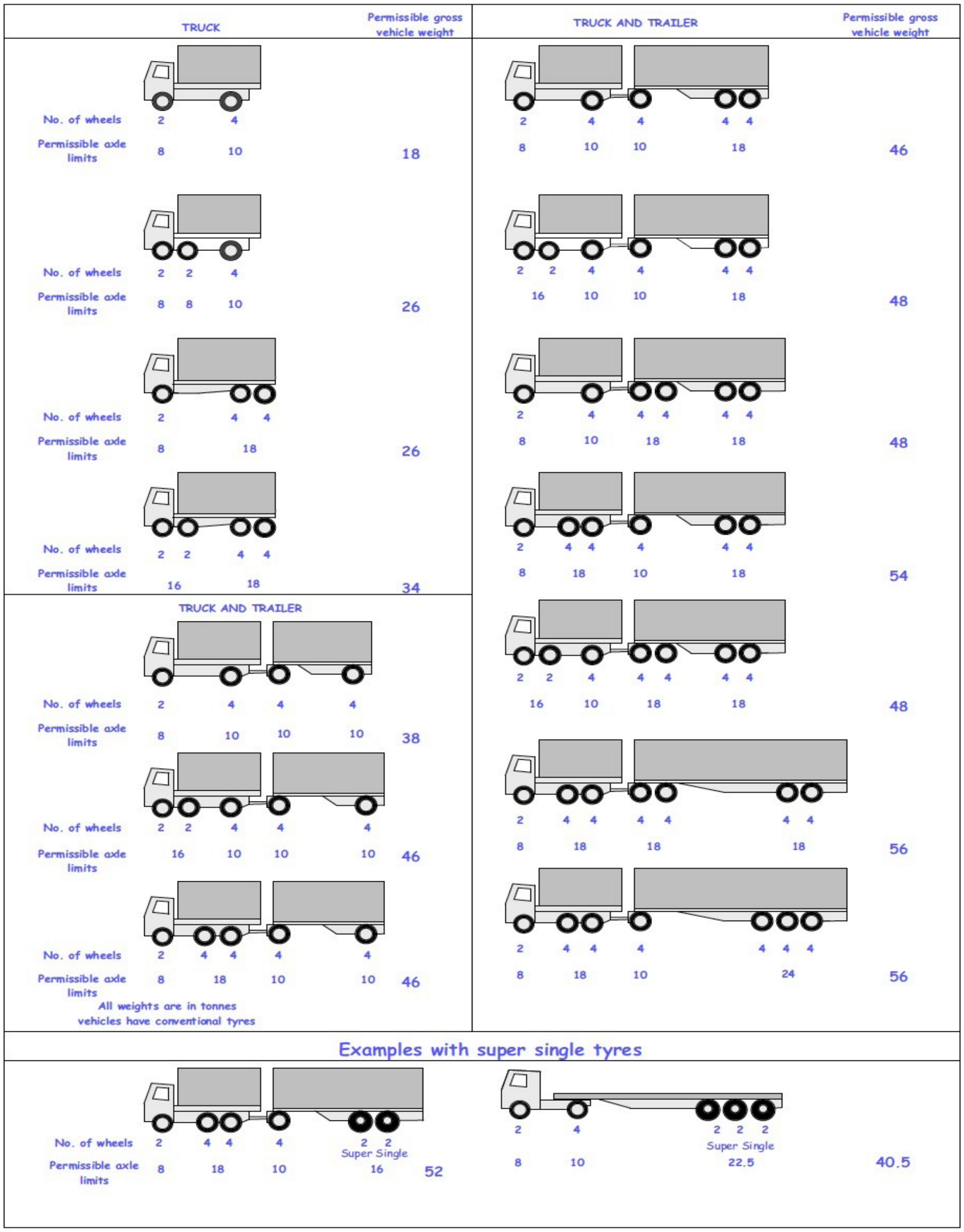
[9] UNRA (Vehicle Dimensions and Load Control) Regulations, 2017

Figure 12-2: Axle weight limit guide



12.5 Axle Load Survey Samples

Similar to OD surveys, the trucks stopped for axle load measurements are considered a sample because all trucks cannot be measured. It is important that an adequate sample be obtained. The objective of the survey was to collect a 10% sample overall. From experience, this is most possible for low trafficked roads but a challenge for roads with high volumes.

Table 12-2 below presents the sample rates for each survey station. The survey on the Kampala – Masaka highway realized a smaller sample due to inclement weather. Nonetheless, the volume of trucks surveyed (147) was deemed large enough to produce a reliable analysis.

Overall, an average axle load sample of 19.8% was obtained.

Table 12-2: Description and location of axle load surveys

DESCRIPTION AND LOCATION OF ATC SURVEY SITES					
Site Ref.	**Highway / Link Name**	**Direction**	**Average 12-hr Sample**	**12-hr Truck Traffic**	**Axle Load Sample, %**
AL1	Kampala – Entebbe Highway	Both	290	900	32.2%
AL2	Kampala – Masaka Highway	Both	147	1,969	7.4%
AL3	Kampala – Mityana Highway	Both	135	931	14.5%
AL4	Kampala – Hoima Highway	Both	197	1,164	16.9%
AL5	Kampala – Bombo Highway	Both	608	2,246	27.1%
AL6	Ishaka – Kasese Highway	Both	100	426	23.5%
AL7	Kampala – Jinja Highway	Both	498	1,966	25.3%
AL8	Mbale – Tororo Highway	Both	280	2,343	12.0%
AL9	Karuma – Arua Highway	Both	83	683	12.2%
AL10	Gulu – Nimule Highway	Both	38	140	27.1%
				Overall sample	**19.8%**

12.6 Axle Load Analysis for Selected Routes

Axle load data analysis was undertaken for all 10 routes. The results of this analysis is included in Appendix K. However, in this section, the results of four key routes is highlighted. The selected routes are: Kampala – Jinja, Kampala – Masaka, Kampala – Hoima and Kampala – Gulu highways.

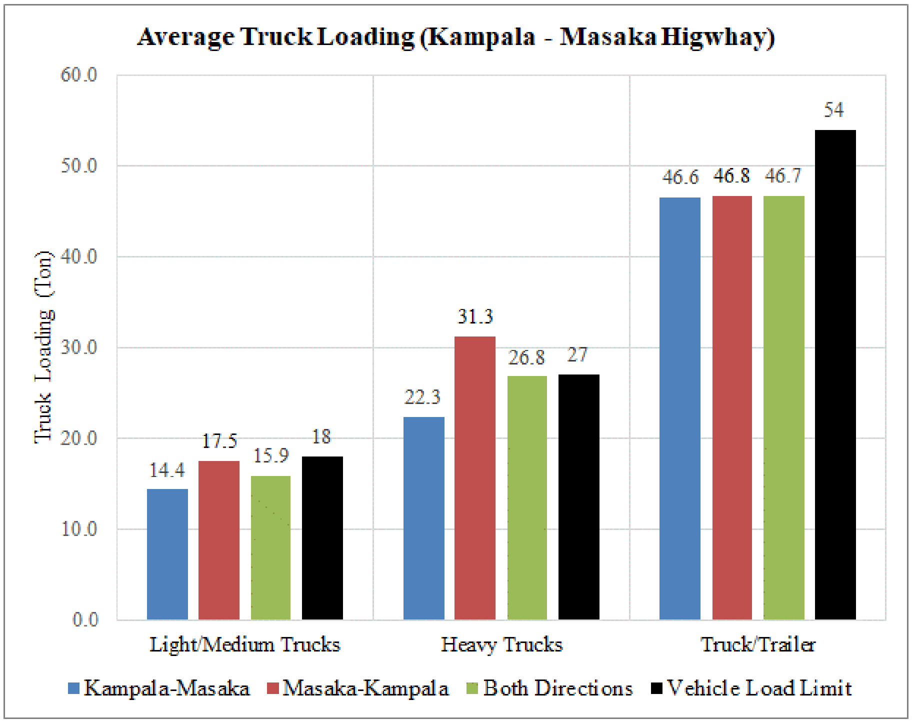
In general, the axle load data was analysed for the following:

- Average total loading per truck type – for this key aspect, the average total loading per truck type was analysed by direction. The vehicle load limit was also plotted to provide context.
- Proportion of overloaded trucks. Any truck is considered overloaded if the weight on any of its axles exceeds the specified load limit by more than 5%. The truck loads were therefore **analysed by axle** to determine the level of loading for each truck, and to determine whether it was overloaded or not.
- Percentage by which trucks are overloaded – from the analysis of loading per axle above, the proportion by trucks are overloaded is provided. This provide a picture of the extent of overloading.

- Description of loads – this analysis determines the most common load type on each of the highway by direction.

12.6.1 Kampala – Masaka Highway

Figure 12-3: Average total truck loading (Kampala-Masaka Highway)



- The following observations can be made with respect average total truck loading (Figure 12-3). Analysis shows that total loading is broadly within the allowable total truck load limits. The exception is with heavy trucks in the Masaka – Kampala direction.
- Figure 12-4 however presents a different picture when the truck loading is analysed by axle. The analysis shows that trucks are highly overloaded. The Masaka – Kampala has the highest overloading with almost all trucks showing overloading.
- In the Kampala – Masaka direction, there is less overloading. Articulated trucks and truck trailers show the highest overloading in this direction.

Figure 12-4: Average proportion of overloaded trucks (Kampala-Masaka Highway)

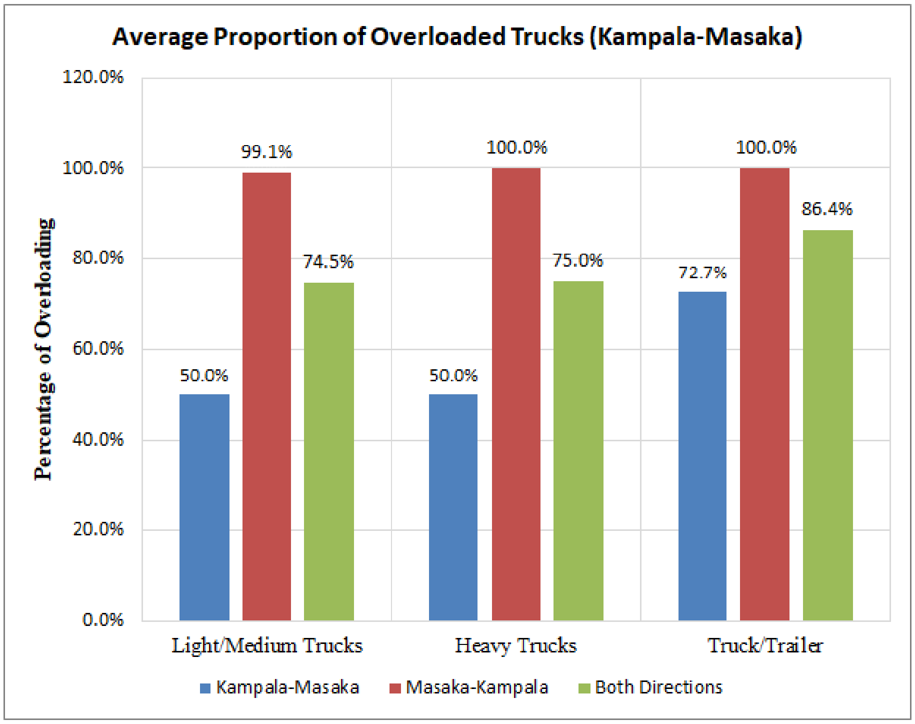
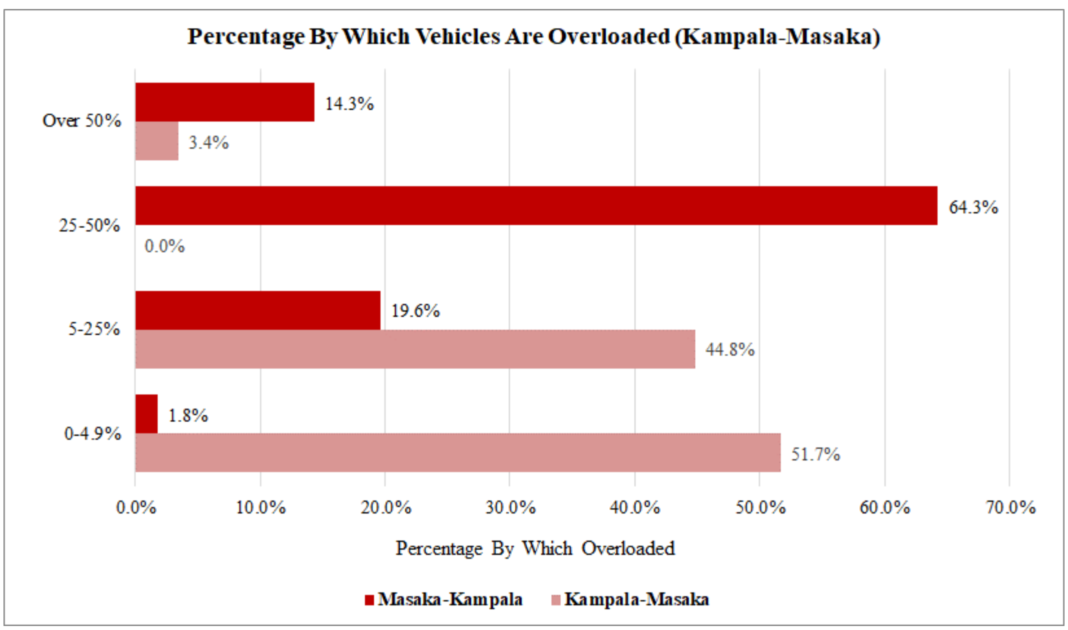


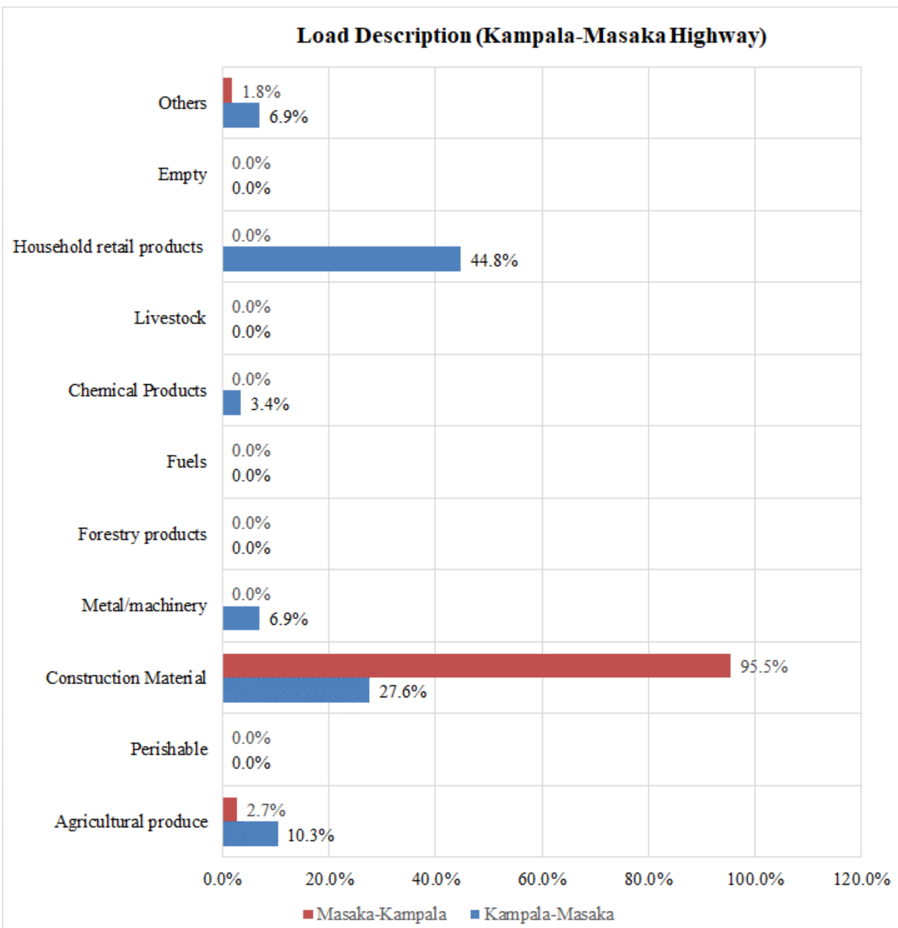
Figure 12-5: Percentage by which trucks are overloaded (Kampala-Masaka Highway)



- Figure 12-5 above shows that 51.7% of all trucks in the Kampala – Masaka direction are within the allowable axle loads. The remainder of the trucks have axle loads between 5-25% above the loading limits (44.8% of all trucks) and a small amount (3.4% of trucks) have higher loads.
- For the Masaka – Kampala direction, only 1.8% have allowable axle loads.

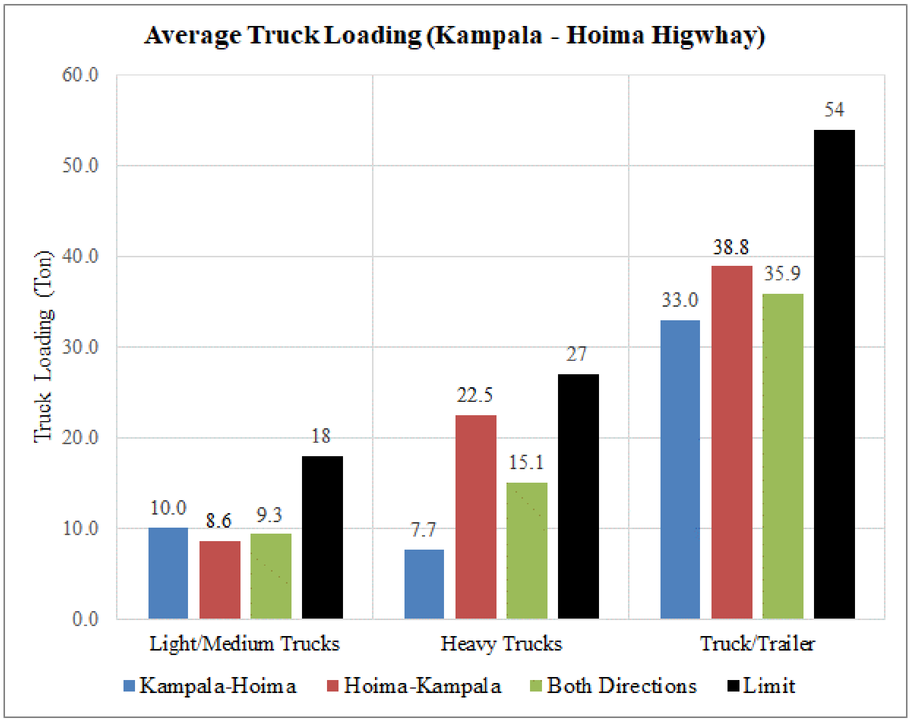
- Most trucks exhibited loading of 25% - 50% above the allowable load limits. 14.3% of all trucks have loads that are 50% higher than the load limits.
- Figure 12-6 shows that in the Masaka – Kampala direction, the most predominant load type is construction material. This was observed to be most sand from the Lwera sand mining area. It is envisaged that the high overloading is due to the absence of a UNRA axle load weighing station between Lwera and Kampala (the destination of these products). There is a designated mobile weighing area at Buwama Town, however, based on the observed results, it may seem that this is not effective.
- The comparison between vehicle loading and axle loading to determine whether a vehicle is overloaded or not shows significantly different results. It may be a policy issue. There is a need to clarify on the basis to determine vehicle overloading.

Figure 12-6: Load description (Kampala-Masaka Highway)



12.6.2 Kampala – Hoima Highway

Figure 12-7: Average total truck loading (Kampala-Hoima Highway)



- Figure 12-7 above shows that with regard to total truck loading, the loading on the highway is below the allowable limit.
- However, with respect to axle loading, some overloading is observed (Figure 12-8). Truck trailers exhibited the highest overloading with an average of over 60% trucks overloaded in the two directions combined.
- However, a further examination of the axle loads (Figure 12-9) shows that most trucks (an average of 77% for both directions) are within the allowable load limits of 0 to 5%.
- Figure 12-10 shows that the predominant load type is currently agricultural produce.

Figure 12-8: Average proportion of overloaded trucks (Kampala-Hoima Highway)

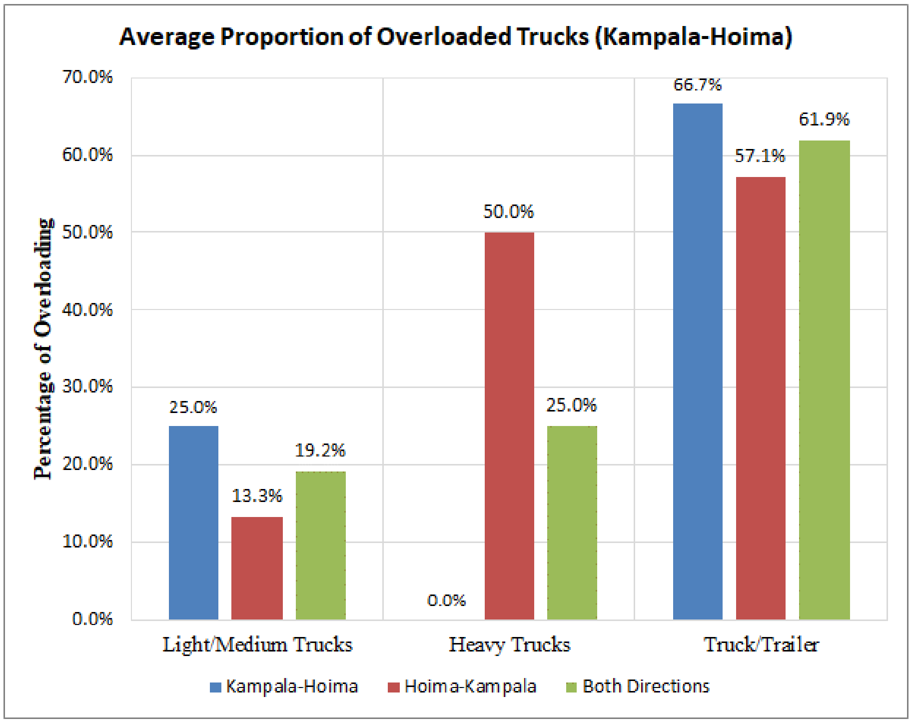


Figure 12-9: Average proportion of overloaded trucks (Kampala-Hoima Highway)

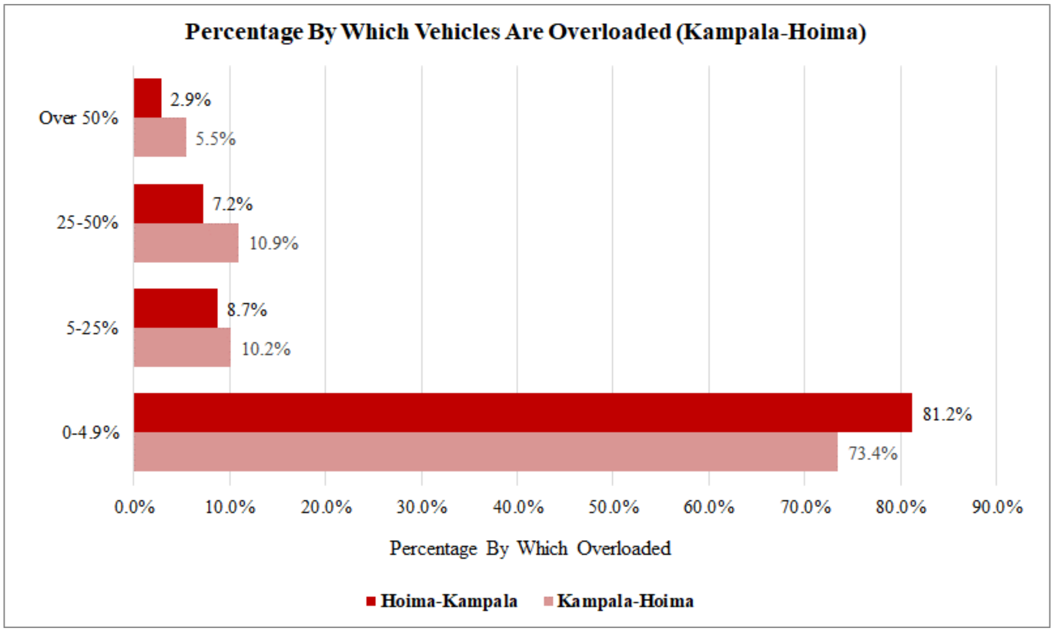
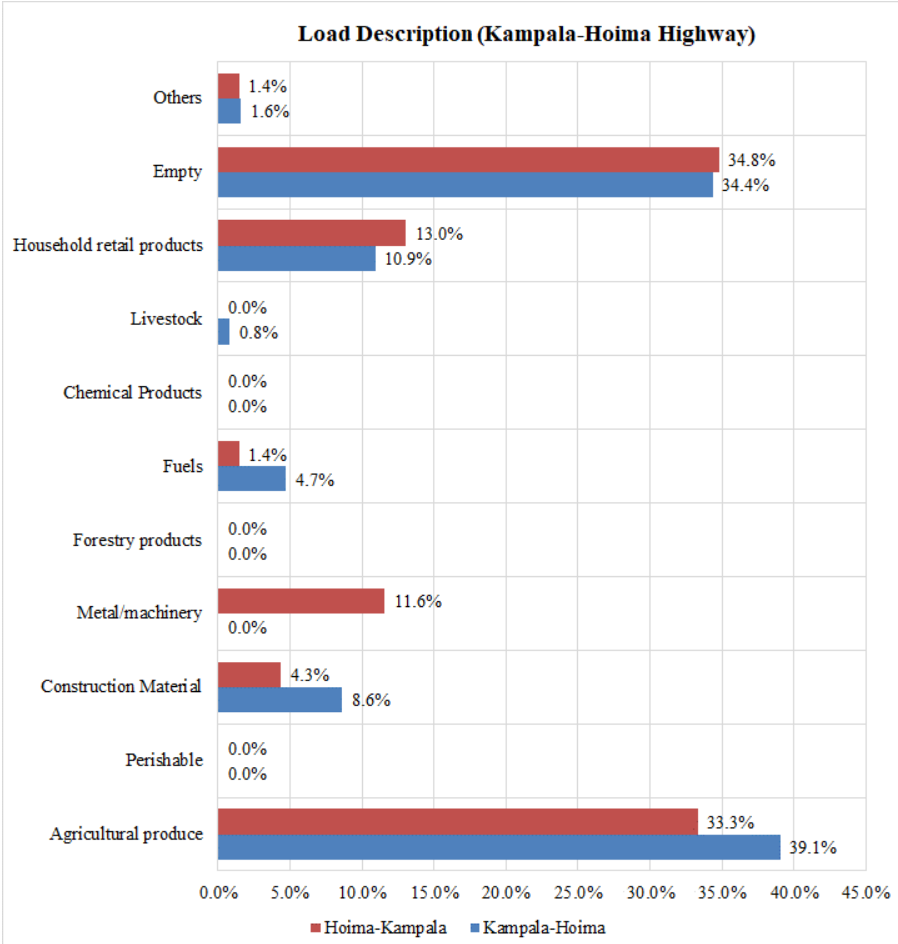


Figure 12-10: Load description (Kampala-Hoima Highway)



12.6.3 Kampala – Gulu Highway

- Figure 12-11 shows that the average total truck loads are below the allowable truck load limits except for truck trailers in the Kampala – Gulu direction.
- When the data is analysed by axle (Figure 12-12), some level of overloading is observed. The figure shows that overloading is highest for truck trailers, with an average of 73.1% overloaded for both directions.
- Figure 12-13 however shows that most trucks are within the allowable axle load limits (60.2% in the Kampala-Gulu and 51.4% in the Gulu-Kampala directions). Nonetheless, a significant amount of trucks are overloaded with 24% on average carrying loads which are 50% heavier than the axle load limits.

Figure 12-11: Average total truck loading (Kampala-Gulu Highway)

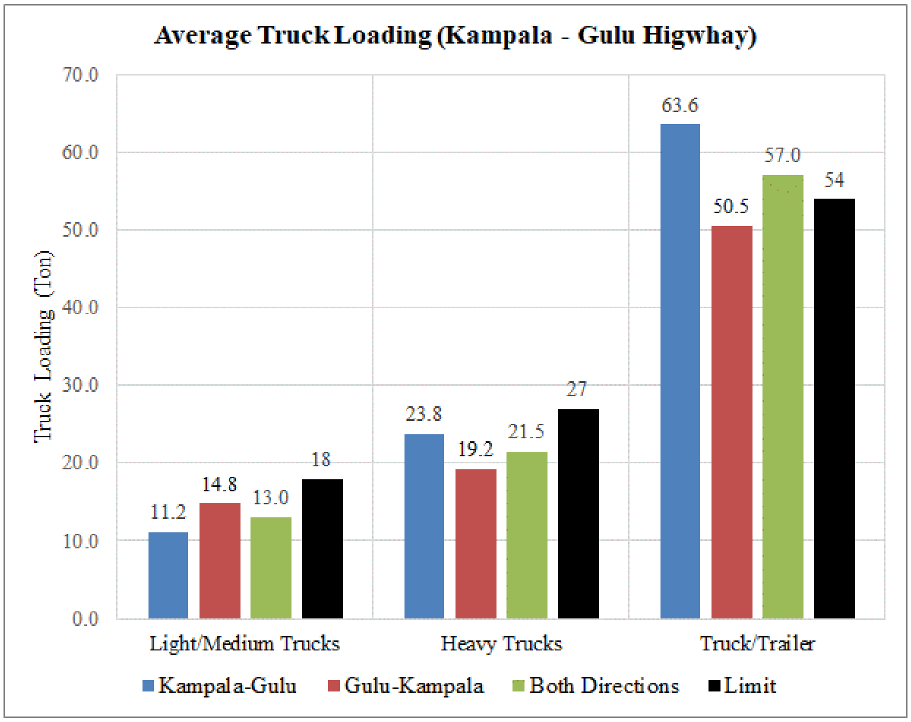


Figure 12-12: Average proportion of overloaded trucks (Kampala-Gulu Highway)

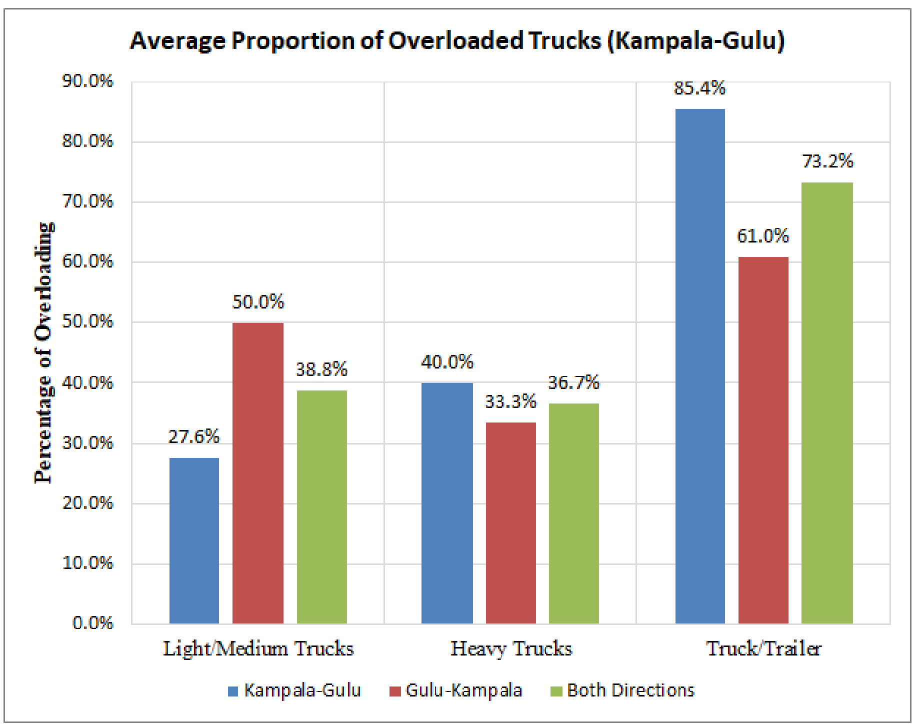


Figure 12-13: Average proportion of overloaded trucks (Kampala-Gulu Highway)

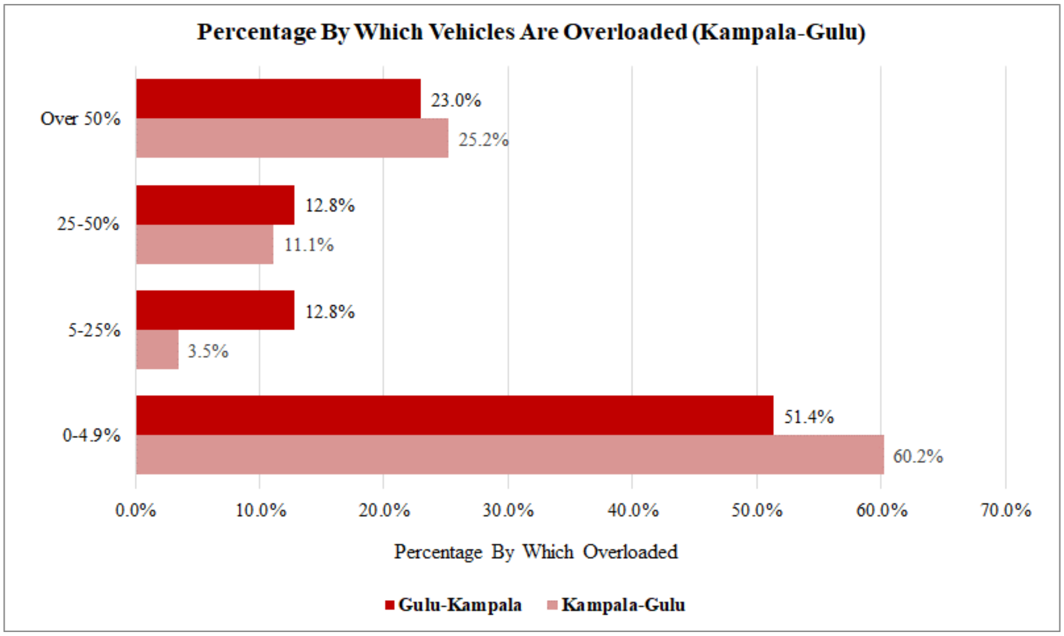
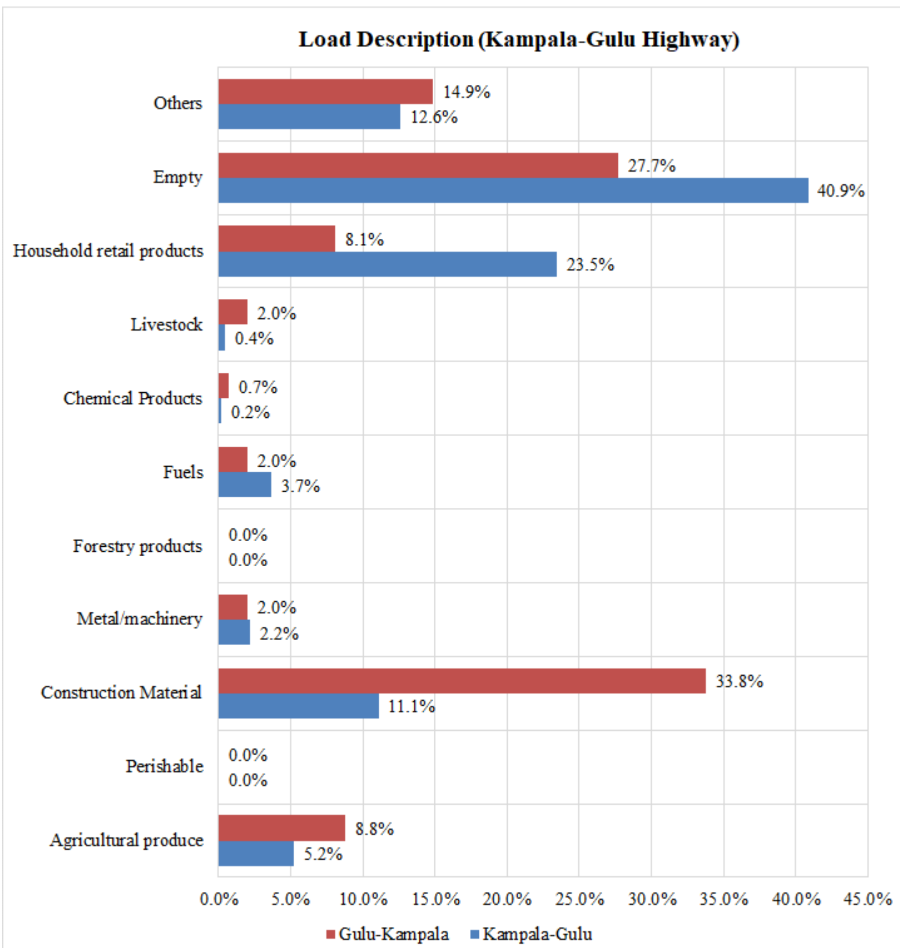


Figure 12-14: Load description (Kampala-Gulu Highway)



12.6.4 Kampala – Jinja Highway

- Figure 12-15 shows that average total truck loading is well below the allowable load limits for all truck types.
- Analysed for axle loading (Figure 12-16), the data shows that there is some overloading especially for heavy trucks in the Jinja-Kampala direction. However, Figure 12-17 shows that most of this observed overloading is within the allowable limits. The most affected direction is Jinja-Kampala, which has over 30% of trucks carrying loads that are above the allowable limits.

Figure 12-15: Average total truck loading (Kampala-Jinja Highway)

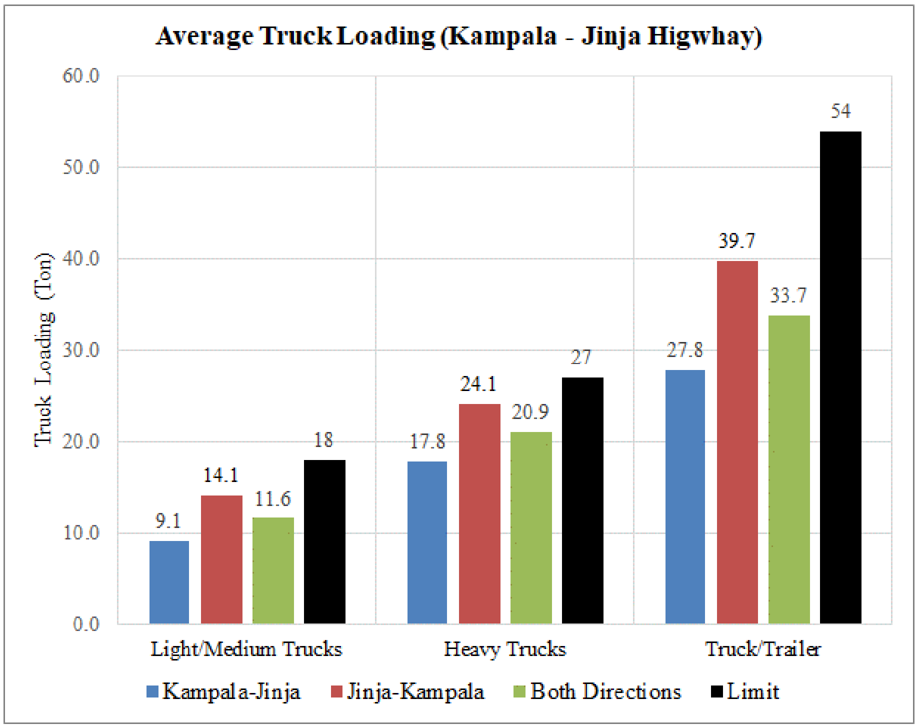


Figure 12-16: Average proportion of overloaded trucks (Kampala-Jinja Highway)

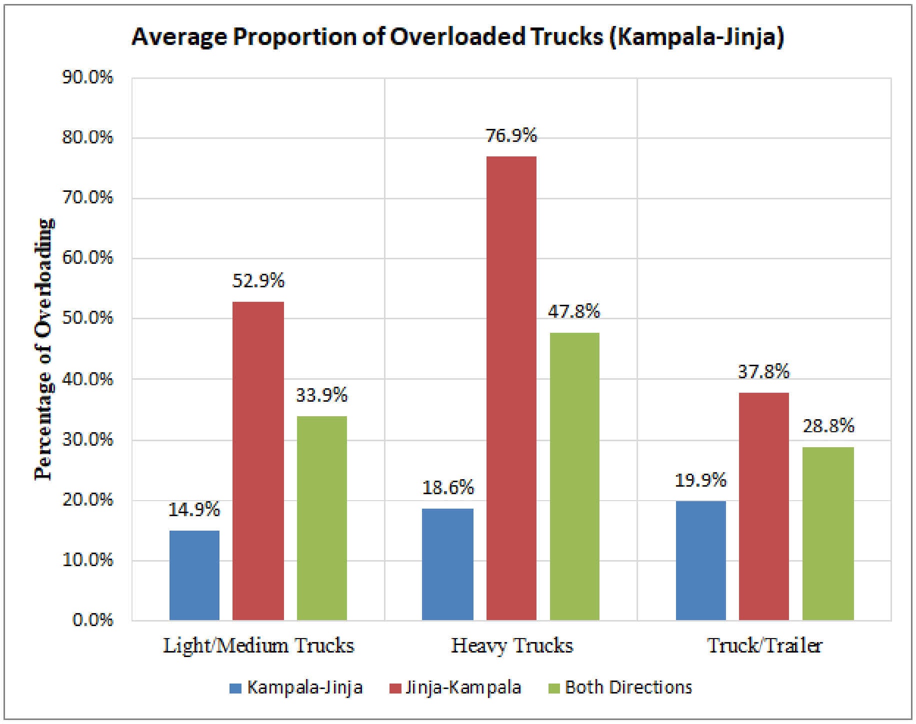


Figure 12-17: Average proportion of overloaded trucks (Kampala-Jinja Highway)

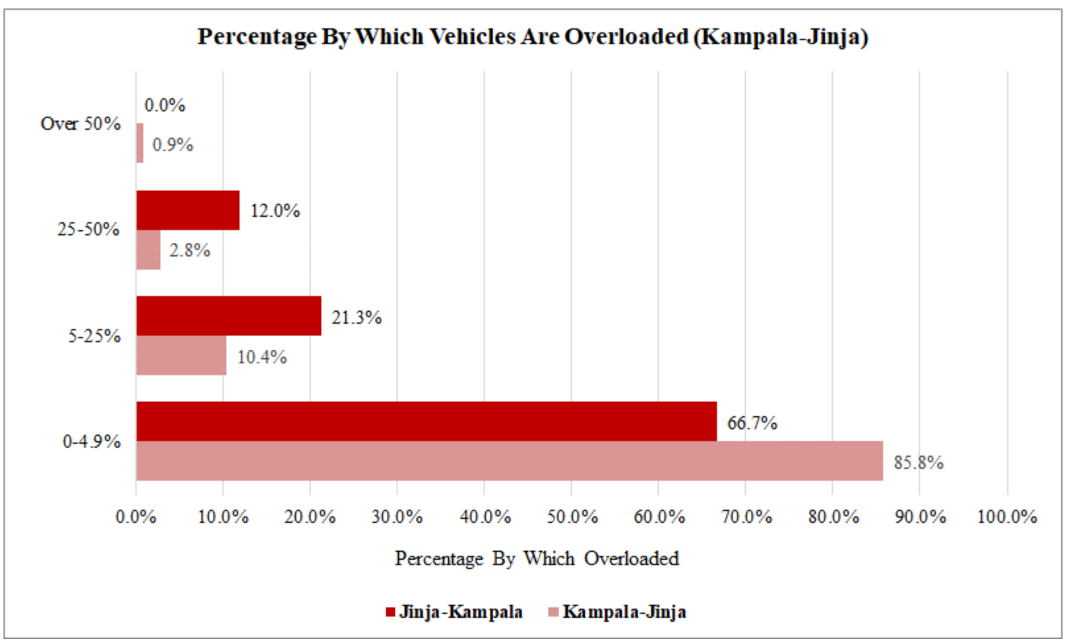
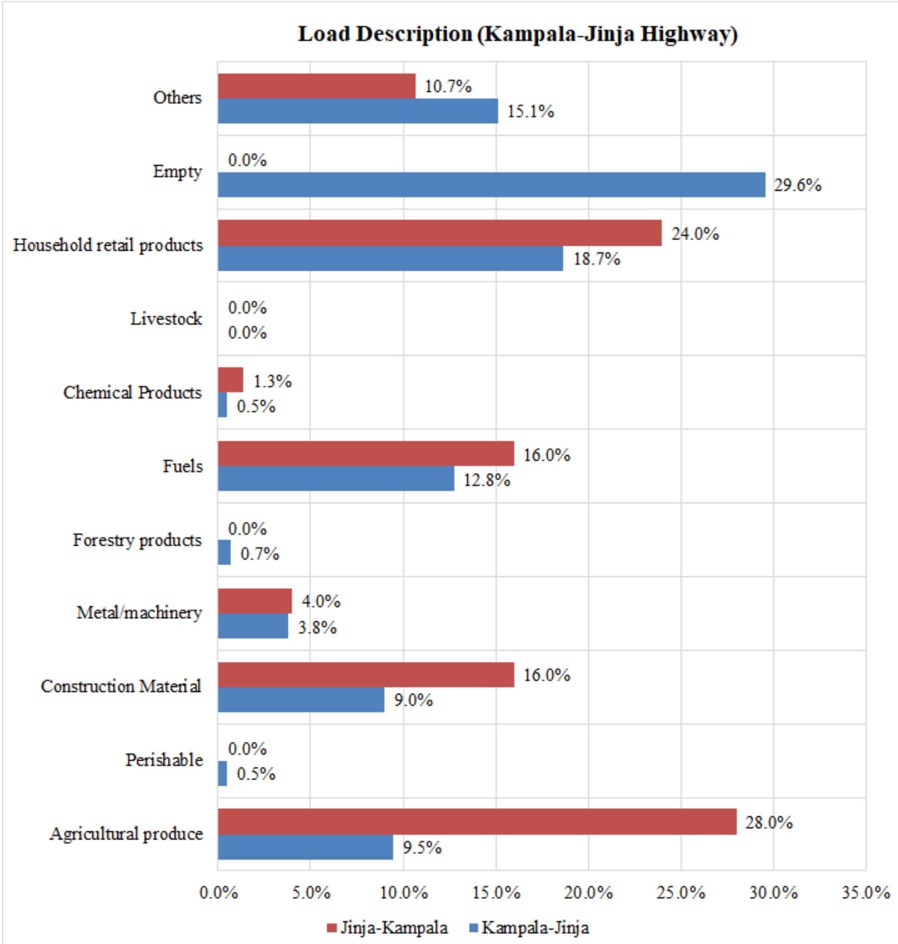


Figure 12-18: Load description (Kampala-Jinja Highway)



13 APPENDICES

13.1 Appendix A – Survey Forms

13.2 Appendix B – MCC Data Analysis Results

MCC Data – Cordon 1

MCC Data – Cordon 2

MCC Data – Cordon 3

MCC Data – Cordon 4

MCC Data – Cordon 5

Vehicle Registration Data (on CD)

13.3 Appendix C – OD Data Analysis Results

OD Data – Cordon 1

OD Data – Cordon 2

OD Data – Cordon 3

OD Data – Cordon 4

OD Data – Cordon 5

13.4 Appendix D – JTC Data

13.5 Appendix E – Journey Time and GIS Data

Journey Time Data

GIS Data (on CD)

13.6 Appendix F – Stated Preference Data

13.7 Appendix G – Public Transport Data (on CD)

13.8 Appendix H – Accident Data (on CD)

13.9 Appendix I – Ferry Data Analysis

13.10 Appendix J –ATC Data

13.11 Appendix K – Axle Load Data Analysis